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TERM ONE
THEME: HUMAN BODY
TOPIC : MUSCULAR-SKELETAL SYSTEM

Topical outline:

- The skeleton.
- Bones in the human body.
- Joints in the human body.
- Muscles in the human body.
- Diseases and disorders of the human skeleton.
- Prevention of the diseases and disorders of the human skeleton.
- Body posture.
- Health habits that help to keep the muscular-skeletal system in proper working condition.

Ref.P.7 standard curriculum page 64.

Advice to teachers.

- *Teach basing on the theme of the topic i.e. Human Body.*
- *Teach according to what is in the curriculum.*
- *UNEB does not set questions outside the curriculum.*
- *Use practical approaches during teaching i.e. use a wooden model to explain the action of joints and antagonistic muscles.*

Lesson vocabulary

skeleton hydrostatic ecdysis cuticle structure organism

Description of the system.

The muscular-skeletal system is the body system that gives an animal support, shape and the ability to move.

Skeleton

A skeleton is the structure that supports the body of an animal.

Types of skeleton

- exoskeleton
- hydrostatic skeleton
- endoskeleton

Exoskeleton

An exoskeleton is the type of skeleton found outside the body of an animal. It is mainly found in **arthropods** such as insects, crustaceans, myriapods and arachnids.

An exoskeleton is made up of a hard structure called **cuticle.**

↘ *Animals with exoskeleton grow bigger in size by moulting.*

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Moulting

- Moulting is the shedding of cuticle by an arthropod.
- It is also called **ecdysis** and is caused by **ecdysone** hormone.
- Moulting promotes body growth.
- It is a biological change because it takes place in the life of organisms.

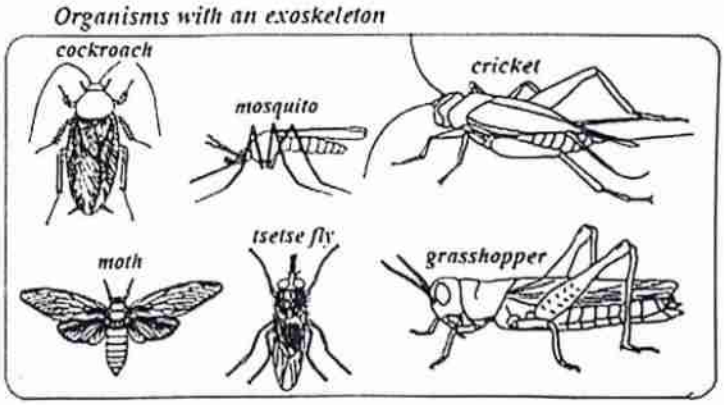
Examples of animals that moult.

houseflies crabs spiders millipedes bees shrim

Besides arthropods, birds and reptiles also moult.
Birds moult by shedding their feathers to enable new feathers grow.
Reptiles moult by shedding their old skin to increase in size..

Examples of animals which have exoskeleton.

- houseflies
- spiders
- crabs
- millipedes
- bees
- locusts
- centipede
- sand hoppers
- scorpion
- ticks



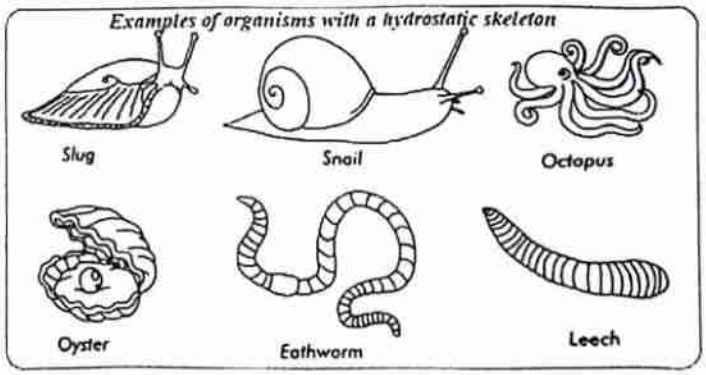
Hydrostatic skeleton

Hydrostatic skeleton is the type of skeleton possessed by an animal whose body i filled with fluid under pressure.

Hydro-static skeleton mainly supports movement and gives the body of an animal shape.

Examples of animals with hydrostatic skeleton.

- snails
- slugs
- caterpillars
- maggots
- worms
- octopuses
- Liver flukes



Endoskeleton

Endoskeleton is the type of skeleton found inside the body of an animal.

It is made up of strong hard tissues called bones.

Examples of animals with endoskeleton.

human-beings lizards snakes tortoises ostriches frogs fis

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Endoskeleton is found in all vertebrates such as fish, amphibians, reptiles, birds and mammals.

Evaluation activity:

1. Name the structure that supports the body of a person.
2. Give the meaning of the term skeleton.
3. Mention any **two** types of skeleton.
1. The list below shows animals having different types of skeleton. Use it to answer questions **5** and **6.**

   tick, hookworm, locust, caterpillar

4. Name any **two** animals from the list above that have hydrostatic skeleton.
5. What body structure make up the skeleton of the locust from the list above?
6. What is meant by the term moulting?
7. In which way is moulting important to a shrimp?
8. What type of change is moulting?
9. How are animals with exoskeleton able to increase in size?

Lesson vocabulary

cartilage ligament fluid tendon calcium vertebral

THE HUMAN SKELETON

The human skeleton is the framework of bones in the human body.
Reason. It is made up of many bones but all working together.
It is an examples of an endoskeleton.

Main parts of the human skeleton.

Part	Composition
Skull	Cranium//maxilla//mandible
Limbs	arm//legs
Trunk	Clavicle//sternum//ribs
Limb girdle	Pelvis//Scapula

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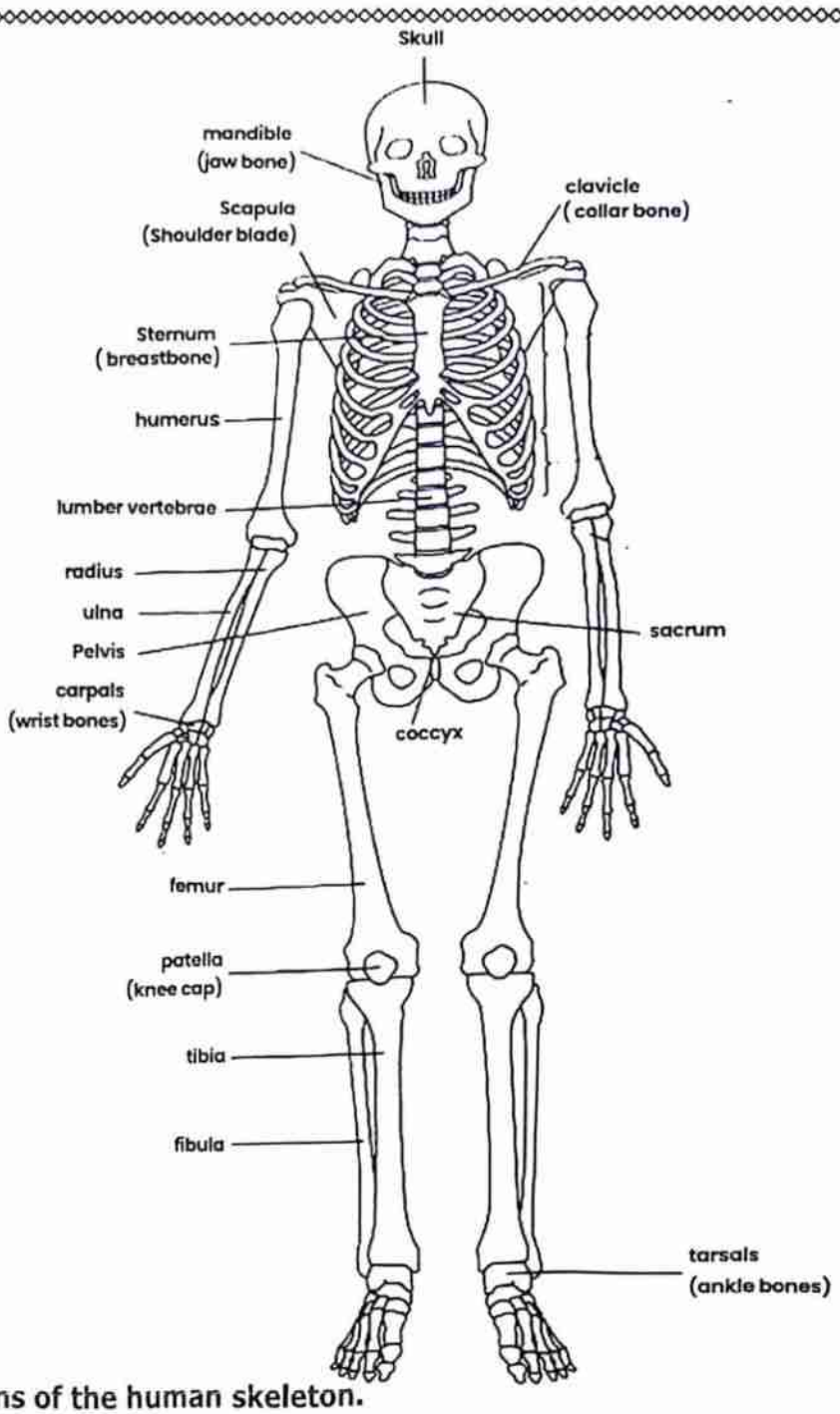

Functions of the human skeleton.

- It protects the delicate body organs.
- It gives the body shape.
- It helps in body movement.
- It provides support to the body.
- It provides point for muscle attachment.
- Bone marrow help in the production of blood cells.

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Parts of the human skeleton used for protection.

Part of the human skeleton	Parts/organs protected
ribcage	heart//lungs
skull/cranium	brain
pelvis/pevic girdle	bladder//uterus//oviducts//ovaries
eye socket/orbit	eyes
backbone/spine/vertebral column	spinal cord
mandibles	tongue

Importance of the human skeleton to the circulatory system.

- The skeleton protects the heart.
- The skeleton helps in the production of blood cells.

Evaluation activity:

1. What type of skeleton do human beings have?
2. State any **two** functions of the human skeleton.
3. Apart from vertebral column, give any **two** other parts of the skeleton that are used for protection.
4. Name the part of the human body that is protected by the vertebral column.
5. State any **one** way in which the skeleton is useful to the circulatory system.
6. The diagram below is of part of the human skeleton. Study and use it to answer the questions that follow.

   a) Name the part marked R.
   b) What type of bone is part marked R?
   c) Give **two** organs in the human body that are protected by the part of the human skeleton above.
7. In which way is the skull important in the skeletal system?
8. The table below shows the parts of the skeleton and the parts of the body they protect. Study and complete it correctly.

Parts of the Skeleton	Body Organ Protected
.....	Heart and lungs
Skull	.....
.....	gonads
orbit	.....

9. Name any **one** human body organ protected by the rib cage.

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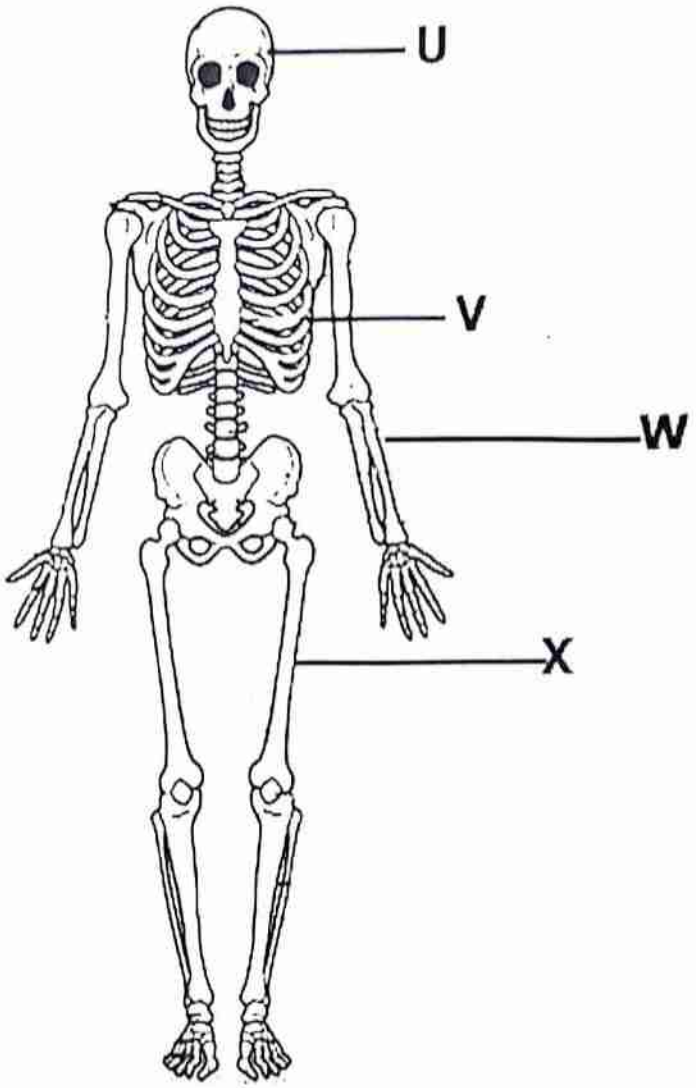
10. State any **two** ways in which the skeleton supports the body.
11. In which part of the human skeleton are the red blood cells produced?
12. Give **two** examples of limbs in the human body.
13. List down the **four** main parts of the human skeleton.
14. The diagram below is of a human skeleton. Study and use it to answer the questions that follow.



a) Name the parts marked **W** and **X**.
b) State the protective function of part marked **V** on the skeleton.
c) Which part in the human body is found in the part marked **U**?

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Lesson vocabulary:

tissue rigid femur collagen ossification phosphorus

BONES IN THE HUMAN BODY.

A bone is a hard and rigid tissue found in the body of vertebrates.

- Bones are formed from cartilages by **ossification.**
- The human skeleton is made up of **206** bones.
- The collagen provides soft frame work of bones.

Note. *Babies have more bones in the body than adults.*
Reason. *Babies are born with many cartilages which later join to form bones.*

Substances found in bones.

blood, blood cells, nerves, collagen, mineral salts

Calcium and phosphorus are the mineral salts found in bones. They make bones hard and strong

Types of bones in the human body.

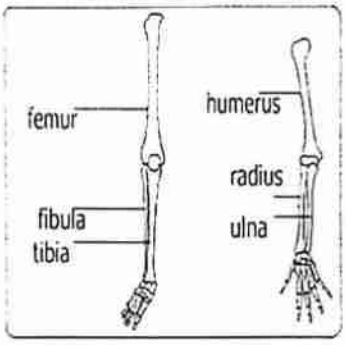
Bones are classified according to their shapes.

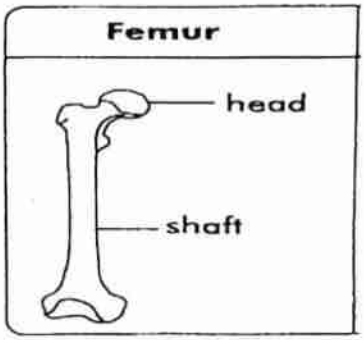
Long bones

They are found in the limbs.
They have yellow bone marrow.
They have thick ends to enable them fit in the bones to which they are connected.

Examples of long bones.

femur humerus radius ulna fibula tibia



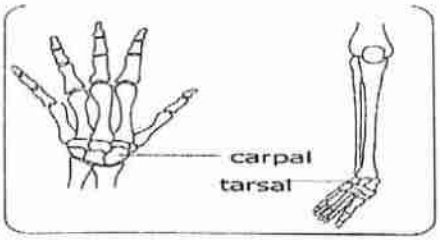


Short bones

They are found in the feet and hands.
They have red bone marrow.

Examples of short bones.

tarsals carpals

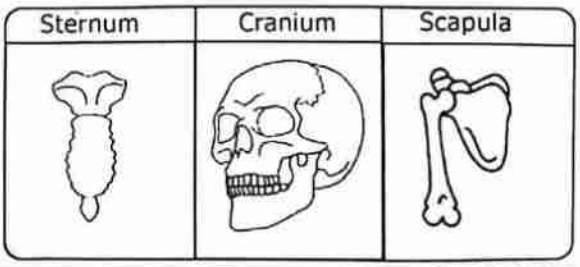


Flat bones

They have flat shapes.

Examples of flat bones.

- cranium
- scapula
- pelvis
- sternum



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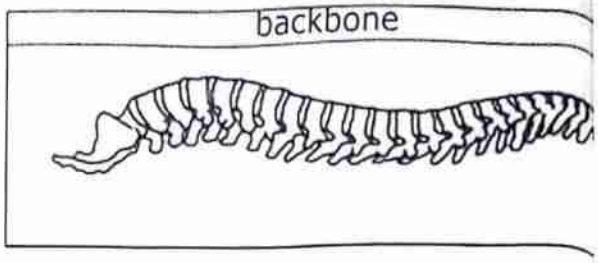
Irregular bones

They are found in the back. An example of an irregular bone in the human body is the **backbone** (vertebral column)



- *The femur is the longest and strongest bone in the human body. It is found in the thigh.*
- *The humerus is the longest bone in the fore/upper limb.*
- *Stirrup/stapes is the smallest bone in the body. The stirrup is found in the*

Bone marrow

A bone marrow is a soft tissue found in the bones.

Types of bone marrows

Red bone marrow

- It is found in short bones.
- It helps in the production of blood cells.

Yellow bone marrow

- It is found in long bones.
- It stores fats.

Bones and their other names.

Bone	Other name
breastbone	sternum
shoulder blade	scapula
kneecap	patella
thigh bone	femur
hipbone	pelvis
Skull	cranium
collarbone	clavicle
backbone	vertebral column/spine/spinal column

Functions of bones in the human body.

- Bones protect internal organs.
- Bones store mineral salts.
- Bone marrow produce blood cells.

This process is called hematopoiesis.

- Bones provide point for muscle attachment.
- Bones enable the body to move.
- Bones provide support to the human body.

Diseases that affect human bones.

• poliomyelitis • tuberculosis of bones • leprosy • bone cancer

Disorders that affect bones in the human body.

• fracture • sprain • cuts • dislocation

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Health habits that keep bones in proper working condition.

1. Doing regular physical exercises.
2. Feeding on food rich in calcium and phosphorus.
3. Immunisation against tuberculosis and polio.
4. Observing good posture.
5. Avoid playing rough games.
6. Drinking boiled water.
7. Going for regular medical checkups.

Evaluation activity:

1. Name the longest and strongest bone in the human body.
2. In which way does calcium help to promote the health of bones in the human body?
3. In the table below, part **A** shows bones and part **B** shows the parts in the human body where they are located

Part A: types of bones	Part B: locations in the body
Tarsals	In the shoulder
Spine	In the feet
Femur	In the back
Scapula	In the thigh

Use the locations in part **B** of the table to match the types of bones below.

a) Tarsals
b) Spine
c) Femur
d) Scapula

4. Mention any **one** disease that affects the human bones.
5. Give any **two** ways in which bones are important in the human body.
6. Sate any **two** ways of caring for bones in the human body.
7. Name the longest bone found in the upper limb.
8. Why do babies have more bones than adults?

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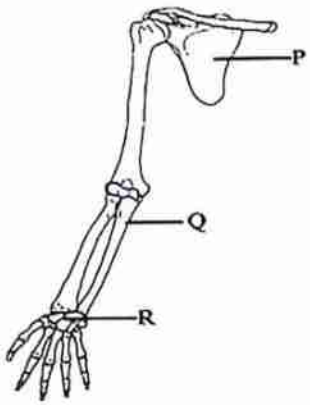
9. In which part of the body is the longest bone found?

11. Below shows the location of bones in the human body. Study and use it to answer questions that follow.



a) Name the bone P and Q.
b) What type of bone is P and R?
c) Apart from calcium, name one other mineral salt that makes bones hard and strong in the human body.

12. The diagram below shows thetype of bone in the human body.study and use it to answer the questions that follow.

a) Identify the bone shown above.
b) What type of bone is the bone shown above?
c) In which part of the human body is the bone above found?
d) How is the bone above different from the sternum in their structure?

Lesson vocabulary:

cartilage	tendon	ligament	synovial
fluid	hinge	lubricate	gliding

JOINTS IN THE HUMAN BODY.

A joint is a place where two or more bones meet in the human body.

Importance of joints in the human body.

- Joints allow body movement.
- Joints provide flexibility to the body.

Types of joints in the human body.

- movable joints
- immovable joints

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Movable joints

These are joints that allow movement in the human body.

Examples/ types of movable joints

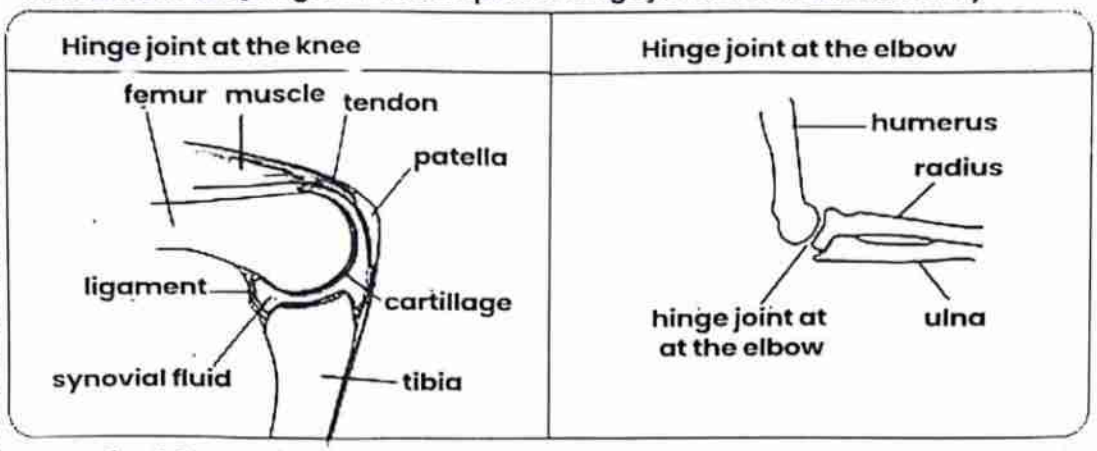
•hinge joints •ball and socket joints •pivot joints •gliding joints

Hinge joints.

Hinge joints are movable joints that allow body movement in only one plane.

- Their movement is similar to that made by door **hinges.**
- Hinge joints are found in the **knees, elbows** and **knuckles.**

Illustrations showing some examples of hinge joints in the human body



Parts of a joint and their functions.

- **synovial fluid**: Reduces friction at the joint.

way: By lubricating bones.

- **ligament:** Attaches bones together at a joint/joins a bone to a bone.
- **tendon:** Attaches muscles to bones at a joint/joins a muscle to a bone.
- The tendon is also called **sinew**.
- **cartilage:** Reduces friction at a joint.

way: By preventing bones from rubbing directly onto each other.

- The cartilage is also called **gristle.**

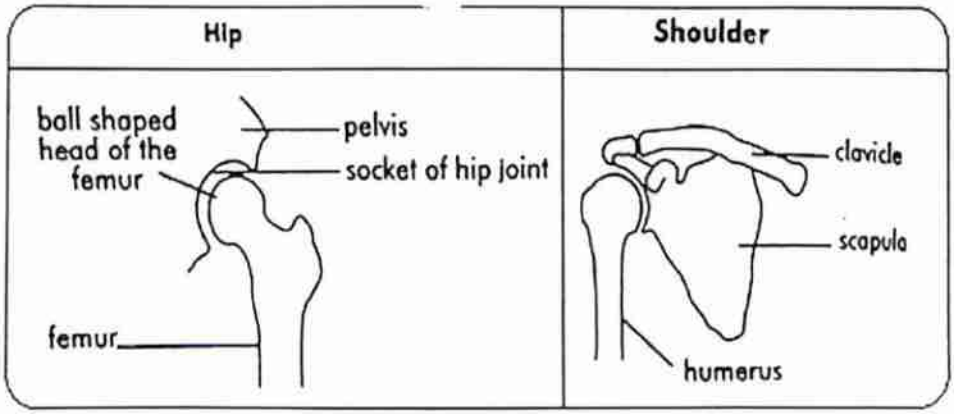
Ball and socket joints

Ball and socket joints are movable joints that allow body movement in three planes.

These three planes are; forward, backwards and sideways.

Ball and socket joints allow all round body movement.

Ball and socket joints are found at the **shoulders** and **hips.**



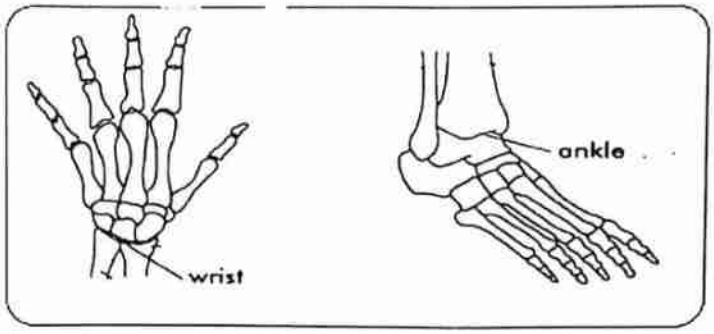
How are joints able to reduce friction?

- They have synovial fluid that lubricates bones.
- They have cartilage that cushions bones.

Gliding joints

Gliding joints are movable joints allow flat bones to slide over one another.

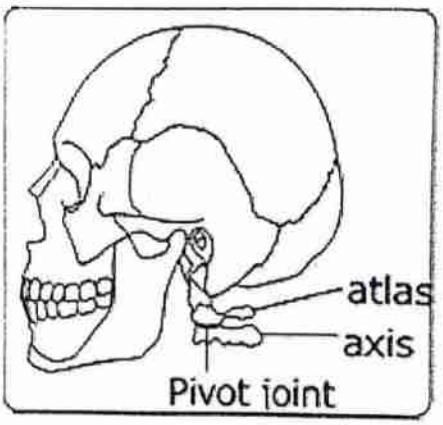
- They are also called **plane/sliding joints**.
- Gliding joint allow movement in two planes.
- Gliding joints are found at the **wrist** and **ankle**.



Pivot joints

Pivot joints are movable joints that allow one bone to rotate on another bone.

- They are also called rotary joints.
- The atlas and axis are bones which make the pivot joint.
- Pivot joint allow the head to nod, rotate or turn around and shake.
- Pivot joint is found in the neck

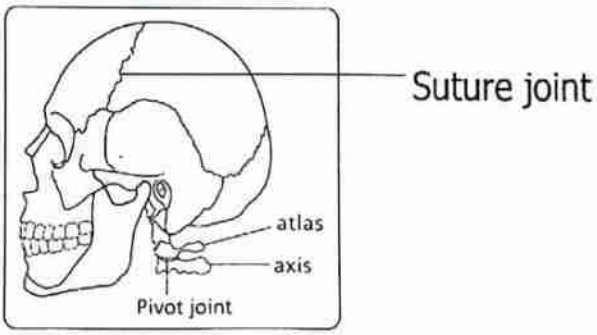


Immovable joints

These are joints that do not allow any movement in the human body.
They are found in the skull.
The bones are fixed together.
They are sometimes called **fibrous joints.**

Reason. The bones are fixed together by tough fibres.

An example of immovable joints in the human body are the suture joints.



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Bones that form joints in the human body.

Bones	location	Joint formed
femur//thigh//tibia//fibula// patella	knee	hinge joint
humerus//scapula	shoulder	ball and socket
femur//pelvis	thigh	ball and socket
humerus//radius//ulna	elbow	hinge joint
tarsals//phalanges//carpals	wrist//ankles	gliding joint
neck vertebrae	neck	pivot joint
cranial bones//jawbone	skull//teeth	suture joint

Disorders of joints.

- Dislocation
- Fracture
- Sprains
- Strains

Evaluation activity

1. Give the meaning of the term joint.
2. In which **one** way are joints important in the body?
3. Name any **one** part of the human body where a gliding joint is found.
4. Mention **two** substances which reduce friction at the joint.
5. a) Name any **two** bones that form the joint at the knee of the human body.
   b) What name is given to the joint at the knee?
   c) In which **one** way is friction reduced at a joint?

The diagram below is of a human skull and a neck vertebra. Use it to answer questions 6 and 7.

6. Name the movable joint at K.
7. Which delicate organ is protected by S?

8. The table below shows the types of movable joints and their locations in the body. Study the table and complete it correctly.

Type of movable joint	Location in the body
.....	Wrists and ankles
Pivot joint	.....
.....	Elbows and knees
Ball and socket joint	.....

9. State the difference between ball and socket joint and hinge joint.
10. Give the function of the following parts at a joint.
    i) Ligament
    ii) Tendon
    iii) Synovial fluid

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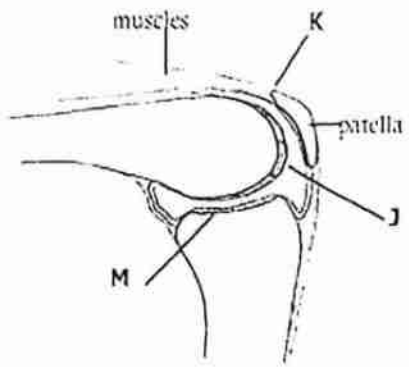
11. The diagram below shows a movable joint.
Study and use it to answer the questions that follow.



a) Name the parts marked J and M.

b) State the function of part marked K a a joint.

c) Identify the movable joint shown above.

Lesson vocabulary:

- muscle
- voluntary
- antagonistic
- involuntary
- biceps
- triceps

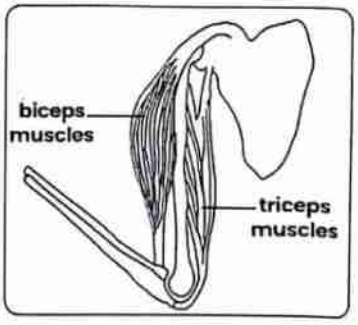
MUSCLES IN THE HUMAN BODY

Muscles are fibrous elastic tissues in the human body.

- Muscles contract or relax to produce movement of a part or the entire body.
- Muscles are attached to bones by tendons.

Characteristics of muscles.

- They can contract and relax.
- They are made up of proteins.
- They are made up of cells with many nuclei.



Types of muscles in the human body.

1. voluntary muscles
2. involuntary muscles

Voluntary muscles

Voluntary muscles are muscles whose movement or action is controlled by one's will.

Examples of voluntary muscles in the human body.

- biceps
- triceps
- hamstring muscles
- quadriceps
- calf muscles
- pectoral muscles
- gluteus muscles
- deltoid muscles
- back muscles
- abdominal muscles

☞ *Points to note: Some muscles work in pairs but have an opposite effect to each other. Such muscles are called* ***antagonistic*** *muscles.*
Examples of antagonistic muscles are ***biceps*** *and* ***triceps***

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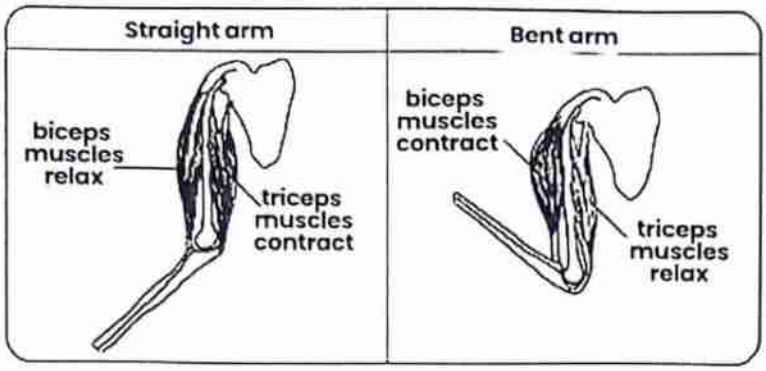
Action of the biceps and triceps muscles



Explanation and points to note.

When the forearm is bent or raised;

- The biceps contracts.
- The triceps relaxes.

When the forearm is straightened;

- The triceps contracts.
- The biceps relaxes.

Biceps and triceps are referred to as;

- Antagonistic muscles.

  Reason. *They work in pairs but have an opposite effect to each other.*
- Voluntary muscles.

  Reason. *Their movement or action is under one's will.*

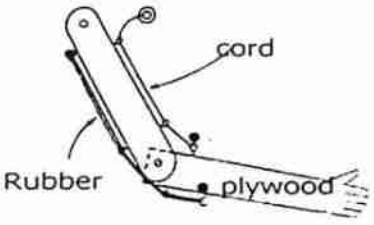
Biceps is called a **flexor.**

Reason. *Biceps bends the forearm when it contracts.*
It decreases the angle formed by the forearm.

Triceps is called an **extensor.**

Reason. *Triceps straightens the forearm when it relaxes.*
It increases the angle formed by the forearm.

A wooden model showing the action of biceps and triceps.



Explanation:

- The cord represents the biceps.
- The rubber represents the triceps.
- The above muscles are voluntary muscles.
- When the plywood is raised, the cord contracts while the rubber relaxes.
- When the plywood is straightened, the cord relaxes while the rubber contracts

Involuntary muscles

Involuntary muscles are types of muscles whose movement or action is not controlled by one's will.

Examples of involuntary muscles in the human body.

- cardiac muscles
- intercostal muscles
- stomach muscles
- eyelid muscles
- intestinal muscles
- diaphragm muscles

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Lesson vocabulary

- leprosy
- cockroach
- tuberculosis
- paralyse
- phosphorus
- calcium

Diseases that affect the muscular- skeletal system.

- poliomyelitis
- tuberculosis of bones
- tetanus
- leprosy
- arthritis
- rickets
- scurvy
- yaws
- tetanus
- bone cancer

CASE STUDY ON SOME DISEASES

Poliomyelitis

- It is a viral immunisable disease of the bones.
- It spread by houseflies and cockroaches.

Polio is spread through;

- Drinking contaminated water.
- Eating contaminated food.

Signs and symptoms of polio are;

- Fever
- Paralysed body limbs.
- Lameness
- General body weakness.

Poliomyelitis is also considered a waterborne diseases and food borne disease.

Polio can be prevented by;

- Getting immunised using polio vaccine.
- Drinking boiled water.
- Eating safe food.
- Spraying insecticides to kill cockroaches.

Tuberculosis of bones

- It is a bacterial immunisable disease that affects bones.
- It affects the backbone/spine.

How tuberculosis of bones is spread;

- Through inhaling contaminated air.
- Through drinking unboiled milk from an infected cow.
- Through saliva droplets from infected people.

Signs and symptoms of tuberculosis of bones.

- Long-lasting painful backache.
- Chest pain
- Paralysis of the legs leading to failure to walk.
- Loss of body weight.
- Swelling or deformity around affected joints.

How to prevent / control tuberculosis of bones.

- Immunizing using BCG vaccine.
- Early treatment of the infected people.
- Drinking properly boiled milk.
- Isolating infected people.
- Encouraging infected people to wear facemasks.

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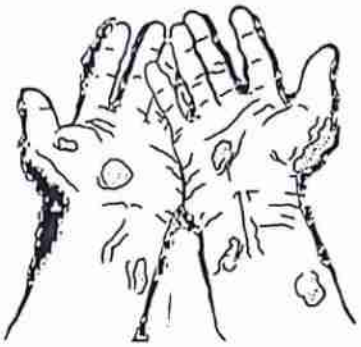
Leprosy

- It is a bacterial disease that affects both the skeletal system and the skin.
- It can lead to loss of toes, fingers or hands and feet.
- It can be spread by cockroaches.

How leprosy is spread.

- Through body contact with an infected person.
- Eating food contaminated by cockroaches.



Signs and symptoms of leprosy.

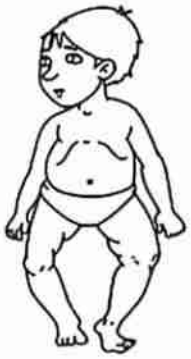
- Pain at the joints.
- Reduced sense of feeling.
- Pale skin sores that last long.
- Deformed toes and fingers.

How to prevent/ control leprosy.

- Early treatment of infected people.
- Isolating infected people.
- Control the breeding of cockroaches.

Rickets

It is a deficiency disease of the bones.



Signs and symptoms of rickets.

- Brittle bones and teeth.
- Bow-shaped legs
- Knock knee legs
- Poor teeth formation in children.
- Weak bones.

How to prevent rickets.

- Eating food rich in vitamin D, calcium and phosphorus.
- Morning sun bathing. Sunshine enables the skin to make vitamin D.

Rickets, arthritis and scurvy are non-contagious diseases.

- *Rickets and scurvy are deficiency diseases that affect the muscular-skeletal system*
- *Leprosy, tuberculosis and tetanus are caused by bacteria.*
- *Polio and yaws are caused by virus.*
- *Scurvy causes sores at the joints.*

Evaluation activity:

1. Name the immunisable water borne disease of the human skeleton.
2. Which communicable disease in human beings affects both the skin and skeletal system?
3. State any **two** signs and symptoms of rickets.
4. How do cockroaches affect the muscular-skeletal system?
5. Name the cattle product through which tuberculosis can be spread to human beings.

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6. Which deficiency disease of the skeletal system is prevented by sun bathing in the morning?
7. Write down **two** diseases of the muscular-skeletal system spread by cockroaches.

The diagram below shows a person suffering from an immunisable skeletal disease. Use it to answer questions 8 and 9.

8. Which skeletal disease is the person above suffering from?
9. State any **one** way in which the disease that the person above is suffering from can be prevented.
10. Mention any **two** non-contagious diseases that affect the muscular-skeletal system.
11. State any **one** way in which the spread of tuberculosis can be controlled.
12. Which vector spreads leprosy to human beings?
13. a) What germ that causes leprosy?
    b) Give any **two** signs and symptoms of leprosy.
    c) State any one way of controlling leprosy in human beings.
14. In which way does scurvy affect the proper functioning of the muscular-skeletal system?
15. In the table below, part A shows the diseases of the muscular-skeletal system and part B shows their signs and symptoms in their wrong order.

Part A: Diseases	Part B: signs and symptoms
Poliomyelitis	Ox-bow legs.
Rickets	Deformed toes and fingers.
Leprosy	Stiff swollen joints.
Arthritis	Paralysed limbs.

Select from the table the correct signs and symptoms of the process and write in the spaces provided below.

a) Poliomyelitis .....
b) Rickets .....
c) Leprosy .....
d) Arthritis .....

DISORDERS OF THE MUSCULAR-SKELETAL SYSTEM.

- sprains
- strains
- dislocation
- fracture
- cuts
- bone deformation
- mytonia
- muscle atrophy
- prolapse
- osteoporosis
- muscle atrophy
- hernia
- muscle wasting
- backache
- muscle cramps

A sprain is a torn or overstretched ligament.
A strain is a torn or overstretched tendon.
A dislocation is the displacement of a bone from a joint.

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Activities that can lead to sprains, strains and dislocation.

- Playing football
- Playing netball
- Wrestling
- Athletics
- Kick boxing
- Walking down the staircases

Signs and symptoms of sprains, strains and dislocation.

- Swelling around the injured part.
- Failure to move the injured part.
- Too much pain around the injured part.

First aid for sprains, strains and dislocation

- Apply cold compress around the injured part.

Fracture

A fracture is a broken or cracked bone in the human body.

Causes of fracture

- Falls
- Knocks
- Rough games
- Playing football
- Kick boxing

Types of fracture.

- simple fracture
- compound fracture

Simple fracture

- This is where the broken bone remains inside the skin.

Signs and symptoms of simple fracture.

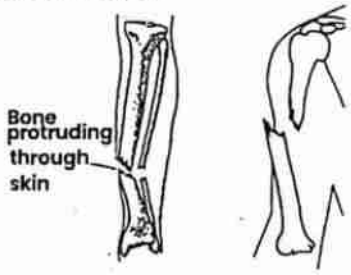
- Swelling of the injured limb.
- Pain at the injured part.
- The broken bone remains inside.
- Internal bleeding.
- Failure to move the injured part.

Compound fracture

- This is where the broken bones tears the skin and comes out.

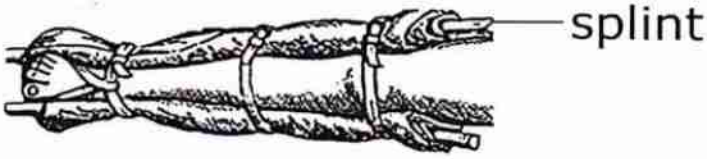
Signs and symptoms of compound fracture.

- The broken bones comes out of the skin.
- Bleeding at the injured part.
- Damaged tissue at the injured part.
- Swelling of the injured part.
- Severe pain around the injured part.
- Failure to move the injured part.



First aid for fracture.

- Tie splints around the injured part.
  Splints keep the broken bone in one position.



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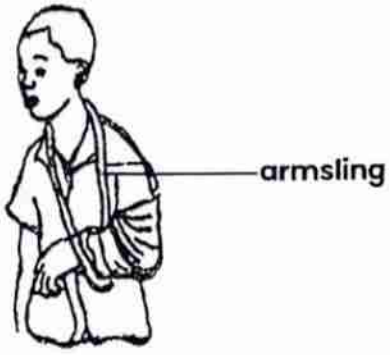
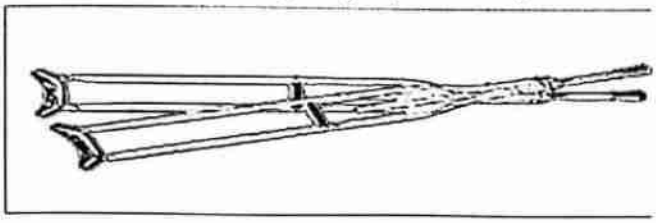
- Tie an armsling around the injured part.
  An arm sling keeps the broken bone in one position.



- Give the casualty crutches.

Crutches support the fractured casualty in walking.
Crutches reduce the body weight put on the fractured leg.



Materials that can be used to give first aid to a casualty with fracture, sprains, strains and dislocation.

- splints
- arm sling
- crutches
- stretcher
- ice
- plaster of Paris

Body parts affected by sprains strains, fracture and dislocation.

- bones
- muscles
- joints
- ligaments
- tendons
- backbone
- hands
- legs
- fingers
- head
- arms
- shoulders
- head

Enrichment exercise:

1. Give the difference between a sprain and a strain.
2. Name the part at the joint that is affected by;
   i) Sprain .....
   ii) Strain .....
3. State the first aid for sprains and strains.
4. Apart from sprains and strains, give any two other disorders of joints.
5. State any **one** sign of compound fracture.
6. Identify the injury that results from tearing of muscles.
7. In which way are the crutches helpful to a fractured casualty during movement?
8. The diagram below shows a type of fracture. Study and use it to answer the questions that follow

   a) Name the type of fracture shown above.
   b) Give any **two** signs and symptoms of the type of fracture above.
   c) Why is the plaster of Paris necessary when treating such a fracture?
9. Name any **two** parts of the skeletal system affected by fracture
10. How are splints useful when giving first aid to a casualty with a fractured leg?
11. What kind of accident requires the use of splints when giving first aid?

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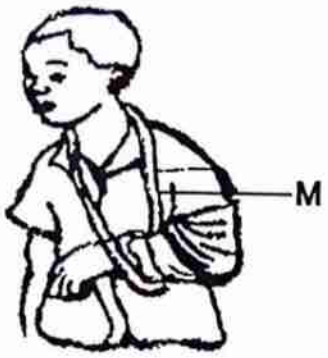
12. Which part of the skeletal system is affected by dislocation?
13. Why are babies commonly affected by greenstick fracture?

The diagram below shows a person with an injury. Use it to answer question 14.



14. How useful is the structure marked M t the injured person?

15. In the table below, part A shows disorders of the muscular-skeletal system and part B shows the meanings of the disorders.

Part A: Disorders	Part B: meanings
Muscle wasting	A torn muscle in the body.
Dislocation	Failure of muscles to support tissues.
Prolapse	The displacement of a bone.
Strain	The decrease in the muscle mass.

Use the meanings in part B of the table to match the terms below.

a) Muscle wasting .....
b) Dislocation .....
c) Prolapse .....
d) Strain .....

16. Regularly overstretched part of his ligament on the ankle and tore it when playing football.
a) What kind of injury did Regularly get?
b) State any two possible signs of the injury given in a) above.
c) What kind of first aid would you give to Regularly?

Lesson vocabulary

• posture • backache • position • deform • indigestion • habit

POSTURE

Posture is the way in which a person positions the body when doing an activity.

Such activities include;

• standing • sleeping • writing • lifting objects
• sitting • walking

Ways of maintaining proper posture.

• Placing both feet on the floor.
• Putting all your weight on both buttocks.
• Standing straight.
• Sitting upright.
• Walking upright.

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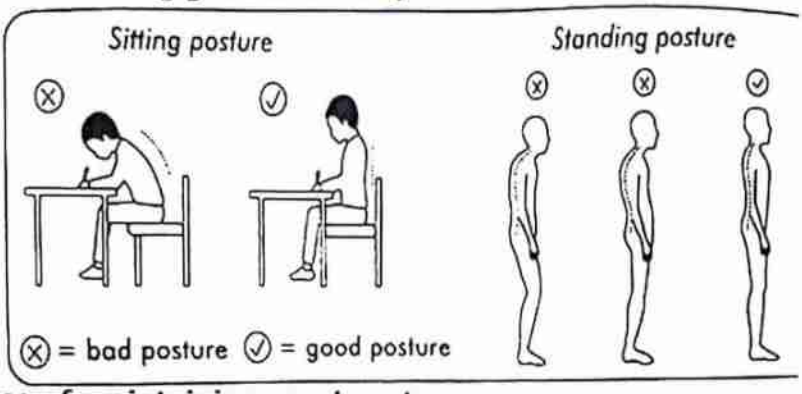
Illustrations showing good and bad posture.



Importance of maintaining good posture.

- It prevents back and chest pain.
- It prevents deformity of bones.
- It helps body organs to function properly.
- It aids proper digestion of food.
- It prevents dislocation.

Dangers of poor posture.

- It leads to deformity of bones
- It leads to backache.
- It leads to indigestion.
- It causes dislocation.

Health habits that help to keep the muscular-skeletal system in proper working condition.

- Doing regular physical body exercises.
- Maintaining proper body posture.
- Going for regular medical checkup.
- Feeding on a balanced diet.
- Avoid playing rough games.
- Taking infants for immunization.

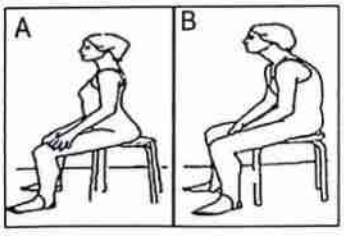
Ways in which regular physical exercises help to maintain the muscular skeletal system healthy.

- They increase flexibility in joints.
- They enable the growth of strong muscles and bones.
- They promote endurance during exercises.
- They improve on the body fitness.
- They reduce fats in the body.

Evaluation activity:

1. Give the meaning of the term posture.
2. State any **two** advantages of maintaining good posture.
3. State any **two** health habits that help to strengthen the human skeleton.
4. Give any **two** ways in which regular physical exercises help to maintain the muscular-skeletal system healthy.

The diagram below shows two girls positioning their bodies when sitting at the chair. Use it to answer questions 5 and 6.



5. Which of the **two** girls is most likely to have deformed bones?
6. Give a reason for your answer in 5 above.
7. State any **two** ways of maintaining good posture.

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MEANING OF KEY SCICENCE TERMS IN MUSCULAR-SKELETAL SYSTEM

Biceps: the flexor muscle in the upper arm which bends the forearm.

Triceps: the extensor muscle in the forearm which straightens the forearm.

Bone marrow: the soft tissue that fills the centre of sponge bones

Muscle: special elastic tissue that contracts or relaxes to produce movement

Ligament: a tough fibrous and elastic tissue that joins bones together at a joint.

Tendon: a tough connective tissue that joins a muscle to a bone.

Cuticle: a protective layer of hard material which covers arthropods.

Cartilage: a soft tough connective tissue in the bondy.

Vertebral column: a flexible series of small bones called vatebrae.

Vertebrae: the bones that make up the backbone.

Skull: the case of bones enclosing the brain.

Skeleton: a framework of the body of an animal.

Girdles: parts of the skeleton where lower limbs are attached.

Joint: a pl;ace where two or more bones meet in the human body.

Dislocation: the displacement of a bone from its normal position..

Strain: an injury on a muscle

Fracture: a broken or cracked bone in the body.

Sprain: an injury on the ligament

Prolapse: failure of muscles to support body tissue

Bone: a hard tissue found in the body of vertebrates

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2

THEME: MATTER AND ENERGY
TOPIC 2: ELECTRICITY AND MAGNETISM

Topical outline:
Electricity:

- Meaning of electricity.
- Electricity as a form of energy.
- Types of electricity.
- Sources of electricity.
- The dry cell.
- Conductors and insulators.
- Electric circuit.
- Short circuits.
- Importance of electricity.
- Advantages and disadvantages of using electricity.
- Safety precautions in handling electricity and electrical appliances.

Magnetism:

- Meaning of magnetism and magnet.
- Magnetic materials.
- Properties of magnets.
- Types of magnets.
- Making temporary magnets.
- Electromagnets.
- Electric bell.
- Electricity and magnetism in the modern world of work.
- Generating electricity using a dynamo

Ref. P.7 standard curriculum page 67 and 68.

Advice to teachers.

- *Use practical approaches in this topic.*
- *Design a model of an electric bell and a traffic light system with learners.*
- *Evaluate learners at the end of each lesson using the exercises in this book.*
- *Irrelevant content in this topic; details on electric plugs, electric motors, types of circuit connection, wet cell and electricity companies in Uganda.*

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Lesson vocabulary:

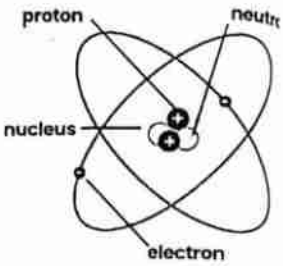
- charge
- molecule
- neutron
- electricity

ELECTRICITY:

Electricity is a form of energy produced by charged particles.
Electricity results from charged particles of an atom.
An atom is a basic unit of matter.
A group of atoms bonded together form a molecule.

Particles of an atom.

- protons
- electrons
- neutrons



Protons are positively charged particles of an atom.
Electrons are negatively charged particles orbiting around an atom.
Neutrons are uncharged particles of an atom.

Electricity as a form of energy.

- Electricity has the ability to do work.
- It can be transformed into other forms of energy.

Energy transformation related to electricity.

electric appliances	energy change
flat iron	Electrical energy changes to heat energy
electric cooker	
water heater	
hot plate	
electric kettle/percolator	
dry cell	Chemical energy changes to electrical energy
wet cell	
car battery	
telephone battery	
solar battery	
electric bulb	Electrical energy changes to heat and light energy
generator	Mechanical energy changes to electrical energy (kinetic energy changes to electrical energy)
dynamo	
alternator	
electric motor	Electrical energy changes to mechanical energy (electrical energy changes to kinetic energy)
electric fan	
washing machine	
juice blender	
solar cell (solar panel)	Light energy from sun changes to electrical energy (sunlight changes to electrical energy)

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loudspeakers	Electrical energy changes to sound energy
electric bell	
microphone	Sound energy changes to electrical energy

☞ *Electricity is not regarded as matter.*
Reason: It has no mass and volume.

Types of electricity

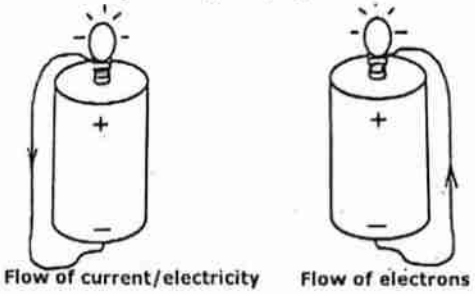
- current electricity
- static electricity

Current electricity

Current electricity is the type of electricity where electrons flow from one point to another through a conductor.

- Electric current is the flow of charges through a conductor.
- It is measured using an **ammeter** in units called **amperes(amps)**

☞ *Electrons flow from negative to positive.(ENP)*
Current /electricity flows flow positive to negative. (CPN)

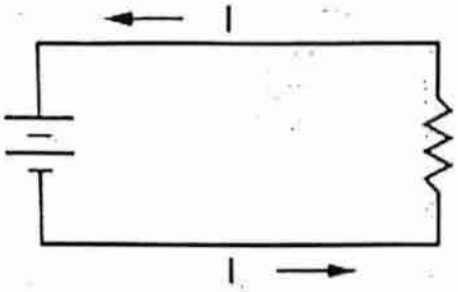


Summary of the flow of current and electrons.

Current	+ve	to	-ve
Electrons	-ve	to	+ve

Types of current electricity.

- direct current electricity
- alternating current electricity

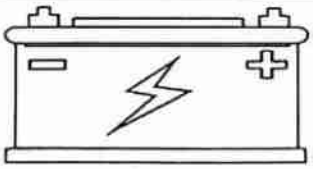


Direct current electricity (d.c)

Direct current electricity is the type of current electricity in which current flows in one direction.

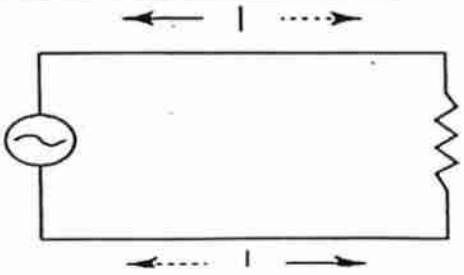
Sources of direct current electricity.

- dry cells
- wet cells
- batteries
- d.c generator
- dynamos
- solar cells

Dry cell	Battery	d.c Generator
		

Alternating current electricity (a.c)

Alternating current electricity is an electric current that periodically reverses its direction as it flows in a circuit.



Sources of alternating current electricity.

- hydroelectric power plants
- nuclear power plants
- tidal power plants

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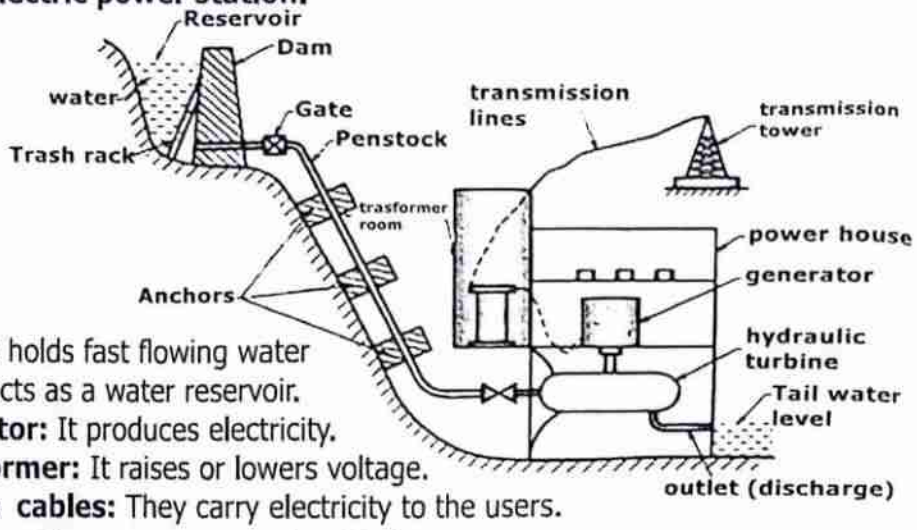
• geothermal power plants • thermal power plants • windmills

Forms /examples of current electricity.

Hydroelectricity: This is a form of current electricity produced by the power of fast flowing water that turns turbines at a dam.

• The source is fast flowing water.

A hydroelectric power station.



Dam: It holds fast flowing water
It acts as a water reservoir.

Generator: It produces electricity.

Transformer: It raises or lowers voltage.

Electric cables: They carry electricity to the users.

Electricity wires are placed high up
Reason. To prevent the risks of electric shocks.

Turbines: They rotate when hit by fast flowing water to produce electricity.

Water: It turns turbines connected to a generator to produce electricity.

How hydroelectricity is carried to the consumers.

• Through the grid. • Through transmission lines. • Through power cables.

◊ These are aluminium metal wires. Aluminium metal wires are light in weight and carry electricity well.

◊ The main cable that brings electricity in our houses from the pole/pylon is called a **solidor/service wire .**

◊ The turbines rotate when hit by fast flowing water to produce electricity.

• *Water in the dam possesses potential energy.*
Falling water that turn turbines possesses kinetic energy.
• *Water turns turbines connected to a generator to produce electricity.*

Energy change in turbines

Mechanical energy to electrical energy

Thermal electricity: This is a form of current electricity produced from burnt fossil fuels in engines.

Such fossil fuel products are;

• petrol • diesel • kerosene/paraffin • coal

Solar electricity:

This is a form of current electricity generated from the sun.

Chemical electricity:

This is a form of current electricity generated from batteries or electric cells.

Atomic electricity:

This is the form of current electricity produced by radiations from stable atoms of uranium.

Atomic electricity is also called nuclear electricity.

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Mechanical electricity

This is the form of current electricity generated from wind energy or dynamo.

Geothermal electricity

This is the form of current electricity generated from hot springs.

Forms of electricity with their sources.

Electricity	Source
hydroelectricity	fast flowing water
thermal electricity	fossil fuels/biogas
geothermal electricity	hot springs
atomic electricity	uranium
mechanical electricity	wind/dynamo
chemical electricity	batteries/electric cell
solar electricity	the sun

Evaluation activity:

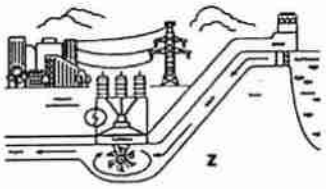
1. Give the meaning of the term electricity.
2. Give any **one** reason why electricity is regarded as a form of energy.
3. Give any **one** reason why electric wires are placed very high.
4. State the role of the following in the generation of hydroelectricity:

a) Turbines
b) Fast flowing water
c) Dam

5. The Uganda Electricity Distribution Company Limited generates its electricity in Jinja.

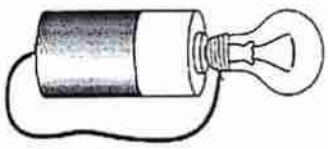
a) What form of electricity is generated in Jinja?
b) How does electricity generated in jinja reach the users across Uganda?
c) Give any **two** ways in which electricity generated in Jinja is useful to people in Uganda.

6. The diagram below shows the production of electricity at a power station. Study carefully and use it to answer the questions that follow.



a) Identify the form of electricity produced from the power station above.
b) State the role of the dam in the production of electricity above.
c) Name the part marked **Z** on the power station above.
d) How are power cables useful at the power station above?

7. Identify the form of current electricity produced by biogas.
8. On the diagram below, show with an arrow the flow of electrons.



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10. Name the form of electricity got from;
a) Coal .....
b) Hot springs .....
c) Batteries .....

11. At a power station of hydroelectricity, a strong wall, turbines and a reservoir are built.
a) Identify the strong wall built at a hydroelectricity power station.
b) What is the importance of a reservoir?
c) State the effect of water on the turbines.
d) Name any **one** other form of electricity produced from water.

Lesson vocabulary

• lightning • static • friction • thunder • charges • insulator

Static electricity

Static electricity is the type of electricity produced by the build-up of charges on the surface of an insulator.

- Static electricity is formed when two insulators rub against each other.
- Static electricity takes place on insulators by the help of friction.
- Static electricity does not involve the flow of electrons.

Ways of producing static electricity.

- Rubbing a plastic pen/ruler against hair.
- Rubbing a balloon in hair.
- Rubbing a plastic comb against hair.
- Rubbing a plastic ruler against woolen cloth.

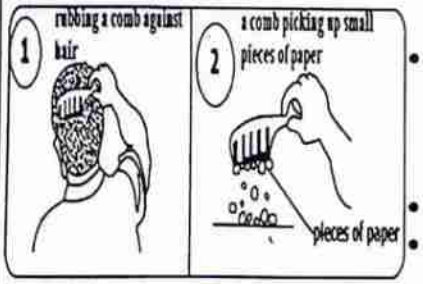
An experiment to produce static electricity.

1. Requirements:
   - a plastic comb /pen/ruler
   - small pieces of paper

What to do:

i. Put the small pieces of paper on the table or desk.
ii. Rub the plastic comb in your hair several times.
iii. Bring the comb very close to the pieces of paper and observe.

Set up experiment



Observation:

- The comb attracts the small pieces of paper.
  Reason: The comb acquires static electricity when rubbed in hair.
- The comb acquires **negative charge.**
- The pieces of paper acquire **positive charge**

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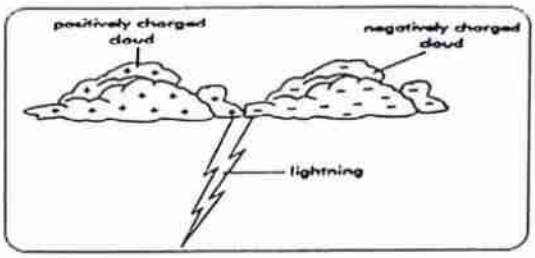
Lightning

Lightning is the flash of light produced when charged clouds meet.

- Lightning is an example of static electricity in nature.
- It is caused by friction.
- Lightning is said to be electricity because it involves electric charges.

Cause of lightning.

When a positively charged cloud meets with a negatively charged cloud.



Thunder

Thunder is the sound produced when heated air around the clouds rapidly expands and contracts.

Cause of thunder.

The sudden expansion and contraction of air between opposite charged clouds.

Lightning is seen before hearing thunder.

- ***Reason:*** *light travels faster than sound in air.*
- *Lightning improves soil fertility.*
- ***Way:*** *Lightning fixes nitrates into the soil.*

Dangers/ effects of lightning.

- Lightning strikes people and other animals to death.
- Lightning sets fire on plants, forests and buildings.
- Lightning destroys people's property.
- Lightning damages electric appliances.

Ways of preventing the dangers caused by lightning.

- Avoid taking shelter under trees during a thunder storm.
- Putting lightning arresters / conductors on tall buildings.
- Avoid swimming in open water when it is raining.
- Avoid standing in open places during thunderstorm.
- Wearing rubber shoes/sandals when walking in rain.

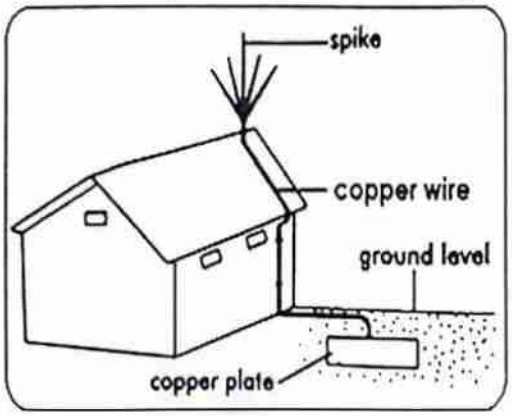
The lightning conductor

It is put on tall buildings to protect them from lightning strikes.

How it protects tall buildings.

It traps static charges and directs them safely to the ground.

Parts of a lightning conductor / arrester.



Functions of parts of a lightning conductor / arrester.

Spike: Conducts excess electric charges from the clouds.

Copper plate: Conducts charges into the ground.

It is buried in the ground to conduct electrons to the soil to be easily neutralised.

Copper wire: Provides route for static charges to flow to the ground

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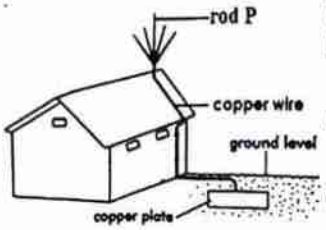
Differences between current electricity and static electricity.

- Current electricity involves the flow of electrons unlike static electricity.
- Current electricity occurs in conductors while static electricity occurs in insulators.
- In current electricity, the charges are inside the conductor while in static electricity, the charges are on the surface of the insulator.
- Current electricity only involves electrons while static electricity involves bot protons and electrons.

Evaluation activity:

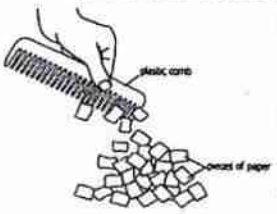
1. Mention any **one** form of static electricity in nature.
2. Which force is involved in the production of static electricity?
3. State what would be observed when a plastic pen rubbed in hair is brought close to small pieces of paper.
4. Give a reason why lightning is usually seen in the sky before hearing the sound of thunder.
5. Mention any **two** dangers of lightning in the environment.

The diagram below shows the structure installed on top of tall buildings. Study and use it to answer questions that follow.



a) Name the rod marked **P** on the structure above.
b) State the function of the copper wire on the structure above.
c) Identify the form of electricity that the structure above protects the building against.
d) Give any **one** danger of building the house with out installing the structure above.

7. Why is it not advisable to stand under a tree during thunderstorm?
8. How does a lightning conductor protect the building from lightning?
9. State any **two** ways in which people can safeguard against the dangers of lightning.
10. Why are lightning conductors put on the top of a tall building?
11. In which way does lightning help to improve soil fertility?
12. In the diagram below, a plastic comb was rubbed in hair and brought nearer to the small pieces of paper. Study and use it to answer the questions that follow.



a) Identify the type of electricity demonstrated using the experiment above.
b) What type of charge is possessed by;
i) Pieces of paper? .....
ii) Plastic comb? .....
c) Why does the plastic comb attract pieces of paper as shown above?

13. What is the importance of using salt and charcoal when earthing the lightning conductor?
14. State any **two** ways in which current electricity is different from static electricity.
15. State any **one** way in which a primary seven pupil can produce static electricity in the class.

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16. Chlorophyll wanted to test for static electricity. He had the following materials;
a plastic ruler, pieces of paper and woolen cloth.

a) Name any one material that Chlorophyll would use instead of a plastic ruler.

b) Why did chlorophyll include a woolen cloth on his materials for the experiment?

c) Describe how Chlorophyll will test for static electricity in this experiment. **(2 marks)**

17. Name the form of electricity in nature that improves soil fertility.
18. Give any **one** way in which static electricity is useful in our daily life.
19. What type of electricity is produced by friction?
20. Explain how lightning takes place. **(2marks)**

Lesson vocabulary:

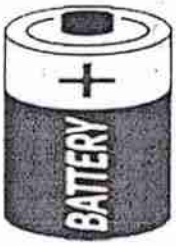
• battery • electrolyte • voltage • electrode • carbon • ammonium

Sources of electricity

- generators
- thermal power plants
- chemicals like lithium
- batteries
- hydroelectric power plants
- geothermal power plants
- nuclear power plants
- electric cells
- windmills
- the sun

The Dry Cell.

A dry cell is a voltaic cell which does not have any free-moving liquid electrolyte.

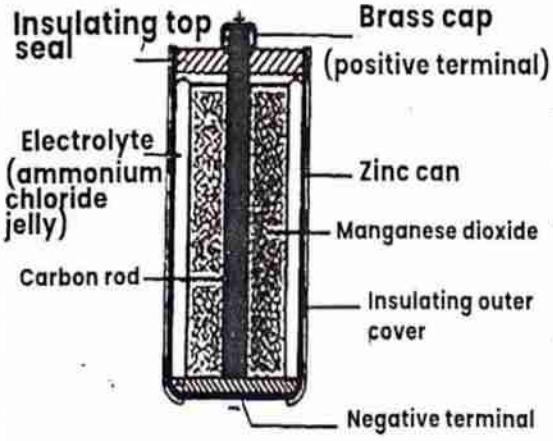


- It produces electricity by changing chemical energy to electrical energy.
- A dry cell stores chemical energy and produces electrical energy.
- A dry cell produces 1.5 volts.

Energy change in a dry cell

Chemical energy to electrical energy.

Parts of a dry cell.



Functions of each part of the dry cell.

Brass cap

The brass cap is the positive contact of a dry cell.

Pitch/top seal

The pitch prevents the electrolyte from drying.

Zinc can

The zinc can holds the electrolyte and depolariser.

The depolariser in a dry cell is manganese dioxide

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- *Manganese dioxide prevents polarisation in the dry cell.*
- *The zinc can acts as the negative terminal of a dry cell.*

Ammonium chloride jelly / paste is the commonly used electrolyte in the dry cell.

- *This conducts electricity in the dry cell.*

Electrolytes

An electrolyte is a substance that conducts electricity.

Examples of electrolytes are;

- ammonium chloride jelly
- lemon juice
- undistilled water
- urine
- dilute sulphuric acid
- salt solution
- dilute hydrochloric acid

Carbon rod

The carbon rod is a non-metallic conductor of electricity in the dry cell.

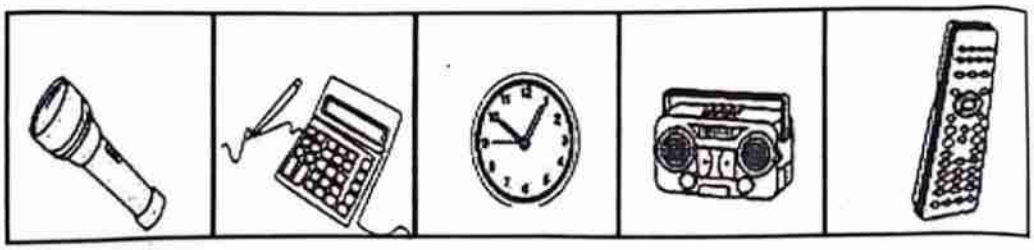
- *Its role is to conduct electricity in the dry cell.*
- *Carbon rod is made from graphite.*

Insulating cardboard

It protects the inside parts of a dry cell.

Devices that use dry cells.

- torches
- radios
- remotes
- wall clocks
- toys
- calculators



Advantages of using dry cells

- They are cheap to buy.
- They can be used in any position.
- They are portable.

Disadvantages of using dry cells.

- They work for a short time.
- They produce less power.
- They can spoil devices when used up.
- They cannot be recharged.

Calculating the voltage of a dry cell in a circuit.

Voltage is the energy needed to drive an electric current through a complete circuit.

- Voltage is measured in volts using a voltmeter
- A dry cell has **1.5 volts**.

Examples.

1. A torch uses 2 dry cells. Calculate the voltage of the torch.

   1 dry cell = 1.5 volts

   2 dry cells = (1.5 X 2) volts

   2 dry cells = 3 volts.

2. How much voltage is in the circuit of 3 dry cells?

   1 dry cell = 1.5 volts

   3 dry cells = (1.5 X 3) volts

   3 dry cells = 4.5 volts

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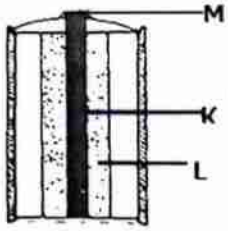
Evaluation activity:

1. Name the non-metallic conductor of electricity in a dry cell.
2. What is the importance of manganese dioxide in a dry cell?
3. Name the form of energy produced by a dry cell.
4. Mr. Dysentery's radio uses 6 dry cells. Calculate its total voltage.
5. The diagram below shows a dry cell. Study and use it to answer questions that follow.



a) Name the parts marked **M** and **K**.
b) State the function of part marked **L**.
c) From which mineral is part marked **K** made?

6. State any **two** disadvantages of using dry cells.
7. Name the form of energy stored in a dry cell.

Lesson vocabulary:

• insulator • aluminium • silver • install • electrical
• fuse

Electric conductors and insulators.

Electric conductors

These are materials that allow electricity to pass through them easily.

Examples of conductors of electricity.

• copper • lemon juice • carbon rod • iron • silver • zinc
• lead • salt solution • aluminium • wet wood • tin
• dilute sulphuric acid • mineral water

Silver is the best conductor of electricity.

- *Wet wood conducts electricity because it contains mineral salts.*
- *Distilled water does not conduct electricity because it lacks mineral salts.*
- *Distilled water does not contain mineral salts because during distillation, mineral salts remain in the heating container as water evaporates.*
- *Aluminium and copper are commonly used to make electric wires.*
  Reasons: ◇ ***They are good conductors of electricity.***
  ◇ ***They are highly ductile.***
- *Aluminium is preferred to copper when transporting electricity over long distances.*
  Reason: ◇ ***Aluminium is lighter and stronger than copper.***
- *Aluminium is commonly used to make frying pans.*
  Reasons. ◇ ***It is light.***
  ◇ ***It does not rust.***
  ◇ ***It is a good conductor of heat.***
  ◇ ***It cools fast.***

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- *Carbon, salt solution, lemon juice, mineral water, dilute sulphuric acid and wet ... are non-metallic conductors of electricity.*
- *Most overhead conductors of electricity are made of copper and not silver.*
  Reason. Copper is light while silver is heavy.

Importance of electric conductors

- They are used for making cooking utensils.
- They are used for making electric wires.
- They are used to transmit electricity.

Electric insulators

These are materials that do not allow electricity to pass them easily.

Examples of insulators of electricity.

- Dry wood
- plastic
- dry cloth
- rubber
- porcelain
- Polythene bag
- paper
- glass
- distilled water

Uses of electric insulators.

- They are used to cover electric wires during electrical installation.
- They are used to cover handles of electric irons.
- They prevent conductors from meeting to cause short circuit.
- They are used to make protective gears like rubber gloves.

An experiment to identify electric insulators.

An experiment to identify electric insulators.

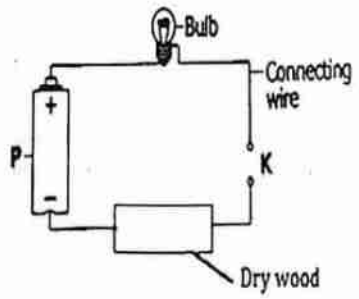
Materials needed:

- bulb
- dry wood
- dry cells
- switch

Procedure:

i. Set up an arrangement of an electric circuit.
ii. Cut the wire in two parts and place a dry wood between the conducting wires and make sure the ends of the dry wood touch the two wires.
iii. Close the circuit and observe what happens to the bulb.

Set up:



P is the dry cell
It produces electricity.

Observation:
The bulb will not produce light when part K is closed.

Reason. Dry wood is an electric insulator
It is a poor conductor of electricity.

Conclusion:
Dry wood is an electric insulator.

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- *People installing electricity wear rubber gloves to protect their hands from electric shock.*
- *Gloves used by electricians are made of rubber which is an insulator of electricity.*
- *Porcelainis a white clay-like substance.*
- *It can be used to make handles of electric kettles and electrical plugs.*

Evaluation activity:

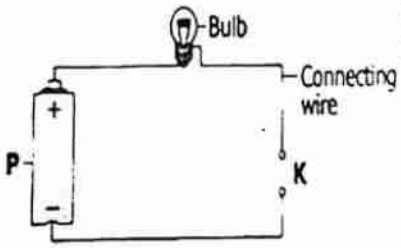
1. What term refers to liquids that conduct electricity?
2. Name any **two** liquids that conduct electricity.
3. Give the difference between conductors and insulators of electricity.
4. Why are handles of electric irons made of plastic materials?
5. Give the reason why most cooking pans are made of aluminium.
6. Why do people installing electricity wear rubber gloves on their hands?
7. The table below shows how different rods **Q, R, S** and **T** were used during a science practical lesson when testing conductors of electricity.

Source of rod	zinc	plastic	carbon	porcelain
Name of the rod	**Q**	**R**	**S**	**T**

a) Identify **two** rods that conducted electricity during the practical lesson.
b) Name any **one** rod when used to connect wires the bulb did not light up.
c) Give a reason to support the answer you have written in (b) above.

8. Why is distilled water unable to conduct electricity?
9. Mention **two** metals that are commonly used to make electric wires.
10. Give any **one** reason why most overhead conductors of electricity are made of copper and not silver.

The diagram below represents a simple electric circuit. Study and us it to answer questions **11** and **12.**

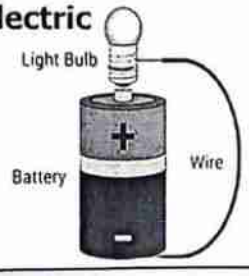


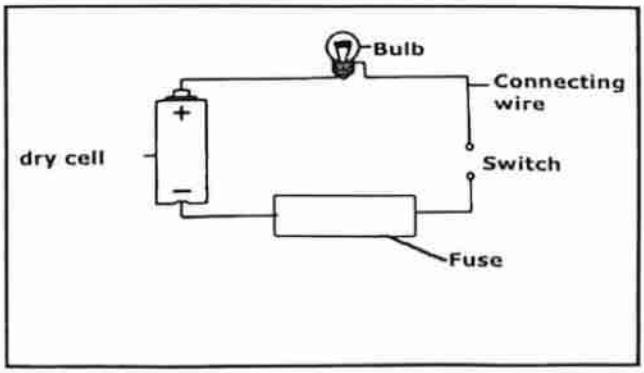
11. Name the part marked **P.**
12. Why would the bulb **not** light when a piece of glass is used to close **K**?

Electric circuit.

An electric circuit is a path for the flow of electric current.

A simple electric circuit.





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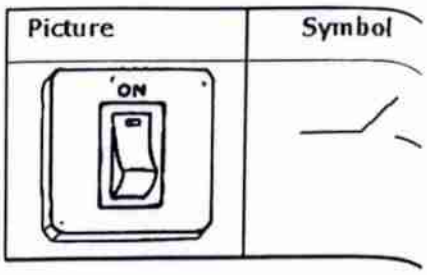
Functions of each part of the simple electric circuit:

Switch: It breaks or completes the circuit.

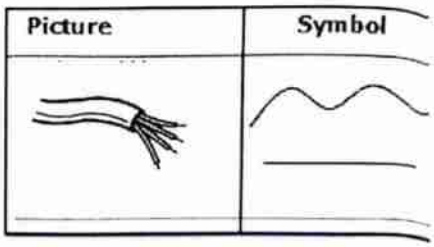
- A switch breaks the circuit by creating a gap in the circuit.
- A switch completes the circuit by closing the gap in the circuit.



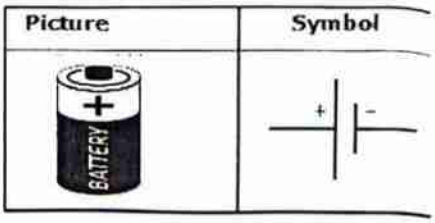
Adaptations of the switch to its function.

- It can open and close.
- It is flexible to be easily pressed on and off to complete or break the circuit.

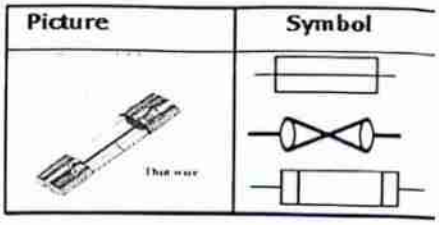
Connecting wire: It carries electricity from the dry cell to the bulb.



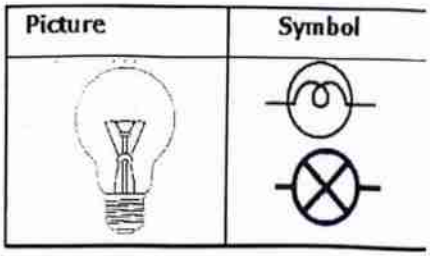
Dry cell: It produces electricity.



Fuse: It breaks the circuit in case there is too much flow of current.



Bulb: It produces light to show that the circuit is complete.



- *The dry cell changes chemical energy to electrical energy.*
- *The bulb stores electrical energy and produces heat energy and light energy when the switch is closed.*
- *Both a switch and a fuse break the circuit.*
- *In order for the bulb to produce brighter light, add more dry cells in the electric circuit*

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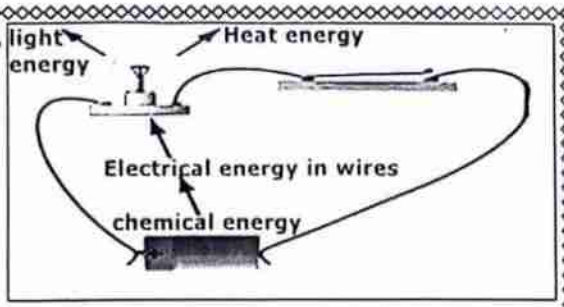
Energy changes in an electric circuit.

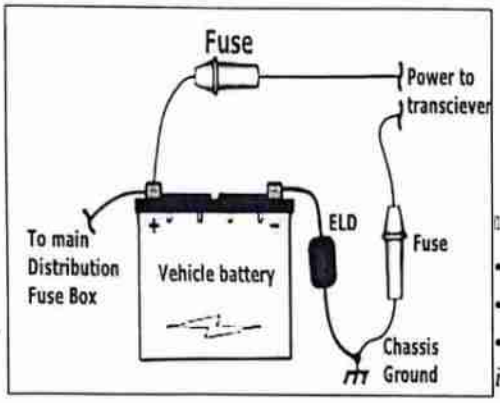


- Chemical energy to electrical energy
- Electrical energy to heat energy
- Heat energy to light energy

The fuse:

- It is a short thin piece of wire which easily melts when there is too much current flowing through the circuit.
- It is a safety device in an electric circuit.
- It works on the principle of melting.
- It melts and breaks the circuit in case of too much flow of current.

A fuse and how it is used in a circuit



Its adaptation:

It is made of a thin wire with a low melting point.

How it works:

By melting and breaking the circuit.

- *A fuse is made of a thin wire for easy melting.*
- *A fuse is made from solder i.e. tin and lead.*
- *A fuse breaks the circuit very fast when there is overheating. This prevents dangers due to short circuit.*

Factors that can make a fuse to blow.

- High voltage in the circuit.
- Overloaded circuit.
- Short circuit.
- Faulty appliance.

Signs of a damaged fuse

- It blows frequently.
- It becomes loose.
- Burnt fuse.

Examples of fused appliances.

- electric iron
- electric cookers
- water heaters
- television sets
- refrigerators

Advantages of using a fuse.

- It reduces the risk of electric fire in the circuit.
- Protects appliances in use from electrical damage in case of short circuit.

Guidance to teachers:

Draw a simple electric circuit with poorly arranged dry cells and ask learners these questions:

1. *What will happen to the bulb when the switch is closed?*
2. *Give reason for your answer in question 1 above.*
3. *What would you do for the bulb to produce light?*
4. *What other factors can make the bulb fail to produce light?*
5. *State any two indicators of a complete electric circuit.*
6. *Draw three dry cells in the space provided below to show how they would be arranged in an electric circuit above.*
7. *What form of energy is produced immediately the switch is turned on?*

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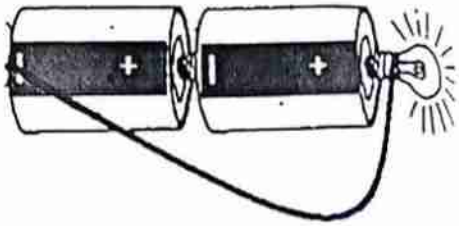
Types of electric circuits.

Complete electric circuit.

- This is one in which electrons/current flows.

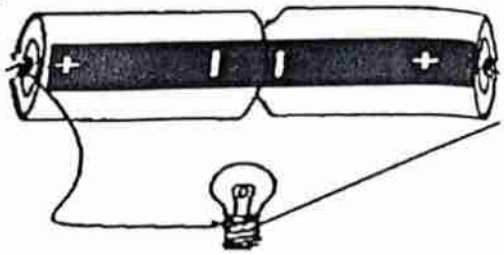


Signs of a complete electric circuit.

- Properly arranged dry cells
- The bulb lights.

Incomplete electric circuit

- This is one in which electrons/current does not flow.



Signs of an incomplete electric circuit.

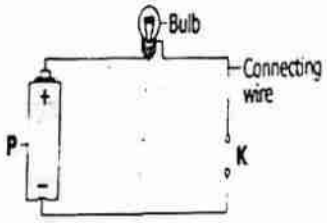
- Failure of the bulb to produce light.
- Poorly arranged dry cells.

Evaluation activity:

1. What is meant by the term electric circuit?
2. Write down any **two** components of an electric circuit.
3. Identify the safety device in an electric circuit.
4. In which way is the function of a fuse different from that of a switch?
5. Why is the fuse made of a thin wire?
6. In the space below, draw an electric symbol of a dry cell.

7. State any **one** factor that can make the fuse to blow.
8. How is the function of a switch similar to that of a fuse in an electric circuit?
9. Which alloy is used to make fuses?
10. What form of energy is stored in an electric bulb?
11. In which way is a fuse adapted to its function?

The diagram below shows a simple electric circuit. Study and use it to answer the questions that follow.



a) Name the parts marked **P** and **K.**
b) State the function of part marked **P.**
c) Give any **one** form of energy produced by the bulb when part **K** is closed.

12. In which way is a switch able to break or complete the circuit?
13. How is a fuse able to prevent dangers due to short circuit?

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14. In the table below, part A shows components of an electric circuit and part B shows their functions.

Part A: components	Part B: functions
Bulb	Breaks or completes the circuit.
Fuse	Transports electric current.
Electric wire	Breaks the circuit in case of high voltage.
Switch	Produces light in the circuit.

Use the functions in part B of the table to match the components below.

a) Bulb .....
b) Fuse .....
c) Electric wire .....
d) Switch .....

15. Arrange the parts of an electric circuit below by drawing to show a complete circuit. **(4 marks)**

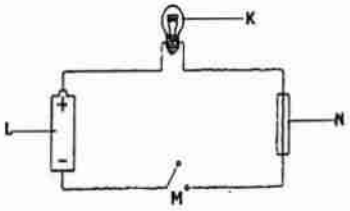
16. The diagram below shows a component found in an electric circuit. Study and use it to answer the questions that follow.

a) Name the component of an electric circuit shown above.
b) Give the function of the component shown above in an electric circuit.
c) Which **other** component of an electric circuit has the same function as the component above?
d) Draw a symbol of the component of an electric circuit above in the space provided below.

17. Distillate wanted to connect a simple electric circuit. He had the following materials for the experiment; **2 dry cells, a bulb and a switch.**

a) Name **one** material Distillate forgot to include on the list of his materials.
b) Why did Distillate include the switch on the list of his items?
c) How will the dry cells enable the bulb to produce electricity in this experiment?
d) What piece of advice would you give Distillate in order for the bulb to produce more light during his experiment?

18. The diagram below shows an electric circuit. Study and use it to answer the questions that follow.



a) Name the parts marked **K** and **N.**
b) State the energy change that takes place at **L** when **M** is closed.
c) Using an arrow, show the direction flow of current in the diagram above.

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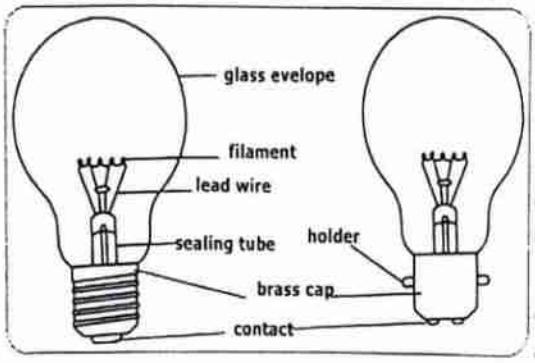
Lesson vocabulary:

• argon • tungsten • wolfram • envelope • filament • coil

Electric bulb

- It is a device that converts electrical energy to heat and light energy.

Parts of an electric bulb



Functions of parts of an electric bul

Glass envelope

- It protects the inner parts of the bulb.
- It prevents argon and nitrogen from escaping.

Reasons why nitrogen and argon ar enclosed in an electric bulb.

- To prevent the coiled filament from burning up.
- To prevent evaporation of tungsten.
- To prevent the glass envelope from be coming black.

☞:

The inside part of an electric bulb is coated with white powder.

Reasons.

- ***To enable the bulb produce visible light.***
- ***To increase the brightness of light produced.***
- ***The powder determines the colour of light produced.***

Oxygen is not used in electric bulbs.

Reason. Oxygen leads to melting and blowing of the coiled filament.

A glass envelope is curved to diverge light and transparent to allow light pass out to the surrounding.

Argon and nitrogen are non-reactive.

Coiled filament

- The coiled filament produces light in an electric bulb.
- This is done by changing electrical energy to heat and light energy.
- It is made of a metal called **tungsten** got from a mineral called **wolframite.**
- Tungsten metal has a high melting point. This prevents it from melting easily.

✎:

The filament of some electric bulbs is coiled.

Reasons:

- ***To increase electrical resistance in the bulb.***
- ***To occupy a smaller space in the glass envelope.***

Occupying a much smaller space greatly reduces heat loss by convection in the bulb.

LED bulbs do not have coiled filaments.

Adaptations of the coiled filament to its function.

- It is coiled to increase electrical resistance.
- It has a high tensile strength.

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- It has a high melting point.
- It has a high tensile strength.
- It is made up of tungsten with low vapour pressure.

Lead wires

Lead wires conduct electricity to the coiled filament.

Sealing tube

It prevents oxygen from entering the electric bulb.

Brass cap

It holds the electric bulb into the lamp holder.

Contact terminal

It conducts electricity into the bulb when the circuit is complete.

Energy changes in an electric bulb.

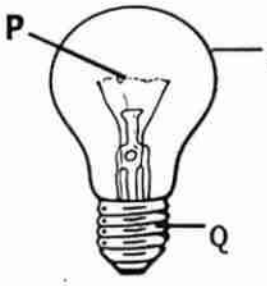
- Electrical energy to heat energy and light energy
- Electrical energy to heat energy
- Heat energy to light energy

Factors/conditions that may make an electric bulb fail to produce light in an electric circuit.

- Poorly placed bulb in its holder.
- Rusty internal metallic plates.
- Poorly arranged dry cells.
- Blown electric bulb.
- Exhausted dry cells.
- Damaged bulb.

Assessment activity:

1. Name the component of a simple electric circuit that changes electrical energy to heat and light energy.
2. Give any **two** conditions that may make an electric bulb fail to produce light in a complete circuit.
3. How do exhausted dry cells affect the working of an electric bulb?
4. State the energy change that takes place in an electric bulb when the circuit is complete.
5. Give any **one** reason why the tungsten filament of some electric bulbs is coiled.
6. Name the mineral used to make the tungsten of the coiled filament of an electric bulb.
7. Why is the inside part of an electric bulb coated with the white powder?
8. The diagram below shows an electric bulb. Study and use it to answer the questions that follow.



a) Name the part labelled **Q.**

b) Why is part labelled **O** made of a transparent material?

c) State the function of part labelled **P** in an electric bulb.

d) Give any one gas enclosed in the part labelled **O.**

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9. Name the material used to make the coiled filament of an electric bulb.
10. Why is oxygen not used in electric bulbs?
11. How is the tungsten filament of an electric bulb able to increase electrical resistance?
12. Why is the tungsten used to make the coiled filament of an electric filament?
13. In which way is the glass envelope able to allow light to the surrounding?
14. Why are argon and nitrogen enclosed in electric bulbs?
15. State the function of the lead wires in an electric bulb.
16. In which part of a bulb is electrical energy converted to heat and light?

Lesson vocabulary

• reflector • enclose • torch • faulty • excess • neutral

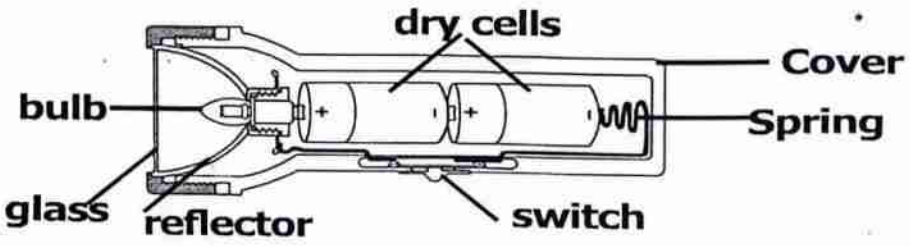
A dry cell powered electric torch.

- An electric torch is a device that uses dry cells in series arrangement to produce light.
- An electric torch works on the principle of a simple electric circuit.
- An electric torch is not fused.

Reasons.

- It uses fixed voltage.
- It uses weak current.

Parts of an electric torch.



Functions of parts of a torch.

Switch: It breaks or completes the circuit.
Electric bulb: It produces light for the user.
Dry cells: They produce electricity in the torch.
Spring: It connects the positive terminal to the negative terminal. It keeps the dry cells tightly fixed and closed.
Cover: It keeps the dry cells tight in the torch.

The cover and spring complete the circuit in a torch.

Reflector: It spreads the light to the glass over a wider area.
Transparent glass: It protects the electric bulb and reflector.

It is transparent to allow light to pass through to the surrounding.

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Factors/ conditions/reasons that may fail an electric torch to produce light.

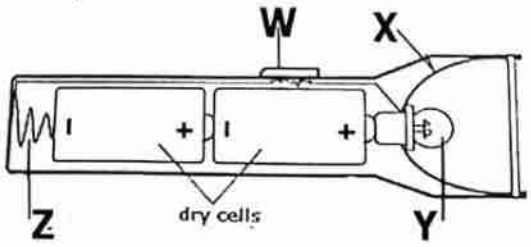
- Used up dry cells.
- Faulty switch.
- Poorly fixed bulb.
- Improper fixing of the cover.
- Poorly arranged dry cells.
- Blown electric bulb.
- Rusty spring.

Energy changes in an electric torch when in use.

- Chemical energy to electrical energy.
- Electrical energy to heat energy and light energy.

Evaluation activity:

1. Name the part in the torch which is not usually found in the torch when it is new.
2. Mention any **two** reasons why a torch may fail to produce light when the switch is turned on.
3. Why doesn't a dry cell powered torch have fuse?
4. On which principle does an electric torch work?
5. The diagram below shows an electric torch. Study and use it to answer the questions that follow.



a) Name the parts marked **X** and **Z**.
b) State the function of part marked **Y.**
c) State the energy change that takes place in the dry cells when part W is closed.

6. Why is the glass of a torch made of a transparent material?
7. State any **two** energy changes that take place in an electric torch when in use.

Lesson vocabulary:

- resistance
- spark
- device
- wiring
- current
- appliance

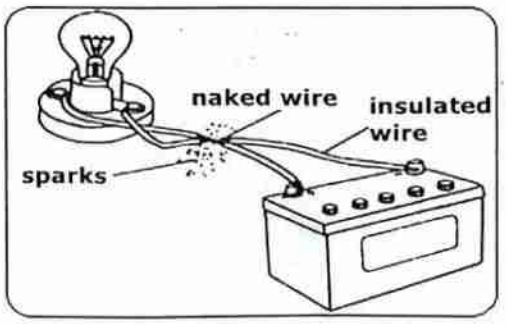
Short circuit.

- A short circuit is a path that gives a lower resistance to the flow of current.

Main cause of short circuit:

- When two bare wires carrying current to opposite directions meet.

✍: *When a spoon or fork is put between two bare conducting wires in an electric circuit, a short circuit occurs.*

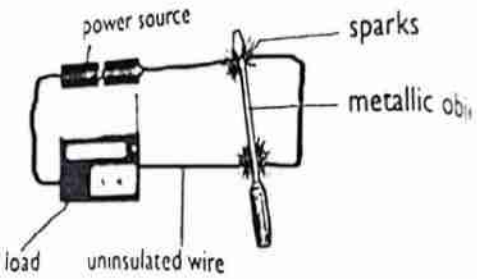


Signs of a short circuit.

- Burnt insulation.
- Smelly smoke.
- Sparks of fire.
- Failure of the appliance to work.

Conditions that may lead to a short circuit.

- Overloading electric sockets.
- Poor wiring when installing electricity.
- Pouring water in sockets and electrical appliances.
- Wrong connection of electric wires in electric appliances.
- Pushing metallic objects in sockets.
- Damaged insulators on electric wires.
- A metallic object falling across uninsulated wires.



Dangers of short circuit.

- Causes electric fires that destroy property.
- May lead to burning of buildings.
- Leads to destruction of electrical appliances.
- May lead to death of people and other animals.

Breaking of power cable lines.

Ways of preventing short circuits.

- Use properly insulated electric wires when installing electricity.
- Avoid overloading the sockets.
- Use fused electrical appliances in homes.
- Replacing electric wires with damaged insulators.
- Electricity should be installed by experts only.
- Repairing faulty/damaged appliances.
- Proper wiring when installing electricity.
- Proper connection of electrical appliances.

Importance of electricity in solving everyday problems.

- Electricity provides heat energy for cooking.
- Electricity provides heat used when ironing clothes.
- Electricity is used to run machines.
- Electricity enables bulbs to produce light.
- Electricity is used in charging phone batteries.
- Electricity is used for powering radios and television.

Advantages of using electricity in solving daily problems.

- Electricity produces clean work.
- Electricity makes work easy.
- Electricity is quick and saves time.
- It reduces the cutting down of trees for wood fuel.
- It does not produce smoke which pollutes air.

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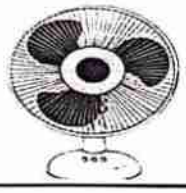
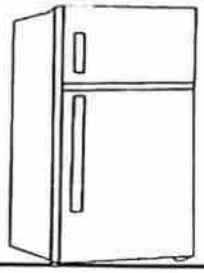
Disadvantages/ dangers of using electricity.

- Installing electricity is expensive.
- It can easily shock and kill people.
- It can lead to fire outbreak.
- It damages appliances in case of short circuit.
- It increases bills when paying for electricity.

Electrical appliances

- These are devices that use electricity in order for them to function

Electrical devices that produce heat	Electrical devices that produce light	Machines run by electricity
electric iron	street lamps	radios
electric cooker	bulbs	televisions
electric heater	torches	printers
hot plate	traffic lights	refrigerators
electric kettle	fluorescent lamps	blenders

Electric fan	Refrigerator	Percolator
		

Electric shock and first aid.

- This is a sudden discharge of electricity through the body.

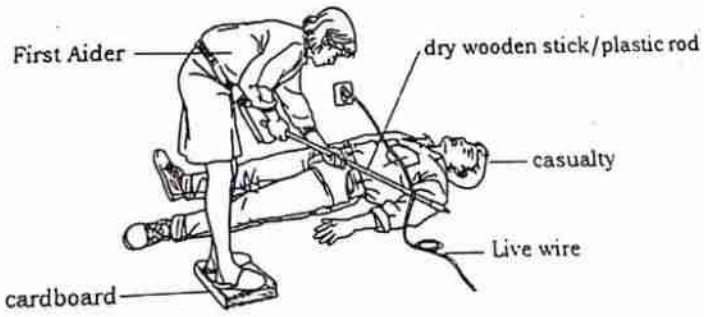
Causes of electric shock.

- Body contact with live electric wires.
- Body contact with a conductor connected to the source of electricity.

✍ ***Electrocution is the sudden injury or death resulting from electric shock.***

First aid for electric shock

- Switch off the source of electricity as soon as possible.
  Reason. To prevent further flow of electric current.
- Use an insulator to move away the victim or to remove live wires from the victim.



☞ *In case the surrounding is wet, the first aider should stand on an insulator.*
Reason. *To prevent him/her from being shocked by electricity too.*

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Safety precautions followed when using electricity and electrical appliances.

- Never touch electric gadgets with wet hands.
- Never touch naked electric wires.
- Never shelter under trees during a thunder storm.
- Never repair electrical appliance when it is still connected to electricity.
- Electrical appliances should be repaired by experts only.
- Never throw objects on power lines.

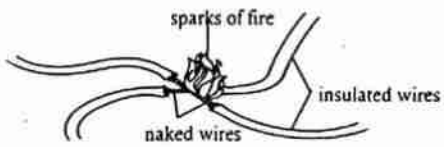
Ways of producing electricity at home.

- Making a simple circuit using dry cells and a bulb.
- Making a wet cell from lemon juice.
- Rotating a dynamo on a bicycle wheel.
- Rubbing insulators against each other.

Electricity is dangerous

Evaluation activity:

1. Give the meaning of the term short circuit.
2. State the main cause of short circuit.
3. Write down any one sign of short circuit.
4. The diagram below shows an electric condition during the flow of electricity along electric wires. Study and use it to answer the questions that follow.



a) Identify the electrical condition above.
b) Give any **two** causes of the electric condition above during the flow of electric current.
c) State any **one** danger of the electrical condition above to people at home.

5. Give any **two** measures you would take to reduce the risks of short circuit at home.
6. Why is it not advisable to touch wires that have fallen off electric poles with bare hands?
7. State the first aid you would give to a victim of electric shock.

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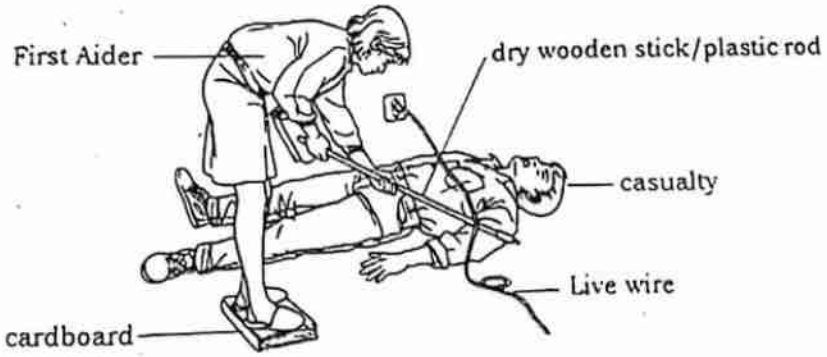
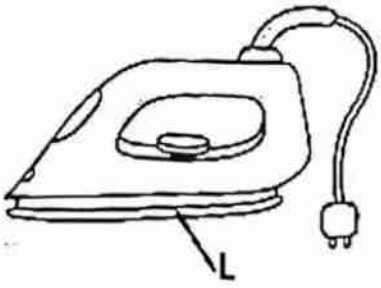
8. The diagram below shows an accident. Study and use it to answer the questions that follow.



a) Name the form of accident shown above.
b) State any **one** cause of the accident above.
c) Why does the first aider using a dry wooden stick to remove the live wire from the casualty?
d) What form of energy causes the type of accident above?

9. Write down any **two** electrical appliances that produce;
i) Heat energy
ii) Light energy

10. State any **four** uses of electricity in our daily life.
11. Give **two** reasons for which electricity is environmental friendly.
12. Mention any **two** sources of electricity.
13. In which way does the use of electricity help to conserve trees in the environment?
14. Give any **two** ways in which the use of electricity is better than the use of wood fuel.
15. The diagram below shows an electrical appliance. Study and use it to answer the questions that follow.



a) What form of energy enables the appliance above to work?
b) How does the electrical appliance above work?
c) In which way is the electrical appliance above useful to people?
d) Name the form of energy produced at part marked **L.**

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MAGNETISM:

Magnetism is a force that makes a magnet to attract magnetic materials.

Magnets:

A magnet is a material which has the ability to attract magnetic substances.

Magnetic material

Magnetic materials are substances that can be attracted by a magnet.

Examples of magnetic materials.

- iron
- nickel
- steel
- cobalt

Non-magnetic materials

Non-magnetic materials are substances that cannot be attracted by a magnet.

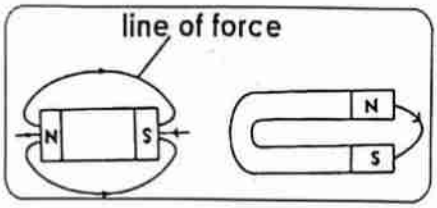
Examples of non-magnetic materials

- copper
- aluminium
- rubber
- wood
- gold
- brass
- glass
- silver
- copper
- rubb
- aluminium
- plastic

PROPERTIES OF MAGNETS

- **Magnet field lines move from the North Pole to the South Pole.**

  Reason. The north pole has a higher concentration of magnetic field lines while the soutl pole has a lower concentration.

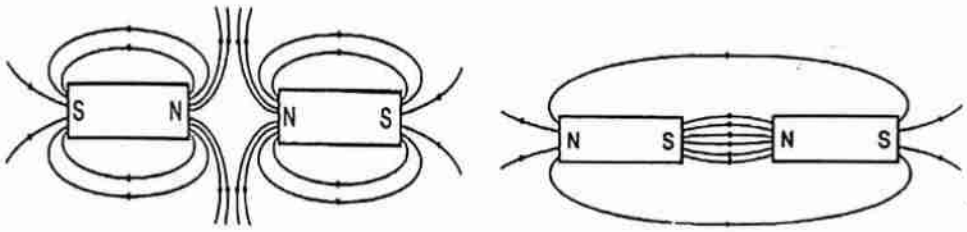


A magnetic field is a region around the magnet where magnetism is detected.

- **Like poles of magnets repel while unlike poles attract each other.**

  ☞ ◇ *Like poles repel becacuse they have similar magnetic fields that are positione in the same direction.*

  ◇ *Unlike poles attract each other because they have different magnetic fields that are positioned in the same direction.*



- **A freely suspended bar magnet rests facing North - South direction.**

  Reason. The north pole of the magnet is attracted by the true south pole of the earth

  ◇ This property of magnets enables compass needle magnets to work.

  ◇ It is used in magnetic compasses to show the direction of places.

  ◇ It is used by pilots, soldiers, sailors, mountain climbers and map readers.

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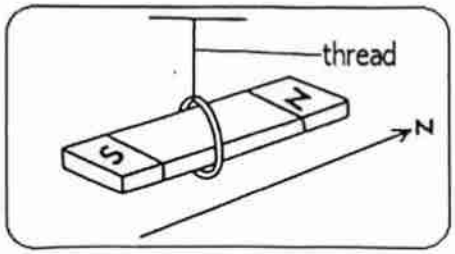


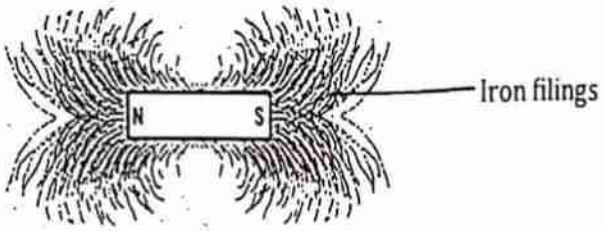
◇ The thread holds the suspended bar magnet freely

- **Magnetism is strongest at the poles of a magnet.**

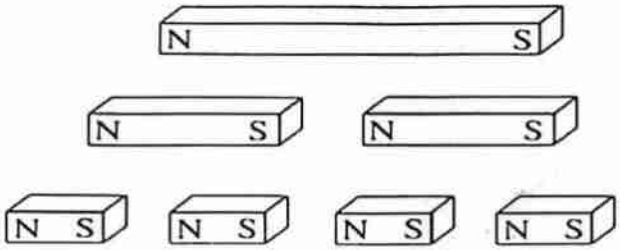
  Reason. *The magnetic field lines are most concentrated at the poles.*

  ◇ *When a magnet is placed in iron filings, it is seen that most of the iron filing are attracted at the poles of a magnet.*

  ◇ *This is because magnetism is strongest at the poles of a magnet.*

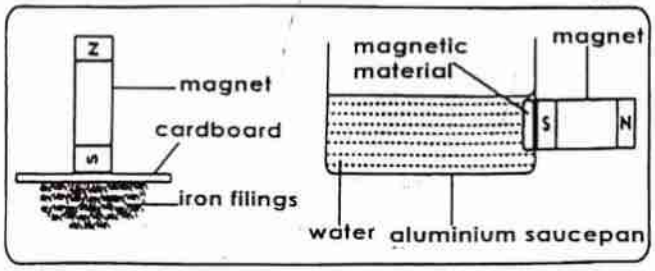


- **Broken pieces of magnets become independent magnets.**



- **Magnetism passes through a non-magnetic material.**

  When a magnetic material is placed under a non-magnetic material, it is attracted by magnet.



Uses of this property of magnets to people.

◇ It helps in separating the mixture of magnetic materials and non-magnetic ones.

These mixtures include;

- *A mixture of sand and iron filings*
- *A mixture of copper and steel*
- *A mixture of flour and iron pins*

◇ It helps in resetting the Six's thermometer.

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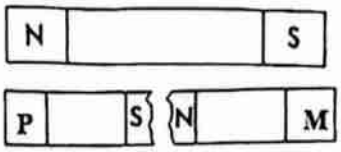
- **Magnets have North Pole and South Pole.**

Evaluation activity:

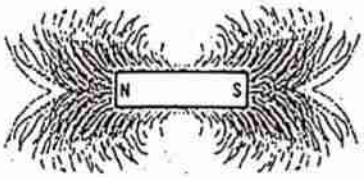
1. In which direction does a freely suspended bar magnet rest?
2. Which property of magnets enables a magnetic compass to work?
3. Give the meaning of the following terms:
   a) Magnet
   b) magnetism
4. State any **two** properties of magnets.

Below is a property of magnets. Use it to answer questions **5** and **6**.

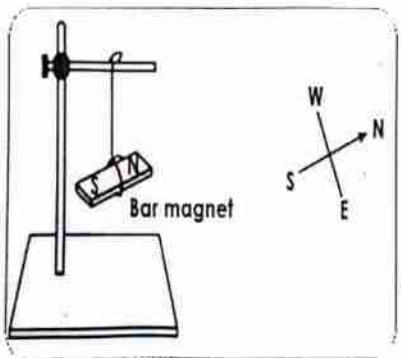


5. Identify the pole marked **M** and **P**.
6. State any **one** property of magnets shown above.

Study the diagram below and use it to answer the questions **7** and **8**.



7. Why are there many iron filings seen at the end points of the magnet?
8. Which property of magnets is shown above?
9. Write down any **one** mixture separated using a magnet.
10. Give the reason why a freely suspended magnet rests facing north-south direction.
11. The diagram below shows a property of magnets. Use it to answer questions that follow.



a) Which property of magnets is shown above?

b) How is the property of magnets above useful to people?

c) Mention any **two** groups of people who use the property of magnets above in their daily work

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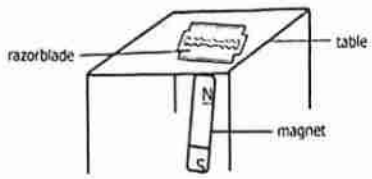
12. The table below shows a list of materials. Study and use it to answer the questions that follow.

Cobalt	glass	iron	copper

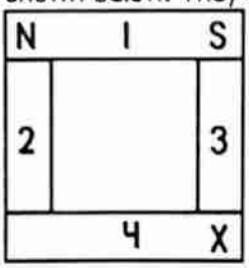
a) Identify any **two** materials from the table above that are;
   i) Magnetic materials.
   ii) Nonmagnetic materials

13. The diagram below shows a property of magnets. Study and use it to answer the questions that follow.



a) State the property of magnets shown in the diagram above.
b) Give any **two** ways in which the property of magnets shown above is useful to people
c) What is the role of the razorblade in the experiment above?

14. Four bar magnets marked **1, 2, 3** and **4** were placed next to each other as shown below. They attracted each other.



Name the pole of magnet **4** marked with letter **X.**

Lesson vocabulary:

• magnetite • magnetic • lodestone • stroking • induced
• electromagnet

Types of magnets.

• natural magnets • artificial magnet

Natural magnets

Natural magnets are magnets that have their own magnetism.
They do not lose their magnetism.

Examples of natural magnets.

• Lodestone / magnetite
• The earth

The earth as a natural magnet.

• The earth is a magnet because it has the North pole and South Pole.
• The earth is a natural magnet because it has its own magnetism.
• The earth has the ability to attract all objects towards its centre by the force called gravity.

Lodestone / magnetite

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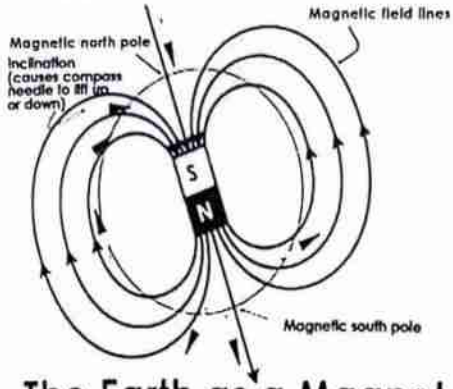


The Earth as a Magnet

Artificial magnets.

Artificial magnets are magnets whose magnetism is influenced by people.

Examples of artificial magnets.

- bar magnet
- needle magnet
- horse shoe magnet

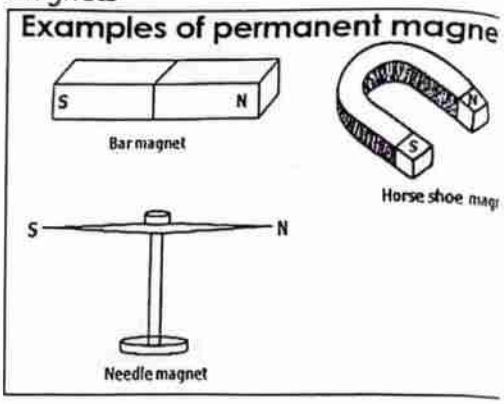
Types of artificial magnets.

- permanent magnets
- temporary magnets

Permanent magnets

- Permanent magnets are magnets
- which retain their magnetism for long.
- Permanent magnets are usually made from steel.

  Reason. Steel keeps magnetism for a long time.
- However, steel is hard to magnetise.



Temporary magnets

- Temporary magnets are magnets which lose their magnetism within a short time.
- These magnets lose magnetism as soon as they are disconnected from a permanent magnet.
- Temporary magnets are usually made from iron.

  Reason. Iron is easy to magnetise.

  However, iron loses magnetism quickly.
- Aluminium cannot be used to make temporary magnets.

  Reason. Aluminium is a non-magnetic material.

Examples of temporary magnets.

- induced magnets
- electromagnets

Magnetisation

Magnetisation is the process of making magnetic materials to acquire magnetism.

Methods of making temporary magnets.

- induction method
- stroking method.
- electrical method

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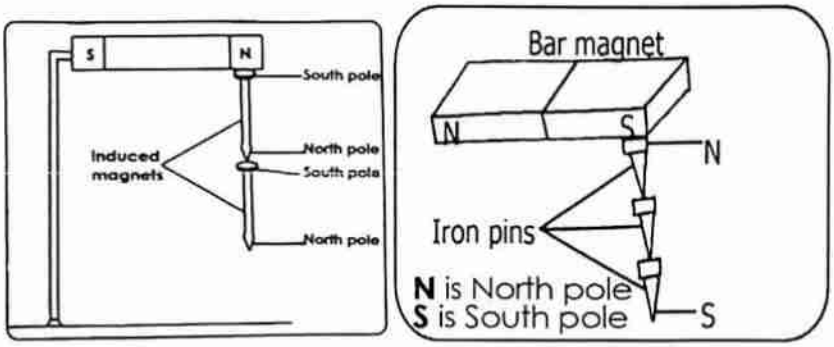
◇ Induction method

Induction is the method of making magnets where a magnetic material is brought closer to the permanent magnet.

- Temporary magnets made by induction method are called **induced magnets.**
- They are temporary because they lose their magnetism easily.
- Induced magnets lose magnetism as soon as they are disconnected from the permanent magnet.
- The polarity of induced magnets is determined by the property; **unlike pole attract each other.**

Magnetisation by induction



Factors that determine the strength of an induced magnet.

- The strength of the magnet used.
- Nature of the material magnetised.
- Number of permanent magnets used.

How to make induced magnets stronger.

- By using a stronger permanent magnet for induction.
- Increasing the number of permanent magnets used to induce the magnetic material.

◇ Stroking method.

This is the method of magnetization where a permanent magnet is passed over a magnetic material severally.

- The magnet produced is an **induced magnet**

  Explanation:

  As you are stroking still the permanent magnet is inducing the magnetic material to gain magnetism hence the magnet formed remains an induced magnet.

  An induced magnet is a temporary magnet made by bringing a magnetic material in contact with a permanent magnet
- Stroking should be done in one direction.

  Reason. To prevent the magnetic material from losing the magnetism it has already gained.

Methods of stroking magnets.

- single stroking method
- double stroking method

Single stroking method.

- This method is also called single touch method.
- During single stroke method, the end of the magnetic substance last stroked becomes opposite to the pole of the magnet used./the end of the magnetic substance first stroked becomes the same as the pole of the magnet used.

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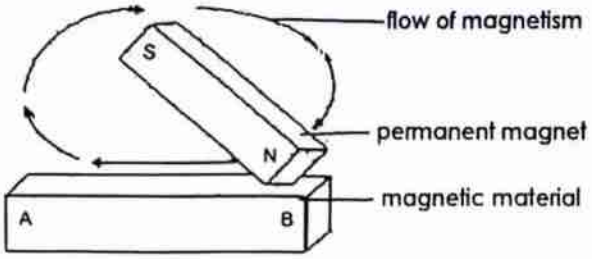
- In the diagram below, **A** is the **south pole** and **B** is the **north pole.**



Factors that determine the strength of the magnet made by stroking.

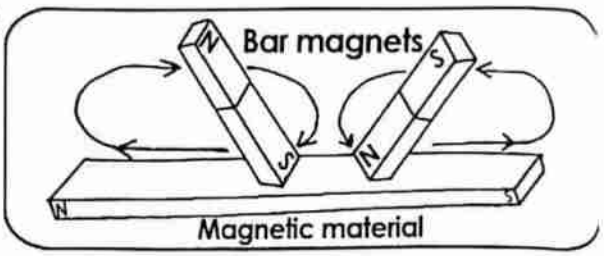
- The number of strokes made.
- Nature of the material magnetized.
- Strength of the material used for stroking.

Disadvantages of using stroking method.

- It is tiring to apply.
- It is time consuming.

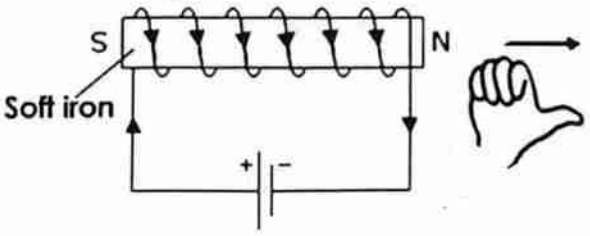
Double stroking method

- This method is also called double touch method.
- In this method of magnetisation, two permanent magnets are stroked the material to be magnetized.
- During double stroke method, opposite poles are produced at the point last stroked/like poles are produced at the poles of the magnetic material first stroked.



Electrical method

- This is the method of making magnets where direct current electricity is passed through a coil of wire wound on a magnetic material.
- The magnet formed is called an **electromagnet.**
- In this method, a magnet is **not** required to induce magnetism in a magnetic material.
- The magnetic material can be a steel meterial or soft iron material.



Explanation:

- **S** is the south pole and **N** is the north pole.
- When the circuit is complete, current flows through the solenoid and the soft iron becomes magnetised (gains magnetism)
- The soft iron loses magnetism when the circuit is broken.

A <u>solenoid</u> is a circular coil of wire in a cylindrical shape.
A <u>coil</u> is a number of turns of insulated wire carrying current

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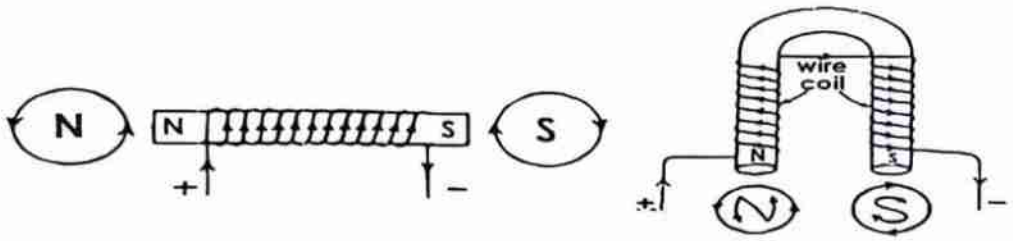
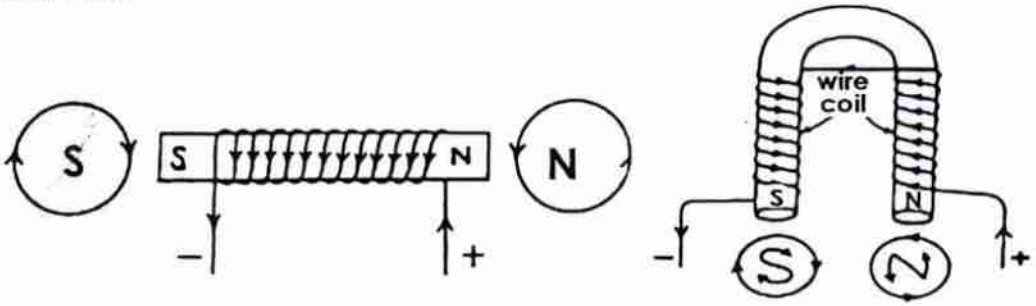
Determining the polarity of an electromagnet using the flow of current:
When current flows clockwise, the point where it enters the solenoid is the South Pole.



When current flows anticlockwise, the point where it enters the solenoid is the North Pole.



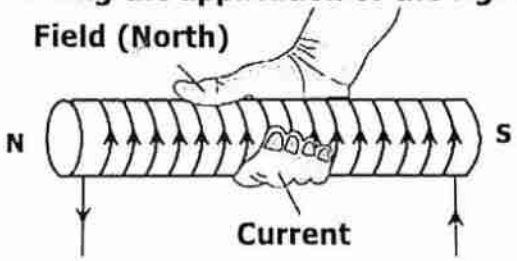
Mnemonic used: **ANCS**
Meaning: **Anticlockwise...Northpole, Clockwise ... South Pole**

Determining the polarity of an electromagnet using the right hand grip rule.

Steps followed

i. Identify the positive and the negative terminals.
ii. Indicate the flow of current from positive terminal to negative terminal through the solenoid.
iii. Fold your fingers of the right hand to move like they are coiling around the solenoid.
iv. Open your right hand thumb wide.
v. Where your thumb points, is the North Pole.

An illustration showing the application of the right hand grip rule.



Demagnetisation

- Demagnetisation is the way of making magnets lose their ability to attract magnetic materials.
- It means weakening magnets.

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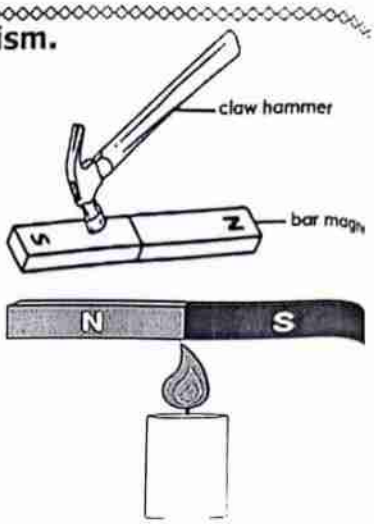
Practices that make magnets lose their magnetism.



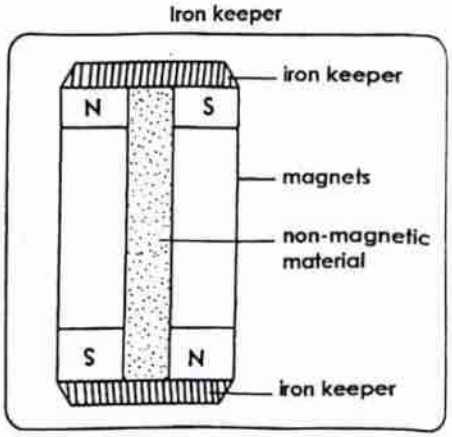
- Hammering a magnet

Hammering the magnet disorganises the magnetic dipoles

- Heating the magnet to red hot.
  - ▹ *Heating the magnet disorganises magnetic domains.*
  - ▹ *It destroys steel.*
- Leaving magnets to rust.
- Hitting the magnet.
- Keeping magnets facing the East - West direction.
- Keeping two magnets with like poles together for a long time.

Ways of preventing magnets from losing magnetism.

- Keeping magnets facing the North South direction
- Keeping magnets with unlike pole together.
- Painting magnets to prevent rusting.
- Keeping magnets away from heat.
- Keeping magnets with iron keepers.



Explanation on the use of Iron Keepers:

- Iron keepers preserve magnetism.
- Iron keeper prevent magnets from losing their magnetism by keeping the magnetism cycle complete.
- The non magnetic material prevents attraction between unlike poles of the two bar magnets.
- A piece of wood can be used as a non magnetic material.

Electromagnets

Electromagnets are temporary magnets made by using electric current that flows through a solenoid.

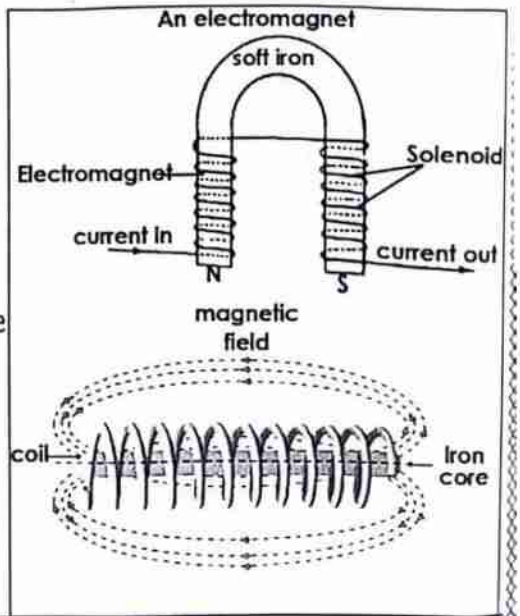
A solenoid is a circular coil of wire in a cylindrical shape.

A coil is a number of turns of insulated wire carrying current.

Electromagnets are made by electrical method.

They are temporary magnets.

Reason. Their magnetism is lost when the current stops flowing through the solenoid.



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How to make an electromagnet.

Materials needed:

- iron bar/ steel bar
- dry cell
- coil of wire
- switch

Procedure:

- Wind a coil of wire around an iron bar or steel rod.
- Make the current flow in the coil.
- When the circuit is complete, the rod will become magnetized.
- When the current stops flowing in the coil, the bar loses magnetism

Devices that use electromagnets:

- electric bell
- sound amplifiers
- electric motors
- generators
- electric alarms
- radio speakers
- electric sirens

Alternating current cannot be used to make an electromagnet.

Reason. Alternating current does both magnetising and demagnetising at the same time.

Diagrams of some devices that use electromagnets.

electric bell	Radio speakers	Sound amplifiers

Uses of electromagnets.

- They are used in electric bells.
- They are used in cranes to lift heavy metallic materials.
- They are used in sound amplifiers and loud speakers.
- They are used in dynamos and generators to produce electricity.
- They are used in circuit breakers.
- They are used in telephone earpiece.

Factors that determine the strength of electromagnets.

- The amount of voltage used.
- The number of turns on the solenoid.
- The material of which the electromagnet is made.
- The distance between the two poles of the magnet.

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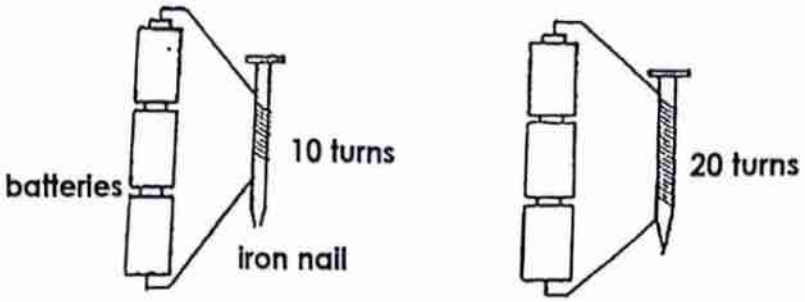
The number of turns on the coil.



Explanation:

The electromagnet in set up **Z** is stronger than the one in set up **Y.**

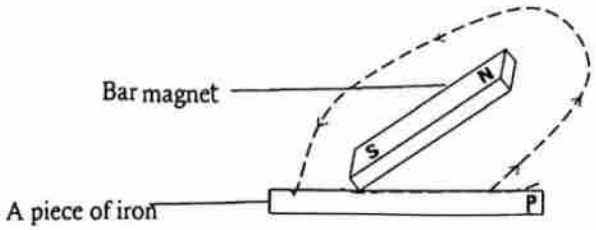
Reason. The number of turns in set up **Z** is more than that in set up **Y.**

Ways of increasing the strength of an electromagnet.

- Increasing voltage in the circuit.
- Increasing the number of wire turns (coils) in the solenoid.
- Placing an iron core inside the coil.

Evaluation activity

1. Name any **two** methods of making temporary magnets.
2. Why is the earth regarded as a magnet?
3. Which method of making magnets does not require the use of a permanent magnet?
4. State any **one** way of increasing the strength of an electromagnet.
5. Why are electromagnets regarded as temporary magnets?
6. Give any **one** factor that determines the strength of electromagnets.
7. Why should a person pass a permanent magnet on a magnetic material in one direction when using stroking method?
8. State any **one** use of electromagnets to people.
9. The diagram below shows a method of making temporary magnets. Study and use it to answer the questions that follow.



a) Name the method shown above.
b) What will be the pole at **P** after magnetization?
c) State any **one** factor that determines the strength of the magnet formed by the piece of iron.
d) Mention any **one** other method of making temporary magnets.

10. State any **two** ways through which a magnet can lose its magnetism.
11. Give any **two** ways of preventing magnets from losing their magnetism.
12. Why can't a metal like aluminium be used to make a temporary magnet?

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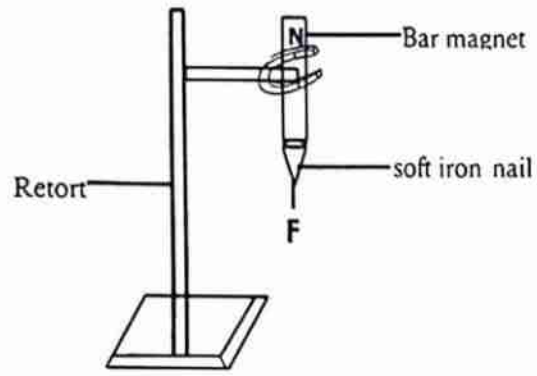
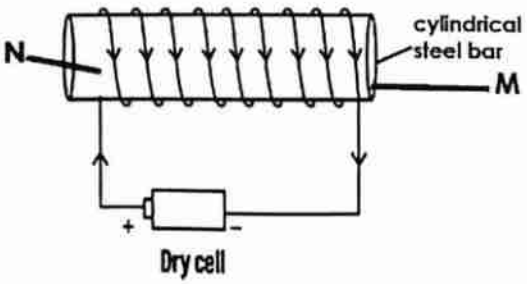
13. The diagram below shows a method of making temporary magnets. Study and use it to answer the questions that follow.



a) Name the method of making temporary magnets shown above.
b) Identify the magnetic pole of the soft iron nail at **F**.
c) Mention any **one** factor which determines the strength of the magnet made by the above method.
d) State the reason why a magnet made using the above method is regarded as a temporary magnet.

14. In which way are the iron keepers useful in the storage of magnets?
15. Name the **two** examples of natural magnets.
16. Write down any **two** examples of magnetic materials.
17. Why don't we use alternating current when making electromagnets?
18. The diagram below shows a method of making temporary magnets. Study and use it to answer the questions that follow.



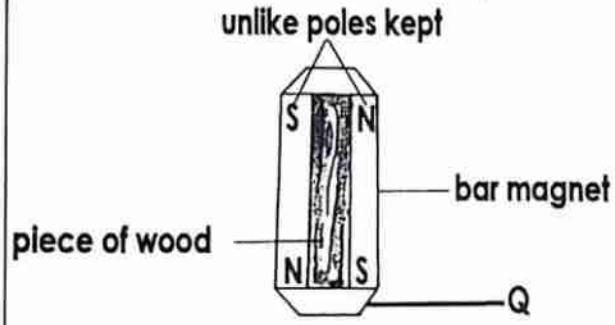
a) Name the method of making temporary magnets shown above.
b) Identify the poles marked **M** and **N**.
c) What happens to the cylindrical steelbar when the circuit is complete?
d) What name is given to the magnet made by the above method?

19. Sternum wanted to make an electromagnet. He had the following materials for the experiment;

an iron bar, a switch, and dry cell.

a) Name **one** material Sternum forgot to include on the list of his materials.
b) What is the use of the dry cell in the experiment?
c) Describe how Sternum will make an electromagnet in this experiment. **(2marks)**

20. The diagram below shows one of the proper ways of storing a bar magnet. Study and use it to answer the questions that follow



a) Name the part marked **Q**.
b) State the importance of the piece of wood on the diagram above.
c) Why is part marked **Q** put on the bar magnets above?
d) Give any one other proper way of keeping magnets.

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Lesson vocabulary:

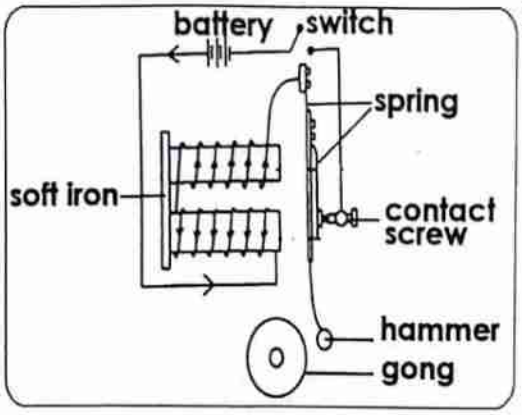
- cutlery
- hammer
- electromagnetism
- dynamo
- rotation
- hit

An electric bell

An electric bell is a device which uses electromagnet to operate a hammer striking a bell.

- It works on the principle of electromagnetism.
- An electric bell is commonly used in schools for communication.

Parts of an electric bell.



Functions of parts of an electric bell.

Battery: It produces electricity.

Soft iron: Becomes a magnet to attract the hammer to hit the gong.

Switch: It breaks or completes the circuit.

An electric bell produces repeated sound when the switch is on.

Reason. *Due to frequent completing and breaking of the circuit.*

Hammer: It hits the gong to produce sound.

Gong: It vibrates to produce sound when hit by the hammer.

Contact screw: It completes the circuit.

Energy change in an electric bell:

Electrical energy to sound energy

Ways in which magnets are important to people.

- Magnets are used to hold cutlery on walls.
- Magnets are used to reset the Six's thermometer.
- Magnets are used to separate some mixtures.
- Magnets are used by doctors to remove iron bits from people's eyes.
- Magnets are used in telecommunication devices.
- Magnets are used to hold doors of refrigerators.
- Magnets are used by mechanics to hold nuts, bolts or tiny pins.
- Magnets are used in cranes to hold scraps.
- Magnets are used in compasses to show direction of places.
- Magnets are used to fasten some bags and wallets.
- Magnets are used in dynamos and generators to produce electricity.

Groups of people who use magnets in their daily activities.

Groups of people	How they use magnets
Doctor	For removing iron bits from people's eyes
Cobbler	For holding tiny nails or pins.
Pilot	For showing direction of places in compass needles.
Mechanics	For holding nuts, bolts or tiny pins.
Cooks/ chefs	For holding cutlery on walls.

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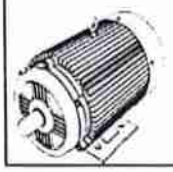
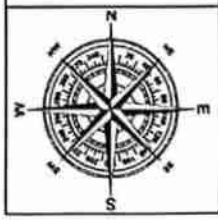
Electricity and magnetism in the modern world of work.

Appliances that use both electricity and magnetism:

- electric bells
- radios
- loud speakers
- electric motors
- microphones
- refrigerators
- sound amplifiers
- generators

electric motors	Sound amplifiers	microphones
		

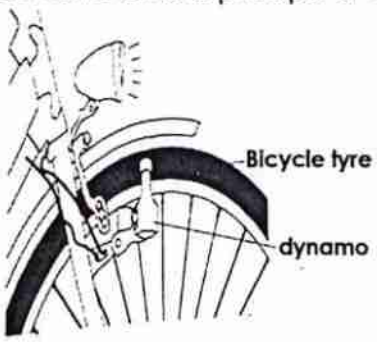
Devices that use magnetism only:

magnetic compass	magnetic tape	telephone receiver
		

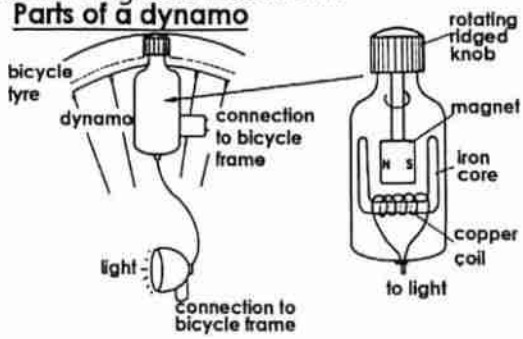
GENERATING ELECTRICITY USING A BICYCLE DYNAMO

A bicycle dynamo is a simple generator which produces electrical energy in the form of direct current.

- A dynamo converts mechanical energy to electrical energy.
- It produces electricity that enables the headlamp to light.
- A dynamo produces electricity by converting mechanical energy to electrical energy/kinetic energy to electrical energy.
- It has a permanent magnet and a coil of wire on the solenoid.
- It works on the principle of electromagnetic induction.



Parts of a dynamo



Functions of some parts:

Dynamo knob: It rotates the magnet in the solenoid.

Bicycle tyre: It rotates the dynamo knob to produce electricity.

Ways of increasing the amount of electricity produced by a dynamo.

- Increasing the speed of rotation.
- Increasing the number of coils in a solenoid.
- Strengthening the magnetic field in the dynamo.
- Strengthening the electromagnet in the dynamo.

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GENERATORS:

A generator is a simple device which produces electrical energy in the form of mechanical energy.

- Generators produce electricity.
- A generator changes mechanical energy into electrical energy.
- Generators produce current electricity.
- Generators can be found at; windmills, hydroelectric power plants, tidal power plants, geothermal power plants, thermal power plants.
- Some generators use petrol or diesel. Such generators produce thermal electricity.

Thermal electricity is a form of current electricity.

A generator

Types of generators.

- a.c generators.
  These produce alternating current electricity.
- d.c generators.
  These produce direct current electricity.

Generators produce large amounts of electricity while dynamos produce small amounts of electricity

Uses of electricity produced by generators.

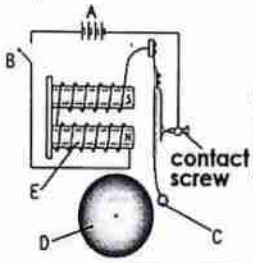
- For lighting places and charging devices.
- For pumping water from underground wells.
- For powering printers and paper scanners.
- For running electrical appliances.
- For powering grinding mills.

How to make a generator to produce more electricity.

- Increasing the speed of rotation.
- Increasing the magnetic field.
- Increasing the number of turns in coils.

Enrichment activity

1. Name the form of energy that enables an electric bell to work.
2. How is an electric bell important at school?
3. Name the temporary magnet used in an electric bell.
4. State the principle on which an electric bell works.
5. What form of energy is produced by an electric bell when the switch is turned on?
6. The diagram below shows an electric bell. Study and use it to answer the questions that follow.



a) Name the parts marked **A** and **C.**
b) What will happen to part **E** when **B** is closed?
c) How useful is part **D** on the electric bell?

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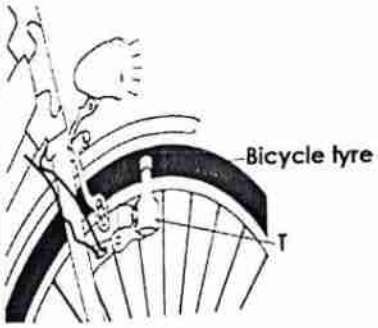
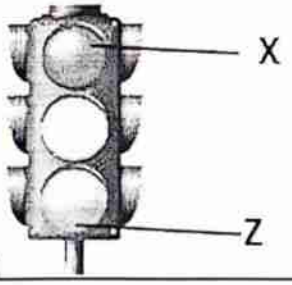
7. Give any **four** ways in which magnets are useful to people.
8. Mention any **two** groups of people that use magnets in their daily activities.
9. Write down any **two** devices that use both electricity and magnetism.
10. In which way are magnets useful to a;
    i) Doctor
    ii) Cobbler
    iii) Meteorologist
11. Mention any **two** groups of people that use magnets to show direction of places.
12. Mention any **two** devices that only use magnetism to function.
13. In which way is a dynamo useful on a bicycle?
14. Name the form of electricity produced by a generator that uses petrol.
15. The diagram below shows an experiment on a bicycle that is used to produce electricity. Study and use it to answer the questions that follow.



a) Name the equipment labeled **T**.
b) Give the function of the bicycle tyre in producing electricity.
c) State the energy change that takes place in the equipment labeled **T** when it is in use.
d) State any **one** way in which the amount of electricity produced by the equipment labeled **T** can be increased.

16(a) Why does an electric bell produce repeated sound during the time the switch is turned on?
(b) What type of artificial magnet is formed when an electric bell is in use?
(c) What energy change takes place in an electric bell when it is in use?
(d) Identify the metal that repeatedly hits the gong in an electric bell to pro duce sound.

17. The diagram below shows traffic light system. Study and use it to answer the questions that follow.



a) Name the colour lights marked **X** and **Z**.
b) What form of energy enables the traffic light system above to operate?
c) State the energy change that takes place in the above traffic light system

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MEANING OF KEY TERMS IN ELECTRICITY AND MAGNETISM

Electricity: a form of energy produced by charged particles.

Fuse: a device that breaks the circuit in case of too much flow of current.

Circuit: a path taken by electricity.

Short circuit: a short path of current flow in an electric circuit.

Solenoid: a long cylindrical coil of insulated wire.

Magnet: a material that has ability to attract magnetic substances.

Electromagnet: a solenoid with core of magnetic material such as soft iron.

Switch: a safety device placed on live wire to turn on or off the circuit.

Grid: the system of overhead wires by which electricity is distributed from the power station to users.

Insulator: a material which allows no electrons (or very few) to pass through.

Conductor: a material through which electrons can flow.

Electric bell: a device which uses electromagnet to operate a hammer striking a bell.

Electrical appliance: a device that uses electricity to function.

Carbon rod: a non-metallic conductor in a dry cell.

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3

THEME: THE ENVIRONMENT
TOPIC 3: ENERGY RESOURCES IN THE ENVIRONMENT

Topical outline:

▹ **Energy resources and their sources.**

- The sun as the major source of energy in the environment.
- Energy resources from water.
- Energy resources from fossil fuels.
- Energy resources from minerals (nuclear fuels).
- Energy resources from plants.
- Energy resources from animals.
- Biogas production.
- Energy resources from wind.

▹ **Importance of energy resources.**

Ref: P.7 standard curriculum page 71.

Lesson vocabulary:

• renewable • hydro • naturally • tidal • satisfy • resource

Energy resources

Energy resources are things which provide useful energy to people.
A resource is anything that people use to satisfy their needs.

Energy resources and their sources.

In this case, a source is a component from which an energy resource is got.

Energy resources	sources
wood fuel	plants
animal energy	animals
Uranium	minerals
tidal energy	water
wind energy	wind
solar energy	sun
Petroleum	fossil fuels

Types of energy resources

▪ renewable energy resources ▪ non-renewable energy resources.

Renewable energy resources

These are energy resources which can be replaced naturally after use.

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Characteristics of renewable energy resources.

- They can be naturally replaced.
- They exist in large amounts.
- They cannot get used up.

Examples of renewable energy resources.

- tidal energy
- wood fuel
- solar energy
- wind energy
- animal energy

Non-renewable energy resources

These are energy resources which cannot be replaced naturally after use.

Characteristics of non-renewable energy resources.

- They cannot be replaced naturally once used up.
- They exist in limited amounts.
- They can get exhausted/finished.

Examples of non-renewable energy resources

- petroleum
- uranium
- coal
- plutonium

Note that all minerals are renewable resources but not all minerals are non renewable energy resources. Refer to PLE 2020 no. 11

The sun as the major source of energy in the environment.

- The sun is regarded as the main and primary source of energy on earth.

  Reason. All forms of energy originate directly or indirectly from the sun.
- The energy resource got from the sun is solar energy.
- Heat energy and light energy are forms of energy obtained from the sun.

Ways in which the sun can be used as an energy resource.

- The sun provides light that enables us to see.
- Heat from the sun dries wet clothes and harvested crops.
- Heat from the sun is used to produce solar electricity.
- Sunlight is used in photography

Harvesting solar energy.

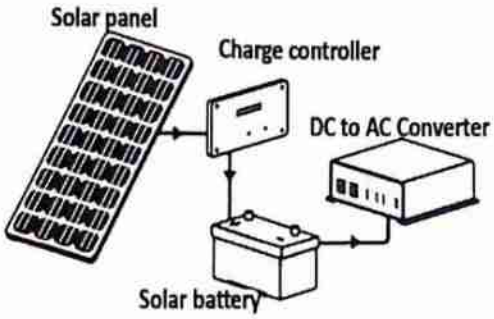
- The solar panels absorb the sunlight.
- Sunlight is changed to solar energy by the solar cells.
- Solar energy is carried and stored in solar batteries in form of chemical energy.
- An inverter changes solar electricity from direct current to alternating current for use.

A solar system

It consists of the;

- solar panel
- solar battery
- solar converter

A solar system



A solar panel

- The solar panel absorbs sunlight.
- It is painted black in order to absorb sunlight.
- It is put on top of the building to access sunlight easily.

The solar cells change sunlight energy into solar electricity.

The solar battery stores solar electricity.

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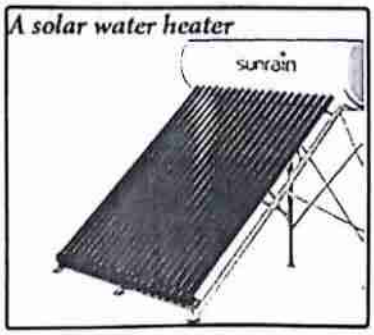
Devices that use solar energy

- Solar lamps
- Solar cookers
- Solar dryers
- Solar water pumps
- Solar water heate

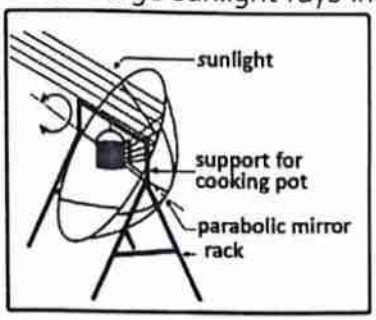
A solar water heater

It produces heat which helps to boil water.



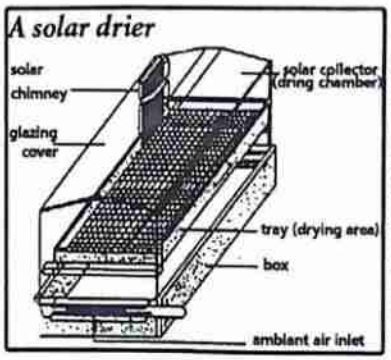
A solar cooker

- It provides heat used to cook food.
- It has a parabolic/concave mirror/reflector to converge sunlight rays into more heat.



A solar drier

It provides heat used to dry some items.
Such items include; grains, vegetables, fruits, meat, fish



Explanation:

- The glazing cover increases the surface area exposed to sunlight.
- The solar drier is raised above the ground to prevent water and small animals from entering it

Points to note:

- *Solar energy is considered as the cleanest source of energy since it does not produce smoke that pollutes the environment.*
- *A solar panel is painted black to easily absorb the sunlight.*
- *A solar panel should be put in an open place to easily access sunlight.*
- *The sun is regarded as a renewable resource.*

Reason: The sun cannot get used up.

Advantages of using solar energy.

- It is generated from a renewable energy resource.
- It does not produce smoke that pollutes air.
- It reduces the rate of deforestation for wood fuel.
- It reduces electricity bills.

Disadvantages of using solar energy.

- It is expensive to buy solar equipment.
- It is a weak source of energy.
- It does not work well during rainy season and weather.
- Installation of solar equipment needs skilled labour.

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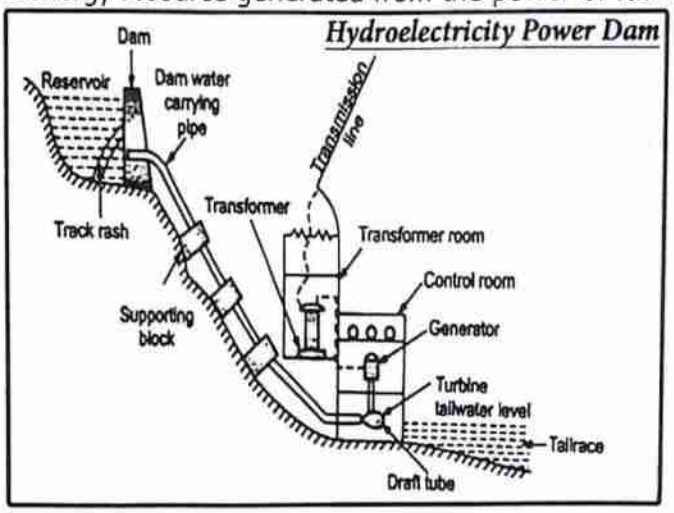
Energy resources from water.

- hydroelectricity
- geothermal energy
- steam energy
- tidal energy

Hydroelectricity

This is the energy resource generated from the power of fast flowing water.



Advantages of using hydroelectricity.

- It is generated from a renewable resource.
- It does not pollute the environment.
- The use of hydroelectricity conserves trees in the environment.

Disadvantages of using hydroelectricity.

- Its generation is not effective in the dry season.
- Dams are expensive to construct.

Factors that affect the production of hydroelectricity.

- Deforestation
- Swamp drainage.
- Presence of water hyacinth.
- Low water levels in the dam.
- Damaged turbines.

Deforestation affects water cycle which reduces the volume of water in rivers

Steam energy

- Steam is the hot vapour from boiling water.
- Steam energy is produced when water is heated at a high temperature.
- Kinetic energy from steam is used to drive turbines for producing electricity.

Uses of steam as an energy resource.

- Steam energy is used to power steam engines.
- Steam can be used to cook food.
- Steam can be used in textile industries to provide heat.

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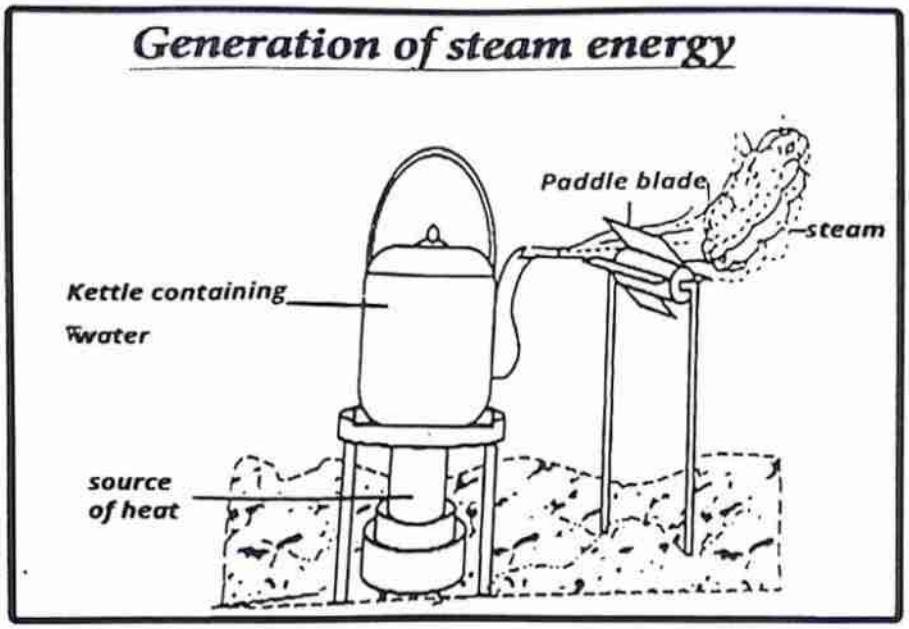
An experiment showing generation of steam energy.



Devices that use steam energy

- Steam boats
- Steam ships
- Steam trains

Tidal energy

- Tidal energy is the energy got from water tides on the sea shores.
- A tide is a regular periodic rise and fall in the level of large water bodies e.g. seas and oceans
- Tides are caused by the attraction of the moon and earth.
- Tides have kinetic energy which helps in turning turbines for producing electricity.

Geothermal energy

This is the energy resource got from hot springs.

Ways in which water can be used as an energy resource.

- Fast flowing water helps to generate hydroelectricity.
- Water tides are used to produce tidal energy.
- Steam energy is used to run steam engines.
- Water from hot springs is used to generate geothermal energy.

Point to note: Water is regarded as a renewable resource.
Reason. Water can be replaced naturally through water cycle

Ways of conserving water as an energy resource.

- Planting more trees.
- Conserving wetlands.
- Controlled cutting down of trees.
- Controlling water pollution.

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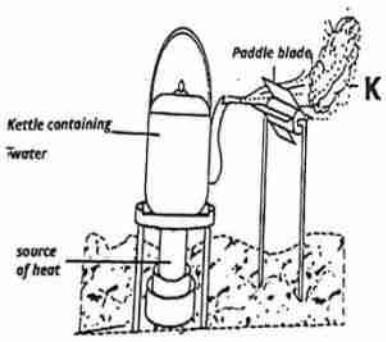
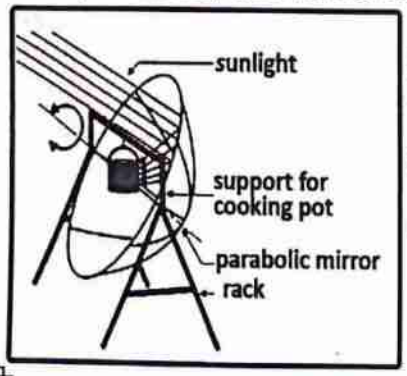
Evaluation activity:

1. Name the source of solar energy.
2. Give any **two** examples of non-renewable energy resources
3. Give the meaning of the following terms.
   a) Resource
   b) Energy resource
4. Why is the sun regarded as the main source of energy on earth?
5. Name any **two** appliances that use solar energy.
6. State any **two** characteristics of non-renewable energy resources.
7. Give a reason why the surface of solar energy is painted black.
8. Mention the energy resource obtained from water tides.
9. Why is water grouped among renewable resources?
10. The diagram below shows an experiment on how a certain energy resource is obtained from water. Study and use it to answer the questions that follow.



a) Identify the energy resource got from the experiment shown above.
b) How is the energy resource obtained from the experiment above useful to people?
c) Name the substance marked **K**.
d) What form of energy helps in the generation of the energy resource got from the experiment above?

11. In which way does the use of solar energy for cooking help to conserve trees in the environment?
12. State any **two** ways in which people use the sun as an energy resource.
13. Give any **two** energy resources got from water.
14. Why are solar panels put on top of buildings?
15. Give any **one** reason why many people do not use solar energy.
16. State any **two** ways in which water can be used as an energy resource.
17. The diagram below shows an equipment that uses an energy resource to operate. Study and use it to answer the questions that follow.



a) Identify the equipment shown above.
b) State the energy resource used by the equipment above.
c) Name the type of curved mirror used in the equipment above.
d) Give any **one** other device that uses the energy resources stated in b) above

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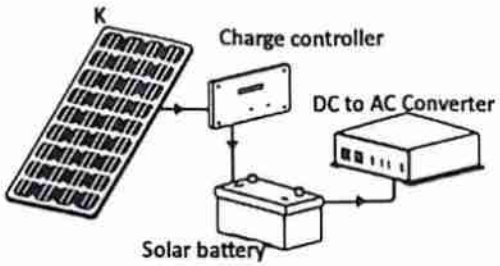
18. Why is the sun regarded as a renewable energy resource?
19. How does deforestation affect the production of hydroelectricity?
20 The diagram below shows devices used in the production of electricity. Study carefully and use it to answer the questions that follow.



a) Identify the device marked **K** in the diagram above.
b) Why is the device marked **K** above usually painted black?
c) State the function of the battery on the solar system above.
d) Name any **one** device that uses electricity produced by the system above.

Lesson vocabulary:

• fractional • petroleum • coal • distillation • fossil • diesel

Energy resources from fossil fuels.

Fossil fuels are energy rich substances obtained from the remains of ancient buried organisms.

They are got from underground **by mining.**

A fuel is any substance burnt to produce heat and light energy.

Energy resources got from fossil fuels are;

- petroleum/ crude oil
- natural gas
- coal
- peat

Petroleum/crude oil

Crude oil is a thick viscous liquid mixture formed from the remains of ancient buried animals.

It is a liquid fossil fuel.

Products obtained from petroleum are;

- petrol
- diesel
- paint
- jet fuel
- grease
- plastics
- polythene
- dye
- fertilisers
- vaseline
- perfume

Note well: Petroleum products are separated by ***fractional distillation/ fractionation.***
Reason: *They have different boiling points.*

Fuels got from petroleum include; petrol, diesel, jet fuel, heating oil and paraffin/kerosene

Uses of petroleum products.

- Paraffin is burnt to produce heat and light.
- Petrol and diesel are used to run engines and generators.
- Jet fuel is used in aircraft engines.
- Vaseline is smeared to make the skin soft and shiny.
- Grease is used as a lubricant.

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Dangers of using petroleum products.

- They are expensive to buy.
- They produce smoke that pollutes air.
- They lead to global warming.
- They lead to competition for oxygen when burnt.
- They can easily catch fire.

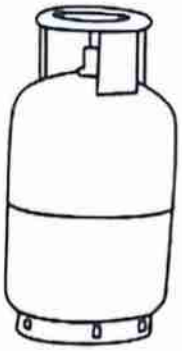
Natural gas

Natural gas is the gas that collects above the underground deposits of crude oil. It usually exists on top of petroleum between rocks.
It is an example of gaseous fossil fuel.

Devices that use natural gas.

- gas cylinder
- bunsen burner
- gas vehicles
- gas lamps

gas cooker



Uses of natural gas.

- It is used as fuel in gas cookers.
- It is used to produce thermal electricity.
- It is burnt in gas lamps to produce light.
- It is used in air conditioning.

Advantages of using gas cookers for cooking.

- It reduces the cutting down of trees for wood fuel.
- It does not produce smoke that pollutes air.

Coal

It is an underground solid fossil fuel formed from the remains of ancient buried plants.

Importance of coal as an energy resource.

- Coal is burnt to produce heat energy.
- Coal is used as fuel in steam engines.
- Coal is used to generate thermal energy.
- Coal is used in blast furnace to melt iron.

Points to note:

- *Coal and peat are solid fossil fuels.*
- *Peat is got from papyrus remains in wetlands.*
- *It can be burnt to produce heat for cooking.*
- *Coal is used to make tar for surfacing roads.*
- *Crude oil is obtained from animal remains while coal is obtained from plant remains.*
- *Fossil fuels produce thermal energy.*

Ways in which fossil fuels can be used as energy resources.

- They are used to produce thermal energy.
- Petrol and diesel from crude oil are used to run engines and generators.
- Paraffin from crude oil is burnt to produce heat and light.
- Natural gas is used as fuel in gas cookers.
- Coal is used in blast furnace to melt iron.
- Peat is burnt to produce heat for cooking.

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Ways of conserving fossil fuel:

- Using fossil fuels sparingly and wisely.
- Using alternative sources of energy.
- Use of renewable energy resources.
- Walking short distances instead of driving.
- Switching off engine after use.

Energy resources from minerals (nuclear fuels).

- uranium
- Plutonium

Importance of uranium and plutonium.

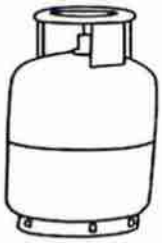
- They are used in nuclear powered submarines.
- They are used to produce atomic energy.
- They are used to make atomic bombs.

Ways of conserving non-renewable energy resources.

- Using fossil fuels sparingly and wisely.
- Using alternative sources of energy.
- Use of renewable energy resources.
- Walking short distances instead of driving.
- Having controlled mining.
- Switching off engine after use.

Enrichment exercise:

1. Name the energy resource obtained from animal and plant remains.
2. Write down any **two** energy resources got from fossil fuels.
3. How is the formation of petroleum different from that of coal?
4. Name the source of thermal energy.
5. Why are petroleum products separated by fractional distillation?
6. In which way is coal useful to road builders?
7. List down any **four** products obtained from petroleum.
8. Which method is used to obtain petroleum products?
9. Give any **two** ways in which fossil fuels can be used as energy resources.
10. State any **two** disadvantages of using fossil fuels
11. The diagram below shows a device used by some people. Study and use it to answer the questions that follow.



a) Name the device shown above.
b) Identify the fossil fuel used in the device above.
c) How is the device above useful to people?
d) In which way is the use of the device above help to conserve trees in the environment?

12. State the source of nuclear energy.
13. Name any **one** example of;
    i) Solid fossil fuel ..... ii) Liquid fossil fuel.....
14. State any **two** ways of conserving fossil fuels.
15. Give any **one** way in which uranium can be used as an energy resource.
15. Give any **one** way in which the following can be used as an energy resource.

(i) Coal .....
(ii) Peat .....

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Lesson vocabulary:

- sawdust
- briquette
- wood fuel
- cart
- biofuel
- ploughing

Energy resources from plants.

- wood fuel
- biofuel

Note well:

Food and biogas are not energy resources from plants.

- *Biogas is obtained from plant residues not directly from plants.*
- *Food from plants does not provide direct energy to people.*
- *Plants are renewable resources because they can be replaced naturally through reproduction.*

Technical advice to teachers:

- What we call energy resource from plants or animals is the direct energy got from plants or animals i.e wood fuel, biofuel from plants and animal energy from animals.
- Therefore, do not teach food as an energy resource from either plants or animals because the concept on this topic does not call for the food value (carbohydrates , energy giving food in the body) got from plants.
  Remember, the theme is **'THE ENVIRONMENT'** not **'HUMAN HEALTH'** where carbohydrates are taught as energy giving food to the body

Wood fuel.

- Wood fuel is any plant material that can be burnt to produce heat and light.
- Wood fuel stores chemical energy.

Forms of energy produced when wood burns.

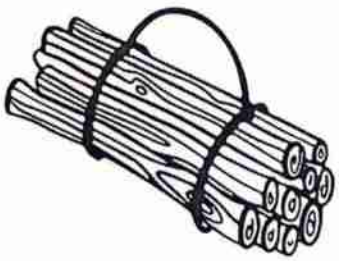
- heat energy
- light energy

Examples of wood fuel.

- firewood
- charcoal
- sawdust
- wood shavings
- wood briquettes

Firewood

This is a form of wood fuel got from trees when they are cut and dried.
Firewood should be split/cut into small pieces **for easy drying**.
Firewood is the commonest and cheapest form of wood fuel in rural area.



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Charcoal

This a form of wood fuel obtained from the logs of wood which are burnt under the limited supply of oxygen in a local furnace.

Formation of charcoal.

- When wood is burnt under the limited supply of oxygen.
- During charcoal formation, chemical energy changes to heat and light energy.
- The logs of wood for burning charcoal are covered with;

Grass. ◇ Grass traps soil to prevent it from falling on burning wood.
◇ Grass allows smoke to escape with minimal heat loss.

Soil. ◇ Soil limits the supply of oxygen to the burning wood.
◇ Soil reduces heat loss from the furnace.

Steps taken when making charcoal

i) Cut trees into logs and pile them into heaps in one place.
ii) Cover the piled logs of wood with leaves/grass.
iii) Cover the wood and leaves with soil.
iv) Set the covered wood on fire.

☞ Note: *Exposing wood to too much oxygen leads to the burning of wood completely to ash.*
Limited oxygen supply prevents wood from burning to ash.

Advantages of using charcoal over firewood.

- Charcoal burns at a higher temperature than firewood.
- Charcoal produces less smoke and ash than firewood.
- Charcoal takes up less space than firewood.

Wood briquettes

- These are compressed bricks or blocks made from combustible biomass.
- They are got from burnt carbonised wood.
- They are used during cooking.
- Briquettes retain heat for a longer time than charcoal.

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Wood briquettes

Materials used to make briquettes.

- charcoal dust
- firewood dust
- clay/mud (For binding to get the required shapes)

Uses of wood fuel.

- It produces heat for cooking.
- It produces heat for artificial brooding.
- Firewood is used provides heat for baking bricks.
- Wood fuel produces heat for smoking or sun drying food.

Disadvantages of using wood fuel.

- Wood fuel produces smoke that pollutes air.
- Wood fuel leads to deforestation.
- Wood fuel is hard to use when wet.
- Wood fuel compete with people for oxygen when burning

Ways of conserving wood fuel.

- Using charcoal saving stoves.
- Using other sources of heat energy.
- Covering food while cooking.
- Use of heat storing equipment.
- Soaking dry /hard food beforecooking e.g. beans.
- Using firewood sparingly.
- Use of heat storing equipment.
- Cooking much food at once.
- Putting out fire after cooking.
- Cutting food into small pieces before cooking.

Reasons for putting out fire after cooking food.

- To save fuel.
- To prevent fire outbreak.
- To prevent burns.

Methods used by cooks to put out fire.

- Pouring water on burning charcoal or firewood.
- Pouring sand on burning charcoal.
- Covering burning charcoal using a saucepan.
- Covering charcoal with ash.
- Using a fire extinguisher.

Covered food during cooking

Banana leaves reduce heat loss from the saucepan. This makes food to get ready faster during cooking. Thus less wood fuel is used.

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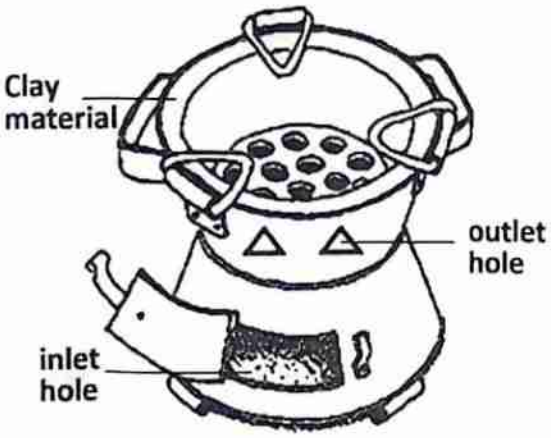
A charcoal saving stove.

It is made of clay lining inside, on top of its sides or the whole body which makes it to use less charcoal..

Advantages of using it.

- It uses less charcoal which helps to reduce the rate at which trees are cut down for wood fuel.
- It retains heat for longer time.
- It produces less carbon monoxide.
- It produces less smoke.



☞ *Points to note:*

- *The air inlet hole allows fresh air to keep charcoal burning.*
- *The inlet hole lets out ash from the charcoal jiko.*
- *The outlet hole lets out warm air from the stove.*
- *It lets out smoke from the charcoal stove.*
- *The clay material absorbs and retains heat for a long time.*

Biofuels

These are fuels obtained from plants or plant wastes.

They are derived directly from living matter.

Examples of biofuels

- bioethanol
- biodiesel
- biogas
- biochar

Bioethanol.

- It is a liquid fuel that is made from sugar cane and other plants such as maize i.e. maize cobs and sorghum.
- It is made through fermentation.
- In this process, carbohydrates are converted into ethanol and carbon dioxide.

$$\text{Carbohydrates} \xrightarrow{\text{fermentation}} \text{ethanol} + \text{carbon dioxide}$$

- The ethanol can be used as fuel in vehicles instead of petrol.

Biodiesel.

- This is a fuel made from vegetable oil or animal fat.
- Biodiesel can be used in vehicle engines.
- It produces less harmful gases to the environment than diesel made from crude oil.

Ways in which plants can be used as energy resources.

- Wood fuel from plants provides heat for cooking food
- Wood fuel from plants provides heat for baking bricks.

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- Wood fuel from plants provides heat for artificial brooding.
- Biofuel from plants is used as fuel in vehicles.
- Wood fuel from plants provides heat for smoking meat or fish.
- Wood fuel from plants provides heat for ironing clothes.
- Biofuel from plants is used to produce biodiesel.

Ways of conserving plants.

- Using charcoal saving stoves.
- Planting more trees.
- Enforcing laws against deforestation.
- Using alternative sources of energy.
- Covering food when cooking. This reduces heat loss from the saucepan.

Energy resources from animals.

Animals provide only animal energy as an energy resource.

Note well:

- *Food and biogas are not energy resources got from animals.*
- *Biogas is obtained from animal wastes but not directly from animals.*
- *Food from animals does not provide direct energy to people.*
- *Animals are renewable resources because they can be replaced naturally through reproduction.*

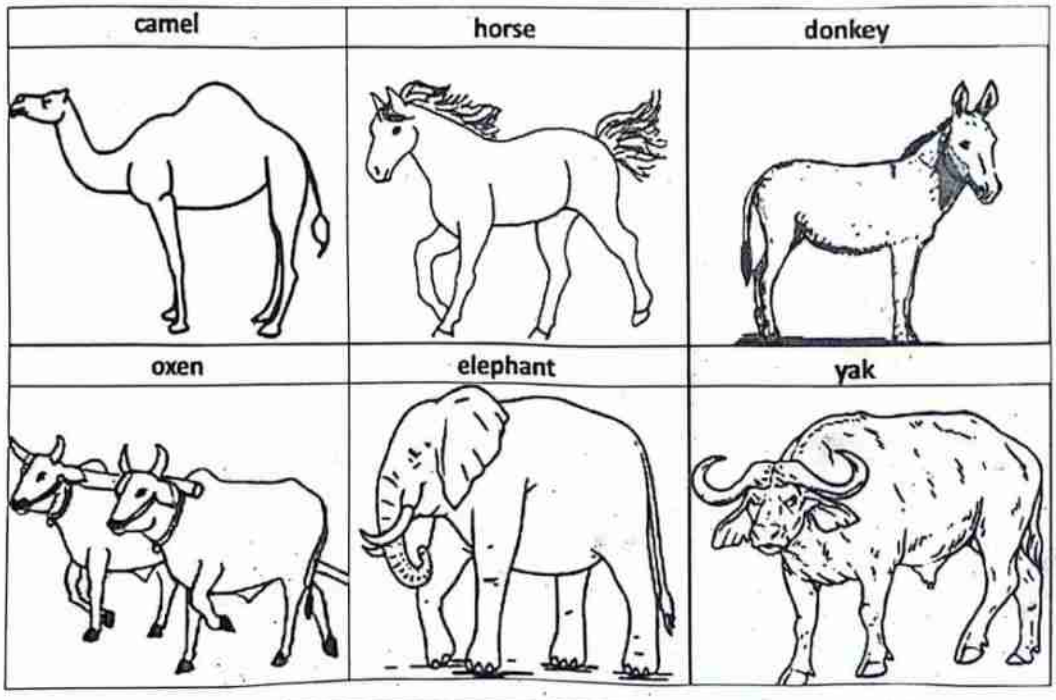
Beasts of burden

These are animals that provide labour to people.

They are used for ploughing, transport, pulling the carts and turning machines.

Examples of beasts of burden.

- donkeys
- camels
- horses
- oxen
- yaks
- elephants



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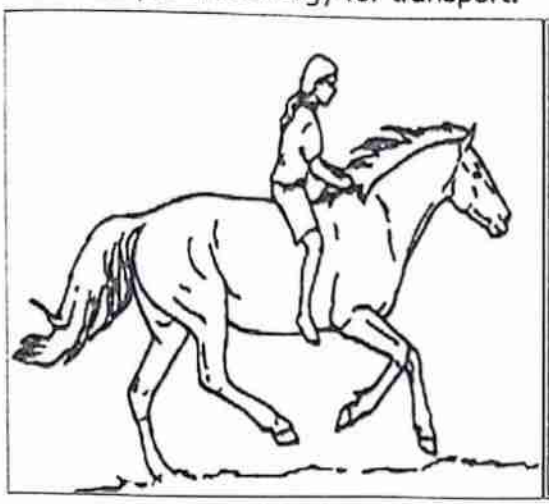
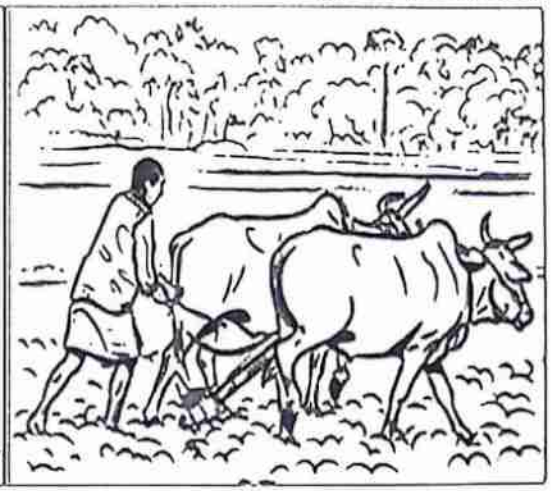
Ways in which animals can be used as energy resources.

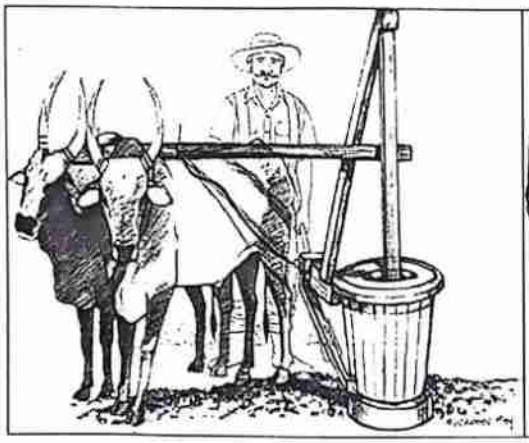
- Animals provide energy for transport.



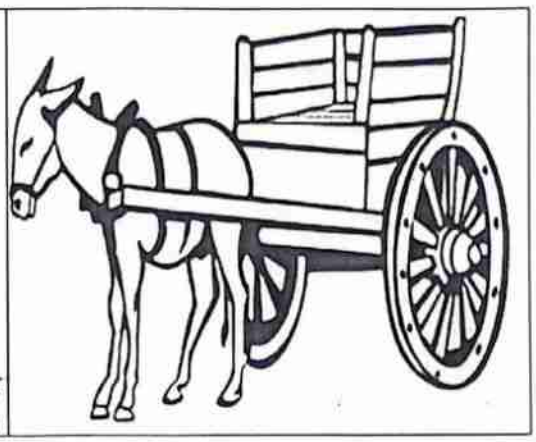
- Animals provide energy for ploughing.



- Animals provide energy for turning machines. e.g. Sugar cane milling on a small scale.



- Animals provide energy for pulling carts.



Reject these points:

- *Animals are source of food.*
- *Animal wastes help in the production of biogas.*

Ref. PLE marking guide and report 2017 no. 32.

Note well: *There is an oversight in the P7 science curriculum on page 71 about some energy resources from plants and animals i.e. biogas and food*

Ways of conserving animals as energy resources.

- Proper feeding of animals.
- Treating sick animals.
- Regular vaccination of animals.
- Controlling poaching.
- Gazetting game parks and zoos.

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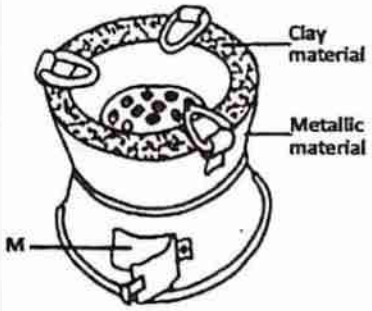
Evaluation exercise:

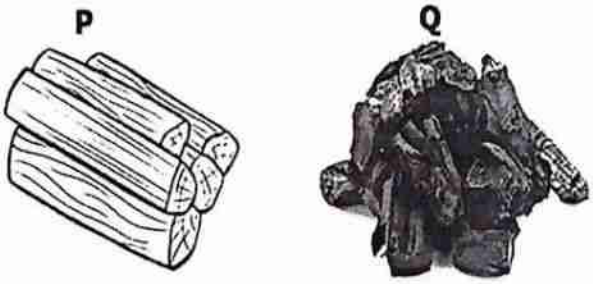
1. (a) What form of energy is stored in wood fuel?
   (b) Name any **one** form of energy produced when wood burns.
   (c) State any **two** practices carried out at home to conserve wood fuel.
2. (a) State the reason why the logs for making charcoal are;
   i) Covered with soil.
   ii) Burnt under the limited supply of oxygen.
   (b) Give any **two** ways in which charcoal burning affects the environment.
3. Give the importance of the clay lining found inside an energy saving stove.
4. How does covering food during cooking help to conserve wood fuel?
5. State **one** advantage of using briquettes over charcoal during cooking.

The diagram below shows a metallic charcoal stove. Study and use it to answer questions **6** and **7**.



6. Give the function of part marked **M**
7. State any **one** advantage of building the inner walls of the charcoal stove with a clay material

8 a) Name the type of fuel obtained when charcoal is burnt.
   b) State any **one** danger of using charcoal for cooking.
   c) Explain how charcoal is formed. **(2 marks)**
9. Apart from charcoal, give any **two** other examples of wood fuel.
10. In which way can plant residues be used as energy resources?
11. Give any **one** reason why most people in rural areas use firewood for cooking instead of charcoal.
12. Name the energy resource obtained when wood is burnt under the limited supply of oxygen.
13. Give any **two** reasons why people in urban areas prefer using charcoal to firewood during cooking.
14. The diagram below shows forms of wood fuel. Study and use it to answer the questions that follow.



a) Name the form of wood fuel marked P.
b) How do people produce wood fuel marked Q?
c) Give any one advantage of using wood fuel marked Q over wood fuel marked P when cooking.
d) In which one way is the use of wood fuel marked P and Q affect the environment?

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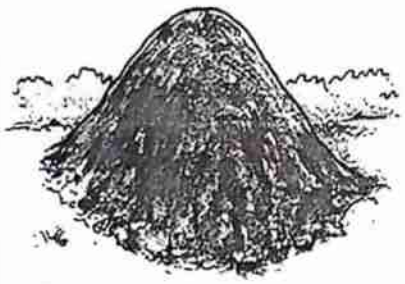
15. What energy change takes place when logs of wood are burnt to produce charcoal?

16. State **one** advantage of using biodiesel over ordinary diesel made from crude oil.
17. Write down any **two** plants which can be used to produce bioethanol.
18. (a) Give any **two** reasons why cooks put out fire after cooking food.
    (b) State any **two** method used by cooks to put out fire after cooking food.
19. Mention any **one** form of biofuel got from plants.
20. Give any **two** ways in which plants can be used as energy resources.
21. Mention any **two** ways in which humans can conserve plant life.
22. Write down any **two** plants from which biofuels can be obtained.

More practice:

1. State any **two** ways in which animals can be used as energy resources.
2. Why are animals referred to as renewable energy resources.
3. Write down any **two** animals that provide labour to people.
4. State **one** way in which animal wastes can be used as energy resources.
5. How is the energy resource got from animals helpful during land preparation?
6. The diagram below shows a furnace used to make wood fuel.
   Study and use it to answer the questions that follow.



   a) Identify the wood fuel made from the furnace above.
   b) Why is the furnace covered with soil?
   c) Give any **one** effect of making the wood fuel identified in (a) above to the environment.
   d) State any **one** way in which people can conserve the wood fuel made from the furnace above.
7. Why are animals referred to as energy resources?
8. The list below shows energy resources in the environment. Study and use it to answer the questions that follow.

   Coal, steam, uranium, charcoal

   a) Name any two energy resources from the list above that cannot be re placed after use.
   b) Identify the energy resource from the list above that is used to produce;
      i) Atomic energy .....
      ii) Thermal energy .....
   c) How is the production of charcoal different from that of steam on the list above?

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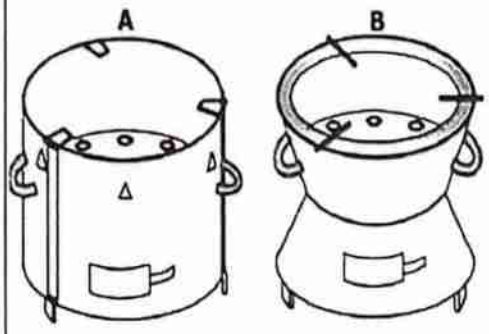
9. The diagram below shows charcoal cooking stoves. Study and use it to answer the questions that follow.



a) Identify the charcoal stove that you would advise people to normally use at home.
b) Why would you advise people to use the charcoal stove you have mentioned in (a) above?
c) State **one** disadvantage of using charcoal stove B.
d) Why is charcoal stove marked A made of clay?

Lesson vocabulary:

• methane • fermentation • biogas • slurry • sludge • biomass

Biogas production.

- Biogas is a gas produced from rotting organic matter.
- Biogas is also called **methane gas**.
- It is produced from **methane** and **carbon dioxide.**
- Biogas is produced by **fermentation** by the action of **bacteria.**
- Bacteria help to decompose slurry in the biodigester.
  Slurry is the mixture of plant and animal matter in the biodigester.
- Decomposition is also involved in the production of biogas.

Biomass is the amount of living matter found in an area.

Plant materials used to produce biogas.

• banana peelings • sweet potato peelings • seed residues.

Animal materials used to produce biogas.

• cow dung • animal urine
• chicken droppings • goat droppings

➢ **Biogas** is produced from an air tight container called a biodigester.

Steps taken when producing biogas.

i) Collect the materials and put them into the biodigester.
ii) Add water to speed up decomposition in the biodigester.
iii) The gas will be produced after decomposition of slurry by the action of bacteria.
iv) The produced gas is trapped by the gas tube to the heating or lighting equipment.
v) Remove the sludge from the biodigester. Sludge is used as manure in gardens.

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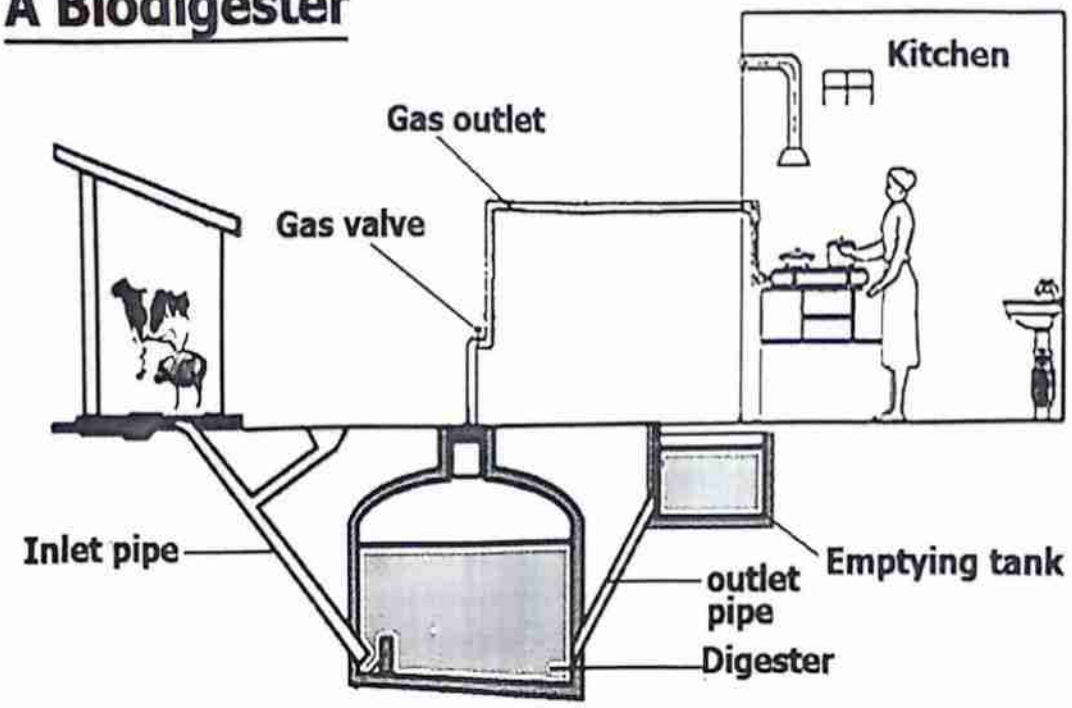
A Biodigester



Functions of parts of a biodigester.

Inlet pipe: For putting wastes in the biodigester.

It should be directed to bottom centre of the biodigester.

Reason. For easy stirring of the wastes in the biodigester.

Outlet pipe: For removing residues from the biodigester.

Gas valve: Controls the flow of biogas.

Biogas digester tank: For holding organic materials.

Gas holder tank: For holding biogas / methane gas before use.

Sludge chamber/ emptying tank/: For holding used up materials from the biogas digester.

Gas outlet/biogas tube: Directs biogas to the heating or lighting equipment.

The water droplets that collect in the biogas tube is removed by using water traps.

Points to note:

- *Water is added in the biodigester.*

Reason. To speed up decomposition in the biodigester.

- *Sludge is the solid residue that remains after the collection of biogas.*
- *Sludge from biogas is used as manure in gardens.*
- *Effluent is the liquid residue that remains after the collection of biogas.*
- *The walls of the biodigester should be made of concrete*

Reason. To prevent leakage of wastes.

- *Materials that would litter the environment can be used in the production of biogas.*

Thus biogas production helps in the promotion of sanitation in an area.

- *Biogas produces thermal electricity.*
- *Biogas is sometimes passed through carbon filters.*

Reason. To remove carbon dioxide.

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Reasons why a biodigester is kept air tight.

- To prevent methane from escaping.
- To prevent the entry of oxygen.

Reasons why the biodigester is buried in the ground.

- To prevent it from exploding/bursting.
- To maintain proper temperature in the biodigester.
- To protect it from physical damage.
- To save space.

Reasons why the biodigester should be built;

A: Far from the kitchen.

- To prevent explosion of biogas.
- To prevent death of bacteria due to heat.

B: Far from trees.

- To prevent roots of trees from damaging the biodigester.

C: Above the water table.

- To prevent slurry from contaminating underground water.

Points to note:

Acids should not be poured in the biodigester.

Reason. Acid kills the bacteria in the biodigester

Uses of biogas to people.

- Biogas can be used for cooking.
- Biogas can be used to produce light.
- Biogas can be used for heating.

Equipment that use biogas.

- biogas cookers
- biogas lamps
- biogas incubators

Advantages of using biogas.

- Biogas does not produce smoke that pollutes the environment.
- Biogas reduces the rate of cutting down trees for wood fuel.
- Biogas is cheaper to use than wood fuel.
- Materials used to produce biogas are readily available.

Disadvantages of using biogas.

- It contains some impurities.
- It is produced on a small scale.
- Biodigesters are less effective in wet seasons.

Ways in which biogas helps to conserve the environment.

- It reduces deforestation.
- It does not pollute the environment.
- Sludge from biogas can be used as manure to support plant growth.
- It is an alternative to wood fuel.

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Energy resources from wind.

- Wind is air in motion.
- Energy resource from wind is called wind energy.
- Wind produces mechanical energy.
- Wind possesses kinetic energy.
- Wind is caused by the differences in atmospheric air pressure.

Ways in which wind can be used as an energy resource.

- Wind energy is used to drive windmills.
- Wind energy helps in winnowing harvested cereals and legumes.
- Wind energy helps to sail boats on waterbodies.
- Wind energy helps to dry wet clothes.

Devices that use wind energy.

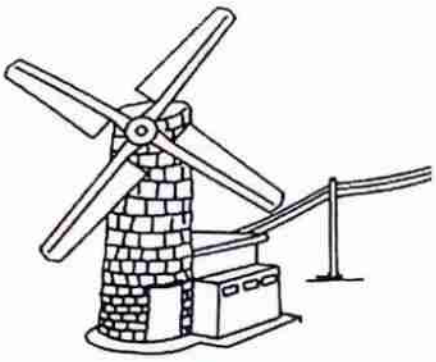
- dhows
- windmills
- air balloons
- parachutes
- weather balloons

dhow	Air balloon	Parachute

A windmill

- A windmill is a machine driven by the force of wind acting on its blades.
- It has propellers/blades which provide kinetic energy in the windmill system by converting wind energy into rotation.
- A windmill is usually raised off the ground.

  Reason. *For easy access to wind.*
- It has a generator that produces electricity.
- A windmill produces electricity used for;
  - Pumping water.
  - Lighting houses.
  - Powering grinding machines.
  - Threshing grains.
  - Cutting wood.



Advantages of using wind energy.

- It reduces the cutting down of trees for wood fuel.
- It does not pollute air.
- It is readily available.
- It is cheaper to use than electricity.
- It is renewable.

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Disadvantages of using wind energy.

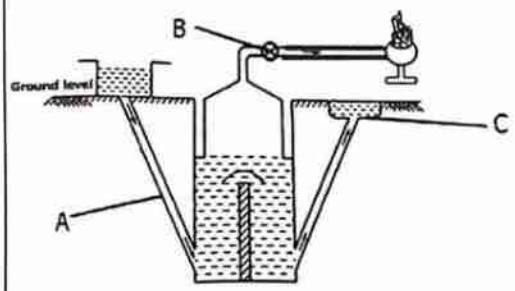
- Wind turbines are expensive to buy.
- Wind energy is only effective on windy days
- Installing windmills needs skilled labour.
- Strong wind capsizes boats on water.

Importance of energy resources.

- Sun heat is used to generate solar energy.
- Fast flowing water is used to generate hydroelectricity.
- Uranium is used to generate atomic energy.
- Plants provide wood fuel.
- Animals provide energy for ploughing land.
- Wind is used to drive windmills.

Evaluation activity

1. Name the energy resource obtained from animal dung and urine.
2. Which process is used in the production of biogas?
3. a) Name any **two** materials used in the production of biogas.
   b) Why should water be added into a biodigester?
   c) State any **two** advantages of using biogas as an energy resource.
4. In which way is biogas production help to promote sanitation of an area?
5. State the role of bacteria in the production of biogas.
6. Give any **two** ways in which people can make use of biogas in their homes.
7. The diagram below shows an equipment used to produce an energy resource. Study and use it to answer questions that follow.



   a) Which energy resource is obtained from the equipment above?
   b) Name the parts marked **A** and **C**.
   c) Give the function of part marked **B** on the equipment above.

8. Give any **one** reason why the biodigester is buried in the ground.
9. Mention any **one** gas used in the production of biogas.
10. How is the production of biogas useful to crop farmers?
11. Why is the biodigester kept tightly sealed?
12. Give **one** way in which the use of biogas is helpful in the conservation of environment.
13. You are provided with the following materials: animal wastes, plant residues, biodigester and water. After collecting animal wastes and plant residues, write **four** other steps you will take to produce biogas for cooking at home.
14. Why is biogas regarded as an environmentally friendly source of energy?

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The diagram below of a device that uses wind to function. Study and use it to answer the questions that follow.

a) Name the device shown above.
b) What source of energy is used to produce electricity in the device above?
c) Give any **two** ways in which electricity produced by the device above is useful to people.

15. Name the energy resource got from wind.
16. What form of energy is produced by wind?
17. State any **two** ways in which wind can be used as an energy resource.
18. Name any **two** devices that use energy resource from wind.
19. The diagram below shows a device that uses an energy resource to function. Study and use it to answer questions that follow

a) Which energy resource enables the device above to function?
b) Give **one** way in which the device above is useful to peaople.
c) Apart from the deveice above, name any **one** other device that uses the energy re sources in (a) above to function

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MEANING OF KEY TERMS IN ENERGY RESOURCES IN THE ENVIRONMENT:

Resource: anything that people use to satisfy their needs.

Biogas: a gas produced from rotting organic matter.

Biomass: the amount of living matter in an area.

Fuel: any substance that can be burnt to produce heat and light energy.

Energy: the ability to do work.

Fossil: remains of dead plants and animals that have been preserved in rocks.

Oxen: cattle kept on the farm purposely for ploughing land

Energy resource: anything that provides useful energy to people.

Tide: a regular rise or fall in the level of a sea or an ocean.

Biofuel: a plant material or animal waste which can be used as fuel waste.

Windmill: a machine used to change wind energy to electrical energy.

Sludge: solid residue that remains after the collection of biogas.

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4

TERM TWO
THEME: MATTER AND ENERGY
TOPIC 4: SIMPLE MACHINES AND FRICTION.

Topical outline:

Friction:

- What it is.
- Friction as a useful force.
- How to increase friction.
- Friction as a nuisance force.
- How friction can be reduced.
- Velocity ratio and efficiency as applied to machines.

Simple machines:

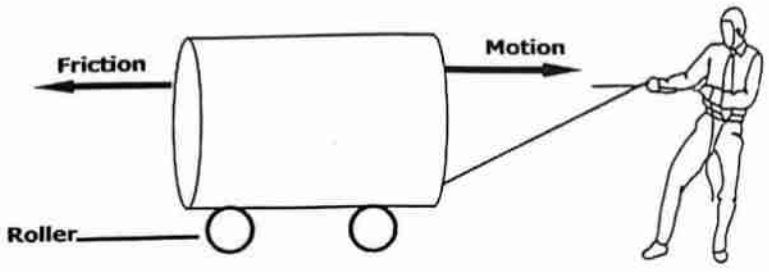
- What they are.
- Characteristics
- Ways in which they make work easier.
- Mechanical advantage of simple machines.
- Levers
- What they are.
- Parts of the lever.
- Classes of the lever.
- Inclined planes (slope/ramp)
- Screws
- Nature of screws.
- Uses of screws
- Pulleys
- Wheels and axles
- Comparison of human body to simple machines

Ref: P.7 standard curriculum page 75 and abridged curriculum.

FRICTION

Friction is the force that opposes motion between surfaces in contact.

How friction acts on a moving object.



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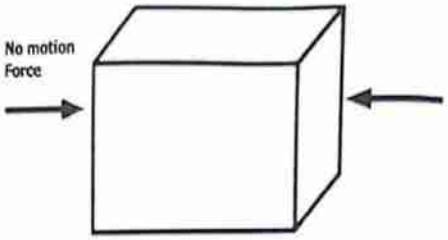
Factors that affect friction.

- The nature of the surface.
- The weight of an object.

Types of friction.

1. Static friction

This is the type of friction experienced by stationary objects.

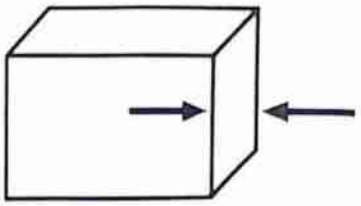


Situations where static friction is experienced.

- A nail fixed in the wall.
- Children standing on the assembly.
- A parked car.
- A book on the table.

2. Dynamic friction

This is the type of friction experienced by objects moving on solid surfaces.
It is also called **sliding friction.**



Situations where sliding friction is experienced.

- Rubbing both palms together.
- Writing
- A moving car, bicycle or motorcycle.
- A rolling stone.

3. Viscosity friction

This is the type of friction which occurs in fluids like gases and liquids.

Situations where viscosity friction is experienced.

- A bird flying.
- An aeroplane flying.
- A fish swimming.
- An insect flying.

Properties of friction.

- Friction increases on rough surfaces.
- Friction generates heat.
- Friction increases with increase in weight of an object.
- Friction occurs in all the three states of matter.
- Friction does not increase with increase in the surface area.

☞ ***Note:*** *Friction is more on rough surfaces than on smooth surfaces.*

Reasons:

Rough surfaces have greater surface area of interlocking than smooth surfaces.
Rough surfaces offer greater adhesion force than smooth surfaces.

Friction as a useful force.

- Friction helps when walking.
- Friction helps when sharpening cutting tools.
- Friction enables bicycles and vehicles to stop when brakes are applied.

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- Friction helps when scrubbing saucepans.
- Friction helps when sweeping rubbish.
- Friction helps when mopping the house.
- Friction helps when brushing teeth.
- Friction helps when lighting match sticks.
- Friction helps when climbing.
- Friction helps when scrubbing the body.
- Friction helps when grinding food or herbal medicine.
- Friction helps when washing clothes.

☞ *Note. Brakes of bicycles are made of rubber blocks. The rubber blocks press on the rim of the wheel when the brake lever is pulled up to stop the bicycle.*

Groups of people who make use of friction in their daily activities.

Group of people	How they use friction.
Pupils / teachers	• It helps when writing.
	• It helps when drawing.
	• It helps when rubbing.
Cooks	• It helps when lighting match sticks.
	• It helps when peeling food.
	• It helps when washing utensils.
	• It helps when mingling posho.
	• It helps when cutting/ slicing food and spices.
Drivers/ cyclists	• It enables them to stop the bicycle or vehicle when brakes are applied.-
Road builders	• It enables them to press the ground to make murram or tarmac stick to the ground.
	• It enables them to hit big stones into small particles.
Builders	• It enables them to break stones for building.-
	• It enables them smooth the floor and walls.
	• It enables them to paint the house.
	• It enables them to mix building materials.
Blacksmith	• It enables them to sharpen cutting tools.
	• It enables them to cut the metals.
Farmers	• It helps when digging
	• It helps when weeding.
	• It helps when pruning.
	• It helps when sharpening the panga and an axe.
	• It helps when harvesting crops.
Carpenter	• It helps when smoothing the wood surface.
	• It helps when cutting timber.
	• It helps when fixing pieces of wood together.

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An experiment to show that friction is a useful force.

Materials needed:

- oil or vaseline
- a piece of paper
- a ballpoint pen

What to do:

- Pour oil or smear vaseline on a piece of paper.
- Use a ballpoint pen to write on the spot where oil has been poured or vaseline has been smeared.
- Record the observation.

Observation and explanation:

- The oil or vaseline makes the piece of paper slippery.
- This reduces friction between the pen and the paper.
- The pen cannot hold onto the paper to write well.

Reason. There is less friction between the paper and the pen.

Conclusion:

This means that friction is needed when writing.

Ways in which friction can be increased.

- Putting treads on car tyres and shoe soles

Treads increase grip between the tyre and the road surface.

- Putting spikes on sports shoes.

Spikes increase the grip between shoe soles and the ground.

- Making smooth surfaces rough.

Roads are tarmacked or murramed to increase friction.

☞ *Note: Car tyres wear out faster on tarmac roads than on murram roads.*

*Reason. **There is increased friction on tarmac roads than on murram roads.***

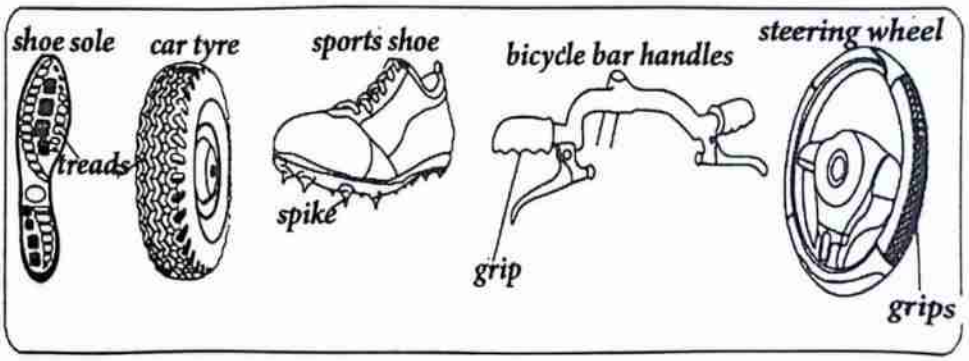
Drivers, cyclists and motorists find it difficult to brake on slippery roads.

*Reason. **There is less friction on slippery roads.***

***A car has treads on its tyres which give it a better grip on the road surface.** This makes a car to move on the road without skidding..*

- Putting grips on bicycle bar handles and steering wheels of vehicles.

Grips increase friction to prevent the user's hands from sliding off.



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Dangers of riding a bicycle with worn out tyres.

- Tyres easily get punctures.
- Tyres easily skid on the road.
- It makes braking of the bicycle difficult.
- It increases the risks of accidents on the road.

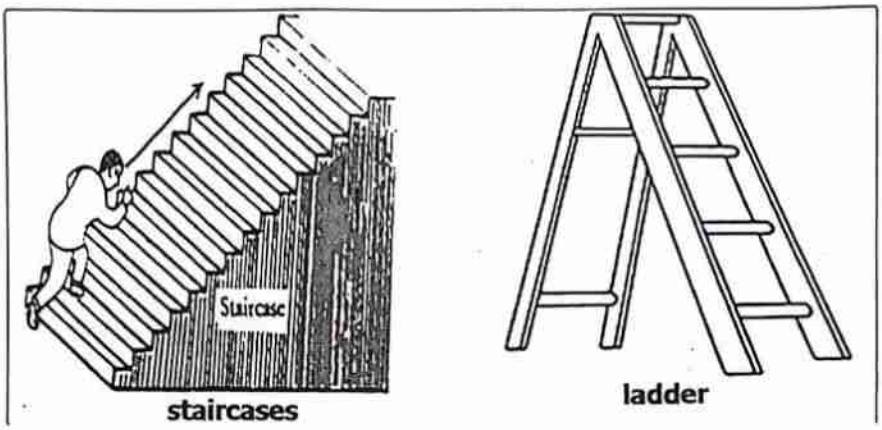
Ways of increasing friction on roads.

- Tarmacking the road.
- Pouring small stones on slippery roads.
- Putting murram on the road.

☞ **Points to note:**

The surfaces of staircases are made rough.

Reason. *To increase friction so that people walking on them do not slide and fall.*



staircases ladder

- *The steps of a ladder have corrugations along its length.*
- *Other ladders are made of rough unplanned timber.*

Reason. *To increase friction and prevent the user from slipping while climbing the ladder*

Materials used to increase friction.

• treads • spikes • grips • tarmac • small stones.

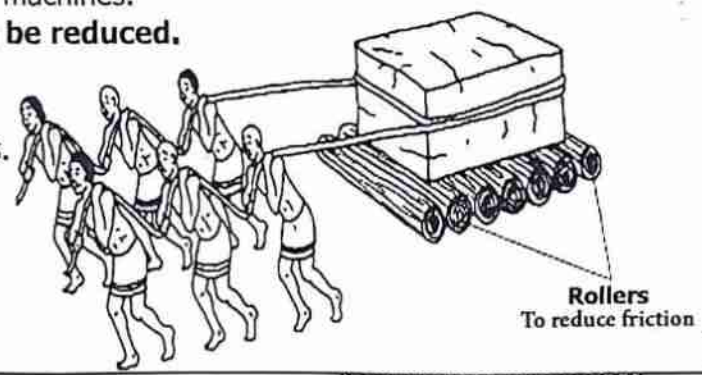
Friction as a nuisance force.

- It causes wear of tyres, wheels and shoe soles.
- It causes unnecessary heat.
- It causes noise in moving parts of machines.
- It reduces the efficiency of machines.
- It retards work done using machines.

Ways in which friction can be reduced.

- Using rollers.

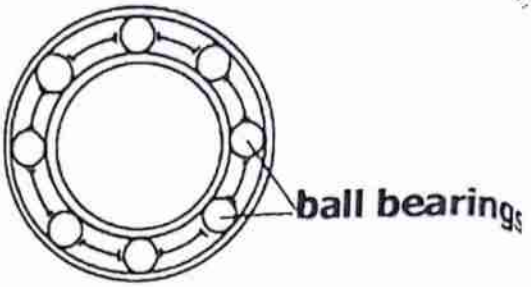
Rollers reduce areas of contact between moving parts.



Rollers
To reduce friction

- Using ball bearings.

They keep moving parts of a machine separate.



- Lubricating moving parts.

This is done by applying oil or grease in moving parts of a machine.

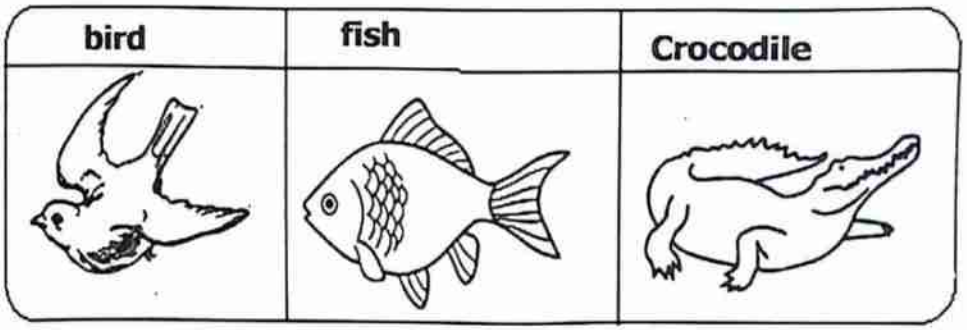
Oil or grease keep the moving parts of a machine slippery.

- Streamlining moving bodies.

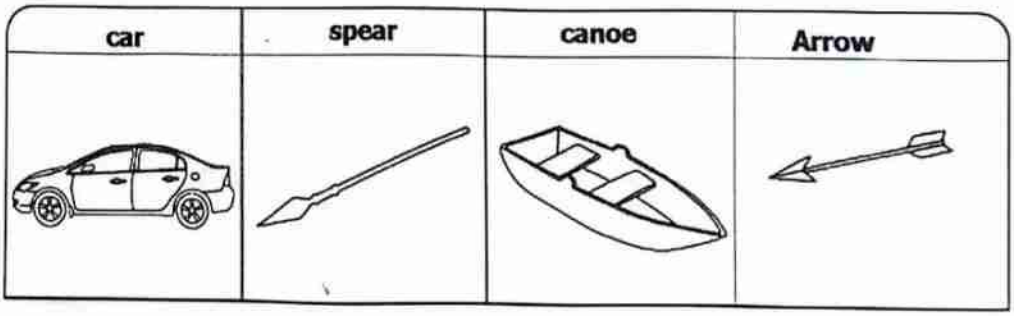
A streamlined body reduces resistance in a fluid.

The body of fish and birds is naturally streamlined to reduce viscosity friction.

Examples of streamlined animals.



Examples of streamlined objects.



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Substance used to reduce friction.

- ball bearings
- grease/oil
- rollers

Substances that reduce friction in the human body.

- synovial fluid
- amniotic fluid
- vaginal fluid
- saliva
- cartilage

Forms of energy produced as a result of friction.

- heat energy
- sound energy
- light energy
- electrical energy

Evaluation activity 1:

1. Why is it difficult for motorists and cyclists to brake on slippery roads?
2. Give any **two** ways in which friction helps when promoting personal hygiene.
3. Give any **two** ways in which friction is helpful to a school cook.
4. How do rubber blocks put near the bicycle rim help in stopping the bicycle when the brakes are applied?
5. Name the force that enables a person to scrub his/her feet using a pumice stone.
6. In which way do ball bearings help to reduce friction in moving parts of a bicycle?
7. Why is grease applied in moving parts of bicycles?
8. Give any **one** danger of riding a bicycle with worn out tyres on the road.
9. Name the force that enables the mother to light fire using a match box.
10. Give a reason why a car is able to move on the road easily without skidding.
11. Whenever Synovial opens his door, the door hinges make noise. What piece of advice would you give Synovial in order to reduce noise in his door hinges?
12. Why are sandpapers made with rough surfaces?
13. Give any **two** ways in which friction is helpful to a P.7 candidate who is writing his/her Primary Leaving Examinations. (P.L.E)
14. Why do road builders pour small stones on murram roads?
15. Give any **two** ways in which friction is a nuisance force.
16. Why do builders make the staircases with rough surfaces when building a house?
17. Give any **one** way in which road builders increase friction on roads.
18. Dr. Viscosity was driving on a rainy day in the morning on a murram road. His car skidded on the road because the road was slippery.

a) Why did Dr. Viscosity's car skid on the road?

b) Name the force which was affected by the slippery road.

19. Write down any **two** pieces of advice you would give Dr. Viscosity in order for his car to move.
20. Why is a boat made of a streamlined body?
21. How are rollers useful in reducing friction?
22. Name any **one** fluid in the human body that helps the body to reduce friction.
23. In which way is the use of the ball bearings in a bicycle similar to that of rollers

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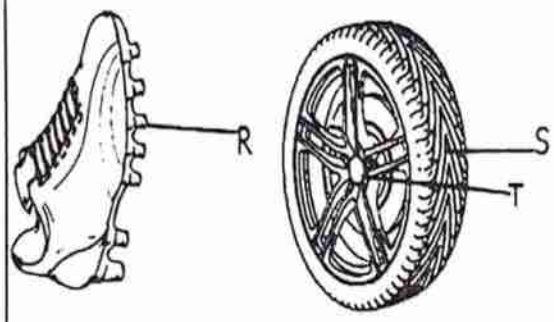
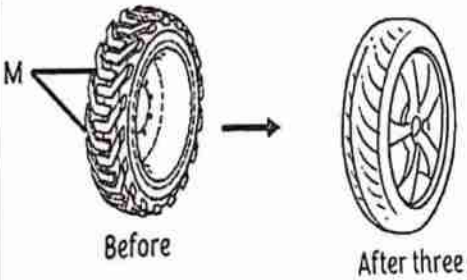
Evaluation activity 2:

1. What is meant by the term friction?
2. State any **one** factor that affects friction in machines.
3. Why does a ballpoint pen fail to write on a piece of paper smeared with Vaseline?
4. Give any **one** way in which friction can be reduced in the moving parts of a bicycle.
5. Write down any **one** example of a lubricant.
6. The diagram below shows a sports boot and a car tyre. Study and use it to answer the questions that follow.



a) Name the structure of the sports boot marked R.
b) State the force which requires putting structures marked **R** and **S** on the diagrams above.
c) In which way is the role of structure **R** on the sports boot similar to that of the structure marked **S** on a car tyre?
d) Give any **one** way in which friction in part marked **T** of the car tyre can be minimised.

7. The diagram below shows a car tyre when new and after three years. Study and use it to answer the questions that follow



a) Name the part marked **M** on the car tyre above.
b) Why are car tyres made with part marked **M**?
c) How is part **M** of the car tyre above similar to the grips on bicycle handles?
d) Why did the car tyre wear out after three years as shown above?

8. Write down any **two** forms of energy produced as a result of friction.
9. Name any **two** substances used to increase friction.
10. Why do car tyres wear out faster on tarmac roads than on murram roads?
11. In which way is an eagle able to reduce friction in air while flying?
12. The diagram below is of a bicycle showing some parts of simple machines. Study and use it to answer the questions that follow.

a) Name the structures on the bicycle handles marked **X**.
b) State the importance of the structures marked **X** to a cyclist.
c) In which way is part marked **Z** able to prevent the cyclist's feet from slipping while riding the bicycle?
d) State any one way of reducing friction in part marked **Y.**

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Lesson vocabulary:

• easier • effort • lever • joule • fulcrum • machine

SIMPLE MACHINES

A simple machine is a device that has few parts and makes work easier.

Characteristics of simple machines.

- They are easy to use.
- They have few parts.

Ways in which simple machines make work easier.

- They reduce the effort needed to do work.
- They increase the speed of doing work.
- They multiply force and distance used to do work.
- They save time used to do work.
- They change the direction of force.

Finding work done by a simple machine.

Work is the product of force and the distance moved in the direction of force.

Work done = force × distance

W.D = F×D

The unit for work done is Joule.

Joule is the work done when force of one Newton moves through the distance of one metre.

Joule = 1N × 1M

Examples:

1. A machine was used to lift a load of 50N through a distance of 4 metres. Calculate the work done by the machine.

Work done = force x distance
= (50 x 4) Joules
Work done = **200 Joules**

2. Find the work done by a man who pushes a wheelbarrow using a force of 30kg through a distance of 7 metres.

1kg = 10N
30kg = (30 x 10)
= 300N
Work done = force x distance
= (300 x 7) Joules
= **2100 joules**

Examples of simple machines.

claw hammer	spade	wheel barrow	hoe

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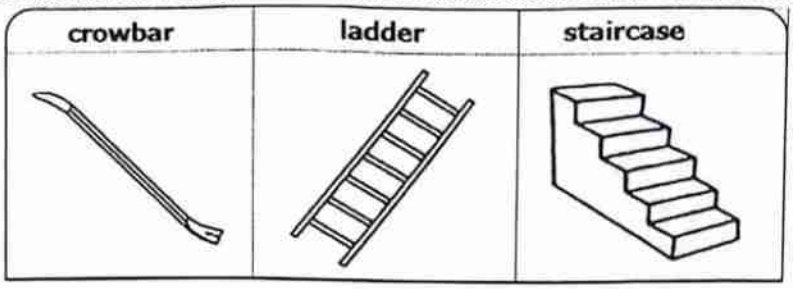


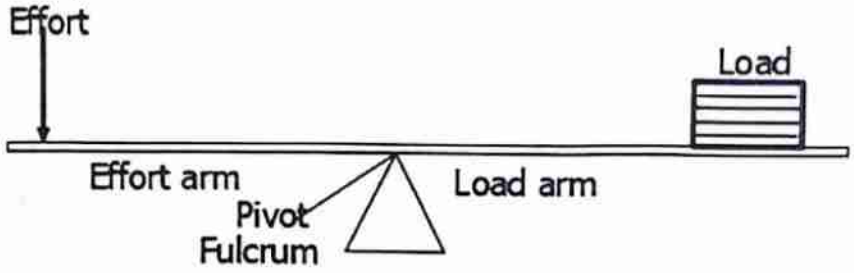
Classes / types/groups of simple machines.

- levers
- inclined planes
- wedges
- screws
- pulleys
- wheels and axles

Levers

A lever is a stiff rod that turns on a fixed point called pivot / fulcrum.
A pivot/fulcrum is the turning point of a machine.

Parts of a lever



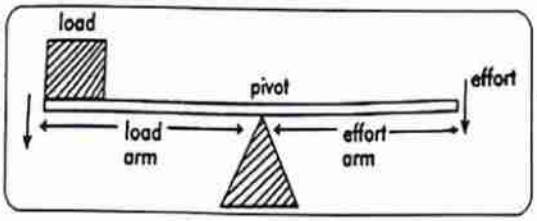
Effort: This is the force applied to lift an object.
Load: This is the object to be lifted when force is applied.
Fulcrum: This is the turning point of a lever.
Load arm: This is the distance of the load from the fulcrum.
Effort arm: This is the distance of the effort from the fulcrum.

Classes of levers

A class of levers is determined by the position of the pivot, load and the effort.
The classes of levers are; first class levers, second class levers and third class levers

First class levers

These are levers in which the fulcrum is in between the load and effort.
Formula: LPE i.e. Load, Pivot and Effort



Ways in which first class levers simplify work.

- By reducing the effort needed to do work.
- By changing the direction of force.

Point to note: *In the first class lever machines, less effort is used when lifting the load.*
Reasons. The effort arm is longer than the load arm.
The load is nearer to the pivot than the effort.

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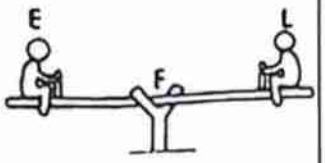
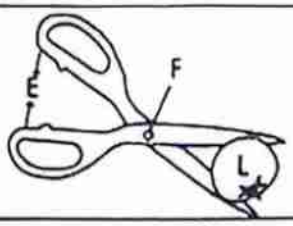
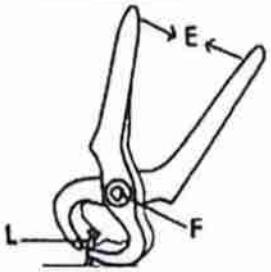
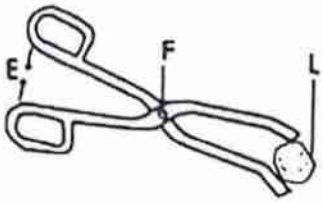
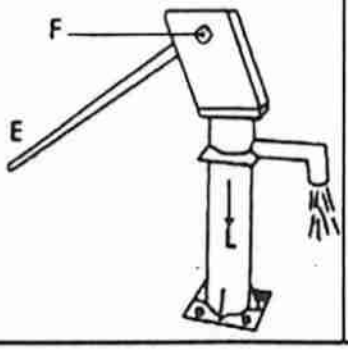
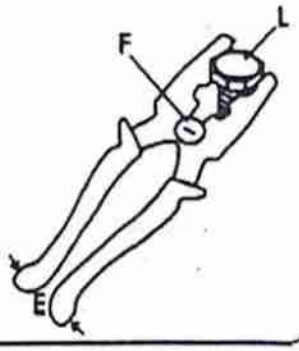
see-saw	pair of scissors	crowbar
		
pincer	**laboratory pair of tongs**	**claw hammer**
		
beam balance	**borehole**	**pair of pliers**
		

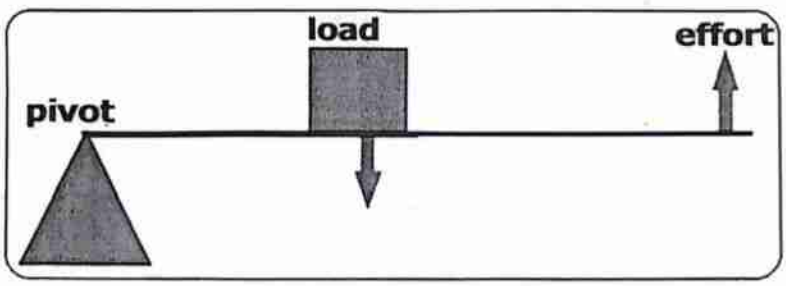
☞*Note: a pair of scissors and a pair of pliers are grouped under double levers.*
<u>Reason</u>. They have two stiff rods that turn on the same fixed point

Second class levers

These are levers in which the load is in between the pivot and effort.

Mnemonic: PLE i.e. Pivot, Load and Effort

- In this class of levers, the load is nearer to the pivot than the effort.
- Second class levers simplify work by making the load and effort move in the same direction.



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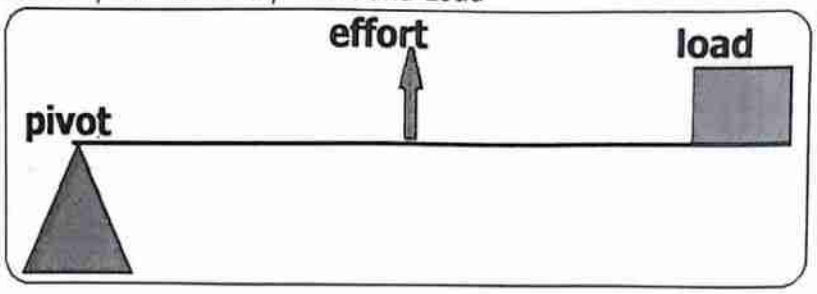
Examples of second class levers

wheelbarrow	human foot
bottle opener	**nut cracker**

Third class levers

These are levers in which the effort is in between the fulcrum and the load.

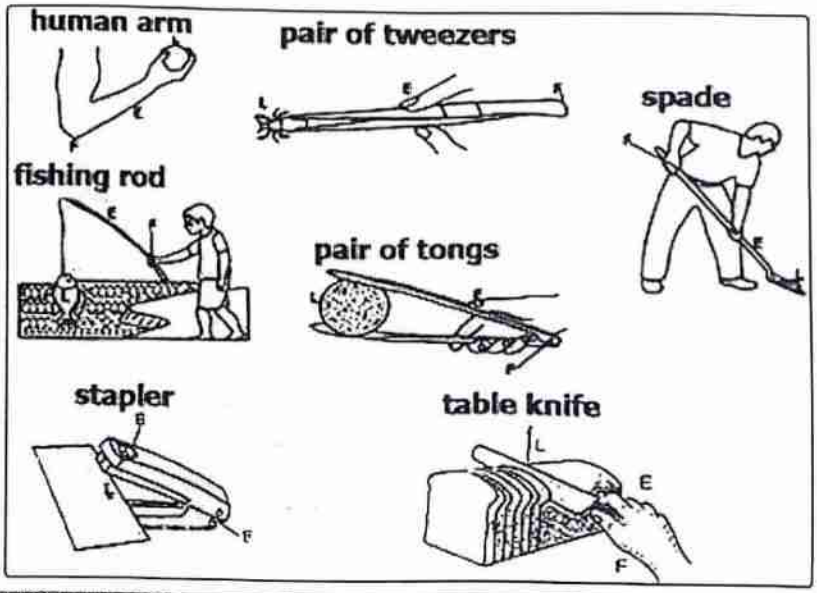
Mnemonic: PEL/LEP i.e. Pivot, Effort and Load



- In this class of levers, the load distance is greater than the effort distance.
- More effort is used to overcome the load but the load moves through a longer distance.
- The advantage of using third class levers is that they move the load over a longer distance.
- They simplify work by increasing the speed of doing work.

Examples of third class levers

- broom
- hoe
- pair of tongs
- fishing rod
- spade
- human arm
- tweezers
- stapler
- table knife



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Note: Third class levers are referred to as distance multipliers.

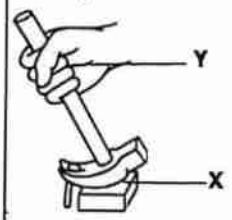
Reason. They move the load over a longer distance

Ways in which levers can be used in our daily life.

- For lifting objects using a crowbar.
- A seesaw is used as a play material.
- For holding burning charcoal using a pair of tongs.
- For cutting papers, bandages and clothes using a pair of scissors.
- For removing nails from wood using a claw hammer.
- For fishing using a rod.
- For opening a bottle using a bottle opener.
- For transporting materials using a wheelbarrow.

Evaluation activity:

1. What name is given to the turning point of a machine?
2. State any **two** ways in which simple machines make work easier.
3. Why is a razorblade referred to as simple machine?
4. A girl applies force of 30N to lift her bag 2 meters up. Calculate the work done by the girl.
5. The diagram below shows a simple machine. Study and use it to answer the questions that follow.



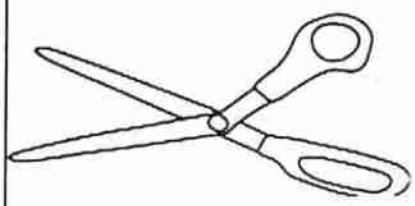
a) Name the simple machine above.
b) Identify the parts of the simple machine above found at **X** and **Y.**
c) How is the simple machine above useful to builders and carpenters?

6. Apart from levers, mention any two other classes of simple machines.
7. To which class of lever do the following belong?

a) Wheelbarrow
b) Crowbar
c) Spade

8. Mention any **two** levers that are force multipliers.
9. Give **one** reason why less effort is applied to move a load using first class levers.
10. Why is a pair of scissors grouped under double levers?
11. State **one** advantage of using third class levers.

The diagram below shows a simple machine. Use it to answer questions **12** and **13.**



12. To which class of levers does the simple machine above belong?
13. Use an arrow with letter **K** to show the fulcrum on the diagram.
14. In which **one** way do nursery children benefit from a seesaw placed in the play ground at school?

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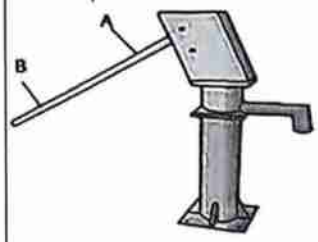
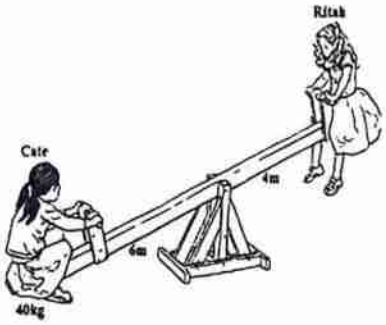
15. How do second class leavers simplify work?
16. Why is a nut cracker classified under second class levers?
17. Give any **two** ways in which people use levers in their daily life.
18. State any **one** reason why you would call a stapler a simple machine.
19. The diagram below shows a borehole. Study and use it to answer the questions that follow.



a) Name the group of simple machines to which the borehole above belongs.
b) Identify the point at which the user will use less effort.
c) Give a reason for your answer in b) above.
d) On the diagram above, show the position of the load.

20. The diagram below shows Cate and Ritah swinging on a simple machine. Study and use it to answer the questions that follow.



a) Name the simple machine above.
b) Which of the two girls is heavier?
c) Give a reason for your answer in (b) above.
d) What should the girls do in order for the machine to balance?

21. In the table below, part **A** shows terms used in levers and part **B** has the meanings of the terms.

Part A: Terms	Part B: Meanings
Fulcrum	An object to be lifted.
Load arm	Turning point of a lever.
Effort	Distance from the lever to pivot.
Load	Force applied to lift an object.

Use the meanings in part B of the table to match the terms below.
a) Fulcrum .....
b) Load arm .....
c) Effort .....
d) Load .....

Moment of force
A moment of force is a turning effect of force on a machine.

The principle of moments
The principle of moments states that; when a body is balanced, the sum of clockwise moments is equal to the sum of anticlockwise moments.

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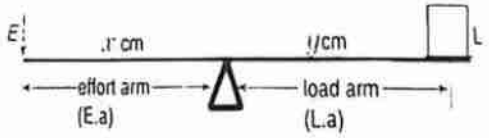
Applications of the principle of moments.

- It is used in a beam balance to measure the weight of objects.
- It used in a seesaw to balance the forces on either side.
- It enables the door to turn the hinges when closing or opening.
- It is used in crowbars when lifting objects.
- It enables mechanics to turn nuts and bolts using spanners.

Calculations on moments.



(E x x(E.a)) = (L x y(L.a))

Load X Load arm = Effort X Effort arm

Correction on concept delivery when calculating forces on moments:

In moments, when asking to calculate/find the force in kilograms, make sure that you ask the learner to find/calculate the mass of an object or body in kg or kgf because it is mass which is measured in kilograms ***not*** *weight. Remember weight is measured in Newtons not kilograms.*

In case you want weight, make sure that the units given are in Newtons (N) ***not*** *in kilograms yet you want the learner to find the weight. i.e.*

1. *Given in the question or on the see-saw:*
   Load is 60kg and effort is K
   The question set should be; Find the mass at K ***not*** *Find the weight at K*
2. *Given in the question or on the see-saw:*
   Load is 40N and effort is Z
   The question set should be; Find the ***weight*** *at Z* ***not*** *Find the mass at Z.*

In summary:

Mass is measured in ***kilograms***

Weight is measured in ***Newtons.***

Clear reference*. PLE 2022 question* ***44b****. i.e. the question was;*

Find the ***mass*** *of Jjuko if the see-saw was balanced* ***NOT***

Find the ***weight*** *of Jjuko if the see-saw was balanced.*

1. Study the diagram below carefully and find the value of **Y.**

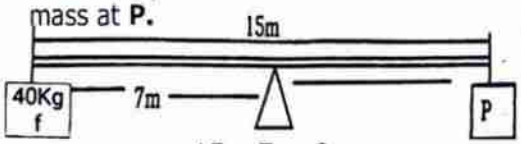
L x L.A = E x E.A

60x 2 = 30 x Y

$\frac{120}{30} = \frac{30}{30}$

Y = 4 metres

2. Use the seesaw below and calculate the mass at **P.**



15 – 7 = 8

L x L.A = E x E.A

40 x 7 = P x 8

$\frac{280}{8} = \frac{8P}{8}$

P = 35kgf

The mass at P is 35kgf

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3.Ligament is 35kgf. If she sits 2 cm away from the fulcrum and balances with Cartilage who sits 5 cm away from the fulcrum, what is Cartilage's mass?

$L \times L.A = E \times E.A$

$35 \times 2 = E \times 5$

$\frac{70}{5} = \frac{5E}{5}$

$E = 14kgf$

The weight of Cartilage is 14kgf.

4. Pupils, Nectar and Propolis sat on a see-saw. Nectar who weighs 40kg sat 2.5 metres away from the pivot on one side of the see-saw. Propolis sat 4 metres away from the pivot on the other side. Find the mass of Propolis if the see-saw was balanced. Let the mass for Propolis be X.

$L \times L.A = E \times E.A$

$40 \times 2.5 = X \times 4$

$40 \times 25 = 4X$

$\frac{100}{4} = \frac{4X}{4}$

$25 = X$

Propolis is 25kg

Mechanical advantage of a machine.

This the ratio of the load to the effort.

$$\text{Mechanical advantage} = \frac{\text{Load}}{\text{Effort}}$$

Mechanical advantage has no units because it is the ratio of forces.

Examples.

1. Calculate the mechanical advantage of a machine used by a boy who to lift a load of 400N using effort of 200N..

$$M.A = \frac{\text{Load}}{\text{Effort}}$$

$$M.A = \frac{400N}{200N}$$

M.A = 2 *M.A does not have units because it is a ratio of two forces*

2. What will be the load if the mechanical advantage of a machine is 2 and the effort is 40N?

M.A = 2 Load = ? Effort = 40	From M.A = Load ÷ Effort 2 = L ÷ 40 *Multiplying both sides by 40* 2 × 40 = L÷40×40	80 = L *Therefore* **L = 80N**	**Check** M.A = L ÷ E *2 = 80N÷40N* 2 = 2

3. Calculate the effort needed if the Mechanical Advantage of a machine is 2 and the load is 40N

M.A = 2 Load = ? Effort = 40	From M.A = Load ÷ Effort 2 = 40 ÷ E *Multiplying both sides by E* 2 × E = 40÷E×E 2E = 40 *Divide both sides by 2*	2÷2E = 40÷2 **E = 20** *Therefore* **E = 20N**	**Check** M.A = L ÷ E *2 = 40N÷20N* 2 = 2

It is usually better to use the same method of M.A=L/E and just keep on substituting

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Efficiency of a machine

This is the ratio of the work output to the work input of a machine.
Work output is the work done on a load by a machine.
Work input is the work done on an effort by a machine.

$$Efficiency = \frac{work\ output}{work\ input} \times 100\%$$

Example:

Calculate the efficiency of a machine whose effort is 30N and moves through a distance of 15m to overcome a load of 120N through a distance of 3m.

$$Efficiency = \frac{work\ output}{work\ input} \times 100\%$$

$$\text{Efficiency} = \left(\frac{120N \times 3m}{30N \times 15m}\right) \times 100\%$$

$$= \left(\frac{36}{45}\right) \times 100\%$$

$$= 80\%$$

The efficiency of a machine is 80%.

Note. The efficiency of a machine is not always 100% due to friction.

How the efficiency of a machine can be increased.

- Oiling the moving parts of a machine.
- Greasing moving parts of a machine.
- Repairing damaged parts of a machine.

Velocity Ratio of a machine

Velocity ratio is the ratio of the distance moved by the effort to the distance moved by load.

$$\text{Velocity ratio} = \frac{\text{Distance moved by effort.}}{\text{Distance moved by load}}$$

$$V.R = \frac{E.D}{L.D}$$

Velocity ratio has no units because it is a ratio of distance of forces.

Example:

Calculate the velocity ratio of a lever whose distance of the effort is 12m and that of the load is 3m.

$$V.R = \frac{E.D}{L.D}$$

$$V.R = \frac{12m}{3m}$$

$$V.R = 4$$

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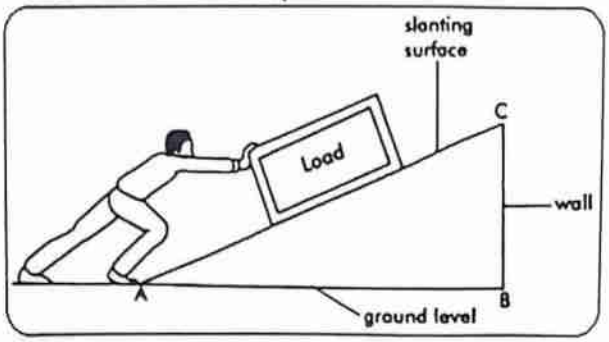
Lesson vocabulary:

• inclined • slope • plane • inclination • gentler • wedge

Inclined planes/slopes.

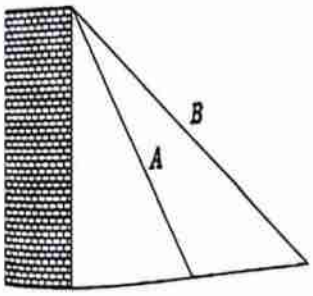
An inclined plane is a simple machine with a flat surface raised at an angle.
An inclined plane is used to raise the load upwards by rolling the load through a slope or slanting surface.

Parts of an inclined plane



BC is the distance moved by the load.
AC is the distance moved by the effort.
AB is the lower level/ground.

How an inclined plane works.



Explanation:

- In the diagram above, A and B are two slopes.
- Less effort is applied to overcome a load using slope **B** than when using slope **A**

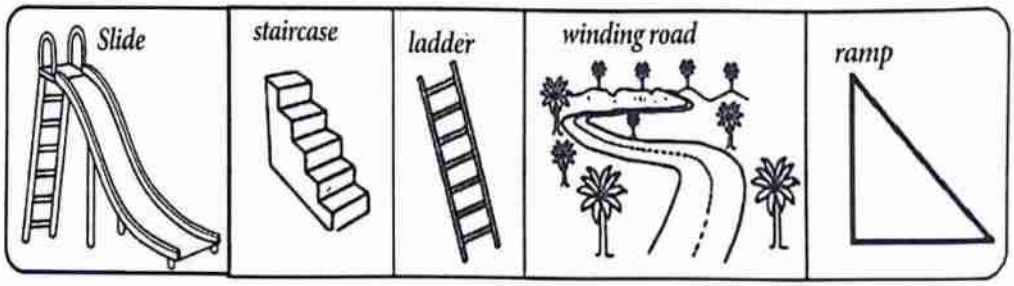
Reason. Slope **B** is longer and gentler than slope **A**.

How to use less effort when using inclined planes.

- Making the slope longer and gentler.
- Reducing the angle of inclination.

Examples of inclined planes.

- plank
- ramp
- winding road
- door steps
- leaning ladders
- a slide
- plank
- staircases
- door steps



Importance of inclined planes.

- Ramps are used to load or offload goods on vehicles.
- Ladders are used for climbing tall buildings or trees.
- Winding roads help people to climb mountainous or steep hills.

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- Staircases are used to climb tall buildings with ease.
- Planks are used to raise building materials during construction.
- A slide is used as a playing material by children.

Areas where inclined planes are applied.

- mountainous areas
- storeyed houses

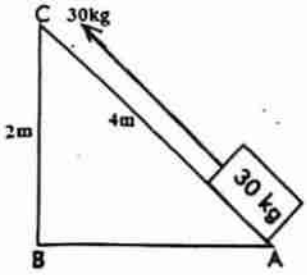
How inclined planes simplify work.

- They increase the length of the slanting surface.
- They reduce the angle of inclination.

Note: It is easier to climb upstairs than to use a ladder.
Reason. Less effort is used to climb upstairs than on a ladder.

Factors that affect the proper working of inclined planes

- Force of gravity
- Friction
- Height of the load.
- Length of the slope.



Calculations on inclined planes.

Explanation:

The distance moved by the effort is 4m.

The distance moved by the load is 2m.

Work done on the load

Work done = force x distance
= (30 x 2m) J
Work done = 60 Joules

Mechanical advantage

$$M.A = \frac{\text{Effort distance}}{\text{Load distance}}$$

$$M.A = \frac{4m}{2m}$$

$$M.A = 2$$

Wedges

A wedge is a double inclined plane with a sharp edge.
It is made by combining two inclined planes.
A wedge is a double inclined plane.
Reason. *A wedge has two slanting surfaces joined together.*

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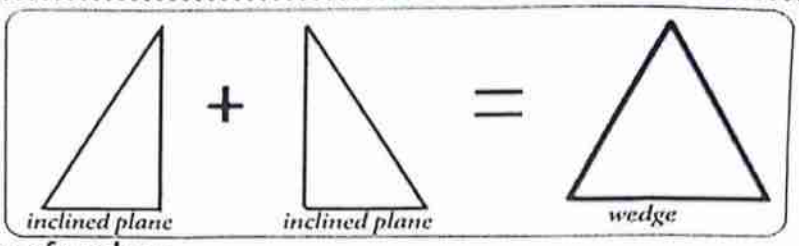
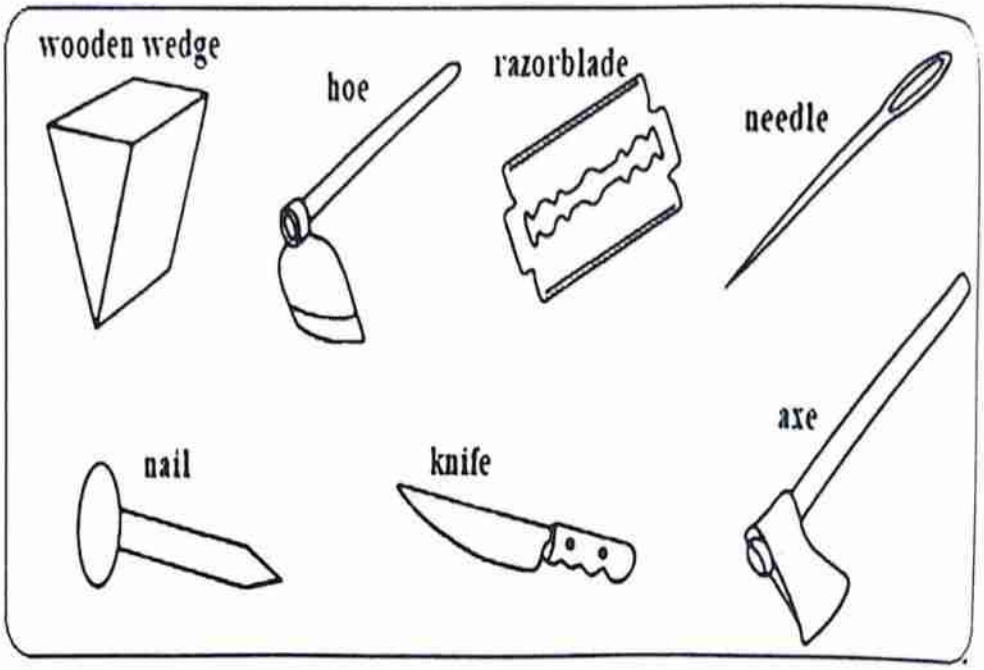


Examples of wedges

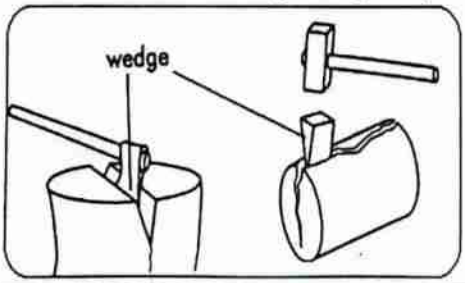


Ways in which wedges can be important in our daily life.

- For splitting wood using an axe.
- For digging using a hoe.
- For sewing clothes using a needle.
- For peeling food using a knife.
- For hunting using a spear.
- For grooming fingernails using a razorblade.
- For cutting trees using a panga or axe.

Friction and rusting affect the proper working of wedges.
Friction reduces the efficiency of wedges.
Rusting makes wedges blunt.
Rusting makes wedges weak

Illustration showing wedges splitting wood.



Ways of improving the efficiency of wedges.

- By sharpening them.
- By protecting them from rusting.

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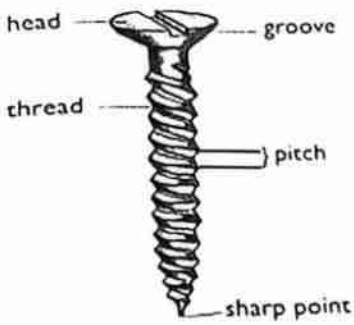
Screws

A screw is an inclined plane wound around a rod/bar.
It is a metal rod with a raised thread wound around it.
Screws work on the principle of inclined planes.



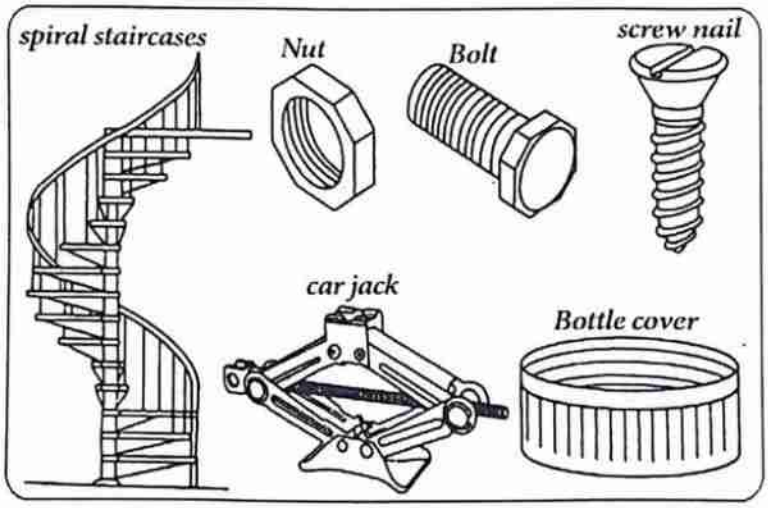
Functions of parts of a screw.

Threads: they hold the screw nail into the material where it is driven.
Pitch: is the distance between two successive threads.
Head: it has a groove where the screw is fitted.
Slot/groove: it is where the screw is fitted.
Shank/rod: it is the smooth part above the threads but below the head.
Tip/sharp point: this is the part of the screw nail that enters the material first.

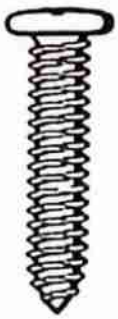
Screws are force multipliers.
Reason. They enable us to use less effort to overcome a bigger load.

Examples of screws.

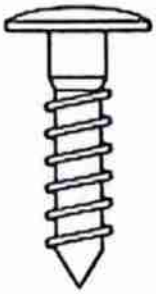
- screw nails
- clamp
- nuts
- bottle lid
- car jack
- driller
- spiral staircase



Points to note:



A screw with many threads moves longer distance. Therefore, less effort is needed to overcome the load



A screw with few threads moves a shorter distance. Therefore, much effort is needed to overcome the load.

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Ways in which screws can be important in our daily life.

- Car jacks are used to lift vehicles during repair.
- Spiral staircases enable people to climb upstairs.
- Bolts and nuts are used for holding parts of some machines together.
- Drillers are used to drill holes in wood and walls.
- Bottle covers are used as lids on tins and bottles.
- Screw nails are used to fasten parts of radios and TV sets.
- Engineer's vise is used to hold metals when filling or cutting.
- Screw caps are used to tighten bottle tops.

Examples of machines that use screws.

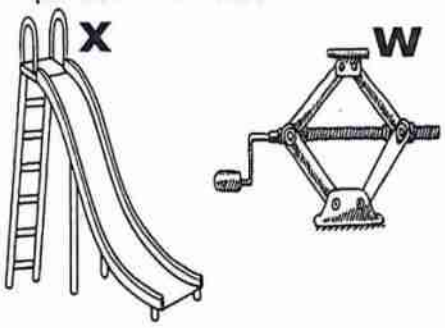
- spiral staircase
- driller
- engineer's vice
- carjack
- geometric compasses

Factors that affect the proper working of screws.

- friction
- rusting

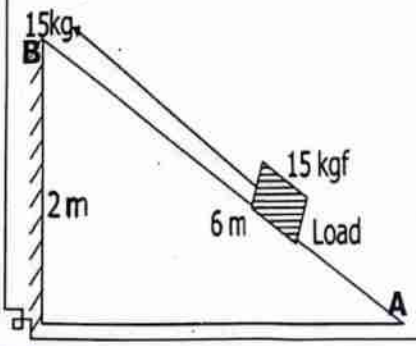
Evaluation activity:

1. a) Write down any **two** examples of wedged garden tools.
   b) State any **two** ways in which a crop farmer uses wedges.
2. a) Give any **two** ways in which inclined planes are useful to people.
   b) State any **two** factors that affect the proper working of inclined planes.
3. State the class of simple machines to which a ladder leaning against the wall belongs.
4. a) Name any **two** examples of screws.
   b) State any **two** ways in which screws can be important in our daily life.
5. The diagram below shows simple machines. Study and use it to answer the questions that follow.



   a) Name the group of simple machines to which **X** and **W** belong.
   b) In which way do nursery children benefit from the simple machine marked **X**?
   c) Give any **one** other machine that can be grouped together with simple machine marked **W**.

6. The diagram below shows a simple machine. Study and use it to answer the questions that follow.



   a) What distance does the effort and the load move?
      i) Effort .....
      ii) Load .....
   b) Identify the simple machine shown above.
   c) Give the advantage of lifting the load along side **AC**

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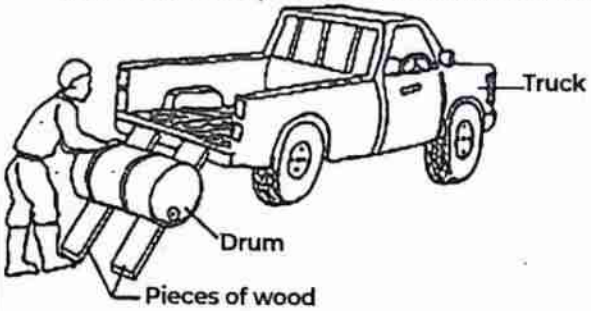
7. The table below shows groups of simple machines in part **A** and their importance in part **B**.

A: simple machines	B: importance
Wedges	For climbing higher grounds.
Screws	For fishing.
Levers	For fastening materials together.
Slopes	For cutting

Match correctly the groups of simple machines with their importance in the spaces provided below.

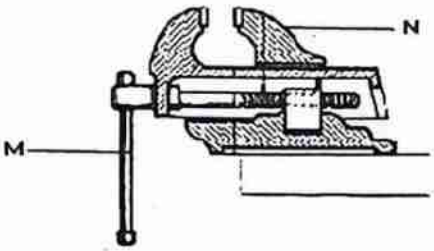
i) Wedges .....
ii) Screws .....
iii) Levers .....
iv) Slopes .....

8. The diagram below shows a man using a simple machine to load a drum on the truck. Study and use it to answer the questions that follow.



a) What type of simple machine is the man using?
b) Give any **two** examples of the type of simple machine you have named in a) above.
c) How can the man improve on the simple machine above in order to make his work easier?

9. The diagram below shows a simple machine commonly used by carpenters. Study and use it to answer the questions that follow.



a) Identify the group of simple machines where the simple machine above belongs.
b) Name the parts marked **M** and **N**.
c) How is the simple machine above useful to a carpenter?

10 a) Why are screws referred to as force multipliers?
b) Apart from screws, name any **one** other group of simple machines that multiply force.
c) Give any **two** forces that screws overcome when in use.

11 a) Name the group of simple machines to which each of the following belongs;
i) a ladder .....
ii) a panga .....
b) Give any **two** ways in which simple machines make work easier.

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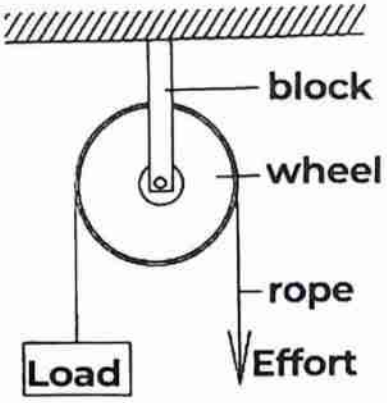
Lesson vocabulary:

- groove
- pulley
- movable
- scaffold
- elevator
- tackle

Pulleys

A pulley is a freely rotating wheel with a grooved rim.
A rope is passed through the groove of a rim.
The rim is grooved to prevent the rope from sliding or falling off the rim.

Parts of a pulley system



Functions of each part.
Block. It holds/supports the pulley.
Grooved rim. It keeps the rope in one position
It prevents the rope from sliding/slippi
Rope/chain. It holds the load.
It is where the effort is applied.

Types of pulleys

- single fixed pulley
- block and tackle pulley
- single movable pulley

Single fixed pulley

A single fixed pulley is a pulley in which the wheel rotates but does not move together with the load.

- It simplifies work by changing the direction of force.
- This makes it easier to raise the load by pulling downwards.

Characteristics of a single fixed pulley.

- The effort applied is equal to the load.
  For instance: *an effort of 20kg is required to overcome a load of 20kg.*
- Mechanical advantage is 1.
  Reason. *The effort applied is equal to the load to be lifted.*
- Effort distance is equal to load distance.
  For instance: *if the load moves 6 meters, the effort will also move 6 meters.*
- It changes the direction of force.
  This gives the advantage of doing work faster.
- The block is fixed and only the load moves.

Examples of single fixed pulleys

- flagpole
- window blind
- ski lift

Advantages of using a single fixed pulley.

- It is faster /it saves time
- It changes the direction of force.

Disadvantages of using a single fixed pulley.

- It does not reduce the effort needed to do work.

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Calculations in single fixed pulley.

Examples

1. Find the mechanical advantage of a single fixed pulley lifting a load of 65N using a force of 65N.

$M.A = \frac{L}{E}$ || $M.A = \frac{65N}{65N}$ || **M.A = 1**

2. An effort of 250kg is required to lift a load using a single fixed pulley. Calculate the mass of the load.

Effort = 1L || L = 250kg

$\frac{250kg}{1} = \frac{1L}{1}$ ||

250 = L || **The mass of the load is 250kg.**

3. Find the effort applied to pull a load of 85N using a single fixed pulley.

Load = 1E || E = 85N

$\frac{85N}{1} = \frac{1E}{1}$ ||

85N = E || **The Effort applied is 85N.**

Applications of a single fixed pulley.

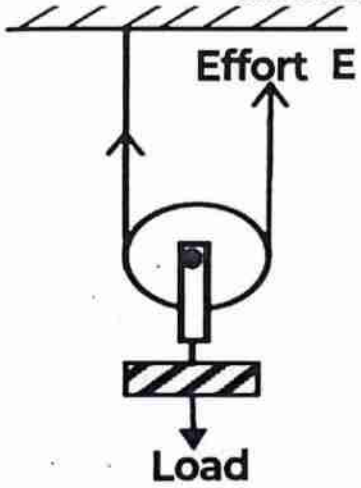
- It used to raise flags on flagpoles
- It is used by builders to lift building materials.

Single movable pulley

A single movable pulley is a pulley where the load is attached to the wheel and moves with a wheel.

It simplifies work by reducing the effort used to overcome the load.

Characteristics of a single movable pulley.



- The effort applied is half the load.
  For instance: if the load is 30kg, the effort needed to overcome the load will be 15kg. This means that the pulley saves energy.
- Mechanical advantage is **2.**
  ***Reason** .The effort is half the load.*
- The load moves together with the pulley.
- The effort distance is twice the load distance.
  ***For instance:** when the load moves through a distance of 8 metres, the effort distance will be (8 x 2) metres giving us **16 metres** as effort distance.*
- No change in the direction of force.
  The load and the effort move in the same direction.
- It is a force multiplier.
- Less effort is used to lift the load when using a single movable pulley.

Reason. The effort distance is longer than the load distance.

This is an advantage. However, it is slow at doing work.

In a single movable pulley, load is twice the effort. Effort needed is ahalf of the load. Therefore its mechanical advantage is 2.

Examples of single movable pulleys.

- cranes
- elevators

Calculations on a single movable pulley.

Example 1

Find the effort required to overcome a load of 60kg using a single movable pulley.

Let the effort be E

Load = 2E
$\frac{60kg}{2} = \frac{2E}{2}$
30kgf = E

E = 30kgf

The Effort applied is 30kgf.

Example 2

An effort of 250kgf is used to lift a load of using a single movable pulley. Calculate the size of the load lifted.

Let the load be L

L = 2E
L = 2×250kgf

L = 500kg

The size of the load lifted is 500kg

Example 3:

Find the effort needed to lift a load of 40kgf using a single movable pulley.

Let the effort be E.

Load = 2E
$\frac{40kg}{2} = \frac{2E}{2}$
20kgf = E

E = 20kgf

The Effort needed is 20kgf.

Differences between a single fixed pulley and a single movable pulley.

- A single fixed pulley changes direction of force unlike a single movable pulley.
- In a single fixed pulley, work is done faster while in a single movable pulley, work is done slowly.
- In a single fixed pulley, effort applied is equal to the load lifted while in a single movable pulley, the effort applied is half the load.
- The mechanical advantage of a single fixed pulley is 1 while that of a single movable pulley is 2.

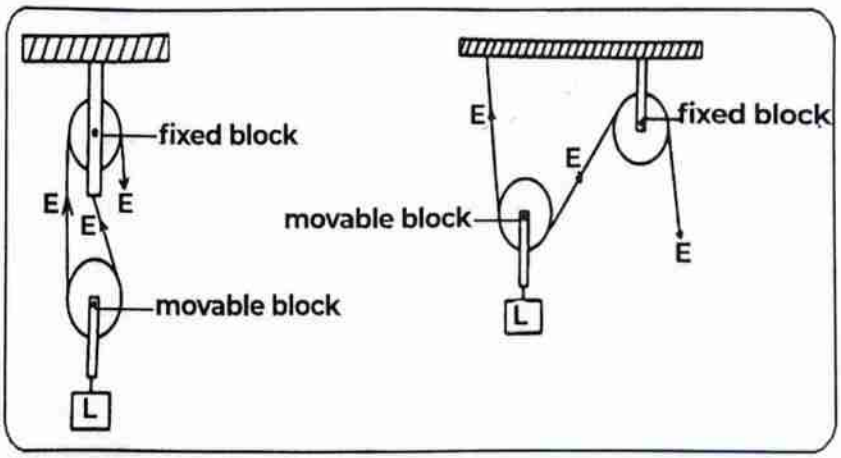
Block and tackle pulley system.

A block and tackle pulley system is pulley which has more than one wheel.

A block and tackle pulley is also called a **fixed movable pulley.**

It simplifies work by changing the direction of force.

Diagrams showing block and tackle pulleys.



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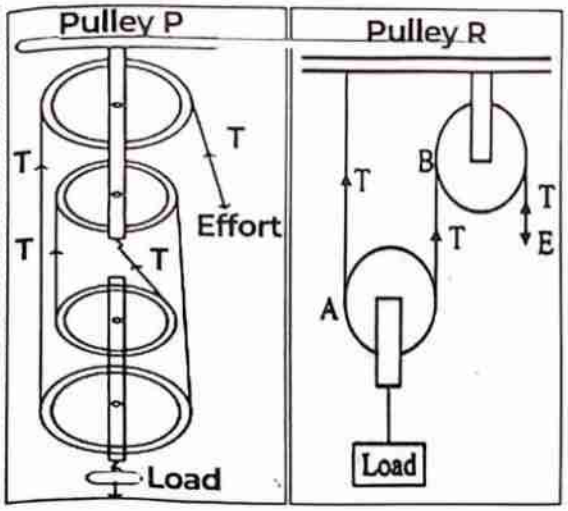
Characteristics of block and tackle pulley system.

- It has both the fixed and movable pulley.

In a block and tackle pulley, the fixed pulley changes the direction of force while the movable pulley reduces the effort need to overcome the load.

- The mechanical advantage of a block and tackle pulley system is determined by the number of pulleys

Refer to pulley systems below



Explanation:

- Pulley **P** and **R** are block and tackle pulleys
- In diagram **P,** there are **4** pulleys joined together (**two fixed and two movable**).
  The mechanical advantage is also **4.**
- This means that the effort required to overcome the load is four times smaller than the load.
- In diagram **R**, there are two pulleys, (**one fixed and one movable**).
  The mechanical advantage is also **2** while the effort required is twice less than the load.

Advantages of using block and tackle pulleys.

- Change the direction of force.
- Reduce the effort needed to do work.

Calculations about block and tackle pulley systems.

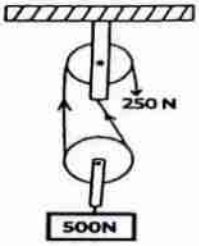
Example 1

A block and tackle pulley system of one wheel in the movable block was used carry a load of 500N using a force of 250N. What is its mechanical advantage?

$M.A = \frac{L}{E}$ $M.A = \frac{500N}{250N}$ **M.A = 2**

Example 2

The diagram below shows a certain pulley system. Study and use it to answer questions that follow.



$M.A = \frac{L}{E}$ $M.A = \frac{500N}{250N}$ **M.A = 2**

Machines that use pulleys

- Scaffolds
- cranes
- elevators/lifts
- breakdown vehicles

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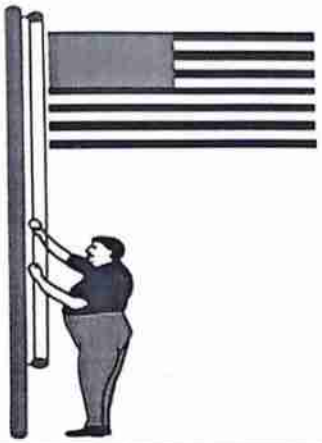
Ways in which pulleys are useful to people.

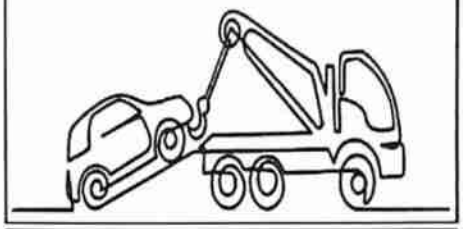
- They are used for raising flags on flagpoles.
- They are used for lifting building materials during construction.
- They are used by breakdowns to tow stranded vehicles.
- They are used in lifts and elevators.
- The used for moving curtains on curtain rods.
- They are used in cranes to lift loads.
- They are used in scaffolds for painting high buildings.

Diagrams showing applications of pulleys.

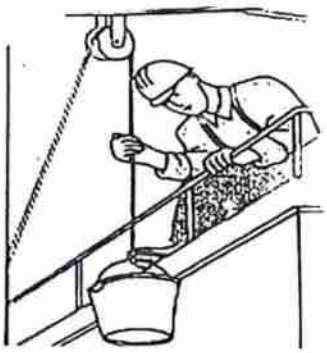
Raising a flag on a flagpole



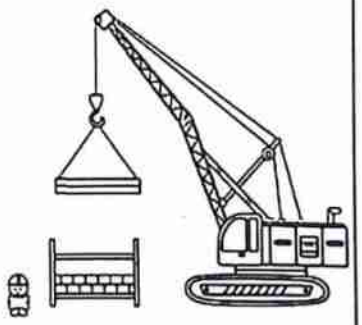
Towing a stranded car



Raising building materials



Lifting heavy materials



Forces that a pulley overcomes when in use.

- friction
- gravity

Ways of improving the efficiency of pulleys

- Lubricating to reduce friction.

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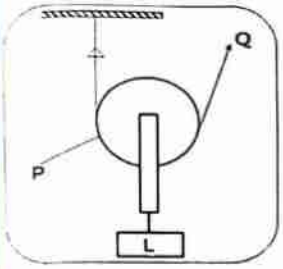
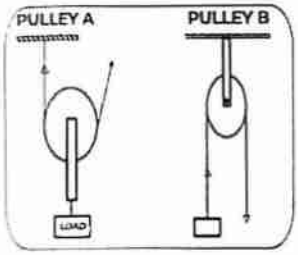
Evaluation activity:

1. Name the type of simple machines with a grooved rim in its wheel.
2. How does a single movable pulley simplify work?
3. Name any **one** force which a pulley overcomes when it is being used.
4. The diagram below shows a simple machine. Study and use it to answer the questions that follow.



a) Name the parts marked **P** and **Q**.
b) Identify the simple machine shown above.
c) Label the rope on the diagram above.

5. Name the type of pulley used to hoist the flag on the flagpole.
6. Give any **two** ways in which pulleys are useful to people.
7. State any **one** difference between a single fixed pulley and a single movable pulley.
8. Why is a single movable pulley referred to as a force multiplier?
9. The diagram below shows pulley systems. Study the diagram below and use it to answer the questions that follow.



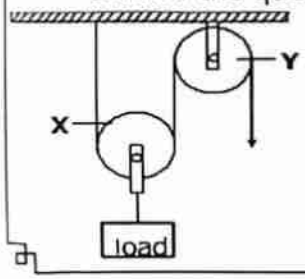
a) Identify the pulley system marked **B**.
b) In which pulley system will one use less effort to lift the load?
c) Give a reason for your answer in b) above.
d) How is pulley system marked **B** useful at school

10. The table below shows pulleys and their mechanical advantages (M.A). Study and use it to answer the questions that follow.

Pulleys	X	Y	Z	Q
M.A	4	1	2	2

a) Which pulley would you use to lift a load of 400N?
b) Give a reason for your answer in (a) above.
c) Identify the type of pulley represented by letter **Y.**
d) Find the effort needed to lift the load of 400N on the pulley **Q.**

11. The diagram below of block and tackle pulley system. Study and use it to answer the questions that follow.



a) Name the pulleys marked **X** and **Y.**
b) Give any **one** way in which the pulley system above makes work easier.
c) State any **one** advantage of using pulley marked **X** over pulley marked **Y**.

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Lesson vocabulary:

• wheel • axle • rod • steering • multiplier • drill

Wheels and axles

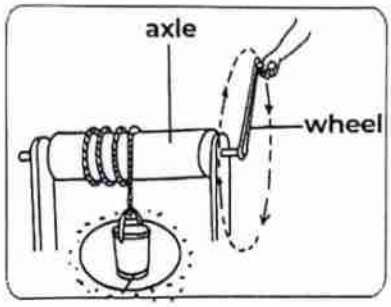
A wheel and axle is a simple machine made up of two cylinders of different diameters which rotate together on the same axis.

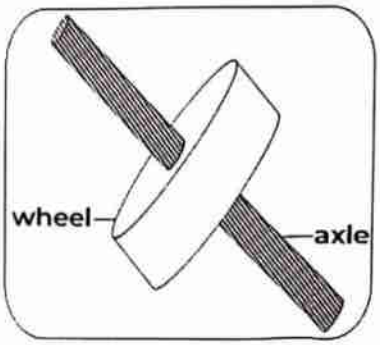
A wheel is a circular rim turning on an axle.

An axle is a rod on which a wheel rotates.

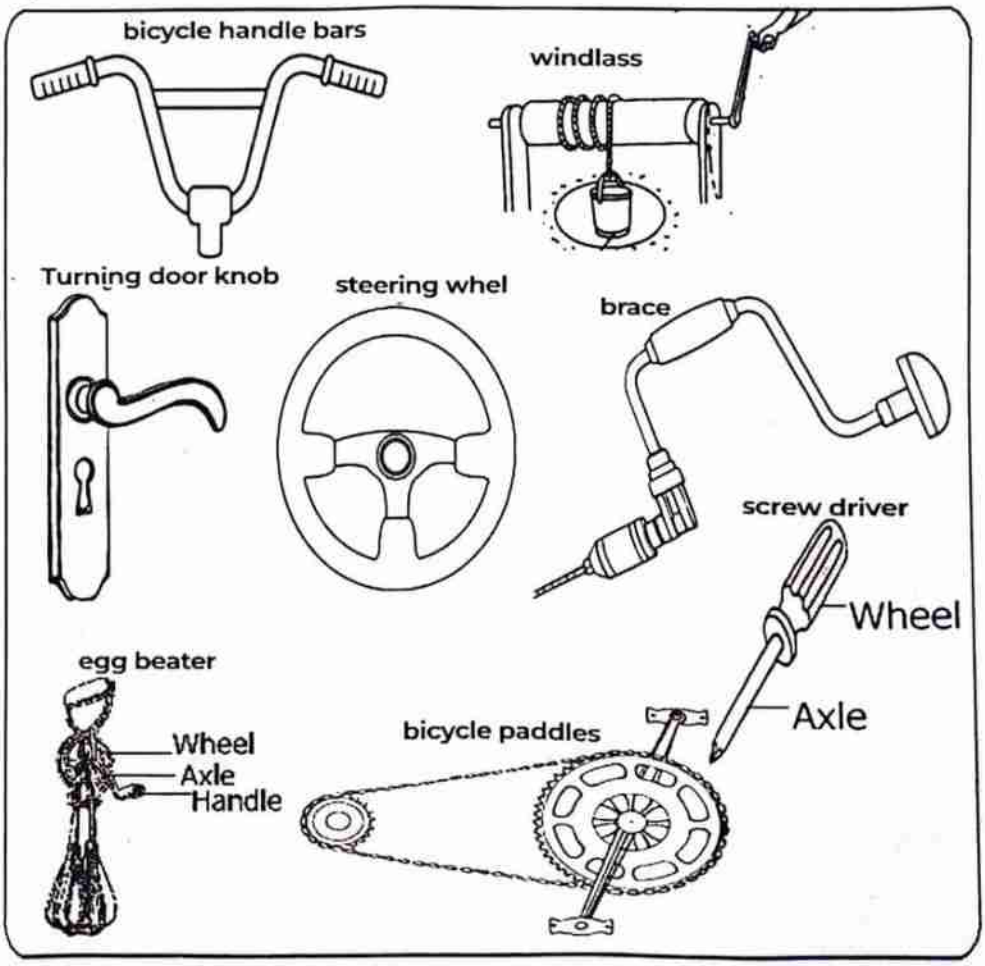
Wheels and axle make work easier by reducing the effort needed to overcome the load

Parts of a wheel and axle





Examples of wheels and axles.



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Points to note:

A pencil sharpener is both a wedge and a wheel and axle.

Both wheels and axles and screws are force multipliers

Ways in which wheels and axles can be important to people.

- For fetching water from deep wells using a windlass.
- For drilling holes into wooden materials using a brace.
- For preparing eggs for frying using an egg beater.
- For raising or lowering building materials using a windlass.
- For controlling direction of vehicles using a steering wheel.

A steering wheel reduces the effort used to turn a vehicle.

Bigger vehicles have larger steering wheels.

Reason. A large steering wheel makes the turning of the vehicle easier.

- For removing soil from pits using a windlass.
- For removing or fixing screw nails using a screw driver.
- For changing the speed of moving cars and motorbikes using gears.

Forces that wheels and axles overcome when being used.

- friction
- gravity
- tension

Comparison of human body parts to simple machines.

- The human arm and foot work like levers.

  The human arm works like a third class lever while the foot is a second class lever.
- The movable joints in the human body work like the pivot of a simple machine.

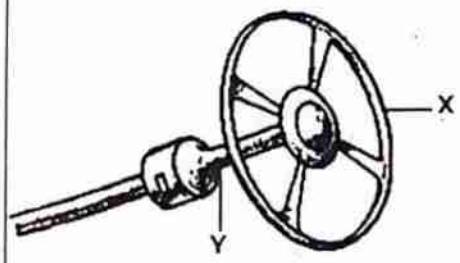
  They are turning points in the human body.
- The incisor teeth work like wedges.

  Their chisel shape enables them to cut food.
- Ball and socket joints can be compared to wheels and axles.

  The bone with ball shaped end acts as an axle while that with a socket acts as the wheel.

Evaluation activity:

1. Name the type of simple machines to which the brace belongs.
2. How are screws similar to wheels and axles in the way they simplify work?
3. The diagram below shows a simple machine. Study and use it to answer the questions that follow.



a) To which group of simple machines does the simple machine above belong?

b) Name the parts marked **X** and **Y**.

c) How does the simple machine above make driving of a car easier?

4. Give any **two** ways in which wheels and axles are useful to people.
5. Name any **one** group of simple machines which are;

   i) force multipliers.....

   ii) distance multipliers.....

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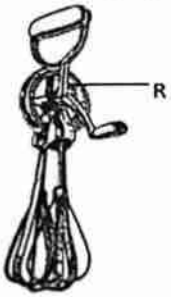
6. The diagram below shows a simple machine used when preparing eggs for frying. Study and use it to answer the questions that follow.

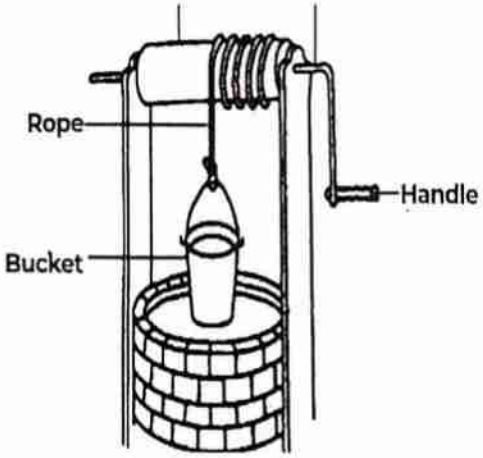


a) Identify the group of simple machines to which the machine above belongs.
b) Name the part marked **R**.
c) State any **one** way in which the machine above makes the preparation of eggs for frying easier.

d) On the machine above, label the axle.

7. Name **one** part of the human body that works like;
   a) wedges.....
   b) levers.....

8. The diagram below shows a simple machine. Study it and answer the questions that follow.



a) To which type of simple machines does the machine above belong?
b) State any **one** use of the simple machine above.
c) Name any **one** force that makes work difficult when using the above machine.
d) Give any **one** other example of a machine that belongs to the above type of simple machines.

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5

THEME: HUMAN BODY
TOPIC 5: EXCRETORY SYSTEM

Topical outline:

Excretion

- Definition
- Excretory organs
- Excretory products

The human skin

- Structure and functions of different parts.
- Functions of the human skin.
- Diseases and disorders of the human skin.
- Care for the human skin.

The urinary system (kidneys, urethra and urinary bladder)

- Structures of the urinary system and the kidney
- Functions of different parts.
- Functions of the human kidney.
- Diseases and disorders of the urinary system.
- Care for the kidneys.

The lungs as excretory organs.

- Structure and functions of parts.
- Diseases and disorders of the human lungs.
- Care for the human lungs.

Ref: P.7 standard curriculum page 79.

Advice to teachers:

Just mention the liver under excretory organs but do not teach the details of the human liver because it is out of primary seven curriculum.

Ref. P.7 standard and abridged curriculums.

Lesson vocabulary:

- excretion
- excess
- metabolic
- lactic
- kidney
- urea

Description:

The excretory system is a body system which removes metabolic wastes from the human body.

Excretion is the removal of metabolic waste products from the human body.

Importance of excretion in the human body.

- It stabilises the internal body temperature.
- It prevents blood poisoning.
- It prevents the damage of body organs like the kidney.

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Excretory organs and their excretory products.

Excretory organ	Waste products	Components
Skin	sweat	• excess water • excess salt • urea • lactic acid
Kidney	urine	• excess water • excess salt • urea • uric acid • ammonia
Lungs	• carbon dioxide • water vapour	
Liver	bile pigments	

Note well:Faeces is not considered as an excretory product.
Reason. Faeces is not produced by excretory organs

Evaluation activity:

1. Name the body system that is responsible for removing waste products from the human body.
2. What is meant by the term excretion?
3. Why is faeces not regarded as an excretory product?
4. Apart from the skin, give any one other human body organ that excretes excess water.
5. Give any **one** way in which excretion is important in humans.
6. Name **one** excretory product removed by the lungs.
7. Name the excretory organ that removes water and carbon dioxide from the human body.
8. Write down any **two** components of sweat.
9. Which excretory product contains uric acid?
10. Apart from excretory system, name **one** other body system to which the human lungs belong.
11. Which excretory waste is removed by the human liver?
12. Give **one** reason why urine is regarded as an excretory product.

Lesson vocabulary:

• epidermis • melanin • vasodilation • vasoconstriction • sensory • malpighian

THE HUMAN SKIN

The skin is the largest organ of the human body.

Reason. It covers most of the parts of the human body.

The skin is the sensory organ for feeling/touch.

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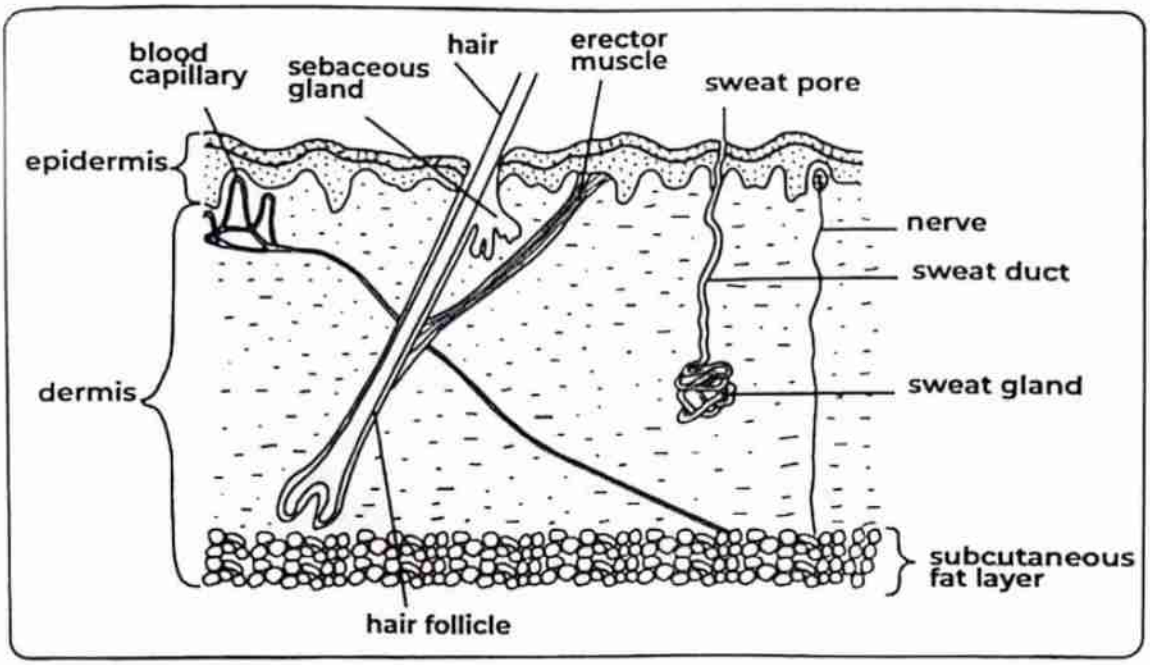
The structure of the human skin.



The main/major layers of the human skin.

- epidermis
- dermis

The epidermis

- It is the outermost layer of the human skin.
- It is made up of three sub-layers namely;
  - cornified layer
  - granular layer
  - malpighian layer

Cornified layer

- It is made up of dead cells and keratin.
- Keratin forms toenails and fingernails.

Granular layer

- It is made up of living cells that gradually die to replace the cornified layer.
- People who do manual work and those who walk barefooted most of the time have thicker skin under their palms and soles of feet.

Reason. To provide protection to the inner layers of the skin.

Malpighian layer

- It contains young cells and melanin.
- Melanin protects the skin against the strong radiations from the sun.
- Melanin also determines the colour of the skin and hair.
- Lack of melanin causes albinism.

The dermis.

It is the inner most layer of the human skin under which a layer of fats called sub-cutaneous layer which stores fats. Fats keep the body warm.

Parts that are found in the dermis and their functions.

Blood vessels:

- They supply food and oxygen to the skin.
- Transport waste products to their organs for excretion.

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Sweat glands
- They produce sweat.

Sweat duct.
- It transports sweat to the skin surface.

Sweat pores.
- It lets out sweat from the skin.

Hair follicle.
- It produces hair.
- The hair insulates the body against heat loss.

Erector muscle:
- It keeps the hair standing.
- The hair stands upright on cold days to trap escaping heat.

Sebaceous gland.
- Produces oily substance called **sebum.**

Functions of sebum in the human skin.
- Sebum lubricates the skin.
- Sebum keeps the skin soft and moist.
- Sebum keeps the skin waterproof.

Hair.
- It controls the body temperature.
- The hair insulates the body against heat loss.

Nerve endings/sensory nerves
- They transmit the impulses for heat, touch and pain to the brain.
- They are the parts of the skin that makes a person to feel pain when pinched.
- Sensory nerves enable the skin to detect change in temperature.

Functions of the human skin.
- The skin excretes sweat.
- The skin regulates human body temperature.
- The skin protects the body tissues from physical damage.
- The skin stores fats which keep the body warm.
- The skin is used for feeling.
- The skin protects the body against germ invasion.
- The skin helps to make vitamin D.

Ways in which the skin regulates the human body temperature.

During cold days/weather:
- Through shivering.
  Muscles contract rapidly to produce heat.
- Through vasoconstriction.
  Vasoconstriction is the narrowing of blood vessels at the skin surface.
  Little blood flows at the skin surface to prevent heat loss.
- Through producing goose pimples.
  Goose pimples are caused by contraction of erector muscles.
  Goose pimples help to reduce heat loss from the body.
  Goose pimples help to maintain internal body temperature.
- The hair on the skin stands upright.
  Reason. To trap escaping heat.

Note: There is less or no sweating on cold days.
Reason. Excess water is lost through urination.

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Practices that people do to regulate body temperature on cold days.

- Taking hot drinks.
- Sitting near fire.
- Putting on thick clothes.
- Doing physical exercises.

During hot days/weather.

- Through sweating.

Evaporation of sweat causes heat loss from the body.
Sweating cools the body.
It helps the body to get rid of some waste materials.

- Through vasodilation.

Vasodilation is the widening of blood vessels at the skin surface.
More blood flows at the skin surface to allow heat loss.

- The hair on the skin lies flat.

This allows heat loss from the body.

Note well:

1. Sweating is similar to transpiration.

Way:

- *Both help in cooling temperature. /both regulate temperature.*

2. Sweating is also similar to panting in dogs and birds.

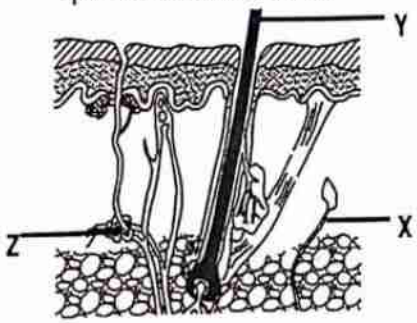
3. People urinate less on hot days.

Practices that people do to regulate their body temperature on hot days.

- Using umbrellas.
- Wearing light clothes.
- Turning on an electric fan.
- Taking cold drinks.
- Moving under the shade.
- Fanning using a piece of cloth.

Evaluation activity:

1. a) Name the outermost layer of the human skin.
   b) State any **one** waste product removed by the human skin.
   c) Apart from removing wastes, mention any **two** other functions of the human skin.
2. The diagram below shows the human skin. Study and use it to answer the questions that follow.



   a) Name the parts marked **X** and **Z**.
   b) Give the function of part marked **Z**.
   c) Why does part marked **Y** stand upright during cold weather?
3. State **one** role of the following substances in the human skin:
   i) Melanin
   ii) Sebum
4. How does the skin cool down the human body temperature on a hot day?
5. Which process in plants is similar to sweating in human beings?
6. Apart from removing wastes materials, how is sweating important to human beings?
7. Why does the body always develop goose pimples during cold weather?

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8. Name the **two** main layers of the human skin.
9. State the importance of fats stored in the subcutaneous layer of the human skin.
10. Mention any **two** practices people do to regulate their body temperature during cold weather.
11. How does shivering help to regulate body temperature during cold weather?
12. The table below shows parts of the human skin in part **A** and their functions in part **B**.

A: Part of the skin	B: Function of the part.
Sweat gland	Transmits impulses to the brain.
Hair follicle	Produces sebum.
Sebaceous gland	Produces hair.
Sensory nerve	Produces sweat.

Match correctly the parts of the human skin with their functions in the spaces provided below.

a) Sweat gland .....
b) Hair follicle .....
c) Sebaceous gland .....
d) Sensory nerve .....

Lesson vocabulary:

- scabies
- fungus
- athlete's
- bleach
- ringworm
- impetigo

Diseases that affect the human skin.

- athlete's foot
- scabies
- ringworm
- impetigo
- leprosy
- measles
- chicken pox
- skin cancer
- scurvy

Athlete's foot

- It is caused by fungi.

How athlete's foot spreads.

- Through body contact.
- Sharing shoes or stocking.
- Wearing wet stocking or shoes.

Signs and symptoms of athlete's foot.

- Peeling of the skin in between the toes.
- Wounds between toes.
- The skin between toes become white and cracks.

Ways of controlling athlete's foot.

- Wearing clean and dry shoes.
- Washing and drying feet.
- Avoid sharing shoes or socks.
- Changing shoes regularly.
- Using fungicides and powder.

Scabies

- It is caused by itch mites.

How scabies spreads.

- Through sharing clothes and bedding.
- Through body contact.
- Through sharing basins or buckets.

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Signs and symptoms of scabies.

- Itchy skin sores.
- Blisters all over the skin.
- Rough pimples on the skin.
- Thick crusts on the skin.
- Swelling of the skin.

Way of controlling scabies.

- Regular bathing.
- Regular washing of clothes.
- Avoid sharing clothes, towels, bedding or combs.
- Ironing clothes and bedding.
- Early treatment of scabies.

Ringworm infection.

It is caused by fungi.

It causes ring-shaped or round patches on the skin.

How ringworm infection spreads.

- Through body contact.
- Through sharing basins or buckets.
- Through sharing clothes and bedding.

Signs and symptoms of ringworm.

- Round white patches on the skin.
- Hair falling off the skin.
- Itching of the skin.

Ways of controlling ringworm infection.

- Regular bathing.
- Avoid sharing clothes and bedding.
- Avoid sharing sponges and towels.
- Early treatment with fungicides.
- Shaving off hair.
- Washing and ironing clothes.
- Avoid contact with an infected person.

Mnemonic for fungal skin diseases: **DR JAB**

Leprosy

It is caused by bacteria.

How leprosy spreads.

- Through body contact.
- Through coughing and spitting.
- Cockroaches carrying germs onto food.
- Through sneezing.
- Blowing the nose without covering it.

Signs and symptoms of leprosy.

- Sores with loss of feelings on the skin.
- Pale skin sores that last long.
- Deformed toes and fingers.

Ways of controlling leprosy.

- Isolating infected people.
- Control the breeding of cockroaches
- Early treatment of infected people.

Disorders of the human skin.

- albinism
- burns
- cuts
- vitiligo
- scalds
- hives
- acne
- skin rash
- corns
- pimples
- boils
- bruises
- shingles (herpes zoster)

Practices that damage the human skin.

- Bleaching the skin.
- Irregular bathing.
- Failure to treat skin infections in time.
- Staying in too much sunshine for long.
- Tattooing the skin.
- Playing rough games.
- Poor nutrition.

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- Playing rough games.
- Poor nutrition.

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Ways of caring for the human skin.

- Regular bathing.
- Feeding on a balanced diet.
- Smearing recommended Vaseline.
  Vaseline keeps the skin soft and moist.
- Doing regular physical exercises.
- Avoid bleaching the skin.
  Bleaching chemicals increase the risks of skin cancer

Evaluation activity:

1. Name the germ that causes ringworm infection.
2. Give any **one** way in which ringworm can spread among people.
3. State any **two** ways of controlling the spread of ringworm among children.
4. Which fungal disease causes round patches on the human skin?
5. Name any **one** human body system affected by leprosy.
6. State any **two** ways of controlling the spread of athlete's foot.
7. Which bacterial disease affects both the human skin and the skeleton?
8. Write down any **two** disorders of the human skin.
9. In which way does lack of melanin affect the human skin?
10. State any **two** practices that can damage the human skin.
11. Why is it important to smear Vaseline regularly?
12. What is the danger of bleaching the human skin?
13. a) What germ causes scabies?
    b) Write down any **two** signs and symptoms of scabies.
    c) State any **one** way in which the spread of scabies can be prevented among children.
14. Which human body organ is affected by scabies?
15. Give any **two** ways through which one can take care of his/her skin.

Lesson vocabulary:

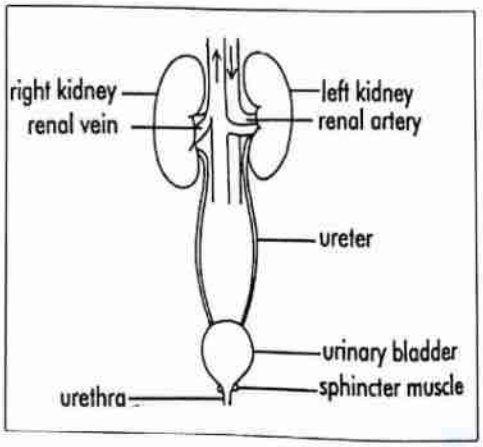
- kidney
- cortex
- medulla
- filter
- urinary
- ureter

THE URINARY SYSTEM

It is the system under the excretory system responsible for the production and removal of urine from the human body.

It is made up of the kidneys, ureter and the urinary bladder.

The structure of the urinary system.



Functions of each part:

Kidneys: They filter blood.

Ureter: It carries urine from the kidney to the urinary bladder

Urinary bladder: It stores urine before excretion.

Sphincter muscles: They control the flow of urine.

Babies often urinate on their bedding.

Reason. Their sphincter muscles are still weak to control the flow of urine.

Urethra: It lets urine out of the body.

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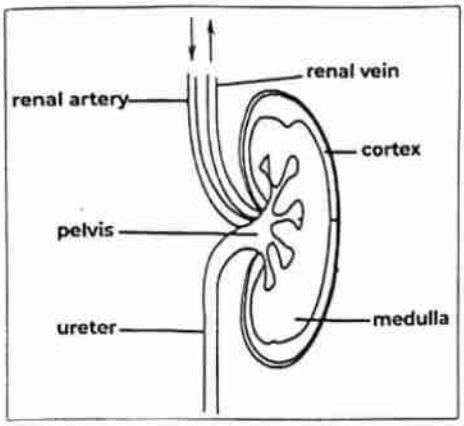
THE HUMAN KIDNEYS

Kidneys are reddish-brown bean shaped organs located at the back in the lower abdomen.
They are enclosed in the transparent membrane called **renal capsule.**
The kidneys are the major excretory organs in the human body.
Reason. The kidneys remove most of the nitrogenous wastes from the human body.
The major function of the kidneys is to filter blood.

Parts of the human kidney



Functions of parts of the kidney

Renal vein:
It carries deoxygenated blood from the kidney to the vena cava.

Renal artery:
It carries oxygenated blood from the aorta to the kidney.

Cortex:
It filters blood in the kidney to remove wastes.
It is where filtration of blood takes place.
Adaptation: It contains nephrons that filter blood.

Medulla:
It reabsorbs useful substances from urine.
These substances include;

- digested food nutrients
- water
- salts
- glucose
- vitamins

It is where selective reabsorption takes place.
Selective reabsorption prevents the loss of useful substances in the body.

Pelvis:
It is where collection of urine takes place.
It leads to the ureters.

Ureter:
It carries urine from the pelvis to the urinary bladder.
The ureter leads to the urinary bladder.

Functions of the human kidney.

- The kidney filters blood.
- The kidney balances the amount of water needed in the body.
- The kidney controls the level of salts in blood.
- The kidney helps in the production of red blood cells.
- The kidney produces renin which controls blood pressure.
- The kidney removes nitrogenous wastes from the body.
  These nitrogenous wastes are;
  ◇ urea ◇ uric acid ◇ ammonia

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Points to note:

- *When the body runs short of water, the kidneys produce very little urine usually dark in colour.*

Reason. Due to high concentration of urea and mineral salts in blood.

- *People who drink a lot of alcohol usually pass out coloured urine.*

Reason. Due to impurities and high concentration of toxins in alcohol.

- *Kidney produce a hormone erythropoietin which helps in the production of red blood cells.*
- *People urinate a lot during cold days.*

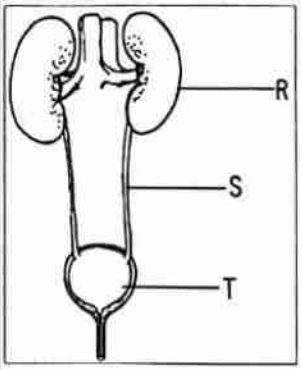
Reason. The sweat pores are closed thus excess water is removed by urinating.

Similarities in function between the kidneys and the skin.

- Both remove waste materials from the body.
- Both regulate the amount of water in the body.

Evaluation activity:

1. Which human body organ balances the amount of water needed in the body?
2. Write down any **two** examples of nitrogenous wastes removed by the human kidneys in the body.
3. What is the main function of the kidney in the human body?
4. Why do people urinate a lot during cold days?
5. Mention any **two** processes that take place in the human kidney.
6. How is the function of the human kidney similar to that of the human skin?
7. The diagram below is of the human urinary system. Study and use it to answer the questions that follow.



a) Name the parts marked **S** and **T**.
b) What is the function of part marked **R**?
c) Where does part marked **S** lead to?

8. Name the human body organ that excretes urine.
9. Mention any **two** substances that are reabsorbed in the kidney.
10. In which part of the kidney does each of the following processes take place?
    i) Filtration of blood.....
    ii) Selective reabsorption.....
11. How is selective reabsorption important in the human body?
12. Apart from filtering blood, in which other way are the kidneys useful to the circulatory system?
13. Why do babies often urinate on their bedding?
14. Give any **two** functions of the kidney in the human body.
15. Why do people who drink alcohol usually pass out coloured urine?
16. State any **one** way in which the human kidney is able to filter blood.

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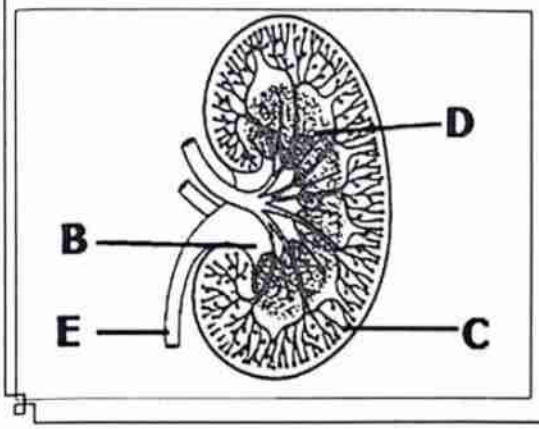
17. The diagram below is of the human kidney. Study and use it to answer the questions that follow.



a) Name the parts marked **B** and **D**.
b) State the function of part marked **C** in the human kidney.
c) Identify the waste product carried by part marked **E**.

Lesson vocabulary:

- bilharziasis
- urinary
- nephrons
- gonorrhea
- cystitis
- fluke

Diseases and disorders that affect the kidneys

Bilharziasis/schistosomiasis

- It is spread by a fresh water snail.
- It is caused by blood flukes.

Signs and symptoms of bilharziasis.

- Blood stained urine.
- Fever
- Difficulty in urination.
- Pain in the lower belly.

Prevention/control of bilharziasis

- Boil water for drinking.
- Proper disposal of human wastes.
- Avoid swimming in dirty water bodies.

Kidney stones

It is the formation of stones by salt in the kidneys.
Kidney stones block the kidney and sometimes the urinary bladder

Signs and symptoms of kidney stones.

- Sharp pain in the lower belly.
- Blood stains in urine.
- Inability to pass out any urine at all.
- Vomiting (nausea)

Prevention/control of kidney stones.

- Drink plenty of boiled water daily.
- Avoid eating raw salt.
- Avoid eating too salty food.

Kidney failure/renal failure

A condition in which kidneys stop filtering blood.
It is also a disorder of the human kidney.

Signs and symptoms of kidney failure.

- Shortness of breath.
- Swelling of the face and limbs.
- Loss of body weight.
- Loss of appetite.
- Sudden increase in heartbeat.

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Treatment of kidney failure.

- Application of kidney dialysis.
- Through kidney transplant.

Nephritis

A condition in which the nephrons swell and fail to filter blood properly.

It is caused by **bacteria**.

Signs and symptoms of nephritis.

- Pus or blood in urine.
- Pain in the pelvis.
- Burning sensation when urinating.
- Swelling of the body i.e. face and legs.

Prevention and treatment of nephritis.

- Drinking plenty of boiled water daily to keep the kidneys filtering.
- Use of intravenous antibiotics.

Cystitis

It is a bacterial infection on the bladder or urethra causing inflammation.

Signs and symptoms of cystitis.

- Passing out cloudy urine.
- Nausea.
- Frequent urination.
- Mild fever.

Treatment of cystitis.

- Treat infected people with antibiotics.
- Carrying out surgery.

Kidney diabetes

It is caused by high level of glucose.

Signs and symptoms of kidney diabetes.

- Pale skin.
- Weight loss.
- Frequent urination.
- Increased thirst.
- Blurry vision.

Treatment and control of kidney diabetes.

- Injection with insulin.
- Avoid eating food with high sugar content.
- Having regular physical exercises to burn down excess glucose.

Other complications of the urinary system

- bladder cancer
- enlarged prostate glands
- gonorrhea
- kidney cancer
- uraemia

Practices that damage the human kidney.

- Drinking alcohol.
- Eating too salty food.
- Eating raw salt.
- Poor nutrition.
- Holding back urine for a long time.

This increases calcium levels in the kidneys.

Care for the kidneys and urinary system.

- Doing regular physical exercises.
- Going for regular medical checkups.
- Eating less salty food.
- Emptying the urinary bladder in time.
- Avoid eating raw salt.
- Drinking plenty of clean boiled water.
- Feeding on a balanced diet.
- Limiting alcohol intake.

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Evaluation activity:

1. Name the condition which makes the kidneys to stop filtering blood in the human body.
2. a) Name the germ that causes bilharziasis.
   b) Which disease vector transmits bilharziasis?
   c) Mention any **one** sign of bilharziasis in humans.
   d) State any **one** way in which the spread of bilharziasis can be controlled among human beings.
3. Which disease of the kidney results from the accumulation of salt in the kidney?
4. Name the sexually transmitted disease that affects the kidney.
5. State any **two** practices that can damage the human kidneys.
6. Why is it not advisable to hold back urine for a long time?
7. Give any **one** disorder that affects the human kidneys.
8. State any **two** signs and symptoms of nephritis.
9. Which condition of the human kidney can lead to dialysis?
10. State any **two** ways of maintaining the urinary system in a healthy working condition.

Lesson vocabulary:

- trachea
- moisten
- alveoli
- diffusion
- choking
- emphysema

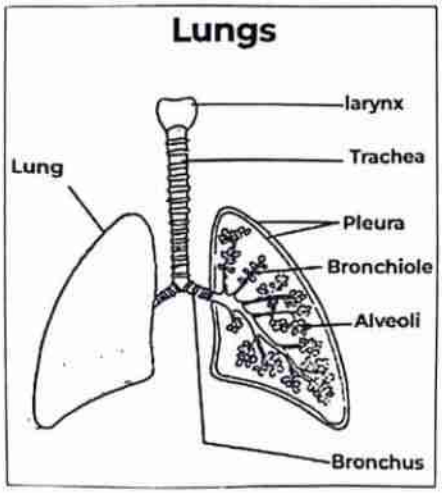
THE HUMAN LUNGS AS EXCRETORY ORGANS.

- The lungs are located in the chest cavity.
- Lung are protected by the ribcage.
- The lungs belong to both excretory system and respiratory system.
- The lungs are excretory organs because they help to get rid of water vapour and carbon dioxide from the body.
- Carbon dioxide and water vapour are by products of respiration.

Respiration is the chemical breakdown of glucose in the body cells to produce energy.

- It takes place in the body cells.

The structure of the human lungs.



Functions of each part:

Trachea/windpipe

- It is the air passage to and from the lungs.
- It is made up of the rings of cartilage

Reason. *To keep it open all the time.*

Alveoli/air sacs

- They allow gaseous exchange to take place.
- Oxygen joins bloodstream in the alveoli through diffusion.

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Adaptations of alveoli/air sacs to their function.

- They are many in number that increases the surface area for gaseous exchange.
- They are thin walled which allows gases to diffuse through easily.
- They are surrounded by a dense network of blood capillaries to supply oxygen

Lungs

- They are used for breathing.
- They excrete water vapour and carbon dioxide from the body.
- They are organs where oxygenation of blood occurs.

Adaptations of lungs to their functions.

- Ability to expand and contract during breathing.
- Contain millions of air sacs for exchange of gases.
- Have thin moist walls for easy diffusion of gases.
- Have network of blood capillaries which supply oxygen.

Diaphragm

- It flattens to create space for the lungs to expand.
- It separates the chest cavity and abdominal cavity.

Bronchioles

- Supply air to the air sacs.

Bronchi

- They take air from the trachea to the lungs.

Things that happen to air in the nose.

- Air is filtered.
- Air is warmed.
- Air is moistened.

Diseases that affect the human lungs.

Non infectious diseases

- lung cancer
- emphysema
- asthma

Infectious diseases.

- COVID 19
- influenza
- pneumonia
- bronchitis

The above diseases are caused by virus.

- tuberculosis
- diphtheria
- whooping cough
- pneumonia

The above diseases are caused by bacteria.

Disorders of the human lungs.

- choking
- hiccups
- sneezing
- yawning
- wheezing
- nasal congestion

Practices that can damage the human lungs.

- Regular smoking.
- Poor nutrition.
- Late treatment of lung diseases.
- Staying in dusty places.
- Failure to do regular exercises.
- Staying in areas where teargas is exploded.

Ways of caring for the human lungs.

- Doing regular physical exercises.
- Avoid smoking.
- Avoid staying in very cold areas.
- Avoid dusty environment.
- Feeding on a balanced diet.
- Immunizing children.
- Going for medical checkups.

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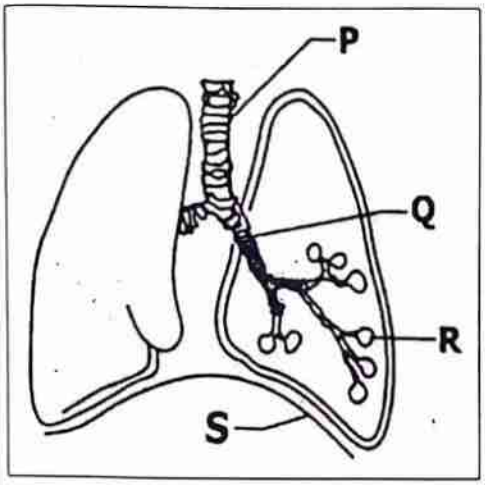
Comparison between the human excretory system and other natural environment or man-made systems.

- Blood filtration to water filtration.
- Sweating to panting in dogs.
- Dialysis machine to human kidney.
- Sweating to transpiration in plants.
- Urine formation to water filtration.

Evaluation activity:

1. Name the excretory organ that removes water vapour and carbon dioxide from the human body.
2. By what process does oxygen join the blood stream?
3. Name any **one** waste product excreted by the human lungs.
4. Why are human lungs regarded as excretory organs?
5. Which part of the human skeleton protects the lungs?
6. The diagram below is of human lungs as excretory organs. Study the diagram and use it to answer the questions that follow.



a) Name the parts marked **Q** and **S**.
b) Why is part marked P made up of rings of cartilage?
c) Give the function of part marked **R**.

7. Write down any **two** things that happen to air in the nose.
8. Name any **one** disease that affects the lungs due to smoking.
9. Apart from smoking, state any two practices that can damage the lungs.
10. Give any **one** reason why air sacs are able to carry out gaseous exchange.
11. What is meant by the term respiration?
12. State the excretory function of the human lungs.
13. Write down **two** by products removed from the body during respiration.
14. Mention any **two** non-infectious diseases that attack the human lungs.
15. Give any **two** ways in which an individual can care for his/her lungs.
16. Write down any **two** disorders that can affect the normal functioning of the human lungs.
17. Name the part in the human body where respiration takes place.

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6

THEME: MATTER AND ENERGY
TOPIC 6: LIGHT ENERGY

Topical outline:

- Natural and artificial sources of light.
- How light travels.
- Rays and beams of light.
- Importance of light in the environment.

Effects of different materials on light.

- Transparent objects
- Translucent objects.
- Opaque objects.

Shadows

- Eclipses (solar and lunar)
- Pinhole camera

Reflection:

- What it is.
- Laws of reflection.
- Types of reflection.
- Characteristics of images formed by plane mirrors.
- Periscope

Refraction:

- What it is.
- Principle of refraction.
- Effects of refraction.
- Refraction through glass, water and air.
- Dispersion of light.

The rainbow

- Formation of the rainbow.
- The glass prism and spectrum.
- Refraction through a rectangular glass prism.

Lenses

- Types of lenses.
- Effects of lenses on beams of light.
- The magnifying glass.
- The lens camera
- Parts of the camera and their functions.
- Characteristics of images formed by the lens camera.

The human eye.

- Structure
- Parts and functions.
- Characteristics of images formed by the human eye.
- Comparison of the human eye and the lens camera.
- Eye defects and their corrections.
- Diseases of the human eye.
- Care for the human eye.

Ref: P.7 standard curriculum pages 82-84.

Advice to the teacher:

The following content is out of primary seven curriculum in this topic of "light energy"; types of mirrors (concave and convex mirrors), classification of colours i.e. primary, secondary, complementary and tertiary colours, calculations on image distance and the solar system (planets /the universe) therefore, teach this topic under the guidance of the curriculum.

Follow the guided outline above to teach the right and curriculum based content.

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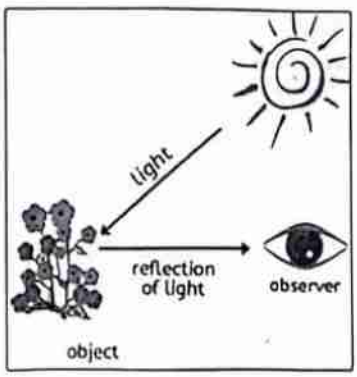
Lesson vocabulary:

• luminous • straight • incandescent • emit planet reflect

Light energy:

Light is a form of energy that enables us to see.
Light stimulates the sense of sight/seeing.
We are able to see different objects
because light which falls on them reflects light into our eyes.



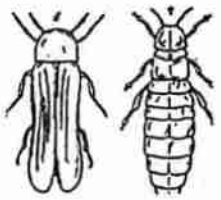
Light as a form of energy.

- Light has the ability to do work.
- Light can be transformed into other forms of energy.

Natural sources of light.

- sun • star • fireflies
- lightning • erupting volcanoes • glowworms

Natural sources of light

Sun	firefly	star	glowworm
			

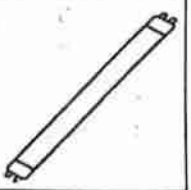
Artificial sources of light.

- lamps • electric torches • electric bulbs • burning candles
- fluorescent tubes • fire from burning fuel

Points to note:

Light is not regarded as matter because light does not have mass and volume.
The sun is the major source of light on the earth.
Reason. The sun provides the greatest amount of light.
The moon is not regarded as the natural source of light.
Reason. The moon reflects its light from the sun
The sun, torches, fireflies, lightning and lamps are luminous objects.
Reason. They emit light on their own.
The moon, plane mirrors, clear water and planets are non-luminous objects.
Reason. They reflect light from other sources.
Some objects produce light when they are hot.
Such objects are called incandescent objects.
Example are; the sun, stars, burning charcoal and the filament of an electric bulb in use.

Artificial sources of light

Electric bulb	fire	Fluorescent tube
		
burning candle	**lamp**	**Electric torch**

How light travels.

- Light travels in a straight line.
- It travels in rays. **A ray** is a path of light.

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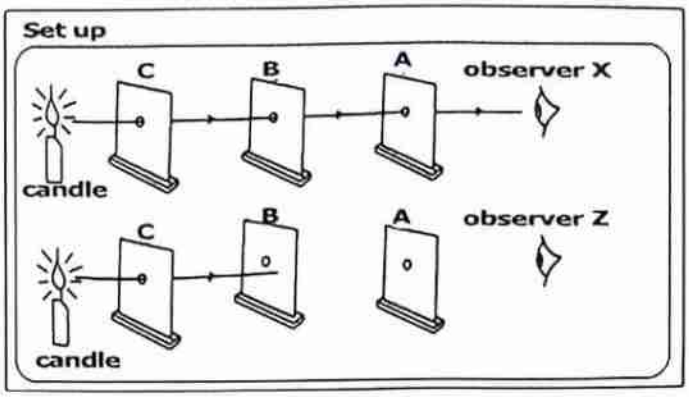
An experiment to show that light travels in a straight line.

A: Using cardboards.

Materials needed:

- lit candle
- thread
- three cardboard with a pinhole



Procedure:

i. Get three cardboards and make a hole at the centre of each one.

ii. Put them on a table in such a way that the holes are in a straight line.

iii. Place a burning candle in front of the hole of the last cardboard.

iv. Look at the candle through the first hole in the first cardboard.

Observations

The observer is able to see light from the candle flame.

Reason. The cardboards are placed with the holes in a straight line.

In case one of the cardboards is displaced so that the holes are not in the straight line, the observer will not see the candle light.

Reason. Light will be blocked by the displaced cardboard.

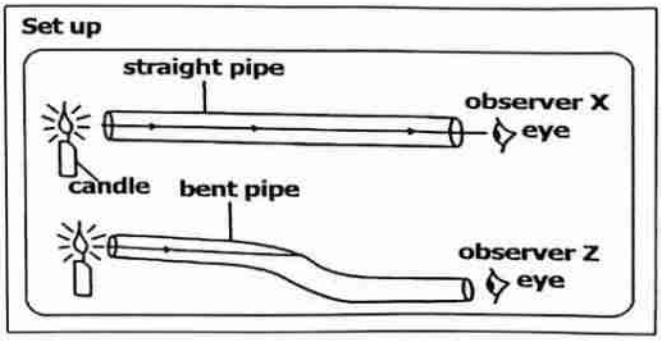
Conclusion.

Light travels in the straight line.

B:Using pipes.

Materials needed:

- one straight pipe
- one bent pipe
- lit candle



Steps:

i. Light the candle and place it on the table.

ii. Observe the candle through the tube as shown by the observer X.

iii. Now bend the pipe and try to observe the candle once again as shown by observer Z.

Observation:

- In the diagrams above, the observer in diagram X will be able to see light from the candle.

Reason. The pipe is placed in a straight line.

- The observer in diagram Y will not see light from the lit candle.

Reason. The tube is bent yet light travels in a straight line.

Conclusion

Light travels in a straight line.

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Note: We cannot see around corners.
Reason. Light travels in a straight line.

Incidences that prove that light travels in a straight line.

- Inability to see around corners.
- Formation of shadows.
- Formation of eclipses.

Rays and beams of light. *A ray of light*

A ray is a path taken by light.
A beam is a group of light rays travelling in the same direction.

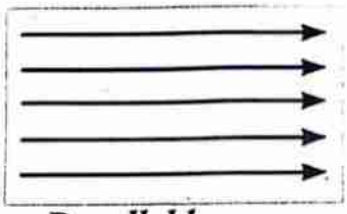
Types of beams of light.

- parallel beam
- convergent beam
- divergent beam

(a) Parallel beam *(b) Divergent beam* *(c) Convergent beam*

Parallel beam

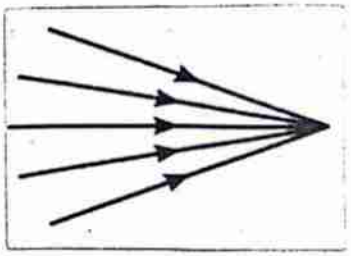
This is a beam in which light rays do not intersect and have the same distance apart.
When light strikes a plane mirror, the rays bounce off in parallel beam.



Parallel beam

Convergent beam.

This is a beam in which light rays from different directions meet at one point.
When light is passing through a convex lens, light rays converge.
Convex lenses and concave mirrors produce convergent beam.



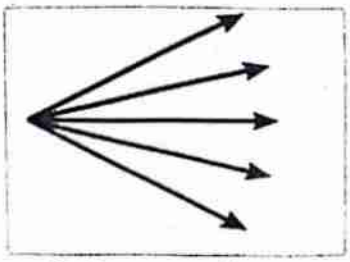
Convergent beam

Divergent beam.

This is a beam where light rays from the same point are spread out in different directions.

Devices that produce divergent beam.

- electric bulbs
- torches
- burning candle
- bicycle head lamps
- concave lens



Divergent beam

Applications/uses of beams of light.

- They help in projecting light in projectors.
- They help in projecting light produced by headlamps of vehicles at night.

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- They help in projecting light produced by torches.
- They are used in communication especially at night by soldiers and disco-theques.

Importance of light in the environment.

- Light enables people to see.
- Sunlight enables plants to make their own food.
- Light is used in cameras for taking photographs.
- Light helps in promoting security in dark places.
- Light is used in paper scanners and photocopiers.
- Sunlight enables the human skin to make vitamin D.
- Strong light rays are used in X-ray machines.

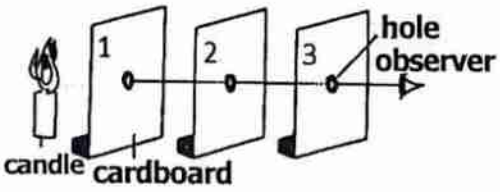
Groups of people who entirely use light in their daily activities.

- security guards
- astronomers
- spectators
- photographers
- stationers
- submariners
- video jokeys

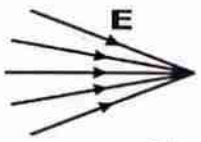
Evaluation activity:

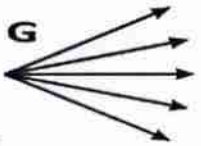
1. Name the form of energy that stimulates the sense of sight.
2. Why is the moon not regarded as a natural source of light?
3. Give the meaning of each of the following terms:
   i) Ray of light
   ii) Beam of light
4. What happens to light rays when they meet a convex lens?
5. Give any **one** reason why light is referred to as a form of energy.
6. Give any **two** natural source light in the environment.

The experiment below was carried out to find out how light travels. Use it to anser questions **7** and **8**.



7. Give the role of the candle in the above experiment.
8. State what would happen to the observer if cardboard **2** was dis placed so that its hole is not in line with that of **1** and **3**.
9. State any **two** ways in which light is useful to people.
10. Why is the sun regarded as the main natural source of light?
11. Give any **two** artificial sources of light in the environment.
12. The diagram below shows rays of light. Study and use it to answer the questions that follow.





a) Name the beam of light marked **G**.
b) Which type of lens affects the beam marked **E**?
c) How does the type of lens given in (b) above affect the beam **E**?
d) Mention any **one** device that produces the beam of light marked **G**.

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13. Aorta wanted to prove that light travels in a straight line. He had the following materials for the experiment; 3 cardboards, string, a wax candle and a table.
a) Name **one** material that Aorta would use instead of cardboards.
b) What is the use of the wax candle in the experiment?
c) Describe how Aorta will prove that light travels in a straight line in this experiment. (2 marks)

Lesson vocabulary:

- opaque
- shadow
- reflected
- diffused
- translucent
- tinted

Similarities between light and sound.

- Both are forms of energy.
- Both can be absorbed.
- Both can be reflected.
- Both can be refracted.

Differences between light and sound.

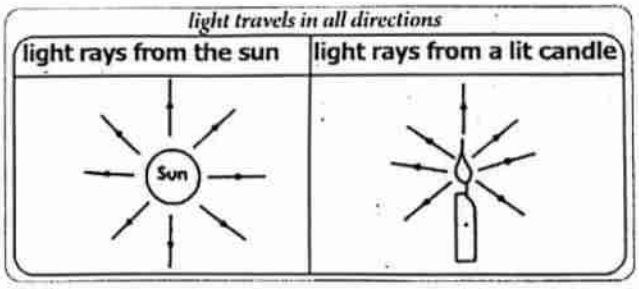
- Light enables us to see while sound enables us to hear.
- Light travels through rays while sound travels through waves.
- Light doesnot require a medium of transmission while sound does.
- Light travels through a vaccum while sound doesnot.
- Light travels faster than sound.

Situations that show that light travels faster than sound.

- During a thunder storm, we see lightning before thunder is heard.
- During a race, we see a flash of light at the starters gun before hearing its sound.
- During a gunshot, we see a flash of light before hearing its sound.

Properties of light.

- Light travels in a straight line.
- Light travels in all directions from the source.
- Light can be reflected.
- Light can be refracted.

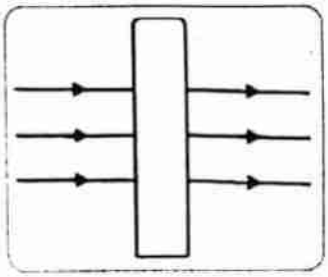


Effects of different materials on light.

- transparent objects
- translucent objects
- opaque objects

Transparent objects

- Transparent objects are materials which allow all light to pass through them.
- Transparent objects do not form shadows because they allow all light to pass through them.



Examples of transparent objects.

- clear glass
- clear water
- clear air
- transpaerent polythene paper

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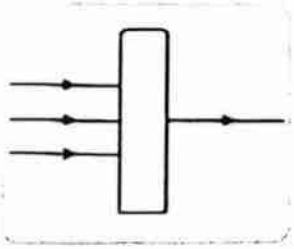
Importance of transparent objects.

- They are used to make vehicle and aeroplane windscreens.
- They are used to make window and door frames.
- They are used in eye pieces of some optical instruments.

Translucent objects



- Translucent objects are materials which allow little or part of light to pass through them.
- Translucent objects form light shadows (penumbra shadows) because they allow some light to pass through them.
- When light passes through translucent objects, it is diffused or scattered.

Examples of translucent objects.

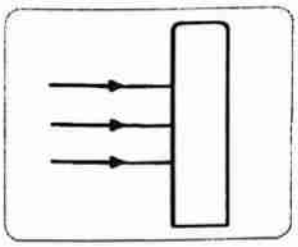
- frosted glasses
- tracing paper
- skylight
- oiled paper
- thin cloth
- waxed papers
- foggy air

Importance of translucent objects.

- Frosted glasses are used in bathroom and toilet windows.
- Tinted glasses are used to make windows of vehicles.

Note: We are unable to see clearly through translucent objects.
Reason. *They diffuse light rays passing through them*

Opaque objects



- Opaque objects are materials which do not allow any light to pass through them.
- Opaques objects block light rays forming a shadow.
- We are unable to see through opaque objects because they do not allow any light to pass through them.

Examples of opaque objects.

- stones
- wood
- hard papers
- metals
- concrete walls
- sand paper
- human body

Importance of opaque objects.

- Opaque walls provide privacy to people in houses, bathrooms and latrines.
- Opaque objects provide shade to people and other animals.

Evaluation activity:

1. State any **two** ways in which light energy is similar to sound energy.
2. Mention any **one** example of each of the following:
   i) Transparent material
   ii) Translucent material
3. State any **one** way in which transparent materials are different from translucent materials.
4. State the effect of opaque objects on a beam of light.
5. Why is lightning seen before hearing thunder during thunderstorm?

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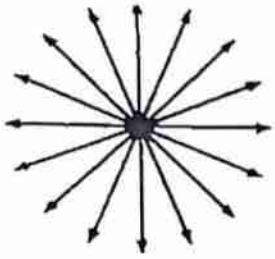
6. The list below is of different materials that affect light. Study and use it to answer the questions that follow.

frosted glass, wood, clear air, metal.

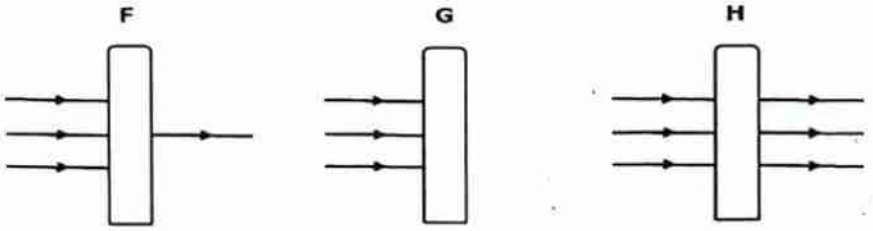
a) Name **two** materials from the list above that form shadows.
b) How is the frosted glass from the list above useful to builders?
c) Which type of objects is clear air from the list above?

The diagram below shows a property of light. Use it to answer question **7.**



7. State the property of light shown above.
8. Give any **two** differences between light and sound energy.
9. What type of objects do not form shadows?

10. The diagram below shows the effect of different objects on light. Study and use it to answer the questions that follow.



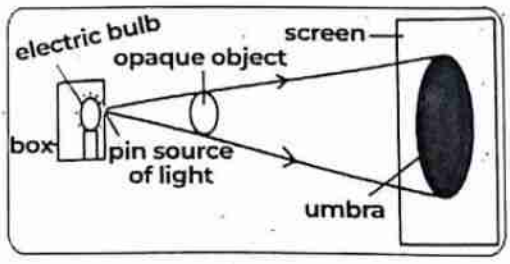
a) Name the type of objects marked **F** and **G.**
b) State any **one** use of objects represented by illustration **H** to people.
c) Give any **one** object that belongs to illustration marked **F.**

Lesson vocabulary:

• shadow • lunar • eclipse • solar • umbra • penumbra

SHADOWS

A shadow is an area of darkness on the surface caused by an obstacle to light rays. A shadow is formed when light rays are blocked by an opaque object.
Shadows are formed on the principle; Light travels in a straight line.



Characteristics of shadows.

- A shadow takes the shape of an object.
- Its size depends on that of an object.
- It has two regions i.e. umbra and penumbra.
- Its size depends on the size of a source of light.

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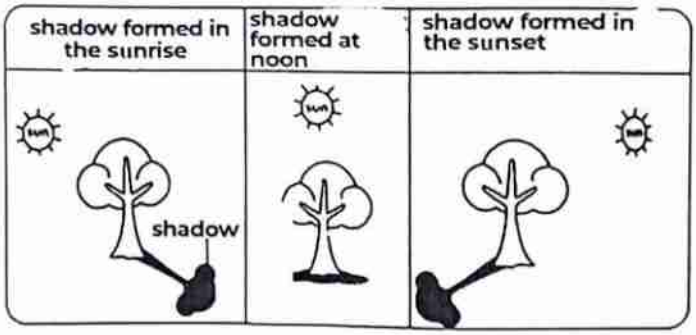
Factors that affect the nature of the shadow formed.

- Distance of an object from the source of light.
- Size of an opaque object used.
- Size of the source of light.

Note: Shadows appear longest at sunset and sunrise because the sun is at its horizon. Shadows appear shorter at midday or noon because the sun is overhead

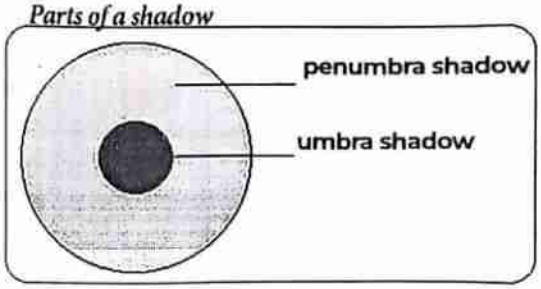
Positions of shadows at different periods of the day.



Parts/types of shadows.

- umbra
- penumbra

Parts of a shadow



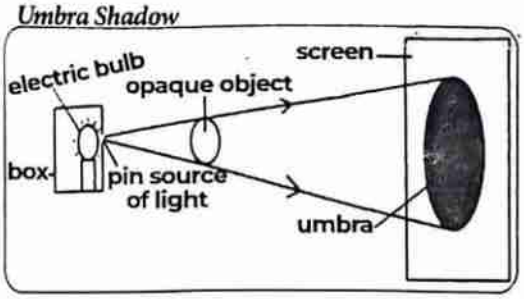
Umbra

This is the dark part of a shadow.

Its formation:

- By total obstruction of light.
- When light from the small/point of source of light is blocked by an opaque object.
- When light from a narrow opening is blocked by an opaque object.

Umbra Shadow



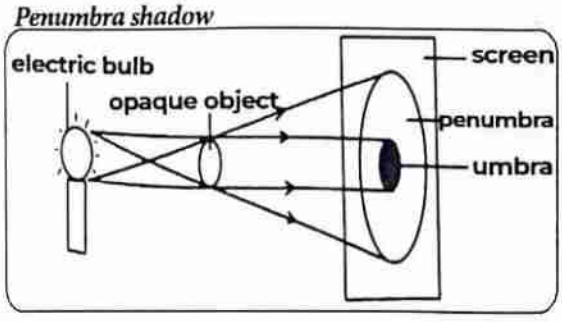
Penumbra

This is the lighter part of a shadow.

Its formation:

- By partial obstruction of light.
- When light from the big source of light is blocked by an opaque object.
- When light from a wide opening is blocked by an opaque object.

Penumbra shadow



Importance of shadows.

- They help to determine time.
- They help to determine direction.
- They provide shade.

ECLIPSES

An eclipse is a natural shadow formed when sunlight is blocked by the moon or the earth.

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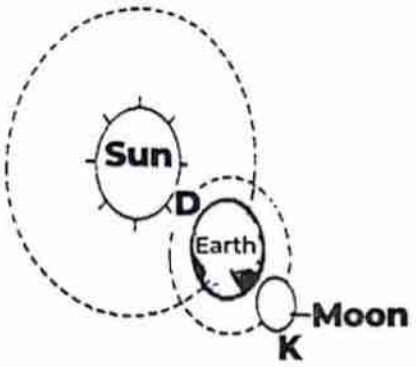
How the moon and the earth revolve to form eclipses.

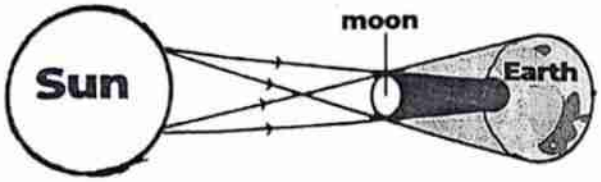


When the moon revolves and reaches point **D**, solar eclipse is formed.
When the moon is at point **K**, lunar eclipse is formed.

Types of eclipses
- lunar eclipse
- solar eclipse

Solar eclipse
A solar eclipse is the eclipse which occurs when the moon comes between the sun and the earth.
A solar eclipse is sometimes described as the **eclipse of the sun**.
Mnemonic used: **SME** ... sun, moon, earth

Solar eclipse



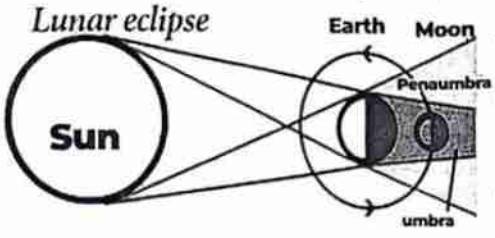
Explanation:
- The moon blocks sunlight and its shadow is cast on the earth.
- The moon's shadow does not fully cover the earth.
  Reason. The moon is smaller than the earth.

- During a solar eclipse, part of the earth is under total darkness.
  Reason. The moon is smaller than the earth.
- A solar eclipse occurs during day time.

Warning! It is dangerous to to expose the eyes directly to the solar eclipse.
Reason. It can damage the retina.

Lunar eclipse
A lunar eclipse is the eclipse which occurs when the earth comes between the moon and the Sun.
The shadow of the earth is formed on the moon.
A lunar eclipse is sometimes described as the **eclipse of the moon**.
Mnemonic used: **SEM** i.e. sun, earth, moon

Lunar eclipse



Explanation:
The whole moon is under total eclipse/ darkness.
Reason. The earth is bigger than the moon.
The moon cannot reflect any light.
Reason. The moon is in total eclipse.

The earth's shadow is able to cover the moon fully.
Reason. The earth is bigger than the moon.
During lunar eclipse, the shadow is cast on the moon.
A lunar eclipse occurs at night.

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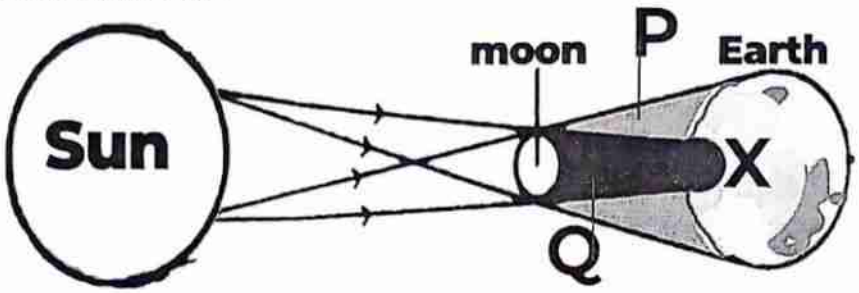
Dangers of eclipses.

- Direct view of the solar eclipse spoils the eyes.
- They lead to darkness whenever they occur.

Evaluation activity:

1. What is meant by the term shadow?
2. Explain how a shadow is formed **(2 marks)**
3. Mention any **one** factor which determines the type of shadow formed by an object.
4. What type of objects form shadows?
5. The diagram below shows a type of eclipse. Study and use it to answer the questions that follow.



   a) Identify the type of eclipse shown in the diagram above.
   b) Name the shadows marked **P** and **Q.**
   c) What happens to a person who would be in part **X** during the eclipse?
6. Give any **two** ways in which shadows are useful to people.
7. Name the type of eclipse formed when;
   i) The earth comes between the sun and the moon.
   ii) The moon comes between the sun and the earth.
8. Why is it dangerous to expose the eyes directly to the solar eclipse?
9. Name any **two** objects that are involved in the formation of eclipses.
10. What type of shadow is formed when light from a narrow opening is blocked by an opaque object?

Lesson vocabulary:

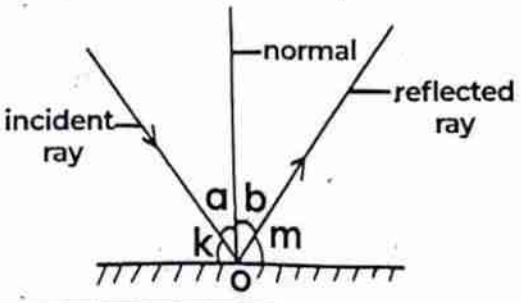
- bounce
- shiny
- reflection
- incidence
- diffuse
- specular

REFLECTION OF LIGHT

Reflection of light is the bouncing back of light rays as they hit a shiny surface.

Examples of shiny surfaces.

- plane mirrors
- polished surfaces
- shiny metals
- clear water



Explanation:

During reflection;

- **Incident ray.** A light ray from the source striking a shiny surface.
- **Normal.** The imaginary line that is perpendicular to the mirror surface.
- **Angle of incidence.** The angle formed between the normal and the incident ray

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- **Reflected ray.** The light ray bouncing off the mirror or shiny surface.
- **Angle of reflection.** The angle formed between the reflected ray and the normal.
- **Glancing angle.** The angle formed between the shiny surface and the reflected or incident ray.
- **Point of incidence.** The point where the incident ray, reflected ray and the normal meet.

∠**k:** glancing angle ∠**m**: Glancing angle ∠**a:** Angle of incidence
Point **O** is point of incidence. ∠**b:** Angle of reflection

Note: We are able to see objects in plane mirrors because of reflection of light. Light, heat and sound are forms of energy that can be reflected.

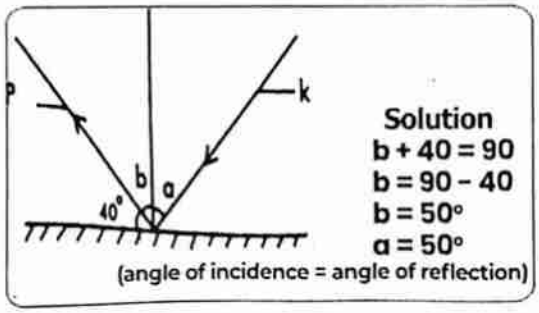
Laws of reflection.

- The incident ray, the reflected ray and the normal at the point of incidence, all lie in the same plane.
- The angle of incidence is equal to the angle of reflection.

Calculating simple problems on reflection of light.

Example:

Calculate the size of angle marked b and a in the diagram below.

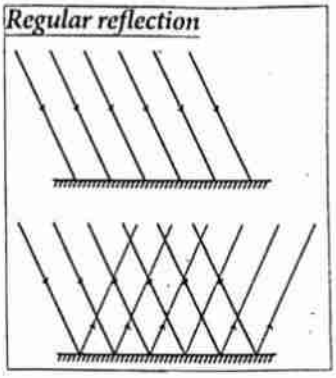


Types of reflection.

- regular reflection
- irregular reflection

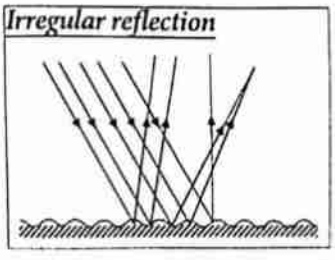
Regular reflection

Regular reflection is the type of reflection which occurs on smooth shiny surfaces.
When a parallel beam of light strikes a smooth shiny surface, the light rays are reflected in a parallel form.
Regular reflection is also called **specular / linear reflection.**
Regular reflection occurs on smooth flat shiny surfaces such as plane mirrors and clear water.



Irregular reflection

An irregular reflection is the type of reflection where light rays are bounced back in different directions.
In this type of reflection, the reflected rays are scattered.
An irregular reflection is also called **diffuse reflection.**
An irregular reflection occurs on shiny rough surfaces such as iron sheets, frosted glasses, muddy water, cracked plane mirror and disturbed water.



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Effects of reflection of light.

- It leads to the formation of images.
- It leads to the formation of eclipses.
- It causes vision.
- It causes scattering of light rays.

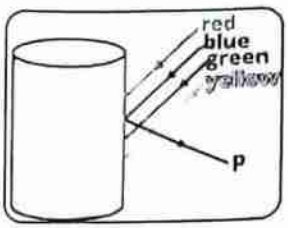
Ways in which reflection of light is useful to people.

- It enables people see their body parts using plane mirrors.
- It enables drivers to see traffic behind them.
- It enables periscopes to work.
- It helps in spreading out light in torches and electric bulbs.
- It enables people to identify colours of objects.

Reasons why objects appear by their colours.

- Objects appear by their colours because they absorb all other colours and reflect their own colour.
- A blue tin appears blue.

Reason. It absorbs all other colours and reflects only blue.



P is the reflected ray

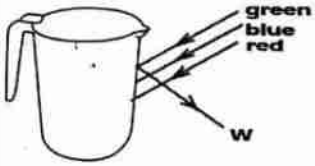
- Objects appear black because they absorb all colours and reflect none.
- Objects appear white because they reflect all colours and absorb none.

Evaluation activity:

1. What is meant by the term reflection of light?
2. State any **one** law of reflection of light.

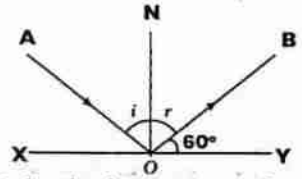
The diagram below shows reflection of light. Study and use it to answer questions **3** and **4**.

3. What type of reflection is shown above?
4. Why are the rays reflected as shown in the diagram?
5. Give any **two** ways in which reflection of light is useful to people.
6. Name any **one** optical instrument that uses reflection of light to function.
7. What type of reflection enables people to see their images in plane mirrors?
8. Why does a green dress appear green in white light?
9. The diagram below shows light rays of different colours striking a plastic jug. Study and use it to answer the questions that follow.



a) Name the ray marked **W**.
b) What is the colour of the jug?
c) Give a reason for your answer in b) above.
d) State the property of light that forms the colour of the jug above.

10. The diagram below shows a ray of light striking a plane surface. Study and use it to answer the questions that follow



Reflection from a plane mirror

a) Name the light rays marked **A** and **B**.
b) What is angle marked ***i*** called?
c) Calculate the size of angle marked ***r*** (2 marks)

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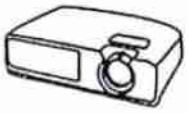
OPTICAL INSTRUMENTS

Optical instruments are devices that use light to form images of objects.
They use light energy in order for them to function.

Examples of optical instruments.

- plane mirrors
- lens cameras
- lenses
- periscopes
- magnifying glasses
- binoculars
- telescopes
- spectacles
- microscopes
- projectors

Microscope	Projector	Telescope	Spectacles	Binoculars
				

Uses of some optical instruments to people.

Optical instrument	Its use to people.
Microscope	For viewing tiny objects which cannot be seen using our naked eyes.
Projector	It is used to form images on the screen. It is mostly used by video jokeys, pastors, teachers, lecturers and doctors.
Telescope	It is used to view distant objects clearly. It is mostly used by astronomers.
Binocular	It is used to view distant objects clearly. It is mostly used by tourists and soldiers.
Spectacles	They are used to correct eye defects. They are mostly used by opticians.

Plane mirrors

A plane mirror is a mirror with a flat reflective surface.
It is polished on one surface in order to reflect light.
A plane mirror produces regular reflection of light.
Reason. A plane mirror has a smooth shiny surface.

Note: Both an image and an echo are results of reflection.

Characteristics of images formed by plane mirrors

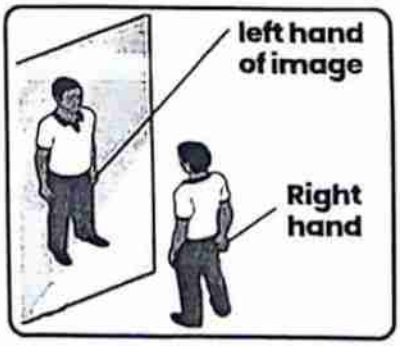
An image is the light picture of an object on the screen or film.

- Images are virtual

A virtual image is an image which cannot be cast on the screen or film.

- Images are laterally inverted.

The right side of the object appears to be the left side of the image formed in the mirror.(They are laterally inverted due to reflection of light by the plane mirror)



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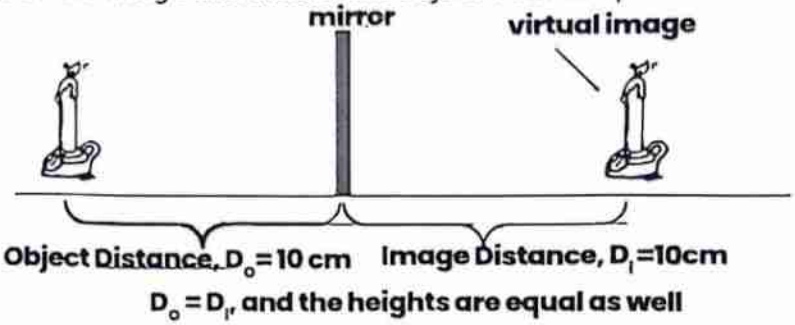
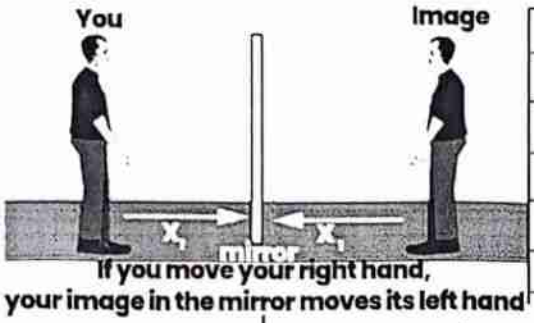
- Images are equal to objects in size.
- Images are erect / upright.
- Distance of the image and that of the object from the plane mirror is equal.

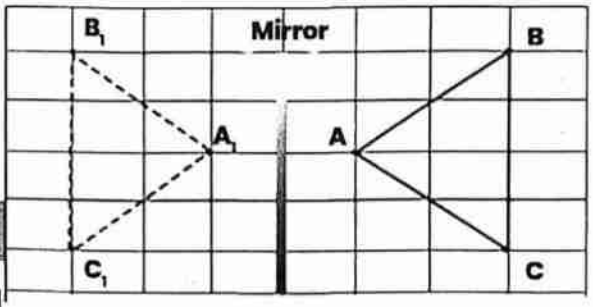


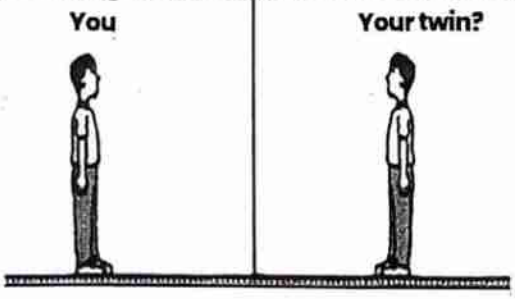
Mnemonic used: **VEELD** i.e. **V**irtual, **E**rect, **E**qual to the object, **L**iterally inverted and distance of image and object from plane mirror are equal.

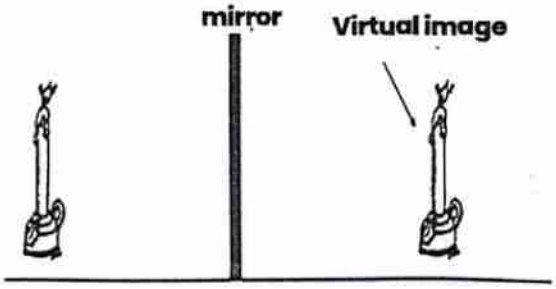
Illustrations showing images formed by plane mirrors.

Uses of plane mirrors to people.









- They are used in an overhead projector.
- They are used in salons and barber's shops.
- They are used as dressing mirrors.
- They are used in periscopes.

Devices that use plane mirrors.

- periscope
- kaleidoscope
- overhead projector

A kaleidoscope is used to produce colourful designs and patterns.

The periscope

A periscope is an optical instrument used to see overhead objects.

A periscope works on the principle of reflection of light/Light can be reflected.

A periscope uses two plane mirrors placed parallel to each other at an angle of 45.

Reason. To reflect light easily at the right angles/ To reflect light from one mirror to another.

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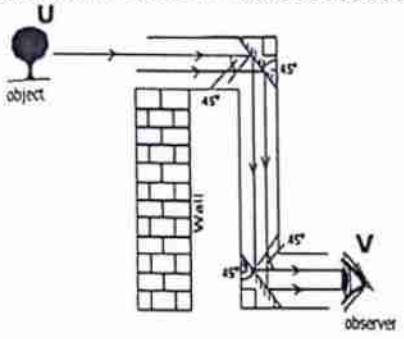


Explanation:

The observer at point **V** is able to see the image of the object at point **U** by reflection of light through plane mirrors.

The plane mirrors are used in the periscope to reflect light.

Characteristics of images formed by a periscope.

- The images are upright.
- The images are virtual.

Uses of a periscope to people.

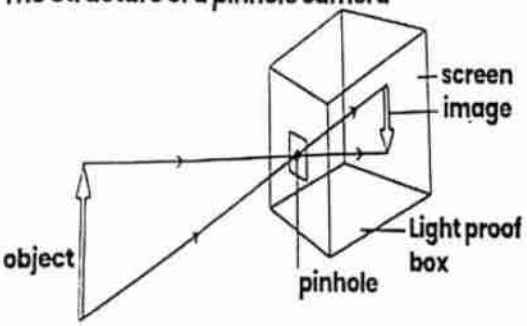
- It is used by submariners to see ships on the surface of the sea.
- It is used by soldiers in trenches to see their enemies on the ground.
- It is used by spectators to watch a match over the heads of the crowd.
- It is used by gatemen to see behind the fences.
- It is used to see around the corners.

Groups of people who use a periscope.

- submariners
- spectators
- soldiers
- security guards

The pinhole camera

The Structure of a pinhole camera



A pinhole camera is a simple camera with a single small hole and a screen.

It works on the principle that; Light travels in a straight line.

Explanation:

- The pinhole allows light into the camera.
- The screen is where the image is formed.
- Light-proof box allows light to only pass through the pinhole.

Characteristics of images formed by the pinhole camera.

- The image is inverted/upside down
  Reason. Light travels in a straight line and light rays intersect at the pinhole.
- The image is real.
  A real image is an image that can be cast on the screen.
- The image is diminished/smaller than the object.
  Reasons.
  - ☛ Light rays converge as they pass through the pinhole.
  - ☛ The image distance is shorter than the object distance.

Points to note: The sharpness of an image depends on the size of the pinhole and distance between the pinhole and image plane.

The image size depends on the distance between the object and the pinhole.

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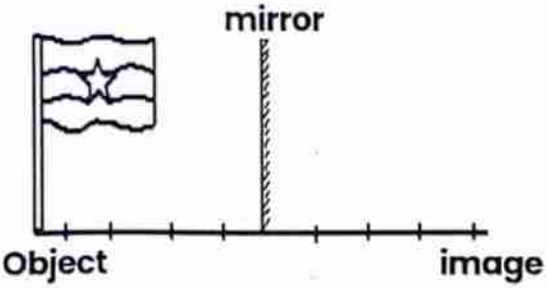
Evaluation activity:

1. What type of reflection is produced by plane mirrors?
2. State any **two** characteristics of images formed by plane mirrors.
3. How is an image similar to an echo?

The diagram below shows an object placed in front of a mirror. Use it to answer questions **4** and **5**.



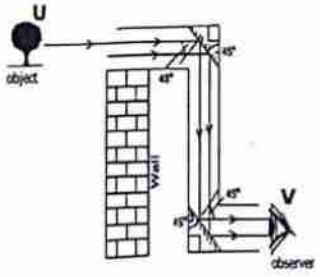
4. What type of mirror is shown in the diagram above?
5. Complete the diagram by drawing the image to show its appearance in the mirror.

6. Why do plane mirrors produce regular reflection of light?
7. Which type of mirrors are used in hair dressing in salons?
8. Give any **two** ways in which plane mirrors are useful to people.
9. Why are plane mirrors in a periscope placed at an angle of 45^{o}?

The diagram below shows a simple optical instrument. Use it to answer questions **10** and **11.**

10. Name the optical instrument shown above.
11. How is the above optical instrument useful to opticians?

12. Mention any **two** groups of people that use a periscope in their daily activities.
13. State the principle on which a periscope works.
14. Mention any **two** examples of optical instruments.
15. Why are plane mirrors polished on one side?
16. The diagram below shows an optical instrument. Study and use it to answer the questions that follow.



a) Name the optical instrument shown above.
b) What type of mirrors are used in the optical instrument above?
c) How is the observe at point **V** able to see the object at point **U**?
d) Give any **one** way in which the optical instrument above is useful to people.

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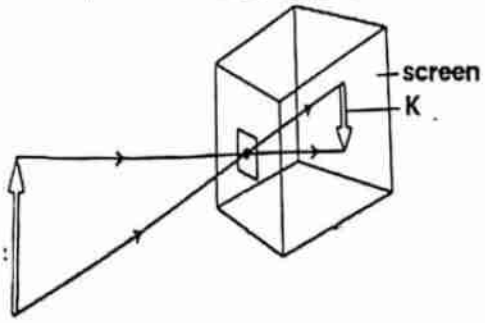
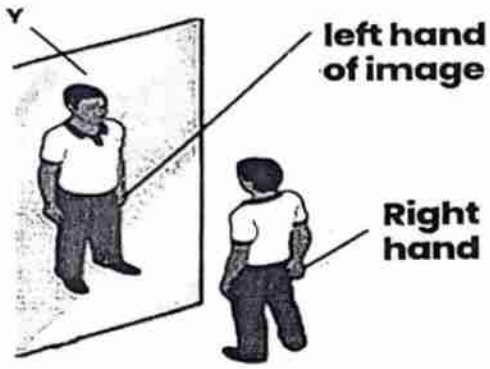
17 a) On which principle does a pinhole camera work?
b) Give any **two** characteristics of an image formed by a pinhole camera.
c) Mention any **one** factor which determines the sharpness of an image formed by a pinhole camera.

18. The diagram below shows a pinhole camera. Study and use it to answer the questions that follow.



a) Name the part marked **K** on the pinhole camera.
b) Give the importance of the screen on the pinhole camera above.
c) State any **two** characteristics of part marked **K** of the pinhole camera.

19. The diagram below shows a man viewing his image in a mirror. Study and use it to answer the questions that follow.



a) Name the type of mirror marked **Y.**
b) State any **two** characteristics of image of the man formed by the mirror marked **Y.**
c) Identify any one optical instrument that uses the mirror marked **Y.**

Lesson vocabulary:

- refraction
- medium
- prism
- spectrum
- emergent
- apparent

REFRACTION OF LIGHT

Refraction of light is the bending of light rays as they move from one transparent medium to another.

Causes of refraction of light

- Change in the speed of light.
- Difference in the density of transparent materials.

Principle of refraction of light.

The incident ray, the refracted ray and the normal at the point of incidence, all lie in the same plane.

Points to note: Refraction is the bending of light rays while reflection is the bouncing of light

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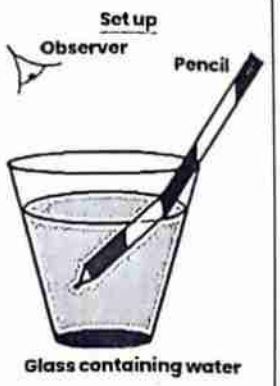
An experiment to investigate refraction of light.

Materials needed:

- a pencil
- a clear glass
- water

Procedure:

- Put water in a clear glass.
- Dip a pencil in water as shown below.
- Look at the pencil in water from different angles.
- The pencil appears bent due to refraction of light.

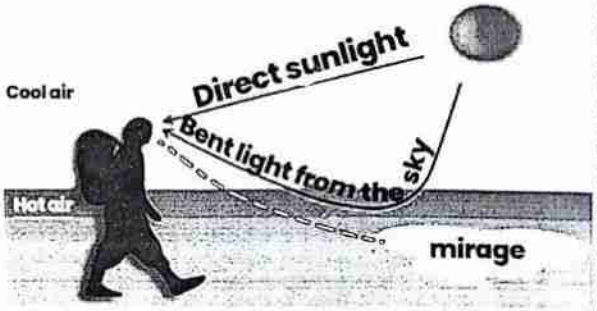


Effects of refraction of light.

- It causes water bodies to appear shallower than the real depth. This condition can lead to drowning or near drowning in water

- It produces **spectrum**.
  A spectrum is a band of seven colours of white light.

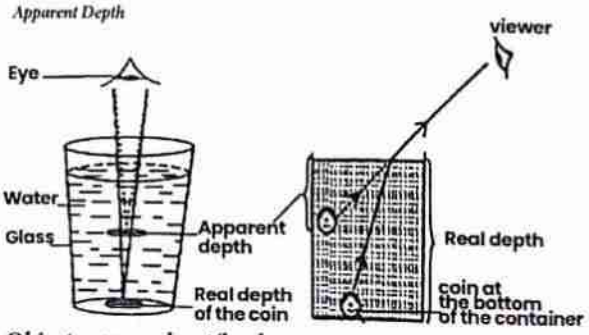
Mirage



- It leads to formation of mirage.
  A mirage is appearance of pool of water in a desert or on roads due to refraction of light by heated air.
  ☛ Mirage occurs on tarmac roads, desert areas and iron sheets.

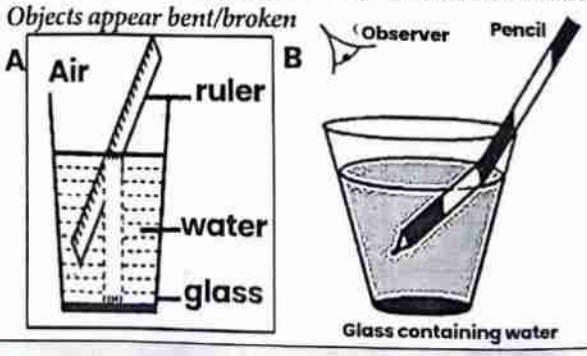
- It increases the width of vision.
  Words under a glass block appear to be raised and bigger relative to those away from the glass.

Apparent Depth



- It causes apparent depth.
  A condition in which an object place in water appears to be nearer than the real depth.

Objects appear bent/broken



- It makes straight objects dipped in water to appear bent or broken.
  A straight ruler/pencil partially dipped in water appears bent or broken.
  Reason. Due to refraction of light.

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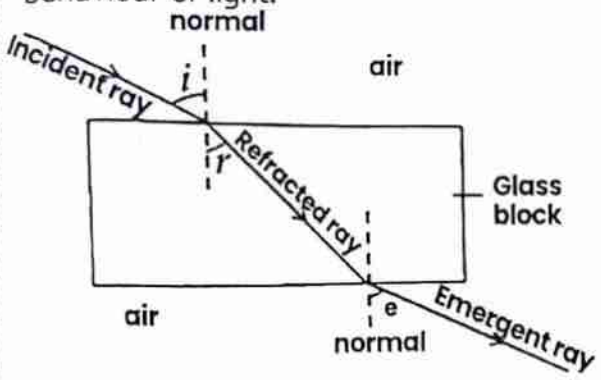
Refraction of light through the glass block.

A glass block is a piece of glass used in experiments or applications to study the behaviour of light.



Explanation:

Incident ray
The ray of light entering the glass prism.

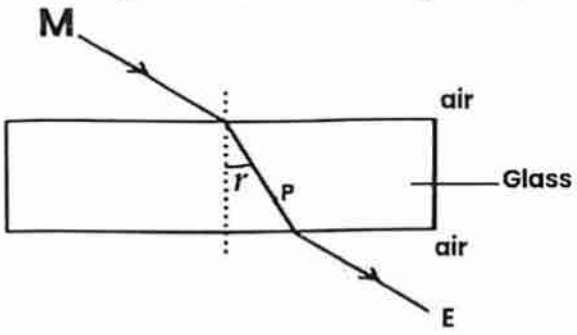
Refracted ray
The ray of light that bends away from the normal.

Emergent ray
The ray of light coming out of the glass prism after refraction.

∠i = angle of incidence.
The angle between the incident ray and the normal.

∠r = angle of refraction.
The angle between the refracted ray and the normal.

∠e = angle of emergency
The angle between the emergent ray and the normal.



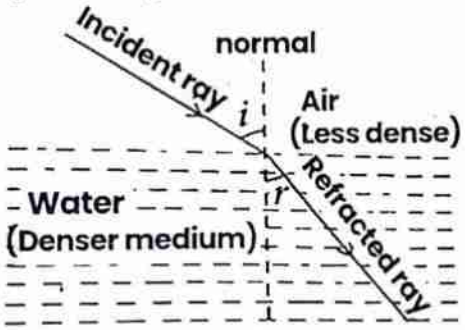
More explanation:

- In the diagram above, the ray marked **M** (incident ray) bends towards the normal.
  Reason. It moves from a less dense medium into a denser medium.
- The ray marked **E** (emergent ray) bends away from the normal.
  Reason. It moves from a denser medium into a less dense medium.

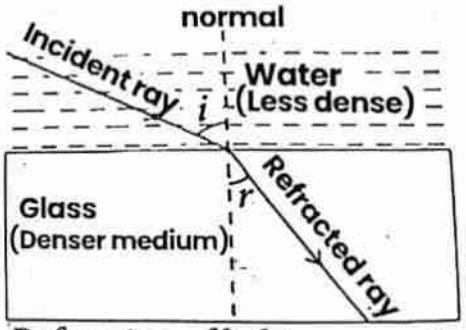
Note the following:

- *When a ray of light moves from air (less dense medium) into the glass prism (denser medium), the speed of the light ray reduces and it bends towards the normal.*
- *When a ray of light moves from a denser medium (glass prism) into a less dense medium, (air or water) the speed of light ray increases and it bends away from the normal.*

Study the diagrams below:



Refraction of light ray moving from air into water



Refraction of light ray moving from water into glass block

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Applications/importance of refraction of light.

- It helps to focus images onto the retina/film.
- It helps to magnify tiny objects using a hand lens.
- It helps lenses to focus light onto a single point.
- It enables images to be formed in a lens camera.
- It helps in correction of eye defects.

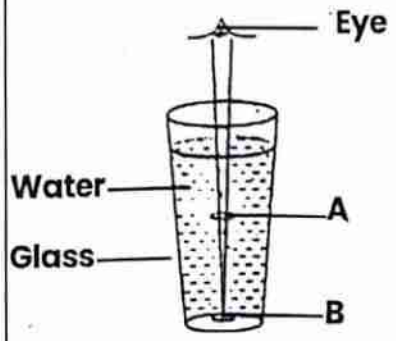
Dangers of refraction of light.

- Refraction of light causes drowning by making water bodies appear shallower than they real depths.
- Refraction of light makes fishing using a spear difficult.
- Refraction of light causes road accidents due to mirage.

Evaluation activity:

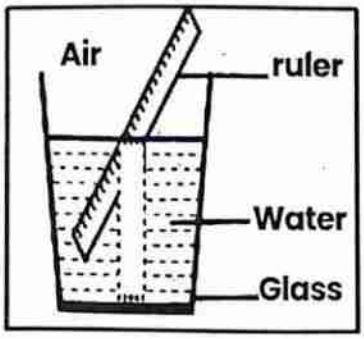
1. What is meant by the term refraction of light?
2. In which way is refraction of light different from reflection of light?
3. State any **one** factor that causes refraction of light.

The diagram below shows an object in a glass of water. Study and use it to answer questions **4** and **5**.



4. Name the point of the object marked **A.**
5. Why does the object seem to appear at point **A** yet it is at point **B**?
6. State any **two** effects of refraction of light.
7. Which property of light leads to the formation of mirage on tarmac roads?
8. State **one** danger caused by the formation of mirage on tarmac roads.

The diagram below shows a ruler appearing bent when placed in a glass of water. Use it to answer questions **9** and **10**.



9. Why does the ruler appear bent as shown above?
10. Name any **one** other material that can appear bent like a ruler above when put in water.
11. Give any **one** way in which refraction of light is helpful in our daily life.
12. Mr. Vacuum wanted to demonstrate refraction of light. He had the following materials for the experiment; a pen and a transparent glass.

a) Name **one** material Mr. Vacuum forgot to include on the list of his materials.
b) Why did Mr. Vacuum use a transparent glass for the experiment?
c) Describe how Mr. Vacuum will demonstrate refraction of light in this experiment. **(2 marks)**

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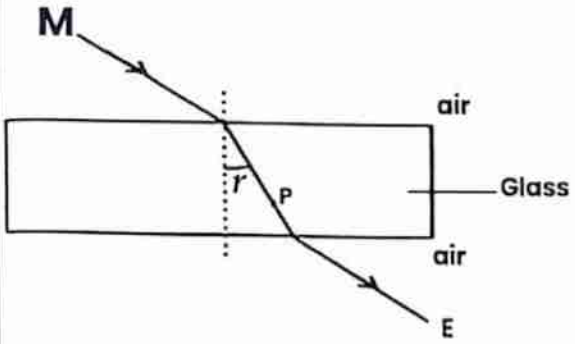
13. The diagram below shows how light passes through a glass block. Study and use it to answer the questions that follow.



a) Name the rays of light marked **M** and **P.**

b) Identify the angle marked ***r.***

c) Give a reason why the ray of light marked **E** bends as shown in the diagram above.

Lesson vocabulary:

• dispersion • spectrum • magnify • convex • converg • lens

DISPERSION OF LIGHT

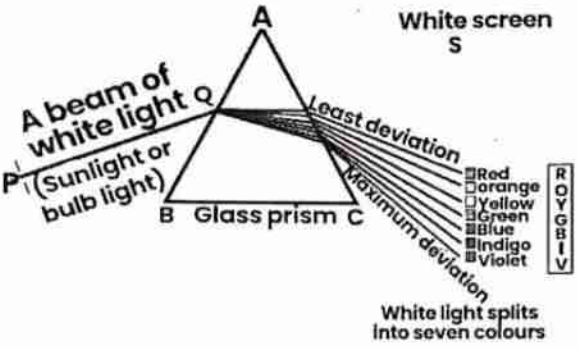
Dispersion of light is the splitting of white light into seven different colours.

- It is caused by refraction of light.
- The seven colours that make up white light is called a **spectrum.**
- These colours include, red, orange, yellow, green, blue, indigo, and violet.

Mnemonic used: **ROYGBIV**

A glass prism is a tranparent glass block with a flat polished surface that refracts light

Explanation:



The spectoral colours bend at different angles through the glass prism.

Reason. Due to difference in speed of wavelength that passes through the prism.

The violet ray of light bends most.

Reason. It travels at the fastest speed compared to other colours.

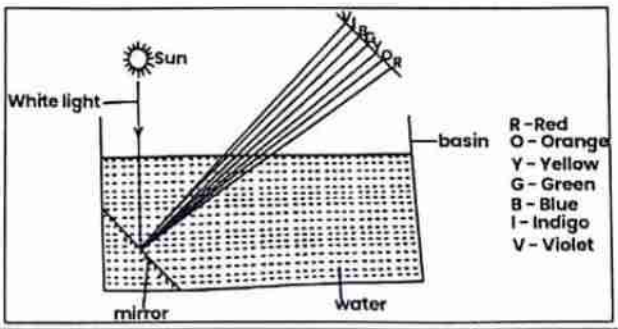
Red light ray bends least.

Reason. Red colour has the least wavelength/It travels at the slowest speed compared to other colours.

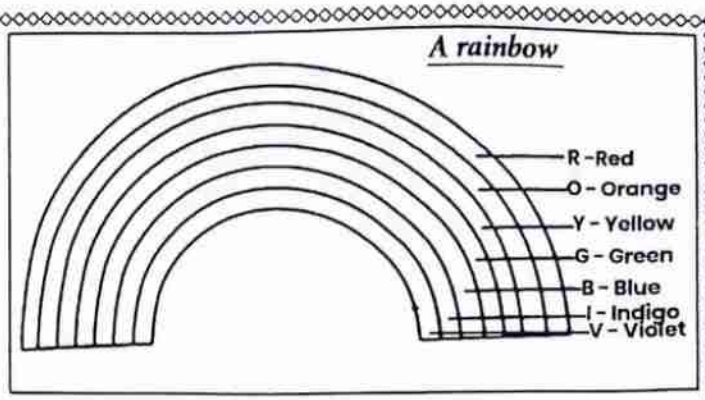
These colours are visible after white light has been passed through a glass prism.

The prism refracts light.

Formation of a spectrum using water and a mirror in a basin.



The rainbow



- A rainbow is a natural spectrum formed in the sky during rain or when it is drizzling.
- A rainbow is formed by refraction of sunlight through tiny droplets of rainwater.
- A rainbow is an example of spectrum in nature.
- It is usually seen in the sky when it is raining lightly.

Reason. Light rays are dispersed by raindrops.
Refraction of light and dispersion of light are processes that form a rainbow.

LENSES

A lens is a transparent curved material capable of refracting light.
It is an optical device that focuses or defocuses light.
Lenses can be made from glass, plastic or tissues.

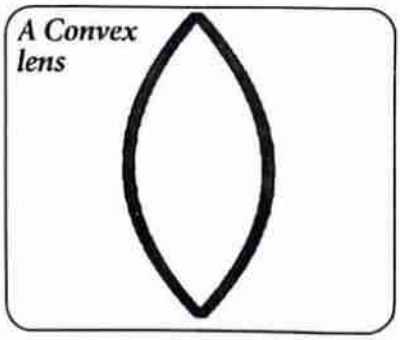
Note: When light rays pass through a lens, they bend towards the thicker surface

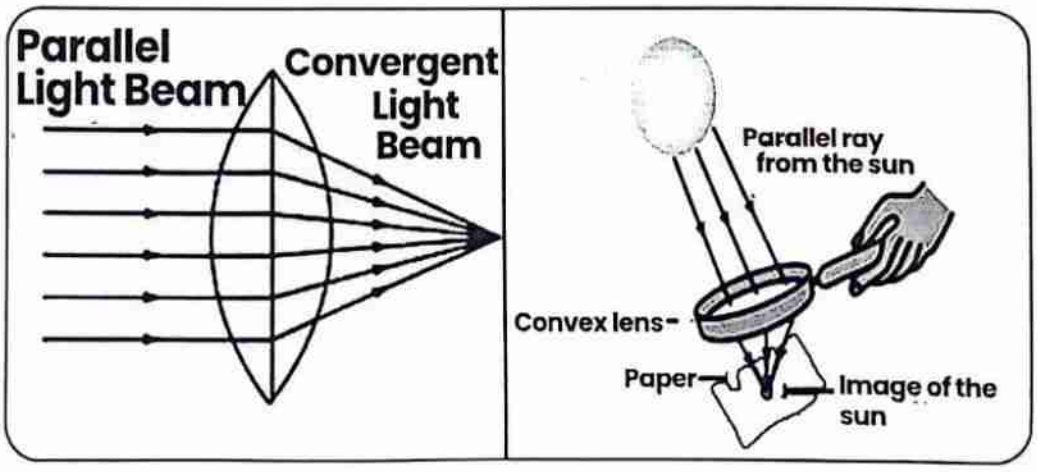
Types of lenses

- convex lens
- concave lens

Convex lenses

- A convex lens is curved outwards. It is thicker at its centre and thinner at the edges.
- A convex lens is also called a converging lens because it converges a parallel beam of light.
- When a parallel beam of light passes through a convex lens, thelight rays converge.
- Images formed by convex lenses are real, inverted and diminished.
- Lens cameras, microscopes and magnifying glasses use convex lenses.





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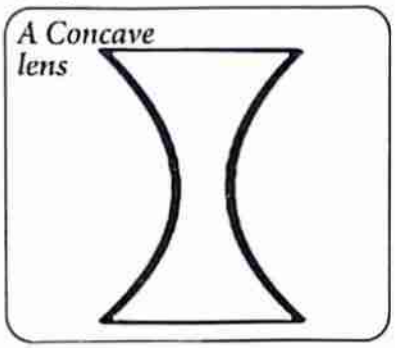
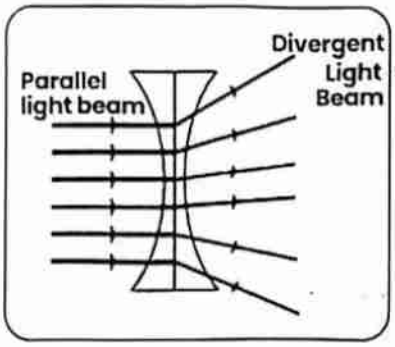
Concave lens

- A concave lens is curved inwards. It is thinner at the centre and thicker at the edges.
- It is also called a diverging lens because it diverges/spreads light rays passing through it.
- When a parallel beam of light passes through a concave lens, the rays are refracted and spread out into different directions.
- Images formed by concave lenses are virtual, erect and diminished.
- A projector uses a concave lens.





Advice to candidates:

When drawing light rays throught the lens, all lines should enter the lens upto the apparent centre because it is where the effect begins from.

The light ray that is moving through the middle since the thickness is the same, it does not bend , it moves straight . Then others when they reach the apparent centre, they move towards the thicker point of the lens.

That is why for the concave lens, they diverge then for the convex lens, they converge since the middle part for convex lens is thick so they converge towards the thicker point.

Additionally, a dotted line(principle axis) can be drawn in the middle of the lens for easy converging or diverging of light rays.

Ways in which lenses are useful to people.

- They are used in telescopes and cameras.
- They are used in film projectors to magnify images.
- They are used in spectacles to correct eye defects.
- They are used in magnifying glasses.
- They are used in binoculars to see distant objects.

Devices /instruments that use lenses.

- cameras
- telescopes
- binoculars
- projectors
- spectacles
- kaleidoscope
- magnifying glass
- microscopes

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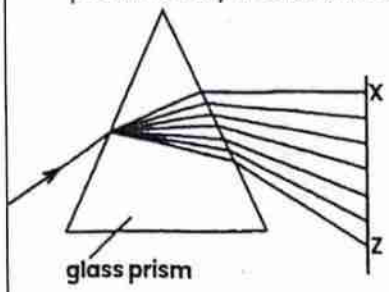
Evaluation activity:

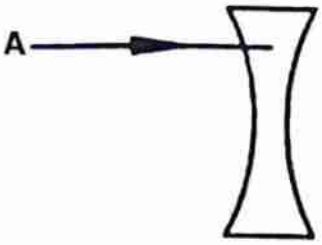
1. Give the meaning of the term dispersion of light.
2. What causes dispersion of light?
3. Write down any **two** colours of the spectrum.
4. The diagram below shows how light is split into seven colours using a glass prism. Study it and answer the questions that follow.



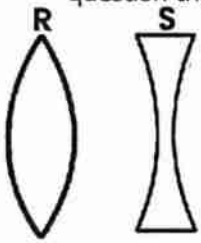
a) Name the colours marked **X** and **Z.**
b) Why do the light rays bend at different angles as they leave the glass prism?
c) Why do we usually see the above band of seven colours in the sky when it is raining lightly?

5. Name the type of lens used in a film projector.
6. What happens to light rays when they meet a convex lens.

The diagram below shows a light ray A onto a lens. Use it to answer questions **7** and **8.**



7. Name the type of lens in the diagram above.
8. Complete the path of ray A as it passes through the lens.
9. Give any **two** ways in which lenses are useful to people.
10. Apart from cameras and microscopes, name any **two** other optical insruments that use a lens.
11. Name the type of lens used in a magnifying glass.
12. The diagram below shows types of lenses. Study and use it to answer the question that follow.

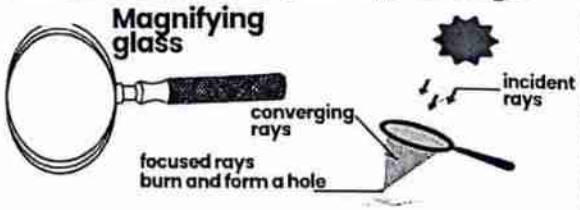


a) Name the types of lenses marked **R** and **S.**
b) How is the lens marked **R** useful in a lens camera?
c) What happens to a beam of light when it passes through lens marked **S**?

13. State the difference in structure of a concave lens and a convex lens.

Magnifying glass

- A magnifying glass is a simple hand lens used to make objects appear larger than their actual sizes.
- It used to magnify small objects.
- It is also called a hand lens.
- It uses a convex lens.



Characteristics of images formed by a magnifying glass

- The image is upright.
- The image is magnified.
- The image is virtual

Mnemonic used: **VUM**

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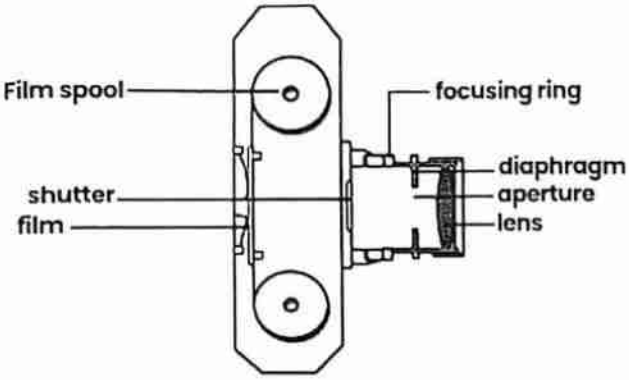
Optical instruments that use a magnifying glass.

- binocular
- telescope
- microscope

THE LENS CAMERA

A lens camera is an optical instrument used to take photographs.
It is painted black inside to prevent internal reflection of light.
It works on the principle of light travels in a straight line.

Parts of a lens camera



Functions of parts of a lens camera

Camera lens

It focuses light onto the film to form an image.

Diaphragm

The diaphragm controls the amount of light entering the camera.
It does so by regulating the size of the aperture.

Aperture

The aperture lets light into the camera.

Focusing ring

The focusing ring adjusts the distance of the lens from the film.

Shutter

The shutter keeps out light when not needed in the camera.
It exposes the film to light.
It opens or closes the aperture.

Film spool

It allows the film to move while taking images.

The film

The film is where images are formed.
It is the light sensitive part of the lens camera.

Characteristics of images formed by a lens camera.

- The images are real.

Reason. The image is formed by real light rays and can be cast on the screen.

- The images are diminished.
- The images are inverted.

Evaluation activity:

1. Give the function of each of the following parts of a lens camera:
   i) Aperture
   ii) Diaphragm
2. In which part of a lens camera are the images formed?
3. State the principle on which a lens camera works.

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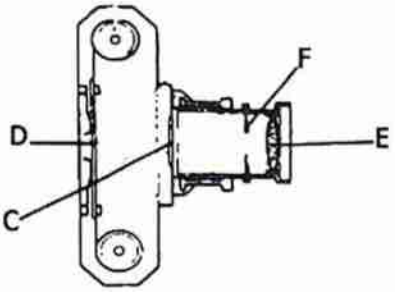
4. The diagram below shows an optical device. Study and use it to answer the questions that follow.

a) Name the optical device shown above.
b) How is the optical device above useful to people?
c) What type of lens is used in the above optical device?
d) State any **one** characteristic of images formed by the device.

5. State any **two** characteristics of images formed by a lens camera.
6. Explain how a rainbow is formed. **(2marks)**
7. What name is given to a transparent glass material that refracts light passing through it?
8. Why is the inner surface of the lens camera painted black?
9. The diagram below shows a lens camera. Study and use it to answer the questions that follow.



a) Name the parts marked C and E.
b) Give the function of part marked F on the lens camera.
c) State any **one** characteristic of an image formed by part marked D.

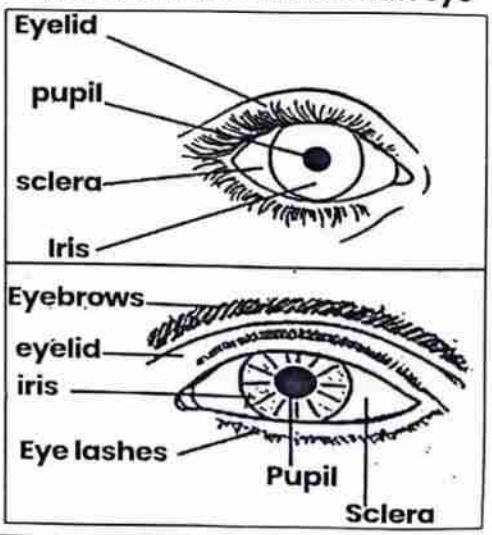
10. The table below shows parts of the lens camera and their functions. Study and complete it correctly.

Part of the camera	Function of the part
Diaphragm	.....
.....	Keeps out light when not needed
Lens	.....
.....	Allows light into the camera.

THE HUMAN EYE

The human eye is a sense organ for sight or seeing.
It is an optical organ of the human body.
Reason. It uses light to form images of objects.
An eye uses a **convex lens** to focus light onto the retina.
The complete shape of the eye is called an **eyeball**.

The front view of the human eye



Functions of each part of the human eye

Eyelashes
They trap dust and insects entering the eye.

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Eyelids

They close to keep out light when not needed in the eye.

Eyebrow

It prevents sweat from flowing into the eye.

Pupil

It lets light into the eye.

Iris

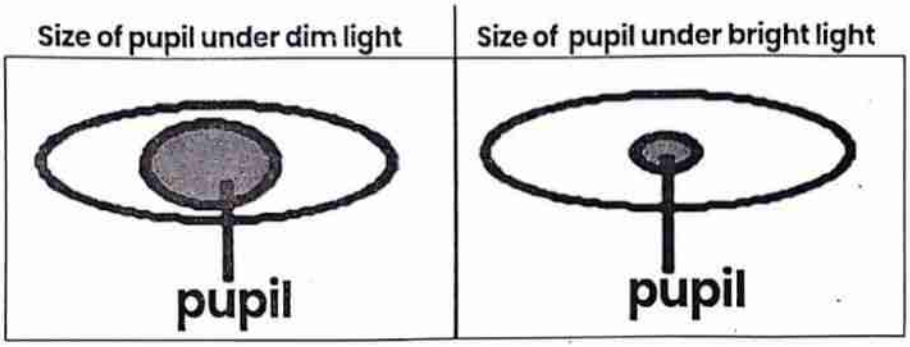
It regulates the amount of light entering the eye.

It does so by contracting and relaxing to alter the size of the pupil.

Note: Under bright light, the pupil narrows to prevent too much light from entering the eye.

Under dim light, the pupil widens to allow enough light into the eye.

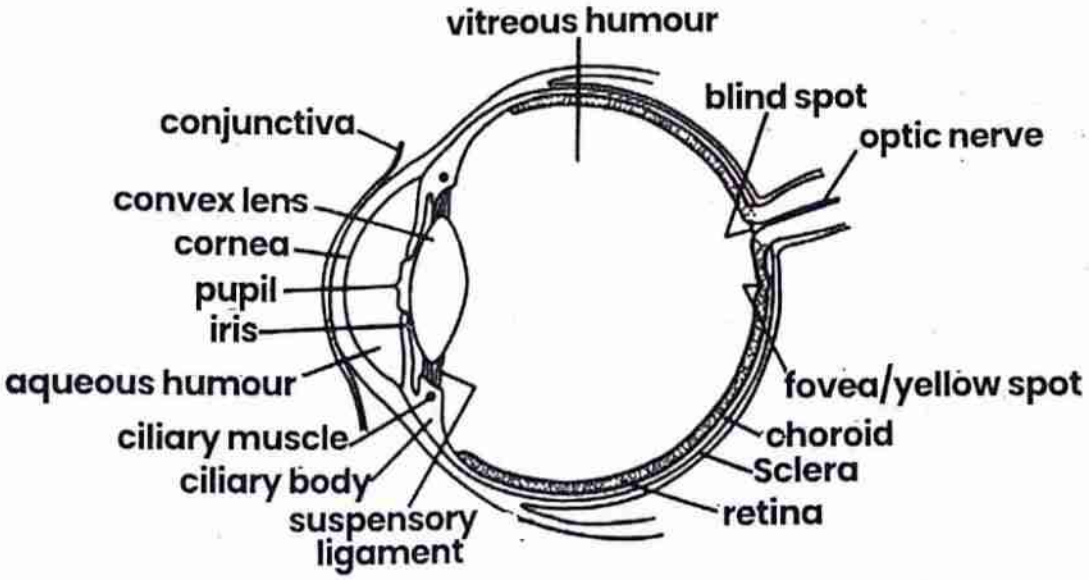
Appearance of the pupil under dim and bright light.



Uses of tears in the eyes.

- Tears lubricate the eyes.
- Tears kill germs in the eyes.
- Tears keep the eye moist.
- Tears wash away dust from the eyes.

The structure of the human eye.



Parts of the human eye and their functions.

Cornea
The cornea refracts light rays onto the lens.

Choroid
It prevents internal reflection in the eye.
The choroid layer has black pigment which controls light to prevent internal reflection.

Vitreous humour
This is the thick liquid between the convex lens and the retina.
It maintains the shape of the eyeball. This done by exerting pressure on scleratic membrane.
It refracts light from the lens to the retina.

Fovea
It is where the sharpest images are formed.
It has the highest concentration of light sensitive cells.

Ciliary muscle
It helps the lens to focus an image onto the retina correctly.
The ability of the eye lens to focus an image onto the retina correctly is called **accommodation.**

Sclera
The sclera protects and maintains the shape of the eyeball.

Suspensory ligament
It holds the convex lens in one position.

Optic nerves
They carry nerve impulses from the eye to the brain for interpretation.

Retina
The retina is where images are formed.
On the retina, there are light sensitive cells called **rods** and **cones**.
The cones are sensitive to bright light. while the rods are sensitive to dim light

Blind spot
It is where the optic nerves leave the eye.
Images formed on the blind spot cannot be seen since there are no light sensitive cells.

Lens
The eye lens focuses light onto the retina.

Aqueous humour
This is the thick liquid between the cornea and the convex lens.

Characteristics of images formed by the human eye.

- The images are real.
- The images are inverted.
- The images are diminished.

Mnemonic: **RID**

Corresponding parts of the human eye to those of a lens camera.

Human eye	Camera
eyelid	shutter
iris	diaphragm
pupil	aperture
convex lens	convex lens
retina	film

Mnemonic used: **DIFRESAP FC** i.e diaphragm,iris//film,retina//eyelids,shutter//aperture,pupil//focusing ring,ciliary muscles

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Similarities between the human eye and a lens camera.

- Both have convex lenses.
- Both form inverted images.
- Both have light sensitive parts where images are formed.
- Both form real images.
- Both form diminished image

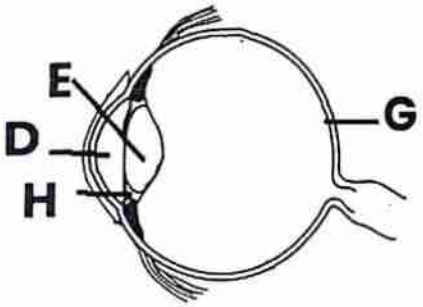
Differences between the human eye and a lens camera.

The images in the human eye are formed on the retina while the images in the lens camera are formed on the film.

- The distance between the lens and the retina in the human eye is fixed while the distance between the lens and the film in the lens camera changes
- The shape of the human eye lens changes while that of the lens camera does not.
- The eye lens is soft and elastic while the camera lens is made of a hard glass.
- The thickness of the lens in the human eye changes while that of the lens camera does not change.
- The iris in the human eye adjusts itself while the diaphragm in the lens camera can be adjusted by the photographer.
- The image in the human eye is focused by making the lens thicker while the image in the lens camera is focused by moving the lens.

Evaluation activity:

1. Name the optical organ of the human body.
2. How is the diaphragm of the lens camera similar to the iris of the human eye in the way they function?
3. In which part of the human eye are the images formed?
4. Give any **one** way in which tears are important in the human eye.
5. The diagram below shows the structure of the human eye. Study and use it to answer the questions that follow.



a) Name the parts marked **E** and **G**.
b) Give the function of part marked **H** in the human eye.
c) How is the function of part marked **D** similar to the aperture in the lens camera?

6. a) Give the function of each of the following parts in the human eye.
   i) Optic nerve
   ii) Cornea
   b) State any **two** characteristics of images formed by the human eye.

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7. The table below shows parts of the human eye in **A** and those of the lens camera in **B**.

A	B
Iris	Shutter
Pupil	Film
Eyelid	Diaphragm
Retina	Aperture

For each of the parts of the human eye, write the part of the lens camera from **B** which perform a similar function.

i) Iris.....

ii) Pupil .....

iii) Eyelid .....

iv) Retina .....

EYE DEFECTS/DISORDERS AND THEIR CORRECTION.

These are abnormalities that affect the proper functioning of the eyes.

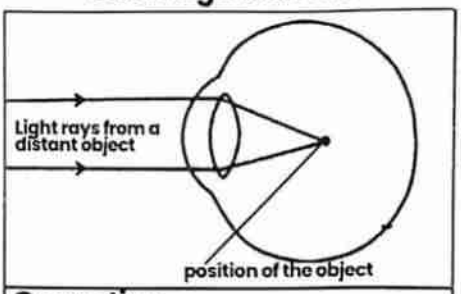
Examples of eye defects.

- short sightedness
- astigmatism
- long sightedness
- presbyopia
- colour blindness

Short sightedness (myopia)

Short sightedness is the inability of a person to see distant objects clearly.
The image of distant objects is formed in front of the retina.

Short sightedness



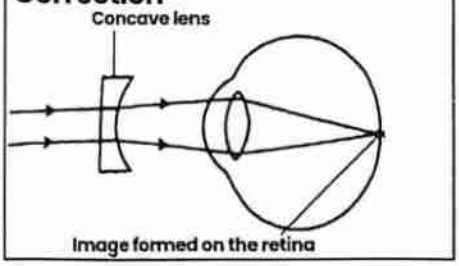
Causes of short sightedness.

- Too long eyeball.
- Too thick eye lens.

Correction of short sightedness

Wearing spectacles / glasses with a concave lens.
The concave lens diverges the light rays from distant objects to focus images on the retina.

Correction



Long sightedness (hypermetropia)

Long sightedness is the inability of a person to see nearby objects clearly.
The images of nearby objects are formed behind the retina.

Causes of long sightedness.

- Too short eyball.
- Too thin eye lens.

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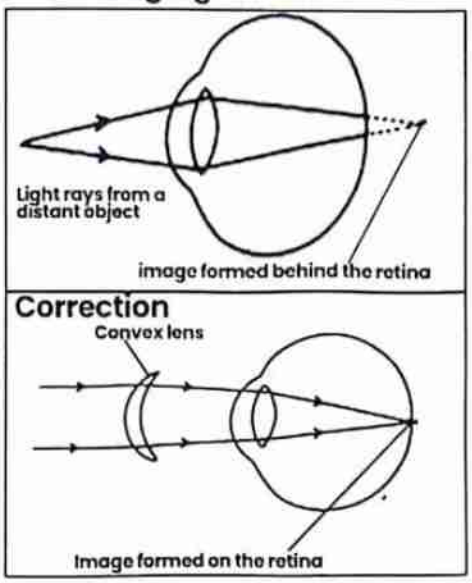
Correction for long sightedness

Wearing spectacles / glasses with a convex lens.

The convex lens converges the light rays from nearby objects to focus images on the retina.

Long sightedness



Astigmatism

Astigmatism is an eye defect caused by unevenly curved cornea.

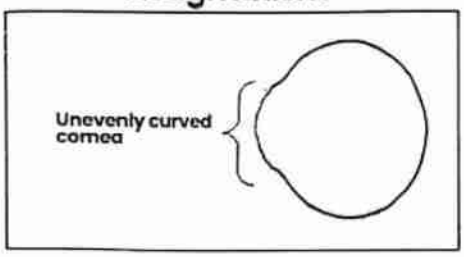
This makes images of objects blurred (unclear)

A person suffering from astigmatism fails to see objects in both vertical and horizontal planes at the same.

Correction of astigmatism.

Wearing glasses with special cylindrical lenses.

Astigmatism



Diseases that affect the human eyes

Eye disease	Signs and symptoms	Prevention, control and Treatment
Trachoma Caused by bacteria. Spread by houseflies	• Watery discharge from the eyes. • Itching of the eyes. • Swollen eyelids • The eyes turn red.	• Treat infected people with antibiotics. • Avoid shaking hands with infected people.
River blindness Caused by filarial worm Spread by blackflies.	• Itching of the eyes. • Blindness • Inflammation of the iris • The eyes turn red. • Swelling on the skin.	• Spraying using insecticides to kill black flies. • Stay away from black fly infested areas. • Early treatment of infected people.

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Conjunctivitis . Caused by bacteria or virus. Spreads by shaking hands with infected people Sharing face towels and handkerchiefs with infected people.	• Watery discharge from the eyelids. • The sclera turns pink. • Burning sensation around the eyes.	• Isolating infected people. • Avoid shaking hands with infected people. • Avoid sharing face towels and handkerchiefs.
Gonorrhoea Causes by bacteria An eye STD that affects newly born babies.	• Blindness in newly born babies. • Pus in the eyes	• Pregnant mothers should go for STD testing during ante-natal care.
Stye An inflammation of the eyelids due to bacterial invasion.	• A lot of pain from the eyes. • Formation of cyst on eye lids	• Practice proper eye hygiene.
Corneal ulcers Caused by bacteria or virus.	• Painful sensitiveness to light.	• Early treatment.
Night blindness Caused by lack of vitamin A in the body. It is non-contagious.	• Poor night vision • Feeling dryness of the eyes. • Low immunity	• Feeding on food rich in vitamin A e.g. onion, pawpaw, liver, carrots..
glaucoma Damage of the optic nerve by too much pressure on the eye.	• Blurry vision • Red eyes • Pain in the eyes	• Going for regular eye checkup in hospitals.

Ways of caring for the human eyes.

- Wash eyes with clean water regularly.
- Do not touch the eyes with dirty hands.
- Avoid reading under too dim or too bright light.
- Avoid wearing glasses with lenses that do not match with your eye lens.
- Feeding on a balanced diet.
- Having regular medical check-ups
- Avoid sharing face towels with people with eye diseases

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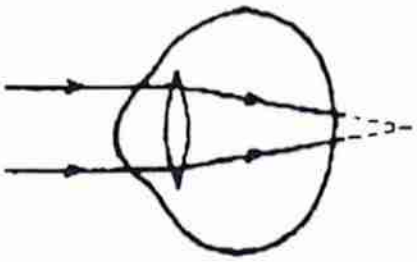
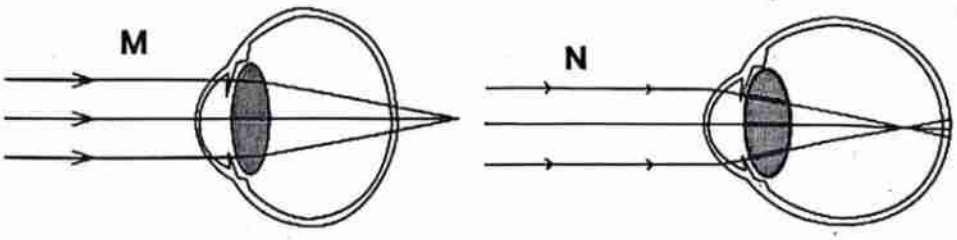
Evaluation activity:

1. The diagram below shows a defect of the human eye. Study and use it to answer the questions that follow.



a) Name the eye defect shown above.
b) State the **two** causes of the eye defect shown above.
c) What type of lens is used to correct the above eye defect?

2. Give any **two** ways in which the images formed by the lens camera are similar to those formed by the human eye?
3. In which way is the cause of river blindness different from that of night blindness?
4. Apart from river blindness and night blindness, mention any two other diseases that affect the human eye.
5. Why is a concave lens used to correct short sightedness?
6. Name the non-contagious disease that affects the human eye.
7. Give any **two** disorders that can affect the normal functioning of the human eye.
8. Name the Sexually Transmitted Infection that affects the eyes of the newly born babies.
9. State any **two** ways of caring for the human eyes.
10. a) Name the vector that transmits river blindness to people.
    b) Which body organ is affected by river blindness?
    c) Write down any **two** signs of river blindness.
11. How are houseflies a threat to human eyes?
12. The diagram below shows eye defects. Study and use it to answer the questions that follow.



a) Name the eye defects marked **M** and **N**.
b) State any **one** cause of eye defect marked **M**.
c) Which type of lens is used to correct eye defect marked **N**?

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MEANING OF KEY TERMS IN LIGHT ENERGY:

Ray of light: a path taken by light.

Dispersion of light: the splitting of white light into seven colours of spectrum

Prism: a transparent glass block with polished surfaces that refract light.

Image: light picture of an object.

Refraction of light: the bending of light rays.

Lens: an optic device that focuses or defocuses light.

Spectrum: a band of seven colours of white light.

Reflection of light: the bouncing back of light rays.

Optical instrument: a device that uses light to form images of objects.

Transparent object: a material that allows all light to pass through it.

Opaque object: a material that blocks light from passing through it.

Periscope: an optical instrument used to view overhead objects.

Plane mirror: a mirror with a flat reflective surface.

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7

TERM THREE
THEME: THE ENVIRONMENT
TOPIC 7: INTERDEPENDENCE OF THINGS IN THE ENVIRONMENT

Topical outline:

Interdependence of plants and animals.

- Animals depend on plants.
- Plants depend on animals.
- Animals depend on other animals.
- Plants depend on other plants.

Interdependence of living things and non-living things.

- Animals depend on non-living things (air, water and soil)
- Plants depend on non-living things (air, water and soil)
- Non-living things benefit from living things.
- Food chain

Agroforestry.

- Growing trees and crops together.
- Rearing animals and growing crops on the same farm.
- Rearing and caring for animals, growing and caring for crops.
- Importance of agroforestry.
- Care for trees in agroforestry.
- Proper harvesting of trees in agroforestry.
- Starting and managing a school/home woodlot project.
- Record keeping.

Ref: P.7 Standard curriculum pages 86-89.

Advice to teacher:

- *Do not teach food web in this topic because it out of primary seven curriculum.*
- *Encourage learners to explore the environment so as to enrich the content with suitable relative examples.*
- *Initiate and maintain a tree and crop growing project in which learners can participate. Help learners to go through the planning process.*
- *Develop skills of protecting and conserving the environment.*

Environment

Environment refers to the surrroundings and conditions in which an organism lives.

Components of environment

Rocks//animals//fungi//bacteria//buildings//water//air//vehicles//land

Interdependence

Interdependence is the way in which things benefit from each other in the environment for survival.

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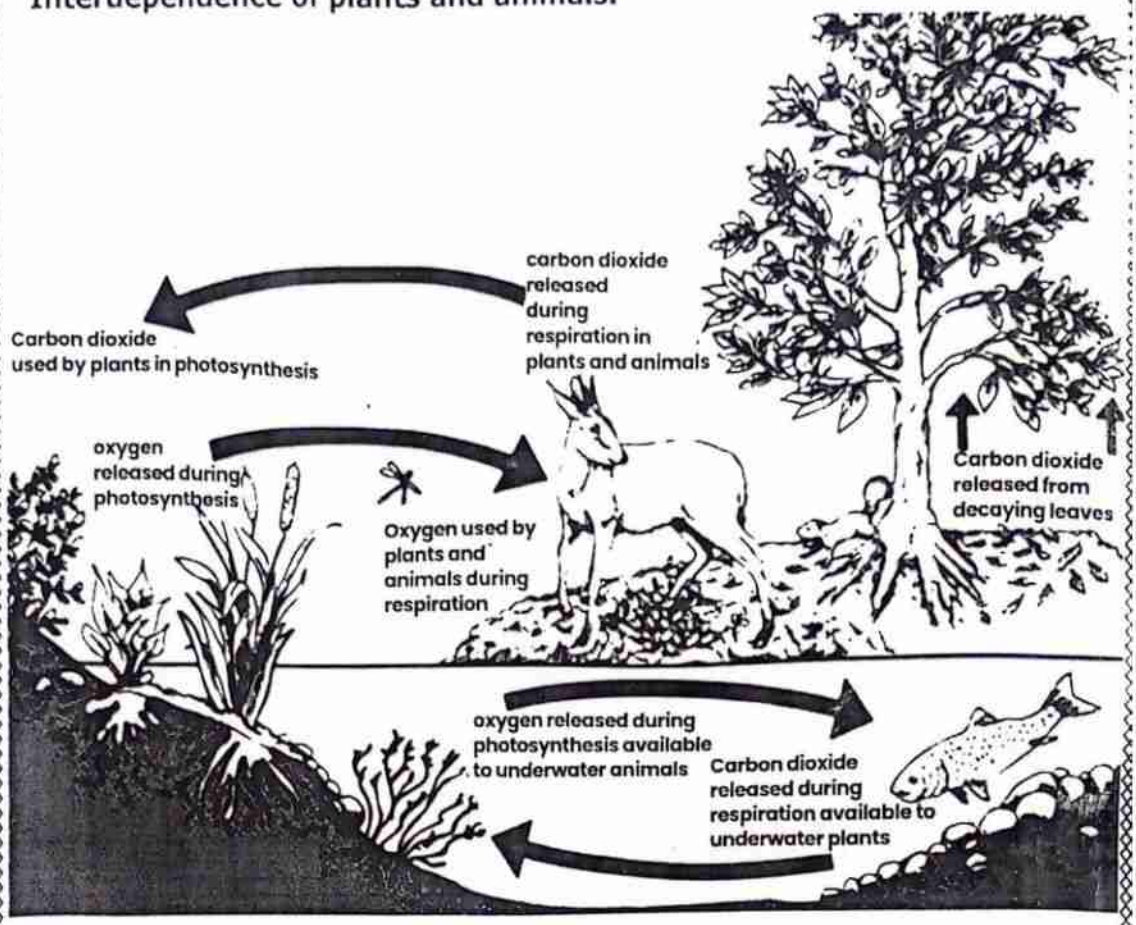
Interdependence of plants and animals.



Ways in which animals depend on plants.

- Some animals get food from plants.
- Animals get herbal medicine from plants.
- Animals get oxygen from plants
- Animals get shelter from plants.

Ways in which animals depend on other animals.

- Some animals get food from other animals.
- Parasites depend on the hosts for survival.
- Animals depend on other animals for protection.
- Some animals use other animals for transport.

Ways in which plants depend on animals.

- Plants get carbon dioxide from animals
- Some animals pollinate plants.
- Some animals aerate the soil for plant respiration.
- Plants obtain manure from animals
- Animals help in seed and fruit dispersal
- Some animals destroy plant pests.

Ways in which plants depend on other plants.

- Big and tall plants protect short plants from strong wind.
- Plants depend on other plants as habitats.
- Some plants provide support to others.
- Some parasitic plants obtain food from the host plants.
- Plant remains are used as manure by other plants for growth.
- Leguminous plants fix nitrogen in the soil which is used by other plants.

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Interdependence of living things and non-living things.

Way in which animals depend on non-living things.

- Some animals live in soil.
- Water is a habitat for aquatic animals.
- Air provides oxygen to animals for respiration.
- Animals drink water.
- Some animals get food from water bodies.

Ways in which plants depend on non-living things

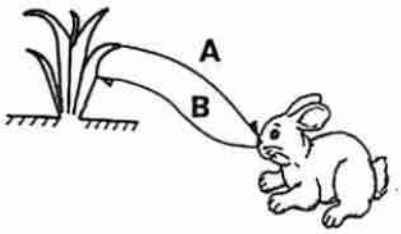
- Plants depend on wind for pollination and dispersal.
- Plants obtain nutrients from the soil.
- Plants use water and carbon dioxide as raw materials for photosynthesis.
- Plants use sunlight as a source of energy for photosynthesis.
- Plants use ~~oxygen~~ nitrogen for protein synthesis.
- Soil holds plant roots firmly.

Ways in which non-living things depend on living things

- Bacteria help in soil formation.
- Plants protect soil against erosion.
- Plants and animals decay to form humus in soil.
- Plants maintain water cycle through transpiration.
- Earthworms aerate the soil.
- Plants purify air by absorbing carbon dioxide from the atmosphere.

Evaluation activity:

1. What is meant by the term environment?
2. Mention any **two** living components of the environment.
3. State ant two ways in which plants benefit from animals.
4. How do plants help to purify air?
5. Give any **two** ways in which plants help to support human life.
6. The diagram below shows a relationship between a rabbit and a sisal plant. Study and use it to answer the questions that follow.



   a) What name is given to the relationship shown above?
   b) Name the gaseous products marked **A** and **B.**
   c) State any **one** way in which the rabbit benefits from a sisal plant above.

7. Give any **two** ways in which animals benefit from plants.
8. In which way do plants help to maintain water cycle in the environment?

FOOD CHAIN

A food chain is the flow of energy from one organism to another.

A food chain shows the feeding relationship among living things.

In a food chain, the arrows (⟶) are used to show the flow of energy.

Components of a food chain.

- producers
- consumers
- decomposers

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Feeding levels in a food chain.

1. Producers.

These are organisms that make their own food.

Plants are the producers in a food chain.

Reasons.

- Plants provide food directly or indirectly to all consumers.
- Plants make their own food.

The sun provides sunlight energy for plants to carry out photosynthesis.

The role of plants in a food chain is; To make food for primary consumers.

2. Primary consumers

These are animals which feed directly on producers/plants.

They are called herbivores because they feed on plants.

Primary consumers provide food to secondary consumers.

They are eaten as prey by secondary consumers.

A prey is an animal hunted and killed by another animal for food.

Examples of primary consumers.

- zebras
- antelopes
- goats
- sheep
- cattle
- giraffe

3. Secondary consumers

These are animals which feed on primary consumers.

They are predators to primary consumers.

A predator is an animal that hunts and kills another animal for food.

Such animals are called carnivores because they feed on meat.

Examples of secondary consumers.

- lions
- leopards
- vultures
- eagles
- kites
- cheetahs
- hyenas
- tigers
- falcons

Humans are omnivores because they feed on both plants and meat.

4. Tertiary consumers

These are animals that feed on secondary consumers.

Examples are scavengers like;

- vultures
- hyena
- crows
- jackals
- foxes
- marabou storks

How consumers get energy from plants.

Primary consumers: By feeding directly on plants.

Secondary consumers: By feeding on primary consumers.

Decomposers

These are organisms that break down dead plants and animals to form humus.

Example are; bacteria and fungi.

They break down dead organic matter.

They return nutrients to the soil.

Decomposers are not a feeding level in a food chain because they obtain their food from various sources.

They have the highest population in food chain because they do not have any organisms that depend on them.

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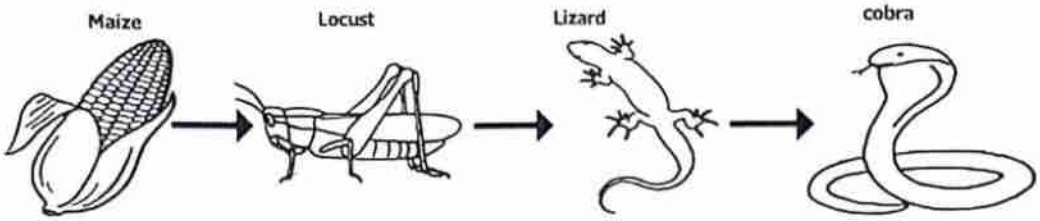
A summary of a food chain.

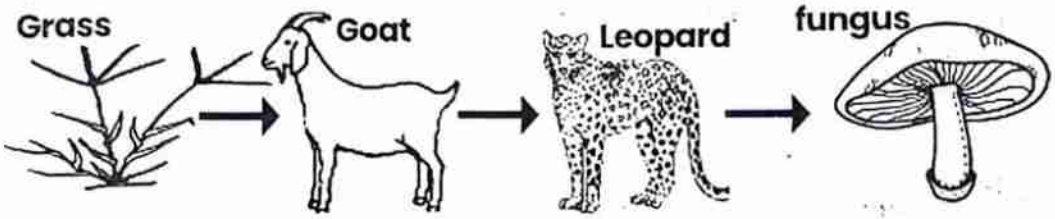
Producer → Primary Consumer → Secondary Consumer → Tertiary Consumer → Decomposer

Study the diagrams below



Explanation:

- In the diagram above, the locust eats maize, the lizard eats the locust and the cobra eats the lizard.
- Therefore, energy flows from maize to locust, from locust to lizard and from lizard to the cobra. This feeding relationship is called a food chain.
- The producer in this food the chain is maize, primary consumer is the locust and the lizard is a secondary consumer while the cobra is a tertiary consumer.
- When the locust is removed from the food chain above, the maize increase in number and lizards reduce in number. When the lizard is removed, the locusts increase in number.



Producer: grass **Primary consumer:** goat **Secondary consumer**: leopard **Decomposer:** fungus

Evaluation activity:

1. What is meant by the term food chain?
2. Give the role of each of the following components in a food chain:
   i) Plants
   ii) bacteria
3. In which way are primary consumers different from secondary consumers?
4. How do carnivorous animals get energy from plants?
5. Why are plants called producers in a food chain?
6. The list below is of living things in the environment. Use it to answer the questions that follow.

maize, lion, duck, zebra

a) Use the given list to construct a correct food chain.
b) Which living thing on the list above is a;
i) Producer?

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ii) Secondary consumer?

c) Name the source of energy for the above producer.

7. Give the importance of plants in a food chain.
8. State the difference between a prey and a predator in a food chain.
9. a) Use the list of living things below to complete the given food chain.

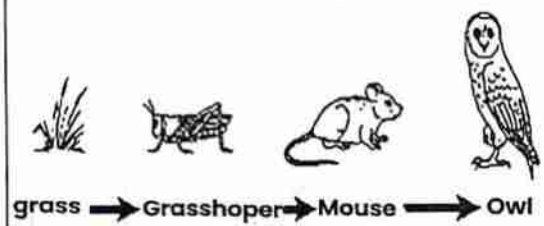
Lion, goat, grass

..... → **goat** → .....

b) Which **one** of the living things in the food chain above is a producer?

c) What is the source of energy for the producer in the food chain above?

10. Why is a sheep called a herbivorous animal?
11. State the role played by the sun in a food chain.
12. State what happens to primary consumers when secondary consumers are removed from the food chain.
13. The diagram below shows the feeding relationship in animals in the environment. Study and use it to answer the questions that follow.



a) Identify the feeding relationship shown above.

b) Name the primary consumer from the diagram above.

c) What will happen to the mouse when the grasshopper is removed from the above feeding relationship?

d) What do the arrows indicate on the feeding relationship above?

AGROFORESTRY

Agroforestry is the growing of crops together with trees on the same piece of land.

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Components of agroforestry.

- crops
- trees
- livestock /animals

Growing crops and trees together.

Ways in which crops benefit from trees.

- Trees provide shade to crops against sunshine.
- Trees protect crops against strong wind.
- Trees protect crops from heavy raindrops.
- Trees provide support to some crops e.g. vanilla and white yams.
- Tree leaves fall and rot to form humus which promotes proper crop growth.

Terms to learn.

Mixed cropping is the growing of more than one crop on the same piece of land at the same time.

Mixed farming is the rearing of animals and growing of crops together on the same piece of land.

Advantages of mixed farming

- Animals make good use of crop remains.
- Animals produce manure that promotes crop growth.
- The farmer gets a balanced diet from crops and animals.
- It is an assurance against total production failure.
- The farmer gets double income from crops and animals.

Caring for animals in agroforestry.

- Proper feeding.
- Vaccinating and deworming animals regularly.
- Treating sick animals.
- Building strong houses for animals.
- Regular cleaning of animal houses.

Growing and caring for trees in agroforestry.

Factors to consider when selecting tree species to grow with crops.

- Faster growth of trees.
- Deep rooted trees.
- Multipurpose tree species.
- Forage and seed pods to provide animal fodder.
- Ability to add fertility to the soil.
- Light canopy.
- Less prone to pests.

Types of trees grown in agroforestry.

Local/indigenous breeds of trees.

- mahogany
- mvule
- acacia
- natal fig (mutuba)
- zebra wood

Characteristics of local breeds of trees.

- They produce hard wood.
- They can withstand poor soil and harsh weather.
- They grow wildly.
- They take long to grow and mature.

Exotic trees include;

- cypress
- cycads
- pine
- cedar

Characteristics of exotic trees.

- They produce soft timber.
- They mature faster.

Characteristics of trees suitable for agroforestry.

- They should be fast growing trees.
- They should be multipurpose.
- Their leaves should decompose easily.
- They should be leguminous.
- They should have light canopy.

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- They should be deep rooted.
- They should not compete for plant ~~nitrients~~. nutrients
- They should be disease resistant.
- They should not interfere with soil moisture.

Importance of agro-forestry.

- Trees provide shade to crops and animals.
- Crops,trees and animals are source of food to a farmer.
- Cover crops and trees control soil erosion.
- Trees protect crops and buildings from strong wind.
- Agro-forestry increases the population of trees in a place.
- Trees help to maintain water cycle.
- A farmer gets poles and wood fuel from trees.

Evaluation activity:

1. (a) Give the meaning of the following terms:
   (i) Agroforestry (ii) Mixed cropping
   (b) State any **two** characteristics of trees suitable for agroforestry.
2. (a) What name is given to the practice of growing crops and trees together on the same piece of land?
   (b) State any **two** ways in which crops benefit from trees when grown together.
   c) Give any **one** factor to consider when selecting tree species to grow with crops.
3. (a) Give any **two** characteristics of exotic trees.
   (b) State any **two** ways in which local trees are better than the exotic trees.
4. (a) State any **two** ways in which farmers can care for animals on their farms.
   (b) Give any **two** ways of caring for trees in agroforestry.
5. The diagram below shows a farm growing practice. Study and use it to answer the questions that follow.

a) Name the farm growing practice shown above.
b) Give any **two** advantages of the above farm growing practice.
c) State any one way in which trees are important in the above farm growing practice.

Lesson vocabulary:

- staking
- pruning
- pollarding
- coppicing
- excess
- lopping

Caring for crops and trees in agroforestry.

- **Watering trees.**

It should be done in the morning and evening because there is low rate of evaporation in the soil.

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It is not recommended during day time due to high rate of evaporation

- **Applying fertilisers.**

To supply the necessary plant nutrients for proper growth of crops and trees.

- Spraying crops and trees using pesticides.
- Timely weeding of crops and trees.
- Mulching the soil.
- Pruning crops and trees.
- Staking crops and trees.
- Fencing crops and trees

Weeding crops and trees in agroforestry

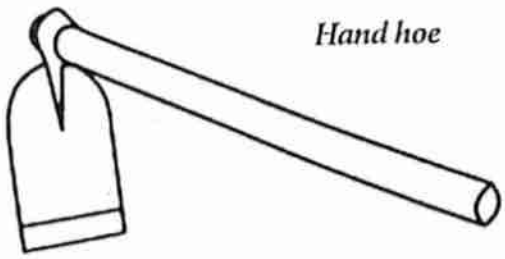
Weeding is the removal of unwanted plants growing in the garden.

It is mainly done using a hoe.

The unwanted crops growing in the garden are called **weeds**.

Examples of weeds.

- spear grass
- black jack
- elephant grass
- pigweeds



Hand hoe

Ways in which weeds can lower the yields of crops and trees.

- They compete with crops for sunlight, space and nutrients.
- They are hiding places for crop pests.
- They spread diseases to crops.

Ways of controlling weeds (methods of weeding)

- Slashing using a scythe /mower.
- Uprooting weeds.
- Mulching the soil.
- Using herbicides.
- Digging weeds.
- Cover cropping.
- Practising crop rotation.

Advantages of timely weeding crops and trees in agroforestry.

- It eases harvesting.
- It eases application of manure.
- It breaks down the life cycle of crop pests.
- It gives crops more strength to fight diseases.
- It destroys the hiding places for crop pests.
- It eases spraying.
- It controls crop diseases
- It eases management of crop diseases by spraying.
- It reduces competition for plant nutrients, space and sunlight.
- It controls crop pests.

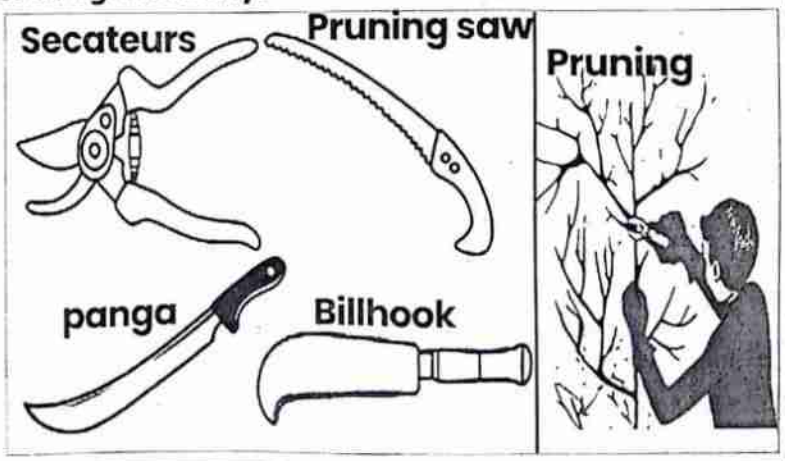
Pruning crops and trees in agroforestry.

Pruning is the removal of excess branches or leaves from the plant.

It can be done using; secateurs, pruning saw, billhook or panga

A billhook is suitable for pruning banana plants.

secateurs Pruning saw pruning



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Reasons why pruning trees and crops in agroforestry is important.

- It reduces the weight of the plant.
- It increases yields of crops and trees.
- It eases spraying and harvesting.
- It eases weeding of crops and trees.
- It reduces the rate of transpiration.
- It reduces hiding places for crop pests.

Staking crops and trees in agroforestry.

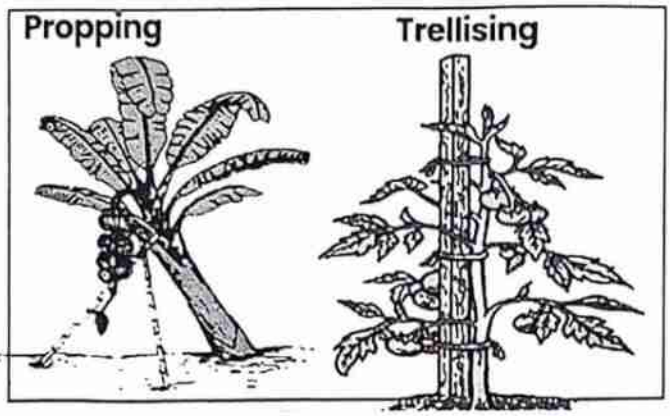
Staking is the provision of support to plants with weak stems.

Crops which are staked are;

- passionfruit plants
- watermelon plants
- irish potato plants
- banana plants

Methods of staking crops and trees.

- Propping e.g. bananas
- Trellising e.g. tomatoes



Reasons why staking trees and crops is important.

- It eases weeding and harvesting.
- It enables plants to get enough sunlight to make food.
- It prevents ground pests from damaging fruits.
- It prevents plants with heavy bunches from breaking.

Mulching in agroforestry

Mulching is the covering of topsoil using dry plant materials. The dry plant materials are called mulches.

Examples of mulches are;

- dry grass
- dry banana leaves
- dry coffee husks
- dry cereal stalks
- dry ground nut shells

Advantages of mulching in agroforestry.

- Mulching controls soil erosion.

Mulches reduce the speed of flowing water.
Mulches trap soil from flowing with water.

- Mulching improves soil fertility.

Mulches decompose and form humus in the soil.

- Mulching conserves moisture in the soil.

Mulches minimise the rate of evaporation of water in the soil.

- Mulching controls the growth of weeds.

Mulches cut off the supply of sunlight for photosynthesis to the weeds.

Disadvantages of mulching.

- Mulching is tiring.
- Mulches are fire hazard.

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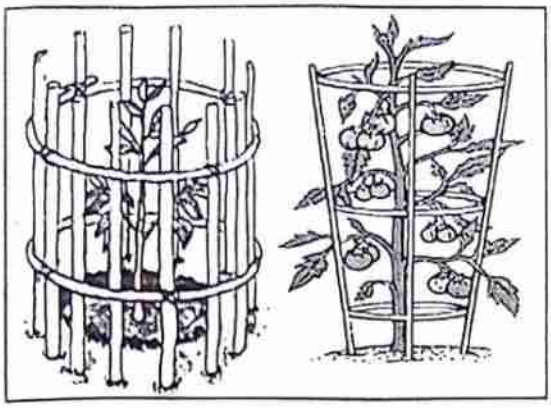
- Mulches hide some crop pests e.g. termites
- Some mulches can grow into weeds.

Fencing trees and crops in agroforestry

Trees need fencing when they are still in young stages.This is done by making a simple wooden fence around each tree



Reasons for fencing trees when still young.

- To protect the trees from damage by animals and young children.
- To protect trees from being destroyed by pests.
- To protect trees from strong raindrops and wind.

When the trees grow strong and tall, the wooden fences can be removed because the trees are able to withstand harsh conditions.

Proper harvesting of trees in agroforestry.

Harvesting trees in agroforestry can be done using;

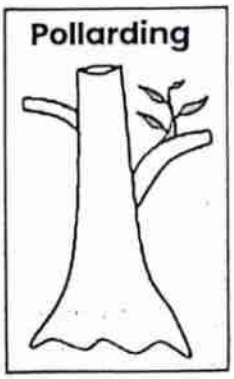
- pollarding
- coppicing
- lopping

Pollarding

Pollarding is the cutting off the top part of a tree.
Cutting should be done at an angle.
Reason. To drain off water from the remaining tree stump.

Advantages of pollarding.

- It encourages thicker growth of the lower branches.
- It helps to increase the yields of fruit trees.

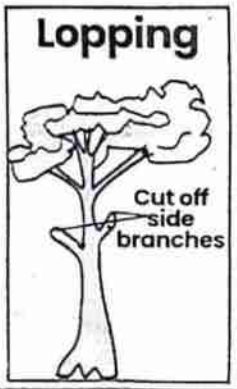


Coppicing

Coppicing is the cutting off a tree at the stump.
This method is suitable for harvesting trees for poles and timber.
Coppicing encourages the growth of new shoot from the stump.

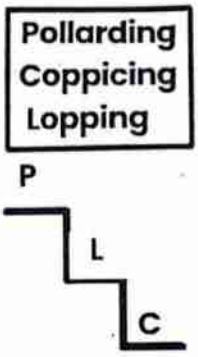
Lopping

Lopping is the cutting off the side branches of a tree.
The side branches can be used as firewood.
This method does not cause damage to the trees



How to recall the above methods and how they are used easily.

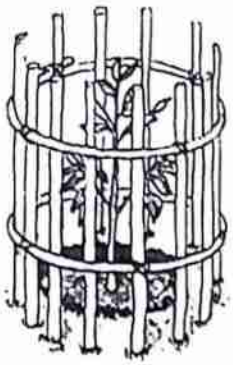
- *List down all the three methods i.e.*
- *Pick the first letter from each method i.e.* **P, C** *and* **L**
- *Draw a ladder with three steps and indicate each letter on each step of the ladder following their alphabetical order as shown below.*
- *You will notice that* **C** *is for coppicing (stump level),* **L** *is for lopping (as side branches) and* **P** *is for pollarding (top branches).*
- *This simplifies the application of coppicing, lopping and pollarding.*



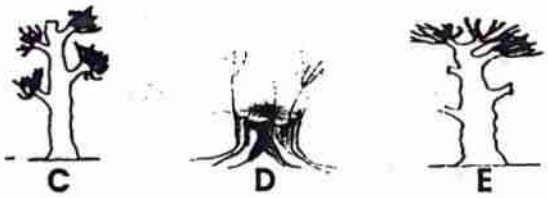
Evaluation activity:

1. State any **two** ways in which weeds can lower the yields of trees in agroforestry.
2. Give any **two** ways in which the growth of weeds can be controlled in agro forestry.
3. State any **two** reasons why pruning of trees is important in agroforestry.

The diagram below shows a practice carried out in tree growing in agroforestry. Use it to answer questions **4** and **5**.



4. Name the tree growing practice shown above.
5. State any **one** way in which the tree growing practice above is important in agroforestry.
6. Give any **one** way in which timely weeding of crops helps to control crop diseases.
7. Give any **two** proper methods of harvesting trees in agroforestry.
8. Which method of harvesting trees in agroforestry is suiatable for harvesting poles for timber?
9. Why is it not recommendable to carry out tree watering during day time?
10. Name the method of harvesting trees that;
    i) Increases yields of fruit trees. ii) Does not damage the tree.
11. Why should trees that grow near electricity power lines be pollarded?
12. How does mulching help to conserve moisture in the soil?
13. Apart from conserving moisture, state any **two** other advantages of mulching a tree garden.
14. The diagram below shows proper methods of harvesting trees in agroforestry. Study and use it to answer the questions that follow.



a) Name the methods of harvesting trees marked **C** and **E**.
b) Which method of harvesting trees encourages the growth of new shoot?
c) State any **one** advantage of harvesting trees using method marked **C**

STARTING AND MANAGING A SCHOOL/HOME WOODLOT PROJECT.

A woodlot is an area of land used for growing trees.

Factors to consider when starting a tree growing project(woodlot)

- Capital
- Land
- Tree species
- Drainage of the soil.
- Security of the places.
- Labour
- Management.
- Water source

Mnemonic used: **LCM** i.e. land//labour//capital//management

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Benefits of a woodlot in the community.

- Trees provide timber for making furniture.
- It is a source of food.
- It provides poles for construction.
- A woodlot can be used for bee keeping.
- Trees help in rain formation.
- It is a source of wood fuel e.g. firewood.
- Trees provide shade.
- Trees help to purify air.
- Some trees are source of herbal medicine.

Selecting crop and tree seeds for planting.

Qualities of good seeds for crop and tree planting.

- The seeds should be viable.
- The seeds should be free from pests.
- The seeds should be free froM diseases.
- The seeds should be obtained from healthy parent trees.
- The seeds should be mature enough.
- The seeds should not be broken.
- The seeds should be of reasonable size depending on the variety.

Reasons why it is important to select seeds before planting them.

- It prevents wastage of time.
- It enables a farmer to get good yields.
- It reduces chances of gap filling.
- It enable a farmer to choose the right variety.
- It enables a farmer to get viable seeds.
- It prevents wastage of space in the garden.
- It promotes proper utilisation of space in the garden

Preparing a nursery bed.

A nursery bed is a small garden prepared for raising seedlings before transplanting.

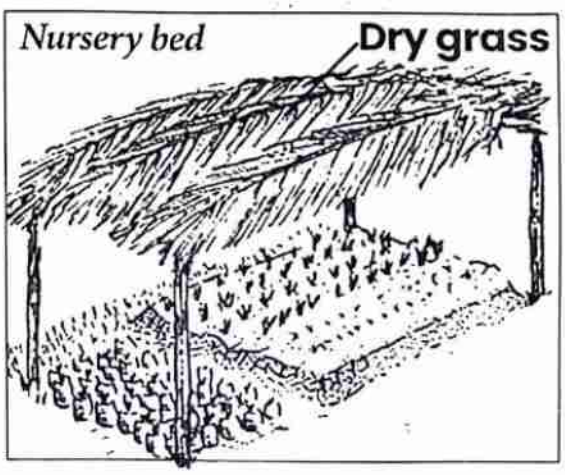
A seedbed is a piece of land prepared to plant seedlings for further growth.

Reasons why farmers first plant some seeds in a nursery bed.

- It give a farmer enough time to prepare the main garden.
- It allows a farmer to select healthy seedlings.
- It protects seedlings from harsh weather conditions.
- It prevents wastage of seeds.
- It allows proper care for seedlings.

Steps taken in preparing a tree nursey bed.

i. Select a good site for the nursery bed.
ii. Clear vegetation and remove the trash.
iii. Dig deep the soil to remove perennial weeds.
iv. Add manure in the nursery bed and mix it well with the soil.
v. Make holes of suitable sizes for planting the seeds.
vi. Plant the selected seeds and cover the holes with light soil.
vii. Raise shade on the nursery bed after planting seeds.



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- Dry grass provides shade and protects seedlings from being damaged by heavy raindrops.
- The grass is spread slightly to allow water pass and reach the seedlings during rain or irrigation.
- The shade should be raised to protect the seedlings from direct sunlight.

Factors considered when choosing a site for the nursery bed.

- Fertility of the soil.
- Nearness to the water source.
- Influence of crop pests and diseases.
- Security of the place.
- Distance from the home.
- Drainage of the soil.
- Topography.

Ways of caring for seedlings in a nursery bed.

- Watering seedlings.
- Spraying seedlings.
- Fencing the nursery bed.
- Regular weeding.
- Thinning

Pricking out.

Pricking out is the practice of removing overcrowded seedlings from a nursery bed to be planted elsewhere.

It is done soon after the sown seeds have germinated.

Importance of pricking out seedlings.

- It reduces competition for water and nutrients.
- It prevents easy spread of diseases among seedling.

Hardening off.

Hardening off is the gradual exposure of seedlings to different weather conditions before transplanting.

Hardening off enables seedlings to adopt external garden conditions.

Activities done to seedlings during hardening off.

- Removing the shade
- Spraying
- Reduced watering

Transplanting

Transplanting is the transfer of seedlings from the nursery bed to the main garden.

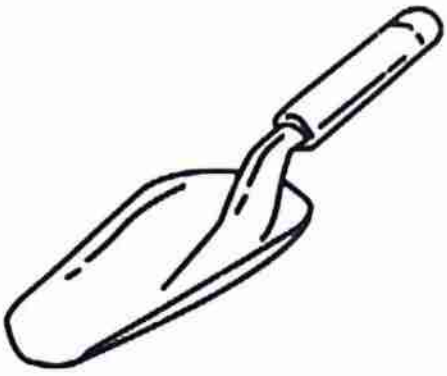
Transplanting should be done in the evening.

Reason. To prevent wilting of seedlings due to high rate of transpiration.

Transplanting is done using a garden trowel.

A garden trowel is ridged to easily scoop seedlings from the soil without damaging the roots.

Garden trowel



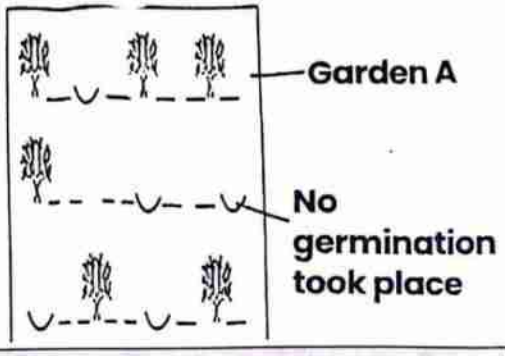
Gap filling

Gap filling is the replanting of fresh seedlings where they did not germinate after planting.

In this garden A, gap filling is needed.

Advantages of gap filling.

- It prevents wastage of space in the garden.



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- It increases crop population for increased yields.
- It allows proper use of land.
- It ensures correct plant population.

Thinning

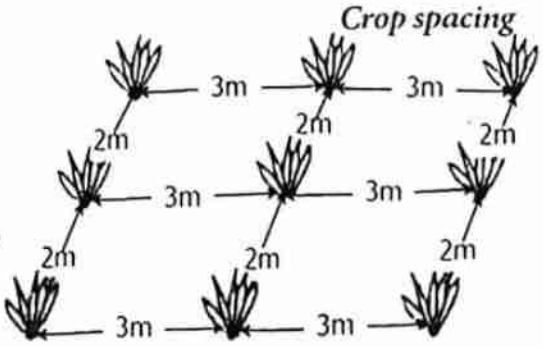
Thinning is the removal of excess seedlings from the garden.

Advantages of thinning.

- It eases weeding, spraying and harvesting.
- It controls easy spread of crop pests and diseases.
- It improves crop yields and quality of harvest.

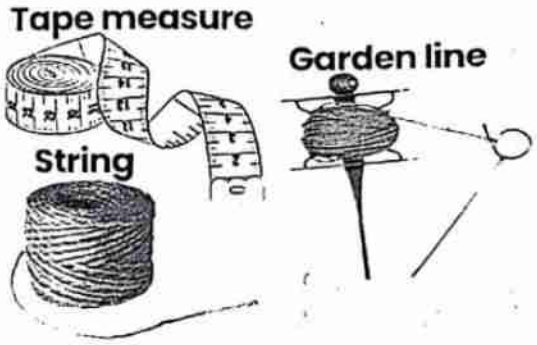
Crop spacing

This is the practice of leaving open gaps between individual crops when planting.
Crop spacing is done in row planting.
A tape measure, garden line or string can be used for crop spacing.



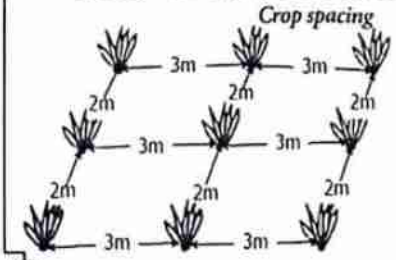
Importance of correct crop spacing in the garden.

- It reduces competition for sunlight, water and nutrients.
- It eases weeding and harvesting.
- It eases movement within the garden.
- It makes pruning of crops easier.



Evaluation activity:

1. What is meant by the term woodlot?
2. State any **two** factors that should be considered when starting a tree growing project.
3. Give any **two** benefits of a woodlot to a community.
4. What is the importance of gap filling in crop growing?
5. Give any **two** reasons why farmers plant some seeds in a nursey bed.
6. The diagram below shows a method of planting crops in the garden. Study carefully and use it to answer the questions that follow.



a) Name the method of planting crops shown in the diagram above.
b) State any **two** advantages of spacing the crops as shown above.
c) Name any **one** material that can be used for spacing crops as shown above.

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7. a) Give the meaning of the following terms:
   i) Nursery bed
   ii) Seedbed
b) State any **two** activities carried out when preparing a tree nursery bed.
8. Write down any **two** factors to consider when choosing a site for a tree nursery bed.
9.a) Give the importance of each of the following activities in tree growing:
   i) hardening off
   ii) pricking out
   b) Write down any **two** activities done to seedlings during hardening off.
10. The table below shows crop growing practices in part A and their importance in part B.

A: Crop growing practice	B: importance
Staking	Prevents overcrowding of plants.
Mulching	Removes unwanted plants.
Thinning	Prevents fruits from rotting.
Weeding	Conserves soil moisture.

Match correctly the crop growing practices with their importance in the spaces provided below.
i) Staking .....
ii) Mulching .....
iii) Thinning .....
iv) Weeding .....

Lesson vocabulary:
- seasoning
- warping
- record
- inventory
- preservatives
- shade

PREPARING WOOD FOR DIFFERENT PURPOSES.
Ways in which wood can be useful to people.
- It is used for making charcoal.
- It is used for firewood.
- It is used as poles.
- It is used for timber.

Wood for electricity poles:
It should be treated using wood preservatives in order to prevent destruction by termites, wood shrimps and carpenter bees.
Used engine oil should be applied on the poles to make them water resistant and prevent rotting.
Wood for firewood Should be cut and split to allow easy drying.

Wood for timber.
It should be put under the shade for drying.
The drying of timber under the shade is called **seasoning of wood.**

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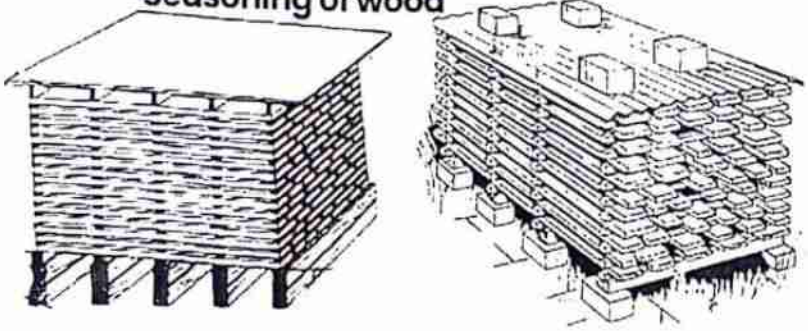
Seasoning of wood



Reasons for seasoning wood.

- To prevent timber from cracking.
- To prevent timber from warping.

Uses of timber

- It is used for making furniture
- It is used for construction

RECORD KEEPING

Record keeping is the gathering and storing of information about various activities and transactions carried out on a farm.

Farm records are written information about different activities and transactions carried out on a farm.

Examples of information on farm records.

- Size of the farm
- Size of paddocks
- Health status of animals and crops.
- Amount spend on feeding and vaccination.
- Date of planning
- Type of fertilisers used for planting
- Date when the animals will give birth.

(Example of a farm record)

JAMIE FARM INC®

FARM RECORD

Date	Crops	Trees	Activities done	Requirements	Expenditure
02/2/2024	cassava	Pine	Spraying, Weeding Manuring	Knapsack sprayer	Shs.50,000
14/3/2024	Tomatoes		Staking	Secateurs	Shs.25000
26/4/2024	Eggplants	Mvule	Buying seeds Nursery bed	Capital Land	Shs. 32000
07/5/2024	maize	ficus	Harvestin Weeding Spraying Hardening off ficus seedlings	Hoes Knapsack sprayer	Shs.47000

Examples of farm records.

Production record:

Provides information about crop yields,

It helps in planning schedules and related activities.

Feeding records:

Feeding records show the amount of feeds given to animals in a specific period of time.

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Health records:

These show cases of diseases, their diagnosis and treatment. It also shows dates for vaccination and health conditions of different animals.

Importance of keeping health records.

- It allows farmers to carry out culling of animals easily.
- It helps to identify sick animals on the farm.
- It helps to know the expenditure for treating animals.
- It helps the farmer to plan for the health of animals.
- It helps to know common diseases and ways of preventing them.

Sales records:

These show the income, daily expenditure and total sales.

Inventory records:

These show detailed list of all the items used on the farm.

Breeding records.

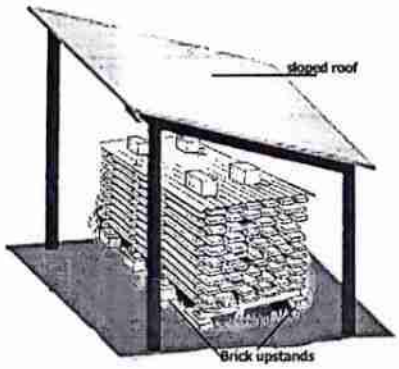
These show the breeds of animals kept and varieties of crops or trees grown.

Ways in which keeping farm records is important to a farmer.

- It helps a farmer to plan for the farm.
- It helps a farmer to know the income and expenditure on the farm.
- It helps a farmer to know the profits and losses made on a farm.
- It helps in decision making regarding the future of the project.
- Records can be used as reference when a farmer wants to get loans.
- It helps a farmer to be taxed fairly by the government.
- It helps a farm to identify areas of development and investment.

Evaluation activity:

1. Give any **two** ways in which wood is useful to people.
2. Why are electricity poles burnt with engine oil before fixing them into the soil?
3. Mention any **two** types of records kept by farmers.
4. Why should firewood be split into small pieces before putting under sunshine?
5. State any **two** ways in which farm records are important to a farmer.
6. What is meant by the term seasoning in wood management?
7. The diagram below shows a method of storing timber. Study and use it to answer the questions that follow.



a) Name the method of storing timber shown above.

b) State the importance of the following on the structure above.

i) Sloped roof.....

ii) Brick upstands .....

c) Give any **one** advantage of storing timber as shown above.

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8

THEME: THE COMMUNITY, POPULATION AND FAMILY LIFE
TOPIC 8: POPULATION AND HEALTH

Topical outline:

Community health and social problems.

- Types of sicknesses in a home and their causes.
- Controlling common sicknesses in a home and community.

Community health and social problems among young people.

- Antisocial behavior
- Sexual deviations
- Ways of avoiding sexual deviations

Activities to address health concerns.

- Health surveys
- Health education.

Collecting information/data on human population.

- Demography
- Avoiding health and social problems.
- Activites of health clubs

Ref. P7 Standard Curriculum page

Guidance to the teacher.

- *Emphasize ways of avoiding sexual deviations and activities that address health concerns.*
- *Ensure that learners do not get exposed to what is beyond their age.*
- *Help learners to evaluate activities of health clubs as they conclude their primary school cycle.*

Lesson vocabulary:

- population
- nutrition
- adequate
- diphtheria
- anaemia
- borne

Community health and social problems.

Population is the number of organisms living in an area at a given time.

Human population is the total number of people living in an area at a given time.

A community is a group of people living or working together.

Health is a person's physical, social and mental state of wellbeing.

Characteristics of a health person.

- Should be physically fit.
- Should be socially upright.
- Should not have any mental problem.

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Community health.

Community health refers to essential health conditions which promote the wellbeing of people in the community.

Examples of health conditions.

- Good nutrition.
- Adequate supply of clean water and safe water.
- Proper personal and famil hygiene
- Good health lifestyles.
- Control of common diseases.
- Safe motherhood.

Common sicknesses in a home and the community.

- AIDS
- malaria
- typhoid
- dysentery
- measles
- mumps
- COVID 19
- influenza
- cholera

Factors that cause common sicknesses.

- Poor sanitation.
- Poor personal hygiene.
- Smoking, alcoholism and drug abuse.
- Drinking contaminated water.
- Poor feeding.
- Lack of physical exercises.
- Inadequate water supply.
- Eating contaminated food.

Ways of preventing/controlling common sicknesses.

- Feeding on a balanced diet.
- Promoting food hygiene.
- drug abuse.
- Promoting proper personal hygiene.
- Boiling water for drinking.
- Avoid smoking, alcoholism and
- Proper disposal of wastes.
- Having regular body exercises.

Evaluation activity:

1. Give the meaning of the following terms:
   i) Human population
   ii) Health
2. State any **two** characteristics of a healthy person.
3. Give any **two** conditions that lead to sicknesses in a home and the community.
4. State any **two** ways of controlling common sicknesses in a home and the community.
5. Write down any two examples of common sicknesses in a home and the community.
6. Give any two examples of health conditions.

TYPES OF COMMON SICKNESSES IN A HOME AND THE COMMUNITY.

A) Immunisable diseases

These are diseases that can be prevented when vaccines are introduced into the body of a person.

Examples of immunisable diseases.

- measles
- tetanus
- diphtheria
- hepatits B
- polio disease
- COVID 19

B) Deficiency diseases

These are diseases caused by lack of certain food values in the body.
They are also called malnutritional diseases.

Refer to the table on page 197

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Deficiency disease	Signs and symptoms	Prevention/treatment.
Night blindness Caused by lack of vitamin A in a diet.	• Poor vision at night. • Dryness of the eyes	• Feeding on food rich in vitamin A e.g. liver, onion, carrots and pawpaw.
Beriberi Caused by lack of vitamin B1 in a diet.	• Retarded growth • Decreased muscle function. • Loss of feelings in the limbs. • Difficulty in speaking. • Mental confusion. • Heart failure.	• Feeding on food rich in vitamin B1 e.g. ground nuts, liver, milk and green leafy vegetables.
Pellagra Caused by lack of vitamin B3 in a diet.	• Skin sores at the mouth and eyes. • General body weakness • Rough scaly skin.	• Feeding on food rich in vitamin B3 e.g. milk, liver, lean meat and fish.
Scurvy Caused by lack of vitamin C in a diet.	• Bleeding gums. • Pain in the joints and muscles. Poor healing of wounds. • Retarded growth.	• Feeding on food rich in vitamin C e.g. fresh fruits like mangoes, oranges and leafy vegetables.
Rickets Caused by lack of vitamin D in a diet.	• Ox-bow legs. • Knock knee legs. • Brittle bones and teeth. • Poor teeth formation.	• Feeding on food rich in vitamin D, calcium and phosphorus e.g. milk, egg yolk, fish oil. • Sun-bathing in the morning.
Goitre Caused by lack of iodine in a diet.	• Swollen neck. • Painful trachea and gullet. • Difficulty in swallowing and breathing.	• Feeding on food rich in iodine e.g. iodised salt, sea fish.
Anaemia Caused by lack of iron in a diet.	• Pale skin • Feeling tired without proper cause. • Constant headache.	• Feeding on food rich in iron e.g. dark green vegetables. • Taking iron supplement tablets.
Marasmus Caused by lack of all food nutrients in a diet. Note: Marasmus is starvation.	• Old man's face. • Child is ever hungry • Growing very thin. • Swollen potbelly. • Loss of body weight.	• Feeding on a balanced diet.
Kwashiorkor Caused by lack of proteins in a diet.	• Swollen moon-like face. • Stunted growth. • Sunken stomach. • Swollen limbs. • Wasted muscles. • Soft brown hair scattered on the head.	• Feeding on food rich in proteins e.g. fish, eggs, meat, beans,peas...

C) Insect borne diseases

These are diseases whose germs are transmitted by insect vectors.

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Ways in which insect borne diseases spread to human beings.

- Through bites
- Carrying germs onto food by the vectors e.g. houseflies.
- Use of hairy body
- Through body contact by the vector.

Examples of insect borne diseases

- malaria
- elephantiasis
- yellow fever
- river blindness
- cholera
- sleeping sickness

D) Sexually Transmitted Diseases (STDs)

These are infections spread through having unprotected sexual intercourse with an infected person.

They are also called venereal diseases.

Examples of STDs are;

- AIDS
- gonorrhoea
- candidiasis
- syphilis

E) Hereditary diseases

These are diseases that are passed on from parents to their children through genes.

Examples of hereditary diseases.

- sickle cells
- asthma
- haemophilia

F) Self inflicted diseases

These are non-infectious diseases which affect people due to their bad health habits.

These bad health habits include;

- Smoking
- Failure to do enough physical exercises
- High alcohol consumption
- Over eating

Examples of self inflicted diseases

- lung cancer
- emphysema

Ways of avoiding self inflicted diseases

- Avoid smoking.
- Do regular physical exercises.
- Limit alcohol intake.
- Eat enough food .

Note the following:

Deficiency and self inflicted diseases are examples of diseases that can be prevented without using drugs. They are non-communicable diseases. Most types of sicknesses that affect people are communicable diseases

Communicable diseases

These are diseases which can be transmitted from an infected person to a healthy one.

They are also called infectious/transmisable diseases.

Characteristics of communicable diseases.

- They spread from one person to another
- They are caused by germs.

Examples of communicable diseases.

- measles
- tuberculosis
- mumps
- chicken pox
- AIDS
- trachoma
- cholera
- ebola
- COVID 19

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Ways in which communicable diseases spread to people.

- Through body contact with an infected person.
- Through droplets of an infected person during sneezing or coughing.
- Through inhaling contaminated air.
- Through drinking contaminated water.
- Through drinking unboiled milk from an infected cow or goat.
- Through eating contaminated food.

Ways in which communicable diseases can be controlled among people.

- Drinking bolied water or milk.
- Proper disposal of human wastes.
- Spraying using insecticides to kill adult insect vectors.
- Promoting food hygiene.
- Isolating infected people.
- Early treatment of infected people.

Evaluation activity:

1. Give the meaning of the following terms:
   i). Deficiency diseases
   ii) Immunisable diseases
2. Mention any **two** examples of vitamin deficiency diseases.
3. State any **one** way in which parents protect their children from getting deficiency diseases.
4. Give any **two** examples of self inflicted diseases that affect human health.
5. State any **two** practices that can lead to self inflicted diseases in humans.
6. Which body system is affected by;
   a) asthma?
   b) sickle cells?
7. State any **two** signs of protein deficiency in children?
8. Why are babies at a higher risk of suffering from protein deficiency than adults?
9. The table below shows deficiency diseases in part **A** and their signs in part **B**.

Part A: Deficiency disease	**Part B: Signs**
Anaemia	Swollen neck.
Pellagra	Poor teeth formation.
Rickets	Pale skin.
Goitre	Rough scaly skin.

Match correctly the deficiency diseases with their signs in the spaces provided below.

i. Anaemia .....
ii. Pellagra .....
iii. Rickets .....
iv. Goitre .....

10. What is meant by the term infectious diseases?
11. Write down any **two** examples of infectious diseases that affect people in the community.
12. State any **two** ways in which infectious diseases can spread to people.
13. Why is it important to cover your mouth with a handkerchief when coughing in public?

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14. Give any **two** ways in which the spread of infectious diseases can be controlled in the community.
15. Name any **two** examples of Sexually Transmitted Diseases in the community(STDs).
16. Which deficiency disease is caused by lack of vitamin B1 in a diet?
17. The diagram below shows a patient suffering from a deficiency disease. Study and use it to answer the questions that follow.

a) Name the deficiency disease which the above patient is suffering from.
b) State the possible cause of the above deficiency disease.
c) Give any **one** sign of the above deficiency disease.
d) How can you protect yourself from contracting the above deficiency disease?

18.a) Give any **one** sign of HIV/AIDS.
b) Write down any **two** effects of HIV/AIDS to an individual.
c) How best can a primary school pupil protect him/herself from contracting HIV/AIDS?

19. a) Name the disease which results from the deficiency of vitamin C in humans.
b) State any **two** signs of the vitamin C deficiency disease in humans.
c) Give any **one** example of food which gives us vitamin C.

20. The table below shows immunisable diseases, vaccines administered and the body sites through which the vaccines are administered. Study carefully and compete it correctly.

Immunisable disease	**Vaccine administered**	**Body site**
Measles	Measles vaccine	.....
Whooping cough	.....	Left upper thigh
.....	BCG vaccine	Right upper thigh
Poliomyelitis	Polio vaccine	.....

Lesson vocabulary:

- cleanliness
- diarrrhoeal
- dysentery
- contaminate
- habitat
- tsetse fly

COMMUNITY HEALTH CONCERNS/PROBLEMS.

Health concerns are health problems that affect people and need immediate solutions.

Examples of community health concerns.

- Poor sanitation.
- Poor water supply.
- Poor food supply.
- Outbreak of diseases
- Anti-social behavior

Sanitation

Sanitation is a way in which an area is kept clean.

Elements of a clean home

- Clean latrines or toilets.
- Clean bathrooms
- Rubbish pit or dustbins

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- Plate stand/rack
- Hand washing facilities

POOR SANITATION

Poor sanitation is a condition in which the environment where people live is unhygienic.

Signs/indicators of poor sanitation.

- Bushy compound.
- Rubbish in the compound.
- Dirty water sources
- Pools of water in the compound.
- Poor disposal of human wastes.
- Dusty classroom
- Presence of houseflies.

Dangers of poor sanitation to the community.

- It leads to outbreak of diarrhoeal diseases.
- It leads to bad smell in an area.
- It leads to water and air pollution.
- It leads to food contamination.
- It encourages the breeding of disease vectors and germs.

Diseases that break out due to poor sanitation.

- cholera
- typhoid
- diarrhoea
- bilharziasis
- sleeping sickness
- polio
- malaria

Vectors that breed due to poor sanitation.

- mosquitoes
- houseflies
- cockroaches
- tsetse flies
- fleas
- lice
- ticks

Ways of promoting proper sanitation.

- Draining stagnant water.
- Scrubbing toilets/latrines.
- Sweeping the compound.
- Dusting tables and chairs.
- Clearing bushes around the compound.
- Mopping classrooms and houses.
- Flushing toilets after use.

Ways in which proper sanitation promotes the health of people in the community.

- It controls easy spread of diarrhoeal diseases.
- It prevents food contamination by houseflies and cockroaches.
- It prevents water and air pollution.
- It controls the breeding of vectors.
- It reduces the breeding of intestinal worms.

POOR WATER SUPPLY

Poor water supply is a condition in which a community lacks clean and safe water.

Dangers of poor water supply.

- It leads to poor personal hygiene.
- It leads to outbreak of certain diseases.
- It leads to poor sanitation.

Causes of poor water supply

- Drought
- Floods
- Over population
- Water pollution
- Swamp drainage

Diseases associated with poor water supply.

Water borne diseases

These are infectious diseases spread through drinking contaminated water.

Examples of water borne diseases.

- typhoid
- cholera
- dysentery
- hepatitis B
- polio disease

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Water habitat vector diseases

These are infectious diseases transmitted by vectors which breed in or near water.

Examples of water habitat vector diseases.

- malaria
- bilharziasis
- yellow fever
- river blindness
- elephantiasis
- dengue fever
- zika disease

Water contact diseases

These are infectious diseases one gets when the body comes into contact with contaminated water

Examples are;

- swimmer's itch
- otitis externa

Water cleaned diseases

These are infections that one gets due to lack of clean water for promoting personal hygiene.

Examples are;

- ringworm
- scabies
- trachoma
- athlete's foot

Solutions to address the problem of poor water supply.

- Constructing valley dams in semi desert areas.
- Drilling boreholes in rural areas.
- Conserving wetlands.
- Fencing community water sources.
- Providing underground piped water.

Poor food supply

Poor food supply is a condition in which the family or community does not have enough food to eat.

Poor food supply leads to deficiency diseases.

Examples of such diseases are;

- rickets
- scurvy
- goitre
- anaemia
- marasmus
- pellagra

Causes of poor food supply

- **Infertile soil (poor soil)**
  It leads to low food production.
- **Floods**
  It leads to destruction of crops.
- **Crop pests and diseases**
  It leads to low food production.
- **Use of poor methods of farming**
- **Drought**
  It dries crops thus lowering yields.
  It retards crop growth leading to poor yields.
  It leads to wilting of crops in the garden.
- **Poverty among people**

Solutions to address the problem of poor food supply

- Growing of enough food.
- Carrying out irrigation farming.
- Growing fast maturing crops.
- Growing crops which are disease resistant.
- Providing food relief to famine affected areas.
- Proper food storage.
- Educating farmers modern farming methods.
- Growing crops which are drought resistant.

Food security

Food security is when the family or community has enough food for eating all year round.

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Ways of promoting food security.

- Growing drought resistant crops.
- Storing excess food
- Growing more food crops.
- Improving soil fertility.
- Proper storage of food.
- Controlling crop pests and dis eases.
- Using good methods of farming.

Importance of food security.

- Prevents malnutrition among children.
- Prevents famine.

Evaluation activity:

1. a) Give the meaning of the following terms:
   i) Sanitation
   ii) Personal hygiene
   b) Identify any **two** activities school children can do to promote good sanitation.
2. a) State any **two** signs of poor sanitation in a school.
   b) Give any **two** diseases that can break out due to poor sanitation in a school.
3. a) Give any **one** way in which each of the following factors lead to poor food supply in an rea.
   i) Drought
   ii) Crop pests
   b) Apart from drought and pests, mention any **one** other natural factor which affects food production.
   c) Name **one** health problem caused by inadequate food supply in a community.
4. Give any **two** ways in which good sanitation helps to promote the health of people in an area.
5. Give any **two** ways of promoting food security in a community.
6. The table below shows information collected from five homes during a health survey. Study it to answer the questions that follow.

Homes	Sanitation items checked			
	Latrine	Rubbish pit	Bathing shelter	Drying rack
A	☒	✓	☒	☒
B	✓	✓	✓	✓
C	☒	✓	✓	✓
D	☒	☒	☒	✓
E	☒	✓	✓	☒

Key☒ item missing ✓ item present

a) Which home had all the sanitation items?
b) Identify the sanitation item missing in most of the homes surveyed.
c) Name any **one** disease that is likely to attack homes **A,C,D** and **E.**
d) Give any **one** other sanitation item which could be added in the health survey.

7. a) Name any one example of;
   i) water bone diseases ii) water cleaned diseases
   b) Give any two ways in which diseases associated with poor water supply can be controlled.

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Lesson vocabulary:

- behaviour
- truancy
- acceptable
- deviation
- awareness
- habits

SOCIAL PROBLEMS AMONG YOUNG PEOPLE.

Social problems are conditions or behaviour which have negative effects on relationships among members of the community.

Examples of social probems in the community.

- Antisocial behaviour
- Sexual deviations
- Drug abuse
- Increased crime rate
- Alcoholism

Antisocial behaviour

Antisocial behaviour are forms of conduct that are not accepted in the society.

Examples/forms of antisocial behaviour

- bullying
- smoking
- rape
- drug abuse
- truancy
- stealing
- deception
- wandering
- fighting
- defilement
- murder
- robbery
- arson

Smoking

Truancy:

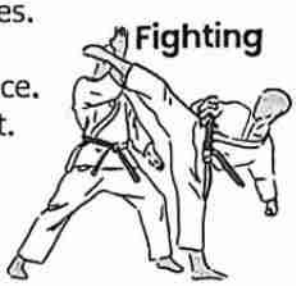
An act of refusing to attend school.

Causes of truancy

- Harsh teachers.
- Lack of scholastic materials.
- Poor teaching methods of some teachers.
- Learning difficulties.
- Bullying by other children.

Causes of antisocial behaviour

- Bad peer influence .
- Over strictness by parents or teachers.
- Failure to enforce rules in a community.
- Disturbed/ broken homes.
- Pampering of children.
- Lack of parental guidance.
- Poor social environment.


Fighting

Effects of antisocial behaviour

- They lead to dropping out of school.
- They may lead to pain, injury and death of an individual.
- They may lead to destruction of property.
- They lead to imprisonment.
- They may lead to injuries.

Ways of controlling antisocial behaviour

- Educating children
- Improving parenting skills.
- Punishing children for wrong behaviour.
- Joining good social and health clubs.
- Providing counselling services.
- Providing equal treatment to all children.

Lifeskills to safeguard children against antisocial behaviour.

- Decision making
- Critical thinking
- Self-control
- Self awareness
- Problem solving
- Assertiveness

Sexual deviations:

Sexual deviations are sexual activities that are not allowed in the society.

Examples of sexual deviations

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- **Homosexuality:** It is the sexual intercourse between people of the same sex or gender.
- **Lesbianism:** It is the sexual activity between a woman and a woman.
- **Incest:** It is a sexual relationship between close relatives. For example; father and daughter, sister and brother, mother and son.
- **Bestiality:** It is the sexual intercourse between a person and an animal.
- **Masturbation**: It is the stimulation of the genitals (sexual organs) by rubbing them. **OR** It is the acting of giving oneself sexual pleasure or satisfaction.
- **Oral sex:** It is a sexual activity involving the stimulation of the genitals of a person by another person using the mouth.

Causes of sexual deviations in the society.

- Alcoholism.
- Watching videos/films with sexual content.
- Lack of proper guidance and counselling.
- Joining bad peer groups.
- Living in a society with poor morals.
- Smoking opium.

Effects/dangers of sexual deviations.

- They lead to easy spread of STDs.
- They may lead to injuries.
- They lead to loss of respect in the society.
- They lead to imprisonment.

Ways of avoiding sexual deviations

- Controlling desires towards members of the opposite sex.
- Keep away from bad peer groups.
- Avoiding alcoholism.
- Sensitising children about body changes.

Evaluation activity:

1.a) Name any **two** examples of antisocial behaviour among people in the community.
b) Give any **two** causes of antisocial behaviour among people in the community.
2. a) Mention any **two** effects of antisocial behaviour to school children.
b) State any **two** ways of controlling antisocial behaviour among school children.
3.a) Write down any **two** causes of sexual deviations among young people.
b) Give any **two** effects of sexual deviations on the health of young people.

Lesson vocabulary:

- survey
- demography
- data
- sensitise
- parade
- habits

ACTIVITIES TO ADDRESS HEALTH CONCERNS.

- Carrying out health surveys.
- Organizing health parades in schools.
- Setting up health clubs at school.
- Carrying out health education.
- Carrying out child to child programmes

Health surveys

A health survey is a way of gathering information about the health status of a family or community.

Pieces of information which are collected in a health survey

- Kind of food people eat.
- Latrine coverage.
- Population status.
- Economic status of people.
- Health facilities in an area.
- Common diseases that affect people.
- Housing status.
- Source of water.
- Family planning services.
- Food security in the area.

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- Sanitation standards in the area.
- Immunisation coverage.

Methods of carrying out health surveys.

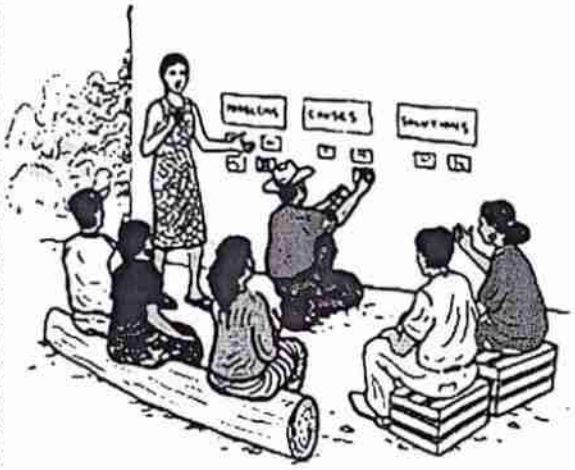
- Using questionnaires.
- Using oral interviews.
- Making direct observations.

Reasons why carrying out health surveys is important in an area.

- It helps in identifying the health needs of people.
- It determines the health status of people.
- It creates awareness about health challenges.
- It provides health services.
- It helps in planning for better health services.
- It helps in monitoring health programmes.
- It helps the government to mobilise health funds.

Health education

Health education refers to sensitising people in a community on matters concerning health.



Issues addressed during health education.

- Proper sanitation.
- Prevention/control of diseases.
- Importance of immunisation.
- Importance of proper nutrition.
- Proper personal hygiene.

Ways of giving health education to people in a community.

- Through songs and poems about health.
- Through plays and storytelling.
- Through radios and television health talk shows.
- Through newspapers.
- Through health assemblies at school.
- Through child to child programmes.

Ways in which giving health education to people is important.

- It teaches people ways of promoting sanitation.
- It makes people aware of certain diseases.
- It educates people on the importance of immunisation.
- It creates awareness on the effects of high population.
- It sensitises people on how to improve personsl hygiene.

Child to child programme

A child to child programme is where older children with more knowledge teach young children good health habits.

Activities carried out under child to child programmes

- Teaching young children toilet habits.
- Teaching young children table manners.
- Teaching young children how to brush teeth.
- Teaching young children how to bathe.
- Teaching young children how to avoid STDs.
- Caring for children who are sick.

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- Teaching young children how to wash their knickers or pants.
- Teaching body changes during growth.

Importance of child to child programme.

- It promotes personal hygiene among children.
- It promotes proper care among children.
- It controls outbreak of diseases like diarrhoea and athlete's foot.
- It promotes proper use of toilets/latrines by young children.

School Health parades

A health parade is an assembly organized to check on the hygiene of school children.

Activities done during a school health parade.

- Checking children's teeth.
- Checking toe and fingernails.
- Identifying children with uncombed hair.
- Checking on the hygiene of school uniform.
- Identifying pupils with hill health.
- Checking children's skin.

Body parts that may be checked during a school health parade.

- teeth • hair • fingernails • skin • toenails
- nose • neck • eyes • ears

Groups of people who carry out health parade at school.

- teachers • school nurse • matrons
- doctors • health prefects

Importance of school health parades.

- They help school chidren to improve on their personal hygiene.
- They help to identify skin diseases among children.
- They help children learn how to prevent diseases like ringworm.

Diseases that can easily be identified during a health parade.

- ringworm • athlete's foot • dental caries • scabies

COLLECTING INFORMATION/DATA ON HUMAN POPULATION.

Ways in which collecting information on human populationis important.

- It enables the government to provide the available health services.
- It enables the government to plan well for for community health.
- It enables the government know and prevent common diseases affecting people in the community.

Demography

Demography is the scientific study of population changes in an area.

Pieces of information collected during demography.

- Birth statistics • Death statistics • Diseases affecting people.
- Available health services. • Housing information.

Information collected on housing.

- The nature of houses people live in. • The number houses in an area.
- The hygienic conditions of the families.

Information on available health services.

- Immunisation. • Family planning
- Common sicknesses and their treatment. • Birth and death rate.

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- Number of available water sources
- Number of available health centre

Importance of demography.

- It provides information about overall health of people.
- It helps the government to plan for health services.

Factors that cause population changes.

- birth rates
- death rates
- natural calamities.
- epidemics

Ways of avoiding health and social problems.

- Boiling water for drinking.
- Promoting food hygiene.
- Proper guidance and counselling.
- Taking children for immunisation.
- Ensuring proper sanitation.

School health club

School health club is an association of members in a school which promote healtl of children.

Other health clubs are village Health Team(VHT) and public health official team.

Activities carried out by a school health club.

- Peer education on HIV/AIDS.
- Organising cleaning sessions in a school.
- Sensitisation of the public on health issues.
- Carrying out debates on healthrelated topic
- Promoting physical education.
- Organising health parades
- Keeping records on different health activitie
- Designing and displaying posters bearing health messages.

Roles of a school health club

- To monitor school sanitation.
- To monitor food preparation
- To monitor children's hygiene.
- Organizes health parades.

Evaluation activity:

1. a) Write down any **two** pieces of information which are collected in a health survey.
2. b) Give any **two** reasons why carrying out a health survey in an area is important.
3. a) Mention any **two** issues addressed during health education.

b) Give any **two** ways of giving health education to a community.

4.a) Write down any **two** pieces of information collected during demography.

) State any **two** ways in which health education is important to the community.

5.a) State any **two** health problems that are common in overpopulated areas.

b) Mention any **two** ways in which collecting information on human population helps to address health problems in a community.

6.a) Name any **two** parts of the body that may be checked during a school health parade.

b) Give any **two** ways in which school health parades are important in the pro motion of health among school children.

7.a) Write down any **two** activities done under child to child programme.

b) State any **two** ways in which child to child programme is important in the community.

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